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A STUDY OF THE TOTAL AND REACTIVE ELECTRONIC QUENCHING OF THE LOWEST METASTABLE STATE OF XENON

by

JOSE ENRIQUE VELAZCO

B.S., Universidad Mayor de San Andres, 1971

A DOCTOR'S THESIS

submitted in partial fulfillment of the

requirements for the degree

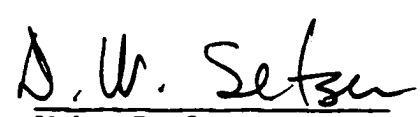
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1977

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A mis padres,

Roger y Lina

y

A mis hermanos,

Rudy y Carlos.

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CHAPTER I INTRODUCTION

"... this is the reward which I have obtained for my labour, and which has occasioned me a true satisfaction that I cannot possibly retain for myself alone. This, and no other, is the reason why I make this work known to my readers."

Carl Wilhelm Scheele
"Luft und dem Feuer", 1777

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a. Introduction. In recent years, interest in the Physics and Chemistry of electronically excited species, in particular, the lowest excited states of the rare gases, has grown extremely rapidly. It is now established that they play an important role in the chemistry of the atmosphere and outer space, are essential intermediates in many photochemical processes, and can induce many new chemical reactions by transferring their excitation energy to added reagents.[1] Among these new reactions, excimer (or exciplex) formation,[2,3] $M^* + AB \rightarrow MA^* + B$, where M^* is the metastable state of the heavier rare gases (Ar, Kr, Xe) and AB a halogen or oxygen atom donor, provides the first example of similar chemical behavior[4] (in the more classical sense) between the excited states of the rare gases and the alkali-metal atoms. The physical similarities have been known for some time.[5-7] In addition to the intrinsic interest in these new examples of excited state chemistry, excimer formation and other reactions involving electronically excited rare gases have been successfully employed to produce efficient high power lasers.[8-22] All this behavior, dramatically contrasts with the unreactivity of the ground state atoms.

The lowest excited configuration of the heavier rare gases is $np^5(n+1)s^1$ (n=2,3,4 and 5 for Ne, Ar, Kr and Xe, respectively) giving four states which, in order of increasing energy, are $^3P_2$, $^3P_1$, $^3P_0$ and $^1P_1$ in the L-S coupling notation.† The resonance states, $^3P_1$ and $^1P_1$ are connected to the ground state by allowed radiative transitions, whereas, electric dipole radiative transitions to the ground state are strictly forbidden from the $^3P_2$ and $^3P_0$

†Since spin is not a good quantum number for Kr and particularly Xe, the J_c-ℓ coupling scheme is more appropriate in these cases; however, the more familiar Russell-Saunders term symbols will be used for all the members of the lowest excited multiplet. Other states will be identified using the most appropriate notation.

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(metastable) states by the $\Delta J = 0, \pm 1$; $J = 0 \not\rightarrow J = 0$ selection rules. For He, the excited configuration is 1s2s, giving two states, ^1S and ^3S, both metastable due to the $\Delta \ell = \pm 1$ selection rule ($S \not\rightarrow S$). Energies vary[23] from 8.32 eV ($Xe(^3P_2)$) to 20.6 eV ($He(2^1S)$). The lifetimes of the resonance states are short,[24,25] ranging from ~3 ns ($Xe(^1P_1)$) to ~30 ns ($Ne(^3P_1)$) while for the metastable states, values between ~0.1 sec ($Xe(^3P_0)$) and 8×10^3 sec ($He(2^3S)$) are expected.[26,31] Study of the metastable states is thus facilitated by their longer lifetimes.

Electronic energy transfer reactions involving excited rare gas atoms were originally studied using electric discharges or radiolysis of mixtures containing the rare gas of interest.[32] In general, these excited species can be produced by deposition of sufficient energy into the rare gas system. The four methods described below (Section 1.b), however, have been shown to be the best means for studying the excited state chemistry of the rare gases.

The bulk of the available data dealing with electronic energy transfer from the lowest excited states of the rare gases pertains to $He(2^{3,1}S)$ and $Ar(^3P_{0,2})$ reactions, and very little is known about the reactivity of $Ne(^3P_{0,2})$, $Kr(^3P_{0,2})$, $Xe(^3P_{0,2})$ or the $^{1,3}P_1$ resonance states. However, the development of new laser systems based on rare gas reactions have underlined the need for basic information (particularly for processes involving Xe* and Kr*), if the mechanisms responsible for laser action are to be identified. Data are needed in the form of rate coefficients for the energy transfer process, identification of products and rate constants for the formation of these products. These same data are needed to better understand the energy transfer process itself, which at present has many puzzling facets.[2b]

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The present work is an attempt to provide a comprehensive data base for the reactions of $Xe(^3P_2)$ metastable atoms. The physical and spectroscopic properties of this state, as well as of the other members of the multiplet, are summarized in Section 1.e.

b. Methods for Studying Excited States of the Rare Gases.

(i) Radiolysis - In this method, the rare gas (or mixture of quencher and rare gas) is subjected to a high dose of radiation (γ- or X-rays) and products of radiation induced ionization are detected with a mass spectrometer. The works of Platzman,[33] and Jesse and Platzman,[34] Jesse,[35] Hummel[36] and Rebbert and Ausloos[37] describe the technique and give some insight into the potential of the method. From these studies, it was concluded that: (1) the metastable states are the main energy carriers; (2) ionization of added gases proceeds via Penning ionization with high efficiency; (3) the ionization efficiency per collision is not necessarily unity;[33] (4) neutral dissociation of highly excited states of added reagents probably takes place, as indicated by the existence of an isotope effect in the ionization efficiency.[34,35]

Although radiolysis experiments give information concerning the total ionization yield of metastable reactions, the complications arising due to the presence of rare gas ions, and the difficulty in identifying the individual metastable states or other neutral species that may also react, limit the applicability of this method. However, recently Bourene and LeCalve[38,39] circumvented some of these problems by using pulsed radiolysis ($\sim$500 keV electrons) in conjunction with a N_2 tracer technique to study the quenching of the metastable states of Ar by a variety of reagents. Metastable concentrations were followed by monitoring the $N_2(C \rightarrow B)$ emission excited by

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$$Ar(^3P_{0,2}) + N_2 \rightarrow N_2(C^3\Pi_u) + Ar(^1S) \quad (1.1)$$

$$N_2(C^3\Pi_u) \rightarrow N_2(B^3\Pi_g) + h\nu \quad (1.2)$$

Other variations in the method include: pulsed excitation - using very high energy electrons or protons - of gas samples at pressures of several atmospheres, high energy u.v. photon excitation, and bulk product analysis.[40-43]

(ii) Electrical discharges. Like the radiolysis technique, electrical discharges generate a large variety of excited species. However, the low cost and the simplicity of the experiments allowed a wealth of data to be gathered, particularly dealing with the interaction of metastables states with electrons, photons and added gases. Early designs used a continuous discharge, and the resulting emission was analyzed spectroscopically.[44,45] Subsequently, after the introduction of pulsed discharges by Phelps,[46] the method was used to measure quenching rate constants and to characterize many decay processes in the afterglow.[47-53] The resonance states are generally followed by emission spectroscopy, and the metastables by absorption spectroscopy. Tha main limitation is the destructive nature of the discharge, which may, under certain circumstances, produce reactive transient species by decomposition of the added reagents.

(iii) Atomic Beams. This technique provides the most sophisticated means for studying the detailed dynamics of the reactions of metastable atoms; beams containing essentially one type of excited species can be produced by electron impact, or by charge transfer from suitable atoms followed by removal of unwanted components by use of appropriate "filters". Thus, ions can be removed by applying deflecting or retarding electric fields; inhomogeneous magnetic fields have been successfully used to selectively remove the $^3P_2$ component out of beams of metastable Ar and Kr;[54] irradiation of the beam with light from an appropriate source has also been used to selectively remove

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one of the metastable states by optically pumping to the resonance states.[55,56] Normally, the detection is based on secondary electron ejection from metalic surfaces, although mass spectrometry is also used, particularly for studies of He* interactions. Secondary electron ejection can also detect ground state particles if their kinetic energy exceeds the work function of the metalic surface.

Most of the work with atomic beams has provided information on collision dynamics and interaction potentials, but recently total quenching rate constants have been estimated for a few systems by analysis of the elastic scattering of the rare gas metastables in the presence of added quencher reagents. However, product, spectroscopic and/or kinetic analysis of these interactions are difficult to perform using this method. In addition to the references given above, the papers by Winicur and Fraites[57a-c], Lee and Martin[57d], Siska et. al.[57e] and references therein should be consulted for descriptions of the technique and theory.

(iv) Flowing Afterglow. This technique is particularly suited for the study of metastable states.[1] Parent gas is flowed through an excitation source to a reactor and, after allowing some time for ion-electron recombinations as well as for radiative decay of higher excited states (and resonance states), the gas flow containing mostly ground state and metastable atoms is mixed with a reagent gas. The "relaxation time" is determined by the flow speed and the distance between the discharge and the quencher inlet. If the metastable concentration is monitored along the reactor (normally by Atomic Absorption Spectroscopy) total quenching rate constants can be measured. Alternatively, analysis of the emission from products excited by the metastables provide information regarding the reaction products. This method was developed by Robertson and coworkers[53] in their studies of the

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He and Ar metastables, and by Schemltekopf and Broida[59] for He*.† Stedman and Setser[1,60] and Setser and coworkers[61] have extended the method to study Kr and Xe metastables. Several excitation sources have been used: low energy electrons,[62] microwave discharges[58,63] and low power hollow cathode discharges.[1,60,61]

The papers by Ferguson, et. al.,[62a] and Bolden, et. al.[62c] comprise an excellent review of the technique, and give a very detailed description of the flow analysis required to extract quenching rate constants from flowing afterglow measurements.

(v) Other methods. Rare gas excited states have been also produced using shock tubes[64] and by resonance photolysis.[65,80c] These methods however, are not widely used because of the complicating presence of several excited species in the first case, and because of problems related to vacuum u.v. photochemistry and radiation imprisonment in the second case.[66]

c. Review of $Xe(^3P_2, ^3P_1, ^3P_0$ and $^1P_1)$ reactions. Bernard[67] appears to be the first one to identify an energy transfer process involving $Xe(^3P_2)$. During his studies of static discharges in rare gas/N_2 mixtures, he observed a dramatic enhancement in the intensity of transitions originating from $N_2(B^3\Pi_g$, v'=5, 11, 12, 13) and concluded that $Xe(^3P_2)$ directly populated $N_2(B^3\Pi_g$, v'=5) from $N_2(X^1\Sigma_g^+$, v''=0); subsequent decay - governed by Franck-Condon factors - populates $N_2(X^1\Sigma_g^+$, v'' $\sim$ 4) which, upon collision with $Xe(^3P_2)$, can be excited up to $N_2(B^3\Pi_g$, v' $\sim$ 14). Harman and Kakotoarijimy[45] repeated the Xe* + N_2 work and also studied Xe* + CO which gave CO* ($d^3\Delta \rightarrow a^3\Pi$) and ($a'^3\Sigma^+ \rightarrow a^3\Pi$) emissions with maximum intensity for v'=5 and v'=11

†In the present work, an asterisk is used to indicate electronic excitation. In the case of the rare gases, it specifically indicates excitation to any level in the lowest excited multiplet.

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respectively. Since then, the reaction of $Xe(^3P_2)$ with N_2 has been studied by at least five other groups[60,68] but no satisfactory explanation has been advanced for the preferential excitation of $N_2(B^3\Pi_g, v'=5)$ which is endoergic by $\sim$480 cm^{-1}. The endoergicity of this reaction was not apparent in the early work[45,67] probably due to the uncertainties in the state energies then available (also, see Section 4.3). Setser and coworkers[60] studied the emissions excited by the reaction of $Xe(^3P_2)$ with N_2, C_2N_2, HNCO, NH_3, H_2O and N_2O; Taylor[69] extended these studies to CS_2 and OCS. Bochkova, et. al.,[70] generated highly excited states of XeII by reaction of $Xe(^3P_2)$ with $He(2^3S)$; Herman and Cermak[71] studied the associative ionization of Xe* with Na and K.

Studies of the reactions of the $Xe(^3P_1)$ state normally are based on Xe photosensitization. Jackson and Sheer[72] used this technique to produce $N_2(A^3\Sigma_u^+)$, and Von Bunan, et. al.,[73a] compared the products of 147 nm photolysis and Xe sensitization of ethane and ethane-d_6. Their results indicated that energy transfer from $Xe(^3P_1)$ slightly enhances the formation of free radicals over direct photolysis. Kawasaki, et. al.,[73b] studied the direct and Xe sensitized photolysis of ethylenimine. Similar comparisons were performed by Slanger and coworkers[74] for CO and NO, and concluded that, (1) the spin conservation criterion (Wigner spin Rule) does not hold strictly; (2) energy matching between the atomic line and specific molecular rotational levels is important in photolysis; (3) the collisional process does not require such specific energy matching. The reaction of $Xe(^3P_1)$ + CO was reinvestigated by Freeman and Phillips,[75] who claimed that it proceeds <u>via</u> a XeCO* complex with a mean lifetime of $\sim$25 μsec.

Several other studies of Xe* interactions have been reported.[76,77] However, the individual states were not identified, or correspond to higher excited states of Xe. Attempts to monitor $Xe(^3P_0)$ in flow reactors[61] and in static pulsed discharges[69d] have not been successful. Apparently collisional

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deactivation to Xe($^3P_{2,1}$) is very efficient (see next Section). Energy transfer from Xe($^1P_1$) has not been reported; production of this state is considerably more difficult than production of the Xe($^3P_1$) state, because the vacuum U.V. output of Xe resonance lamps is almost exclusively the 147 nm line.

d. Review of Xe* decay processes in Xe, Ar and Kr. The excited states of the rare gases are normally prepared in an excess of the parent gas as a buffer[1] and their decay is governed, by two and three body collision processes, as well as, radiative transitions and diffusion to and subsequent destruction at the walls.

$$M^* + M \rightarrow M^{\dagger} + M \qquad (1.3)$$

$$M^* + M \rightarrow M_2^* \rightarrow M + M + h\nu \qquad (1.4)$$

$$M^* + M + M \rightarrow M_2^* (^1\Sigma_u^+) + M$$

$$\rightarrow M_2^* (^3\Sigma_u^+) + M \qquad (1.5)$$

The excited molecules, M_2^*, decay by radiative transitions to the repulsive ground state, although at high pressures can be efficient energy carriers[78] in pulsed systems. Process (1.3) indicates collisional activation or deactivation of the excited state under consideration (see below), whereas process (1.4) refers to collision induced emission which probably involves molecule formation (either the $^1\Sigma_u^+$ or $^3\Sigma_u^+$ states). Processes (1.5) are the three body molecule formation. These molecular states are the lowest of the Rydbergs that converge on the strongly bound M_2^+ ions. The calculations of Mulliken[79] for the interactions of Xe* with ground state Xe atoms, predict a bound $^3\Sigma_u^+(0_u^-, 1_u)$ state (see Figure 1.1), a weakly bound $^3\Sigma_g^+(0_g^-, 1_g)$ and two repulsive states, 2_g and 2_u correlating with Xe($^3P_2$) + Xe(^1S_o); whereas, the interaction of Xe($^3P_1$) with ground state Xe gives the bound $^1\Sigma_u^+(0_u^+)$ and $^1\Sigma_g^+(0_g^+)$ states and the repulsive 1_g and 1_u states. All of the states

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correlating with $Xe(^1S_o) + Xe(^3P_0)$, 0_g^- and 0_u^-, and $Xe(^1S_0) + Xe(^1P_1)$, 0_g^+, 0_u^+, 1_g and 1_u, are repulsive, with the possible exception of 0_u^+ which might be very weakly bound. The qualitative features of these potential curves should be the same for all the rare gas excimers. Emission from the lowest molecular states has been observed for all the rare gases[80] as a continuum extending to the red of the first resonance line ($^3P_1 \rightarrow {}^1S_0$). This continuum consists of two major "peaks", the first continuum near the resonance line, and the second continuum at longer wavelengths. According to Mulliken,[79] the first continuum consists of emission from high vibrational levels of the $^1\Sigma_u^+$ (near the dissociation limit); whereas, the second continuum arises from transitions from low v' levels in $^1\Sigma_u^+$. Contributions to both continua from the $^3\Sigma_u^+$ state can not be ruled out -particularly for Xe- because $^3\Sigma_u^+ \rightarrow {}^1\Sigma_g^+$ transition is allowed due to the strong spin-orbit coupling in Xe_2. In fact, it has been proposed[80] that the second continuum is due mainly to transitions from low v' levels of the $^3\Sigma_u^+$ state. Recent experiments designed to compare the formation rate of M_2^* with the decay rate of specific states of M* have given contradictory results, see below.

If the decay of the excited state M* is followed as a function of total parent gas pressure, it can be expressed, including a diffusion term by (see Section 4.d):

$$1/\tau_d = \frac{D_o}{\Lambda p} + k_2 p + k_3 p^2 \qquad (1.6)$$

where τ_d^{-1} = decay rate constant

D_o = diffusion coefficient

Λ = characteristic length of reactor

k_2 = total two body deactivation coefficient

k_3 = total three body deactivation coefficient.

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The rate coefficient k_2 corresponds to the net effect of processes (1.3) and (1.4). The relative importance of these processes is not known. New evidence seems to indicate that collision induced emission is the dominant process for the $^3P_2$ metastables,[1,77] while the resonant states and the $^3P_0$ metastable may, in addition, undergo activating ($^1P_1$) or deactivating ($^3P_{0,1}$) collisions. The fact that $Ar(^3P_2)$ has the lowest total two body deactivation rate of all the Ar* multiplet components[81] tends to give additional support to this view. The work of Black, et. al.,[82] on the collisional deactivation of $O(^1S)$ and $S(^1S)$ suggest that the two body deactivation process might in fact, be a three body collision with formation of an excimer followed by fast (collisional) dissociation:

$$M^* + M + M \rightleftarrows M_2^* + M$$
$$M_2^* \rightarrow M + M + h\nu \qquad (1.7)$$

Excimer formation from three body collisions is thought[78a] to give $M_2^*(^1\Sigma_u^+)$ when the excited atomic state is $^3P_1$ and $M_2^*(^3\Sigma_u^+)$ for the deactivation of $^3P_2$. However, this view is by no means unanimous.[77,83] Michaelson and Smith[84] analyzed the first and second continua in Ar_2^* and concluded that the same molecular state was responsible for both emissions. This state was identified as the $^3\Sigma_u^+(0_u^+, 1_u)$ with $Ar(^3P_2)$ as the precursor. Their more recent study[77] however, favors the $^1\Sigma_u^+$ (0_u^+) as the emitting state although the precursor was again found to be $Ar(^3P_2)$. Since $^1\Sigma_u^+$ does not correlate to $Ar(^3P_2) + Ar(^1S_0)$, they suggest that the approach might be along the $^3\Sigma_g^+$ (see Figure 1.1) curve followed by collision induced transition to the $^1\Sigma_u^+(0_u^+)$ state. Contrary to these results, Wieme and Wieme-Lenaerts[85] found that the first continuum is associated with the destruction of $Ar(^3P_1)$. Similarly, $Kr(^3P_1)$ has been identified as the precursor of the Kr_2^* first continuum.[83a] As mentioned above, the second continuum is

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thought to originate from lower v' levels of the bound states but in general, this emission is understood less well. Table 1.1 summarizes the total two and three body rate coefficients for the decay of $Xe(^3P_2)$ in Xe. No such data is available for the other excited states.

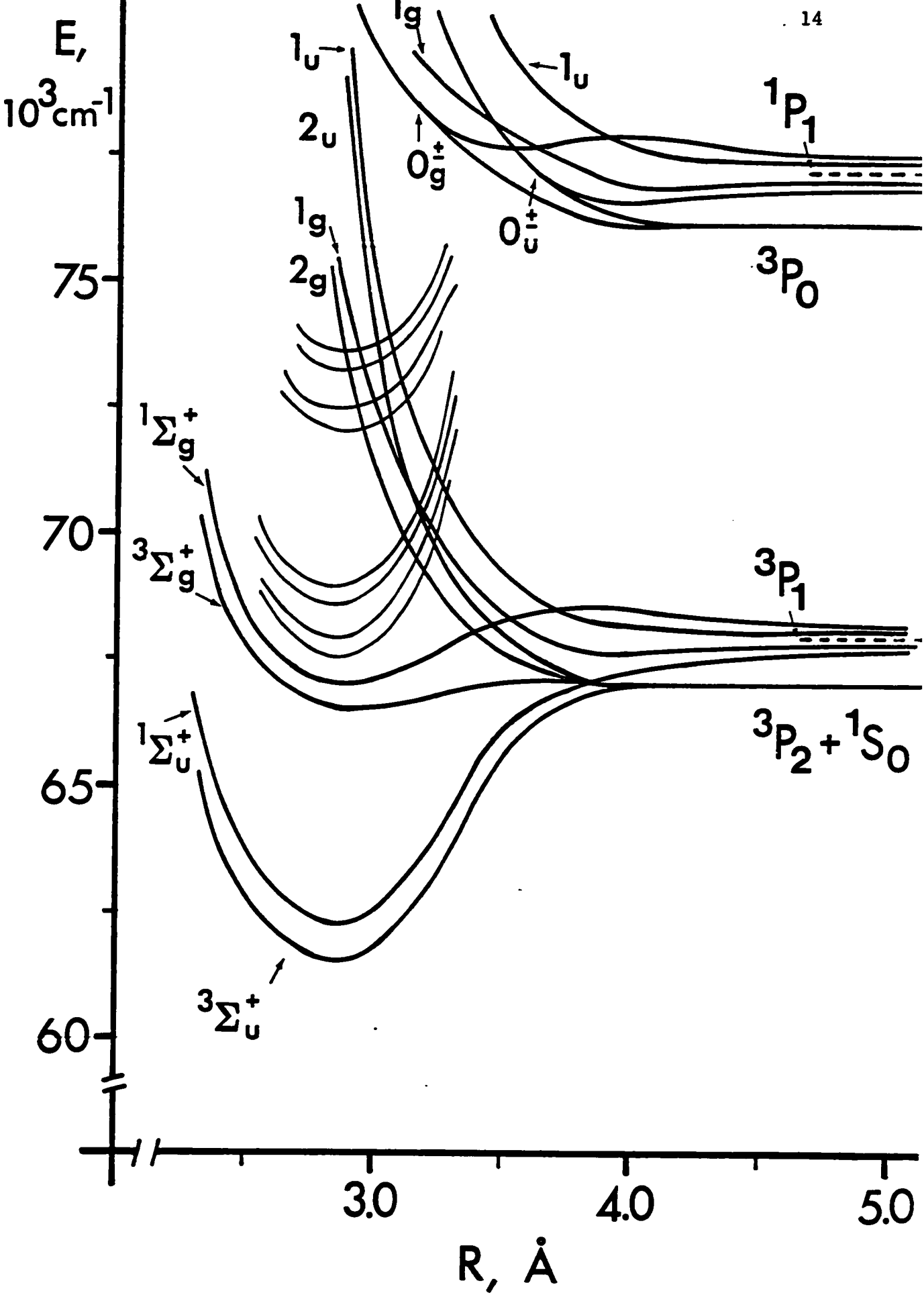
If the excited states under investigation are produced in a buffer gas other than the parent (as in the case in the present study), the overall system may still be complicated, but one grave problem, radiation trapping, can be minimized, which allows processes like (1.3) to be studied. Total two and three body quenching rate constants for Xe* in Ar have been determined in the present work (see Section 3.d) and these results are also listed in Table 1.1. Jortner and corworkers[78] have studied the intramultiplet transitions in Xe/Ar and Xe/Kr systems (with [Xe] of the order of ppm). Their results can be summarized as follows: (i) In the Xe/Ar system, collision-induced deactivation of $Xe(^3P_1)$ produces $Xe(^3P_2)$ which recombines (in a three body collision) with a ground state $Xe(^1S_o)$ to give $Xe_2(^3\Sigma_u^+)$.[78b] (ii) At high Ar pressures, $Xe(^1P_1) \rightarrow Xe(^3P_1)$ conversion becomes important: this is a consecutive collision-induced process involving the collisional excitation of $Xe(^3P_1)$ to $Xe(^3D_1)$ (Xe $6p[1/2]_1$) by collisions with $Ar(^1S_0)$ followed by radiative transitions to the $^3P_2$ or $^3P_1$ states. (iii) The [Xe] dependence of this process is taken as an indication of increased $Xe(^1P_1)$ lifetime due to radiation trapping.[78b] (iv) Collisional electronic quenching of $Xe(^3P_1)$ to $Xe(^3P_2)$ by $Ar(^1S_0)$ is quite efficient.[78c] (v) The reverse reaction, collisional activation of $Xe(^3P_2)$ to $Xe(^3P_1)$ is negligible at 300°K which agrees with Wieme and Mortier's[86] findings. (vi) Finally, for the $Xe(^3P_0)$ state, collisional quenching to the $(^3P_1)$ or $(^3P_2)$ states is quite efficient.

The results of the Xe/Kr system were in general agreement with findings for Xe/Ar, the main difference being that there is evidence to support a (XeKr)*

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Figure 1.1 Estimated potential curves (from Ref. 79) for some for the lower excited states (Rydberg states) of Xe_2^*. The unmarked bound states correlate with Xe(6p) + Xe(^1S_o).

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excimer, whereas the corresponding (ArXe)* emission has not been identified.[78b] Intramultiplet collision-induced transition rate coefficients are listed in Table 1.1 for the few transitions for which they are available.

Finally, under certain circumstances collisions of metastable atoms with electrons may be an important metastable loss mechanism via the reaction

$$M(^3P_2) + e^- \rightarrow M(^3P_1) + e^- \qquad (1.8)$$

followed by radiative decay to the ground state. It has also been suggested that some reactions attributed to the metastable states might in fact involve the resonance states, with reaction (1.8) as the main resonance state-populating step. It is expected that the importance of process (1.8) will decrease as the energy gap between the states involved increases. This expectation is realized[46] for He, Ne, Ar and Xe for which the rate constants vary from $\sim 3.5 \times 10^{-7}$ cc molec^{-1} sec^{-1} (He(^{1}S)) to undetectable for Ar($^3P_2$) and Xe($^3P_2$).[87] However, Smith and Turner[48b] measured the rate constant for Kr($^3P_2$) + e^- → Kr($^3P_1$) + e^- and obtained $\sim 1 \times 10^{-11}$ cc molec^{-1} sec^{-1}. In the present study, contributions from (1.8) to the depletion of Xe($^3P_2$) should be negligible because: (1) The energy gap between the $^3P_2$ and $^3P_1$ states is large and the rate constant for electron collision-induced transitions is very small.[87] (2) Electron densities in the flow reactor are expected to be very small (if not zero) in the interaction region (see Chapters 2I and 3), and only thermal energies are expected.

e. Summary of the Properties of Xe*. Available data for the physical and spectroscopic properties of the Xe($^3P_{0,1,2}$) and ($^1P_1$) states are summarized in Table 1.2. Emphasis is placed on optical parameters, i.e., transition probabilities, oscillator strengths or lifetimes for most the Xe(5p6s) - (5p6p) and-(5p7p) transitions. These states of XeI are shown in Figure 1.2 and the main transitions are indicated.

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Table 1.1 Rate constants for Xe* decay processes in the afterglow.

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Table 1.1 - Summary of rate constants for processes involving Xe* in the afterglow.

Level or transition[a]	Bath	D_o[b]	k_2[c]	k_3[d]	k	Ref.
$Xe(^3P_2)$	Xe	67±7	22±10	7.6±1		53
		52±8	35	8.6		52
		42±4				88
			37	8.7		87
				5.0		89
				1.5		48a
	Ar	270±12	3.6±7	2.2±0.9		e
$Xe(^3P_1) \rightarrow Xe(^3P_2)$	Ar				1.7×10^{-14}	78c
$Xe(^3P_2) \rightarrow Xe(^3P_1)$	Ar				$<< 10^{-14}$	78c
$Xe(^1P_1) \rightarrow Xe(^3D_1)$	Ar				$\sim 10^{-10}$[f]	

a. The indicated transition is induced by (bimolecular) collision with the bath gas. The corresponding rate constant is given under k, with units cm^3 $molec^{-1}$ sec^{-1}.

b. Units are 10^{16} cm^{-1} sec^{-1}.

c. Two-body collision rate constant; units are 10^{-16} cm^3 $molec^{-1}$ sec^{-1}.

d. Three-body collision rate constant; units are 10^{-32} cm^6 $molec^{-2}$ sec^{-1}.

e. Present work, see Chapter 3, Section d. The values listed for k_2 and k_3 are probably too large; preliminary results at lower [Xe] (J. H. Kolts, Kansas State University, 1976) gave much smaller values, suggesting that, even at the concentrations used in the present work, the main contribution to $Xe(^3P_2)$ decay are collisions with ground state Xe atoms. See Ref. 78.

f. Estimated value. See Ref. 78b.

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Table 1.2 Physical and spectroscopic properties of the lowest excited states of Xe.

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Table 1.2 Physical and spectroscopic properties of the lowest excited states

Level[a]	Energy[b]	α	IP, cm^{-1}	τ	Transitions				
					upper state[c]	λ,nm	f_{1k}	A_{1k}[d]	τ,ns[e]
$1s_5$, $6s[3/2]^o_2$, $^3P_2$	67068 cm^{-1} (8.32 eV)	62.6 $Å^3$	30766 (3.82 eV)	149.5 s[f]	$2p_2$, $6p'[1/2]_1$	450.1		0.39[g]	35±2[h]
								1.9[i]	43.5±1.5
									29.5±3[k]
					$2p_6$, $6p[3/2]_2$	823.2		38[g]	48±6[h]
							0.29[l]	23.0[i]	33±20[j]
					$2p_7$, $6p[3/2]_1$	840.9		0.83[g]	
								2.1[i]	
					$2p_8$, $6p[5/2]_3$	881.9	0.5[l]		
					$2p_9$, $6p[5/2]_2$	904.5	0.17[l]		
					$2p_{10}$, $6p[1/2]_1$	980.0	0.23[l]		
					$3p_6$, $7p[3/2]_2$	462.4		0.42[g]	203±15[h]
								2.2[i]	148±11[j]
									107±9[k]
					$3p_7$, $7p[3/2]_1$	461.1		0.26[g]	
								0.51[i]	
					$3p_8$, $7p[5/2]_3$	467.1		0.61[g]	141±10[h]
								2.9[i]	183±10[k]

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Table 1.2 Physical and spectroscopic properties of the lowest excited states (cont.)

Level[a]	Energy[b]	α	IP, cm^{-1}	τ	Transitions				
					upper state[c]	λ,nm	f_{ik}	A_{ik}[d]	τ,ns[e]
$^3P_2$					$4f[5/2]_{2,3}$	419.3		0.11[g]	
					4Y $4f[3/2]_2$	420.4		0.021[i]	
$1s_4$, $6s[3/2]^o_1$, $^3P_1$	68046 cm^{-1}			5.5 ns[m]	$2p_1$, $6p'[1/2]_0$	458.2		0.39[g]	30±3[h]
				3.8 ns[n]					38.5±1.5
				5.1 ns[m]					30.7±1[k]
					$2p_3$, $6p'[3/2]_2$	473.4		0.089[g]	34±2[h]
								1.5[i]	39.0±7[k]
									30.5±3[k]
					$2p_5$, $6p[1/2]_0$	828.0		10.0[g]	30±4[h]
								36.0[i]	40±12
					$3p_5$, $7p[1/2]_0$	480.7		1.52[g]	115±9[h]
								2.6[i]	87±5[j]
									107±8[k]
					$3p_6$, $7p[3/2]_1$	483.0		0.19[g]	140±8[h]
								2.1[i]	101±6[j]
									100±7[k]

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Table 1.2 Physical and spectroscopic properties of the lowest excited states (cont.)

Level[a]	Energy[b]	α	IP, cm^{-1}	τ	Transitions: upper state[c]	λ,nm	f_{ik}	A_{ik}[d]	τ,ns[e]
$^3P_1$					$2p_4$, $6p'[3/2]_1$	491.6		0.25[g]	
								0.03[i]	
					$3p_9$, $7p[5/2]_2$	492.3		0.19[g]	200±12[j]
								1.8[i]	168±15[k]
					4Y $4f[3/2]_2$	438.4		0.08[g]	
					4X $4f[3/2]_1$	438.6		0.23[g]	
					$4p_5$, $8p[1/2]_0$	407.9		0.49[g]	171±9[k]
$1s_3$, $6s'[3/2]^o_o$, $^3P_0$	76197 cm^{-1}		32174 (3.99 eV)	0.08 s[f]					
$1s_2$, $6s'[1/2]^o_1$, $^1P_1$	77186 cm^{-1}			5.6 ns[m]	$2p_2$, $6p'[1/2]_1$	788.7		10.4[g]	
				3.2 ns[n]				14.0[i]	
				4.4 ns[m]	$2p_3$, $6p'[3/2]_2$	834.7		6.36[g]	
								35.0[i]	
					6Y,X $6f[3/2]_{2,1}$	569.6		0.16[g]	

a. The different notations correspond to Paschen, Racah and the more familiar Russell-Saunders.

b. From Ref. 23.

c. Paschen and/or Racah notation.

d. Units are 10^{-6} sec^{-1}.

e. Lifetime correspond to upper state of transition.

f. Theoretical, see Ref. 31.

g. Ref. 90.

h. Ref. 91.

i. Ref. 92.

j. Ref. 93.

k. Ref. 94.

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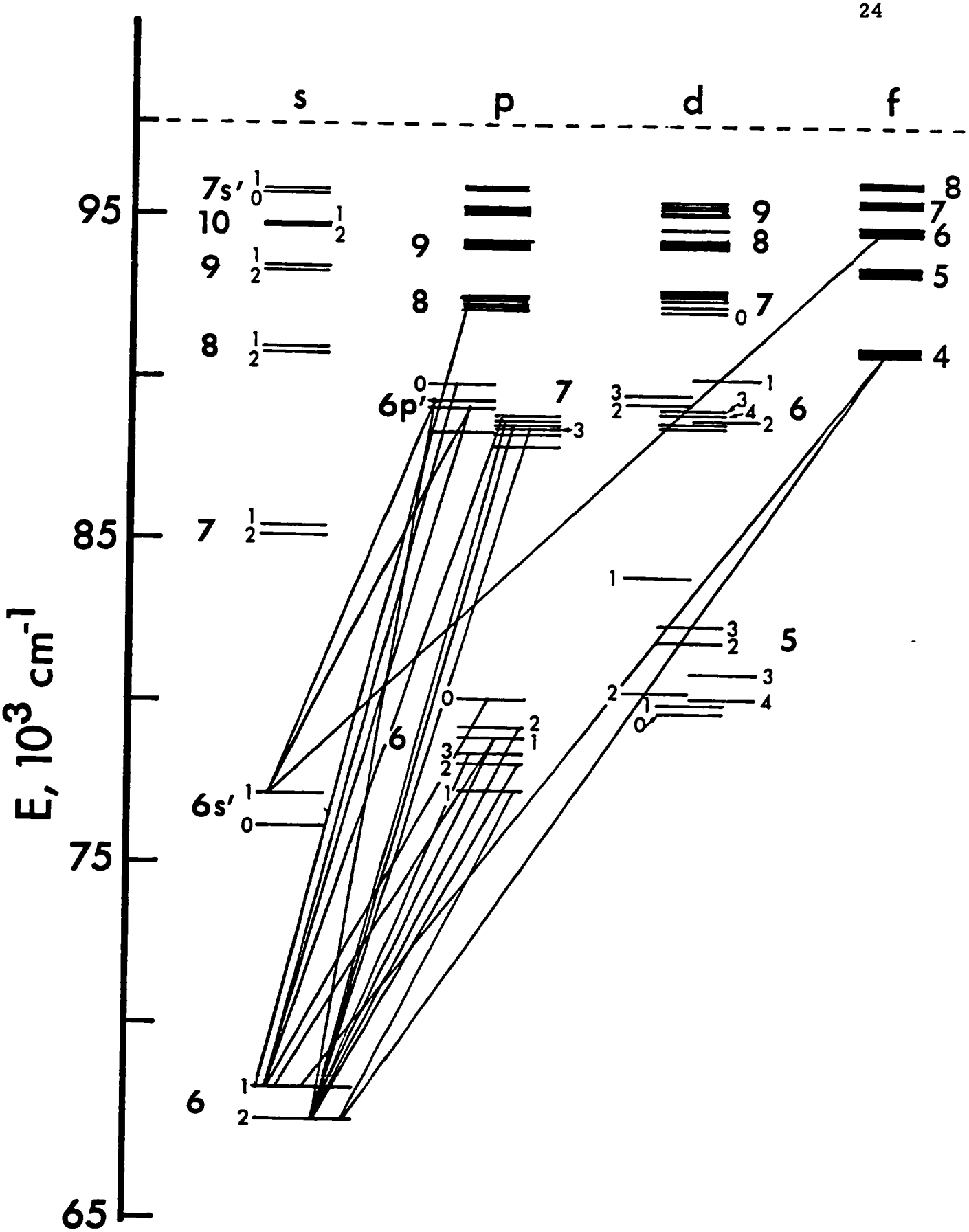
m. Theoretical, Ref. 95.

n. Ref. 25.

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Figure 1.2 Energy level diagram for Xe. The transitions listed in Table 1.2 are shown.

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CHAPTER 2 EXPERIMENTAL

I. Total Quenching Rate Constants

"...facts, observations, experiments - these are the materials of a great edifice, but in assembling them we must combine them into classes, distinguish which belongs to which order and to which part of the whole each pertains."

Antoine Laurent Lavoisier
"Memoires", Sept. 5, 1975

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a. Description of the flow reactor. A schematic diagram of the reactor is shown in Figure 2I.1. The reactor was made out of 3.1 cm I.D. Pyrex tubing with an overall length (including the discharge region) of 78 cm. A 20 cm piece of 3.5 cm I.D. copper flex-tube, not shown, connects (via an "O"-ring seal) the reactor to the 8 cm I.D. steel "T" section that, depending on whether the by-pass valve (2" Veeco Bellows type) in front of the pump is open or closed, directs the flow to the pump (Precision Scientific, PS 1000) or to the blower. Use of the blower (MGD Pneumatics, Inc., Model G70) increases the flow velocity by a factor of ∿4; the flow speed and the pressure are controlled by a 3" gate valve (Airco Temescal, ST-1) located between the "T" and the blower. Whenever possible, 3.5 cm I.D. copper flex-tube was used in the connections to dampen the vibration of the pump and blower. Installation of the larger diameter pipes improved the flow speed by a factor of 2 over earlier reports.[1,2]

Tank argon, at a delivery pressure of 760 torr, was passed through a 5A molecular sieve trap at room temperature on the high pressure side of the main flow meter, and through two such traps at 77°K on the low pressure side before ∿200 μMoles min^{-1} of Xe were added to the flow a few centimeters upstream from the entrance to the reactor.[3] A fritted disk prevented the gas stream from jetting out of the small diameter entrance tube. Xe metastables were produced by a low power, hollow-cathode discharge, both directly (see Section 2.f) and by radiative cascade following the $Ar(^3P_{0,2})$ + Xe excitation transfer reaction.[4] The cathode was made from a 3 cm wide tantalum foil and had a diameter of 2.5 cm; the anode was a 1 mm O.D. tungsten electrode. The discharge was intiated with a Tesla coil and was sustained by applying 250 V across the electrodes. A 10 kΩ ballast resistor in series limited the current to ∿1 mA. The discharge under these conditions was very stable, and in fact, after a warming up period of ∿1 hr., it could be turned off and on

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without any detectable variation in the metastable concentration. Excited species, other than $Xe(^3P_2)$, decayed before reaching the reagent inlet some 15 cm downstream. The right angle bend and light trap (Woods horn) were necessary to avoid the formation of excited states by resonance absorption in the gas that viewed the discharge; such absorption can lead to metastable regeneration.[1,2]

The reagent was introduced to the reactor through four "shower head" type inlets (see insert, Fig. 2I.1) that provided very efficient mixing and little disturbance of the flow (see Section 2I.e), as verified by visual inspection of the chemiluminescent flame produced by certain reagent gases i.e., N_2, Cl_2, F_2, etc., after quenching the metastable atoms. The Xe metastable atom concentration was followed by Atomic Absorption Spectroscopy by monitoring the 881.9 nm line emitted by the Oriel Optics Pen-Ray lamp, at the six observation windows located along the tube at ∿7 cm intervals (center-to-center). The windows were quartz discs fixed in place by Epoxy resin. The lamp and associated detection system, lenses, mirror, mechanical chopper, monochromator and photomultiplier tube, are placed on a table mounted on a rail, which allowed the operator to freely move the lamp and detection system along the axis of the reactor. Pressures in the flow tube were measured with a differential pressure transducer, with the reference side at ∿5 x 10^{-5} Torr, as measured with a McLeod gauge.

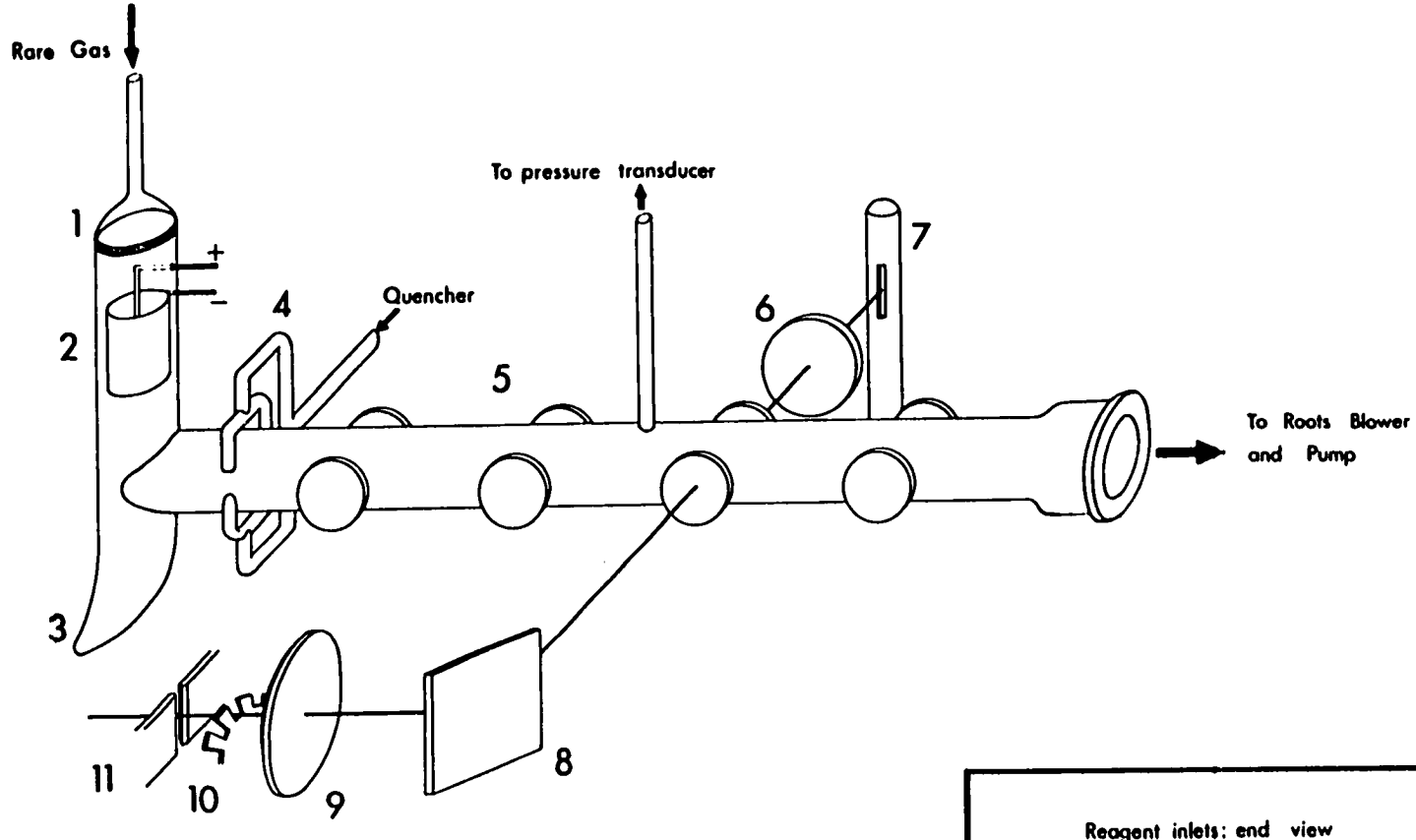
The gas handling side of the rack consisted of four 5-liter and one 12 liter storage bulbs. The ultimate vacuum on the main line was 10^{-5} Torr and pressure could be measured with a McLeod gauge (10^{-5} to 10^{-1} Torr), pressure transducer (0.1 to 15 torr), oil manometer (∿0.3 to 45 Torr), spiral guage (1 to 180 Torr) or mercury manometer (1 to 500 Torr). Finally, three different loading ports were available, terminating on 1/4" Swagelock connector, 10/30 standard taper and 19/24 ball joint, each connected to a

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Figure 2I.1 - Schematic diagram of the fast flow reactor. Tube diameter is 3.1 cm and the overall length (including the discharge) 78 cm. The actual reactor had 6 observation windows instead of the four shown here. The lamp and detection system can be freely moved along the axis of the reactor.

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1 Fritted disc

2 Hollow Cathod discharge

3 Light trap

4 Reagent inlets (four "Shower Head" type)

5 Observation windows

6 Focusing lens

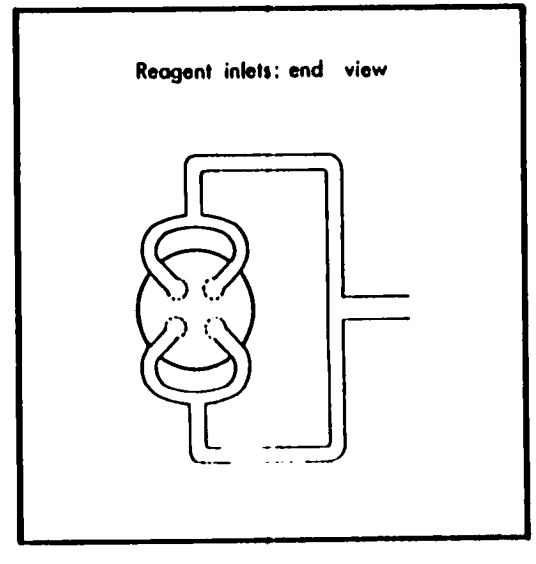
7 Pen–Ray Rare Gas resonance lamp

8 Plane mirror

9 Lens

10 Mechanical light chopper

11 Monochromator and associated signal detection system



series of distillation traps used for vacuum purification of the reagent gases.

During the latter part of this investigation, a small stainless steel section was added to handle the fluorinated reagents. It consisted of a 2.5 liter storage cylinder, a Nupro ss fine metering valve and 1/4" ss tubing leading to the pressure transducer, reagent inlet to the flow reactor and loading port.

b. Detection System. As mentioned above, the metastable concentration was monitored by Atomic Absorption Spectroscopy. The light source was an Oriel Optics Xenon Pen-Ray lamp. The collimated light from this lamp crossed the flow tube perpendicularly (see Fig. 2I.1) and impinged on a flat mirror that directed it to the entrance slit of the monochromator. Between the flat mirror and the entrance slit, there was a second focusing lens and a mechanical chopper (PAR 125) that modulated the signal at 600 Hz. The monochromator, McPherson 218, selected the line of interest (881.9 nm for $Xe(^3P_2)$) and it was detected by an RCA 7102 photomultiplier tube cooled to 240°K (Products for Research, Te-104-RF) which reduced the dark current by a factor of 10^4 relative to room temperature. The output from the PM tube is amplified by a lock-in amplifier (PAR HR-8) tuned to the chopper frequency and finally the signal is recorded on a strip chart.

Under optimum conditions (see following Section), $Xe(^3P_2)$ absorption (at the second window) was ∿9% for one pass of the light across the reactor. In order to increase the detection sensitivity, plane mirrors were placed on either side of the flow tube to obtain three traversals across the tube. This method gave absorptions of ∿25% at the second window (∿25 cm downstream from the discharge). With the monochromator slits set at ∿60μ, the intensity of the 881.9 nm line, in the absence of absorption was $I_o \simeq 20$ mV,

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with a signal to noise ratio better than 10,000. Absorption of this line is related to $Xe(^3P_2)$ concentration in Section 2I.g. Using the back-off mode of the amplifier, signal changes of less than 0.5% could be recorded.

When using the source lamp in the a.c. mode, the recorded signal displayed the line modulation (60 Hz), as could have been anticipated considering the chopper frequency. Therefore, a d.c. power supply was used operating at 1.1 kV with enough ballast resistors in series with the lamp to limit the current to ∿6 mA.

c. Calibration of apparatus. In a flow reactor, the reaction rate constant depends on the reactant (quencher) gas flow, f_Q, and on the square of the bulk flow velocity, v, which is calculated from the reactor pressure and the buffer gas flow, f_B (see Chapter 3). It is of uppermost importance to accurately measure flows and pressures if reliable rate constants are to be obtained. Quencher flows and reactor pressures were measured using a pressure transducer; whereas, the buffer gas (Argon) flows were metered with a pre-calibrated commercial flow meter. When dealing with gases that react with mercury and/or dissolve in oil, gas mixtures had to be prepared using a spiral gauge to measure pressures. In this section, the calibration of the flow meter, pressure transducer and spiral gauge is described as well as the general procedure for measuring reagent flow.

i) Calibration of main flow meter. The main flow meter was a pre-calibrated Fischer-Porter (Cat. No. 449703) with a stated accuracy of ± 1%. This calibration was spot-checked by W. G. Richardson[5] by monitoring the pressure drop in a known volume. Figure 2I.2 is a plot of scale reading vs flow, in Moles min^{-1} at 21°C and 1 atm back pressure for this flow meter using the data supplied by the manufacturer. For work under different P.T. conditions, flows should be multiplied by $[(P/760)(294.2/T)]^{1/2}$ where P is the new back pressure in Torr, and T is the new absolute temperature (°K).

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ii) Calibration of the pressure transducer. The pressure transducer PD 7 (range, $\pm$ 0.1 PSI), was purchased from Celesco Industries, along with the Carrier Demodulator CD25. The CD is used to display the PT output on a scale (-100 -0- +100) and also to provide an output (as $\pm$ 10V) to drive a recorder. The PD 7 is a differential transducer, and hence, absolute pressures can be measured only if the residual pressure on the reference side is negligible compared to the accuracy of the measurement. The CD allows reproducible measurements to be made to three significant figures; in the neighborhood of one Torr; therefore, absolute pressures can be measured if the reference side is kept below 5×10^{-4} Torr. As mentioned earlier (see Section a), the ultimate vacuum for the gas handling side of the rack was $\sim 10^{-5}$ Torr, and typically 5×10^{-5} Torr which is indeed negligible compared with pressures of the order of 1 Torr.

Figure 2I.3 shows a diagram of the PT and standard volume installed in the system. The flow in moles per unit time, is given by

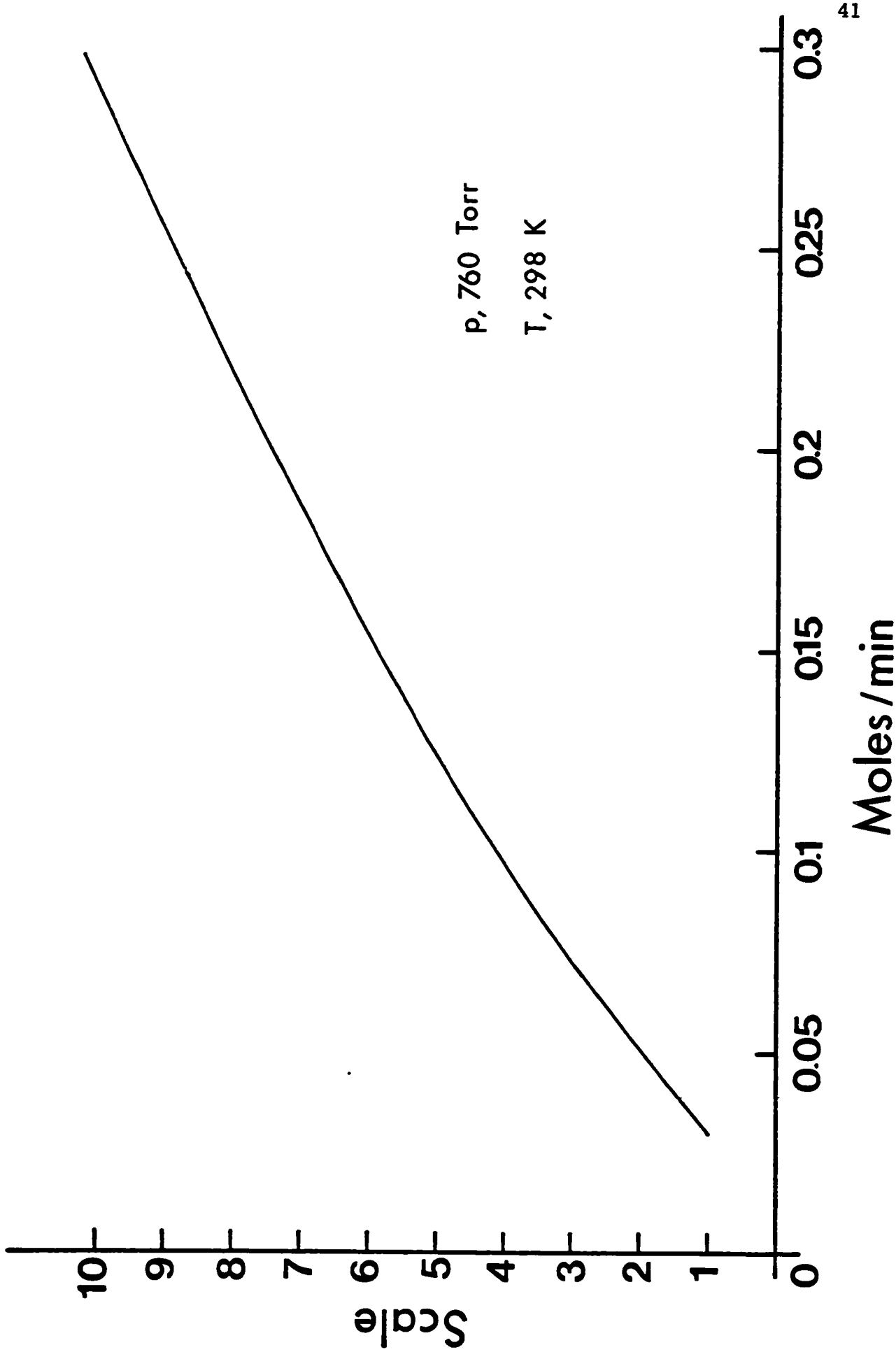
$$\frac{dn}{dt} = \frac{V}{RT} \frac{dp}{dt} \qquad (2I.1)$$

R,T have their usual meaning. V is the "standard volume" (SV), in which the change in pressure dp/dt is monitored, and it must be known very accurately. The standard volume had two contributions; first, the constant or "inert volume" consisting of the connecting tubing and limited as shown in Fig. 2I.3 by: (1) the on-off valve to the Xe line, (2) the stopcock connecting both sides of the PT, (3) the stopcock to the reactor, (4) the positive (+) side of the transducer, (5) the 10/30 standard taper connecting to the working volume, (6) the fine metering valve used to regulate quencher flows. The inert volume was measured using a gas burette with silicon oil as the fluid. The average of 24 determinations gave V_i = (28.70 $\pm$ 0.04) ml. The second and largest contribution to the SV is the variable or "working volume", V_w.

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Figure 2I.2 - Calibration curve for the main flow meter calculated from manufacturer's data. The back pressure is 760 torr.

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This volume determines the range of flows that can be measured for a given pressure increment. Two glass bulbs were calibrated first, by filling the bulbs with distilled water of known density (measured with a Westphal Balance) and weighing the water required and second, by expanding a volume of Ar at a given pressure into the unknown volume. The average of two water determinations and 4 gaseous expantions gave (541.3 $\pm$ 0.3) ml for V_w1 and (216.80 $\pm$ 0.15) ml for V_w2.

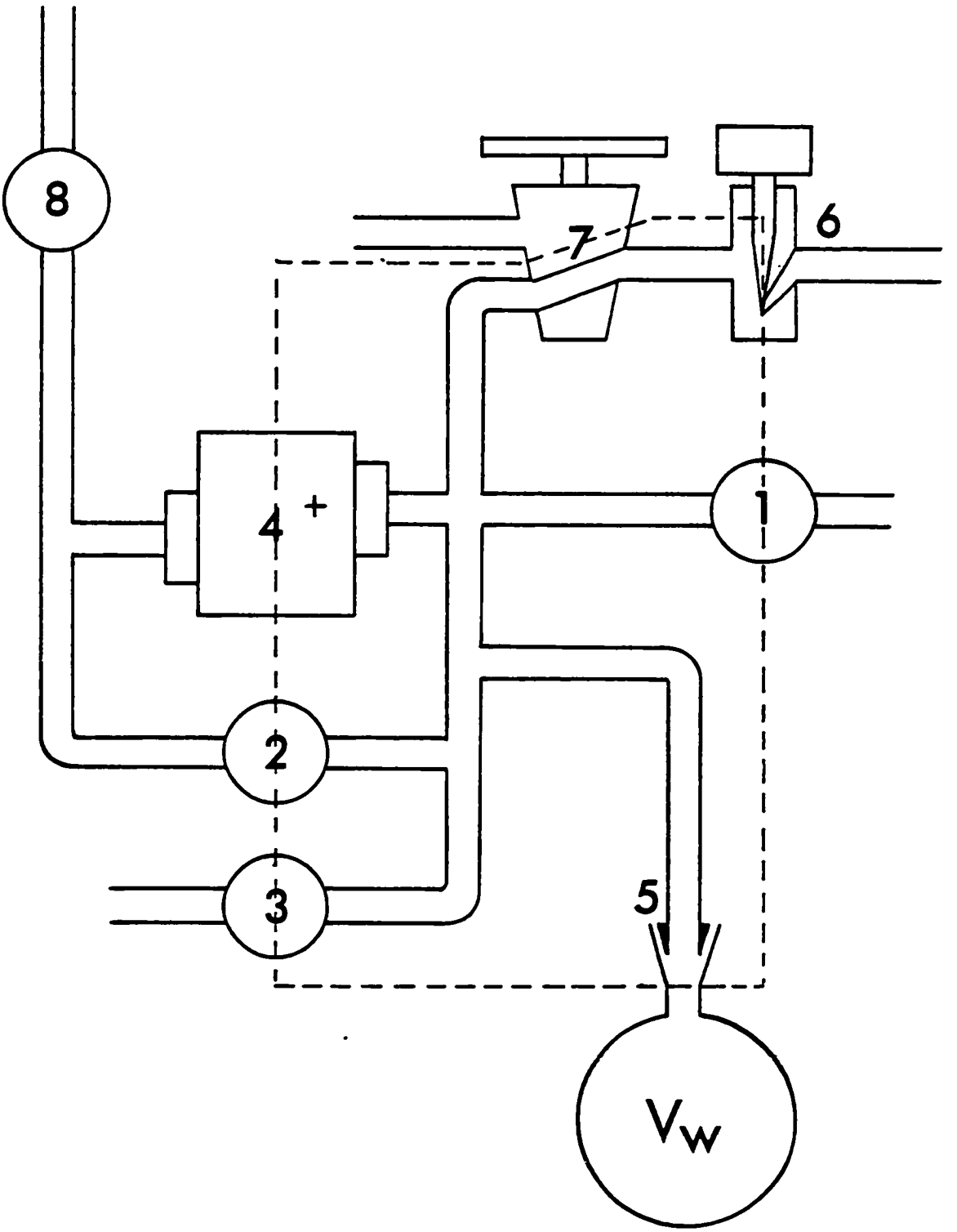
The PT response was determined as follows. With both sides of the transducer at the same pressure (stopcock 2 open, Fig. 2I.3) the CD indicator was set to zero with the Zero Adjust Control. Stopcock 2 was closed and the reference side of the PT was pumped down to $\sim 10^{-5}$ Torr. With the Span Control (sensitivity) set to a fixed value, a number of different pressures (as measured by the oil manometer, ρ_{oil} = 1.067 gr cc^{-1}) were applied to the positive (+) side of the PT, and the readings of the CD indicator were recorded. This procedure was repeated for a number of different span settings; typical data for one calibration session are listed in Table 2I.1. Using these data, a series of straight lines were obtained by plotting pressure vs. scale reading for each span value, see Figure 2I.4. At this point, one can proceed in one of two ways. If it is desirable to obtain the span setting needed for full scale deflection for a predetermined pressure, the scale reading at this pressure for the different span settings is plotted vs. span. A straight line is obtained from which the span value necessary for full scale can be read. This procedure was followed in constructing Fig. 2I.5. Alternatively, from plots like Figure 2I.4 it is possible to obtain the pressure for full scale deflection for each of the span settings. Plotting the inverse of these pressures vs. span, a line is obtained from which the span setting needed can be obtained for any desired full scale pressure. This method was used in Figure 2I.6 and in Section 2II.b.

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Figure 2I.3 - Diagram of the installed pressure Transducer showing the "inert" volume (dashed line) and the "working" volume, V_w.

1. On-off valve, Xe line.
2. On-off bypass valve.
3. On-off valve for pressure measurements in the reactor.
4. Pressure transducer, positive side.
5. Standard Taper.
6. Fine metering valve.
7. Three way stopcock.
8. Stopcock, high vacuum for reference side of transducer.

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Table 2I.1 - Experimental data for the calibration of the pressure transducer. Pressures were measured with an oil manometer, $\rho_{oil} = 1.067$ gr cc^{-1}.

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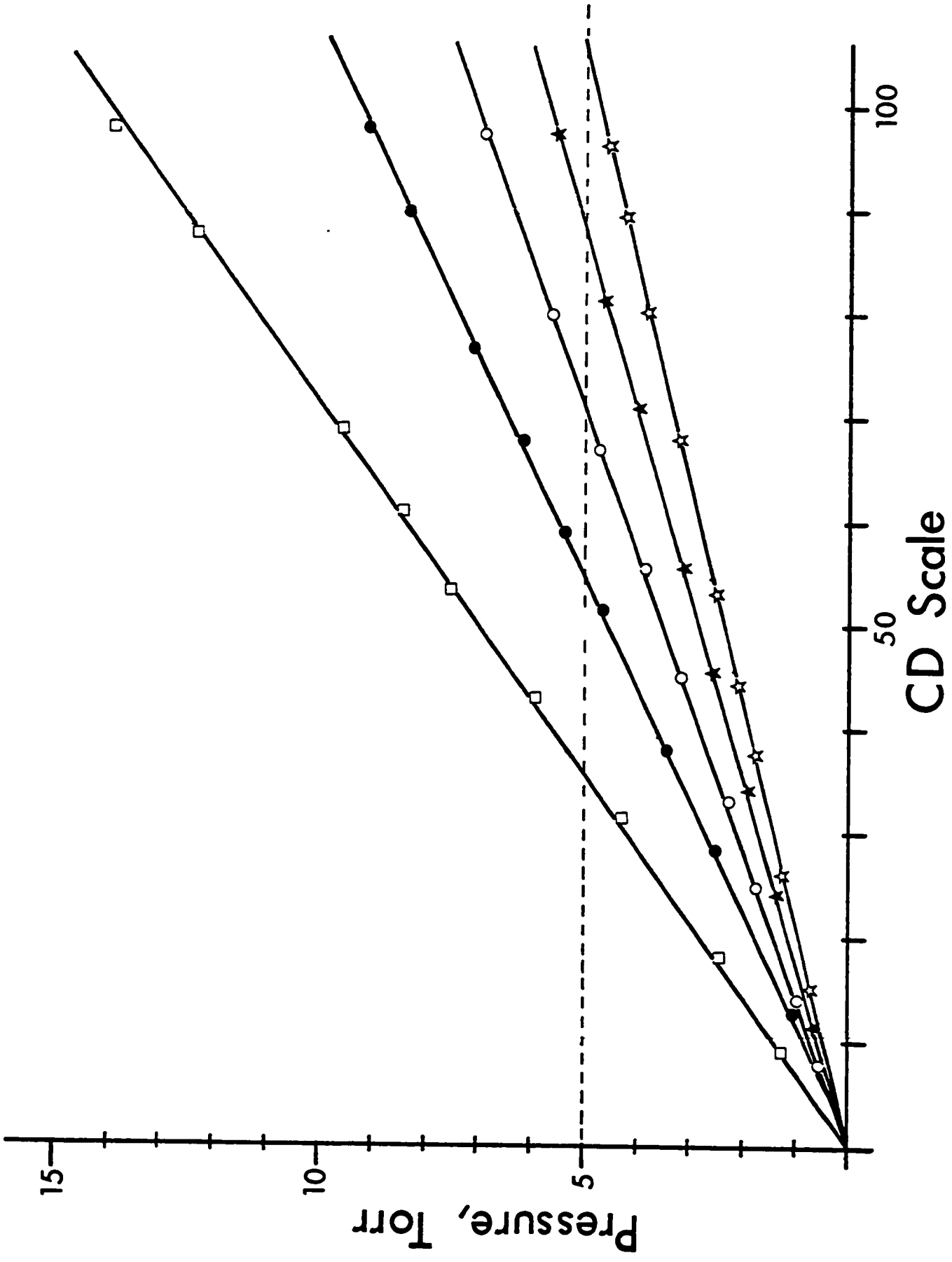
Table 2I.1 - Data for the calibration of Pressure Transducer

Span	Pressure Torr	Scale Reading	Span	Pressure Torr	Scale Reading
600	4.57	96.5	500	5.56	97.5
	4.22	89.0		4.62	81.4
	3.82	80.5		3.99	71.0
	3.22	68.0		3.13	55.7
	2.51	53.0		2.58	45.5
	2.08	44.0		1.88	34.0
	1.78	37.5		1.33	24.0
	1.27	27.0		0.63	11.5
	0.71	15.0			
400	6.92	97.5	300	9.15	98.2
	5.63	80.0		8.37	90
	4.73	67.0		7.12	76.8
	3.87	55.5		6.18	68.0
	3.17	45.0		5.40	59.0
	2.27	33.0		4.69	51.5
	1.72	24.8		3.44	38.0
	0.94	14.0		2.50	28.5
	.47	7.8		1.02	12.8
200	13.93	98.0			
	12.36	7.8			
	9.62	69.0			
	8.45	61.0			
	7.59	53.5			
	5.95	43.0			
	4.30	31.5			
	2.43	18.0			
	1.25	9.0			

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Figure 2I.4 - Pressure transducer calibration curves. Each line represents the PT response for different span settings. The scale reading for 5 Torr at each span setting is plotted vs. span in Fig. 2I.5.

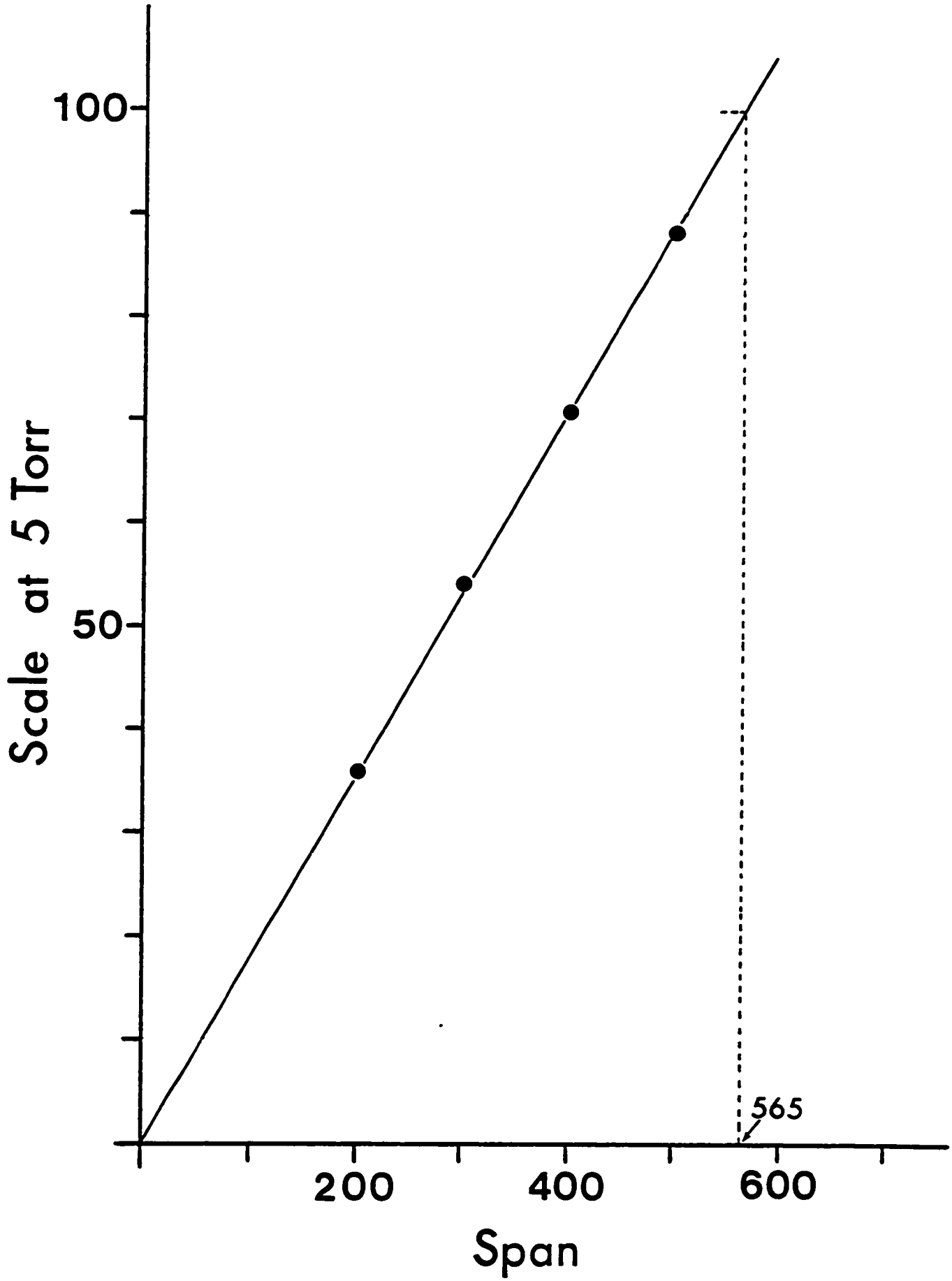
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Figure 2I.5 - Determination of span setting for full scale deflection at 5 Torr.

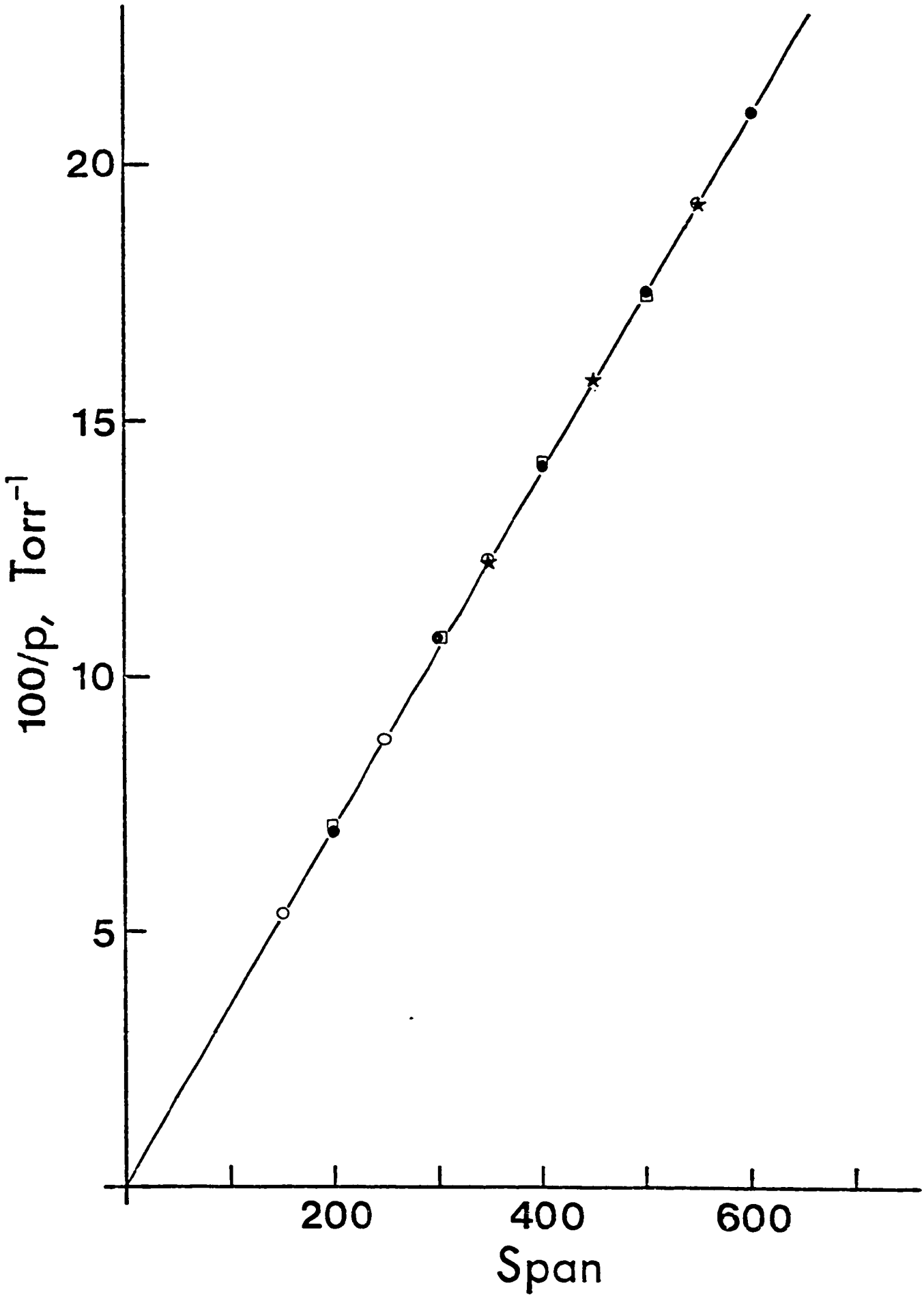
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Figure 2I.6 - Span setting vs reciprocal of full scale pressure for data of Table 2I.1 (●) and several other calibration checks. Each symbol represents a different test.

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This pressure response calibration was frequently checked (∿ every other month), and even though the zero setting varied ± 5% (particularly after accidental overloadings of the transducer) the response was practically unchanged, as can be seen from inspection of Figure 2I.6.

As installed, the PT could be used to measure reagent and Xe flows as well as to monitor the pressure in the flow tube. In conjunction with the Clevite Mark 250 recorder, flows in the 0.5 μMol min^{-1} to 2.5 mMol min^{-1} range could be reliably measured by appropriately selecting the standard volume and the span setting.

iii) Calibration of the Spiral gauge. The spiral gauge was calibrated against the oil manometer from 1 to 45 Torr and against the mercury manometer from 1 to 170 Torr. The plot of pressure vs. deflection is an excellent straight line with slope of (1.643 ± 0.002) Torr cm^{-1}. Although the linear fit is very good, accurate readings of light spot demanded a lot of patience because vibrations from the rack tended to blur the image.

d. Gas handling and purity. Tank argon used in this study was >99.9% pure. It was further purified by passing it through molecular sieve traps at room temperature and a 77°K as described in Section 2I.a. Research grade Xe was purchased from Matheson Gas Products and was used without further purification. Reagent gases were obtained from a variety of sources with purities better than 99.0% in most cases; notable exception were the fluorinated reagents most of which were quoted as >95%, although some of these must be regarded as suspect. The more corrosive gases, particularly HCl, HBr, HI and the mixed halogen were found to contain significant amounts of impurities, in spite of being labeled as more than 98% pure. Reagent gases were diluted in argon with mole fractions varying from 0.05 to ∿0.25, except for a few quenchers that were run pure, i.e., N_2, CO, H_2. As a general

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rule when loading reagent gases, enough material was introduced into the system to allow for at least one flush of the intended storage reservoir and connecting lines after the trap to trap distillation. After the low boiling point impurities (O_2, N_2, H_2, etc) were separated, the sample was "divided" into at least 3 fractions. The first was used to flush the system the second to prepare the gas mixture and the third, containing most of the high boiling point impurities was discarded. In some instances, particularly with the most corrosive gases, extensive gas handling may introduce as many impurities as are removed; therefore, the extent of the "purification" procedure varied widely from reagent to reagent. Liquid reagents required in addition, a lengthy thaw-pump-freeze cyclic process to eliminate dissolved gases.

Some of the reagent gases used are extremely hazardous with regard to both, their toxicity and oxidizing properties, and should be handled with care. No attempt was made to trap these chemicals during or after the experiments, and reaction with hot pump oil was considered the best way of disposal, after making sure that the pumps were properly vented to the hoods.

e. Flow characteristics. Quantitative rate determinations in a fast-flowing afterglow apparatus depend almost entirely on the tractability of the hydrodynamics of the system. For cylindrical reactors, the relevant flow equations have been solved for the ideal flow case[7,8] which presumes no visous pressure drop, no axial diffusion, no wall reactions and maximum radial diffusion. When these conditions are fulfilled, there are no radial concentration gradients and all the elements of the fluid move with a bulk velocity v. The exact solutions have been generalized and used to derive approximate solutions for cases including unit deactivation at the walls, pressure drop along the reactor, axial diffusion, slip flow and the effects of incomplete radial mixing and of reagent inlet geometry.[7-14] Recently Ogren[15] reported

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the exact solution of the continuity equation for first order kinetics in flow tube reactors with wall reactions. In all cases, however, the results apply for small deviations from the ideal viscous flow, and in general, for specific experimental arrangements. A full mathematical description of the flow in the reactor used in this study was not attempted, particularly because the geometry of the system (right angle bend and reagent inlets) would pose formidable computational problems. Instead, it was felt that a careful experimental characterization of the flow would permit the selective application of the flow analysis results in the literature to our system.

A flow is considered viscous (or Poiseuille flow) if in the absence of turbulent motion of the gas, (1) the fluid is incompressible, (2) the flow is fully developed that is, the flow-velocity profile is constant throughout the length of interest, and (3) the flow velocity at the tube walls is zero. Broadly speaking, these conditions are met if the Kundsen number[16] of the flow is <0.01 or the Reynolds number[17] is <1000. Under typical operating conditions of our reactor using argon as buffer gas ($\eta = 2.27 \times 10^{-4}$ poise), the mean free path L, at 298°K and 0.85 Torr is 6.3×10^{-3} cm which gives $N_{Kn} = L/a = 4.0 \times 10^{-3}$ cm; a is the radius of the tube. For the Reynolds number we obtain,

$$80 < N_{Re} = \frac{a\rho v}{\eta} < 100$$

where ρ is the gas density and v the average bulk flow velocity, $7 \times 10^{-3} < v < 8 \times 10^{3}$. The values obtained for N_{Kn} and N_{Re} indicate that the flow is nearly viscous and nonturbulent.[9,18] Deviations from the three conditions above can be estimated as follows. According to Schlichting[19] a fluid is incompressible if $(M^2/2) << 1$, where M is the Mach number of the flow, defined as the ratio of the bulk flow velocity and the velocity of sound in the gas, which for Ar at 25°C is 3.3×10^{4} cm sec^{-1}. For $v = 7.5 \times 10^{3}$ cm sec^{-1},

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$M^2/2 = 0.026$. The practical upper limit[19] for this quantity is 0.06 ($M = 1/3$).

When a fluid enters a tube, the original velocity profile continually changes until, at some distance from the entrance, a new velocity profile is established which remains constant for the rest of the passage through the tube. Langhaar[20], has shown that the flow will become fully developed after traversing a distance $d = 0.0227aN_{Re}$ where a is the radius of the tube. Since N_{Re} varies between 80 and 100 for our flow tube, the distance required for complete flow relaxation varies between 28 and 35 cm.

The flow velocity at the walls can be calculated as a percent of the velocity at the center of the tube using the expression derived by Kennard;[21]

$$\frac{\text{velocity at the walls x 100}}{\text{velocity at the center of tube}} \propto \frac{2sa(100)}{a^2 + 2sa} \qquad (2I.2)$$

where s is the coefficient of slip, equal to $1.39Lp^{-1}$ for a variety of gases in glass flow tubes.[22] At 0.85 torr, it follows that the velocity at the walls is 0.8% of the bulk flow velocity.

From this analysis, the flow is the reactor should be very close to laminar (viscous) with no more than few percent slip. The right angle bend however, introduces a large uncertainty in the distance required for development of parabolic velocity profile. Assuming no perturbation at all, the flow would be fully developed at the second observation window. On the other hand, if the perturbation is large enough that the right angle bend should be considered as a new entry point, relaxation of the flow would not occur much before the last observation window. For this reason, the velocity profile throughout the observation region is considered in transition between "plug" and parabolic.

Finally, the pressure drop between the first and last observation points (second and fifth windows) was calculated using the Poiseuille equation[7]

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and a more refined expression[9,18] that allows for slip contributions. The maximum calculated pressure drop between the first and last windows was ∿10% but due to the location of the transducer connection, the difference between the measured pressure and the calculated pressure for the first observation point does not exceed ∿8%.

For the flow characteristics found in our reactor, the analysis of Ferguson, et. al.,[7] should be valid. They found that if a rate constant is measured for parabolic flow conditions, but under the assumption of plug velocity profile, the rate constant should be multiplied by a factor of 1.6 to give the value of the rate constant for parabolic flow. However, since the flow throughout the observation region in our system is in transition between plug and parabolic, a full correction is not justified, and an intermediate value[23] of 1.3 is more reasonable.

Two other problems need to be considered. First, the perturbation introduced by the inlet effects and the mixing efficiency and second, the fact that our detection system samples an average concentration across the tube, rather than the axial concentration. The first problem was also treated by Ferguson, et. al.[7] They analyzed the effects on the flow of a ideal point (central) inlet, and a coaxial cylinder. In both cases the correction to the rate constants tend to reduce the original factor of 1.6 and in the case of the point source, this correction could lead to an overall correction factor of less than unity. Bolden, et. al.[24] considered the problem of radial sampling vs. axial sampling and concluded that the mean (radial) concentration can be used as a measure of the axial density to a fair degree of accuracy. Therefore, considering that our "shower head" inlets should provide better mixing and less perturbation than either, the ideal point source or the coaxial cylinder, and that radial sampling is quite comparable to axial sampling, the correaction factor of 1.3 was further reduced[3] to a final value of 1.15. It is believed that the absolute uncertainty due to

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flow characteristics for the rate constants obtained with this apparatus should be less than ± 20%.

f. Determination of the Atomic Absorption Law. The absorption of a single non-reversed "line" of radiation by an optically thin medium is described by[25]

$$A_\alpha = \frac{I_o - I}{I_o} = \sum_{n=1}^{\infty} (-1)^{n+1}(k_o \ell)^n / [n!(1 + n\alpha^2)^{1/2}] \qquad (2I.3)$$

where n is an integer, α is the ratio of the breadth of the emission line to the breadth of the absorption line. The optical depth, $k_o \ell$, is given by (2I.4).

$$k_o \ell = (2/\Delta\nu)(\ln 2/\pi)^{1/2}(\pi e^2/mc)Nf\ell \qquad (2I.4)$$

In this expression, N is the concentration of the absorbing species, f the oscillator strength of the transition, e and m are the electron charge and mass respectively, c the speed of light, ℓ the length of the path of the incident radiation in the absorbing medium and $\Delta\nu_o$ is the Doppler half-width of the absorption line,

$$\Delta\nu_D = 2(2R \ln 2)^{1/2}(\nu_o/c)(T/M)^{1/2}. \qquad (2I.5)$$

In order to use eq. 2I.3, α must be determined experimentally. This was done[2] for Ar taking advantage of the fact that the ratio of the optical depths for any two lines diagnostic of the same species should be a constant and equal to the ratio $\lambda_1 f_1/\lambda_2 f_2$, see eq. 2I.4, if the proper value of α is used to relate absorption to optical depth (see Equation 2I.3). Four lines were used, all diagnostic of the $[4s(3/2)]_2$ or $Ar(^3P_2)$ level of argon, and it was found that for α = 3.5 the ratio of the optical depths was constant over a wide range of concentrations.

The line widths for some of the emission lines were also measured directly at the NRC facilities in Ottawa, Canada and, assuming that the

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width of the absorption lines is dominated by pure Doppler broadening at 300°K, the results of the measurements were in reasonable accord with $\alpha = 3.5$. This direct analysis of the emission lines also showed that the emission lines were non-reversed. The commonly used method for determining α is to vary the optical depth $k_o \ell$ by changing the optical path ℓ. This could be achieved by comparing absorptions for a single pass of the light through the reactor with those for multiple pass, but this method was not used because of the difficulty in insuring that the same section of the reactor was being sampled by the several passes of the light. For weak absorptions ($A_\alpha < 0.25$), A_α was rather insensitive to changes in α between 3.0 and 4.0. Once α was determined, it was possible to evaluate Equation 2I.3 as a function of $k_o \ell$ values. The calculated curve of A_α vs $k_o \ell$ for $\alpha = 3.5$ is shown in Figure 2I.7 for $A_\alpha \leq 0.25$. Curves for $\alpha = 3.0$ and 4.0 are also shown for comparison; Appendix A contains calculated A_α values for several α's.

Since Eq. 2I.3 is mathematically too cumbersome to be used in the kinetic analysis of the system, it was desirable to express it in a more compact way. A modified Beer-Lambert law,

$$I = I_o \exp[-a(k_o \ell)\gamma]; \ \gamma = 0.95, \qquad (2I.6)$$

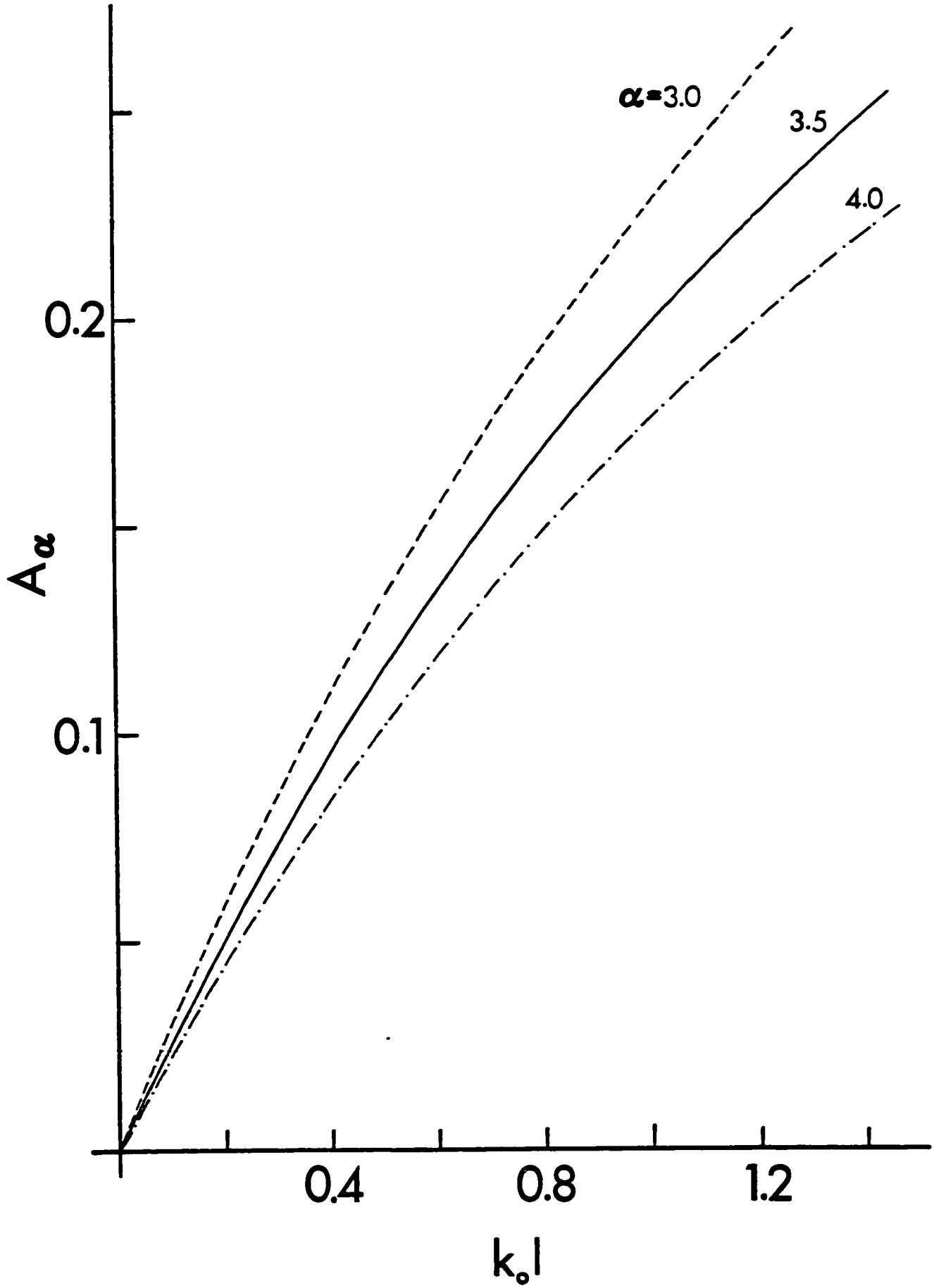
was finally chosen. The exponent γ was determined from the slope of a plot of $-\ln\ln(I_o/I)$ vs $\ln(k_o \ell)$ with $k_o \ell$ calculated from Equation 2I.3 (or Fig. 2I.7), using experimental A_α values. This plot gave good linearity for absorptions up to 30%.

A similar analysis was not possible for $Xe(^3P_2)$ absorption, particularly due to the limited output of wavelengths from the lamp used and the range of sensitivity of the detection system. It was found, however, that the line widths of the Ar 811.5 nm line and the Xe 881.9 nm line were very similar, see Figure 2I.8, and since the Xe metastable concentration was not expected

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Figure 2I.7 - Calculated A_α vs. $k_o\ell$ from eq. 2I.3 for $\alpha = 3.5$. Although the series rapidly coverges, 40 terms were included in the summation (Equation 2I.3). For comparison, curves for $\alpha = 3.0$ (---) and $\alpha = 4.0$ (-·-) are also plotted.

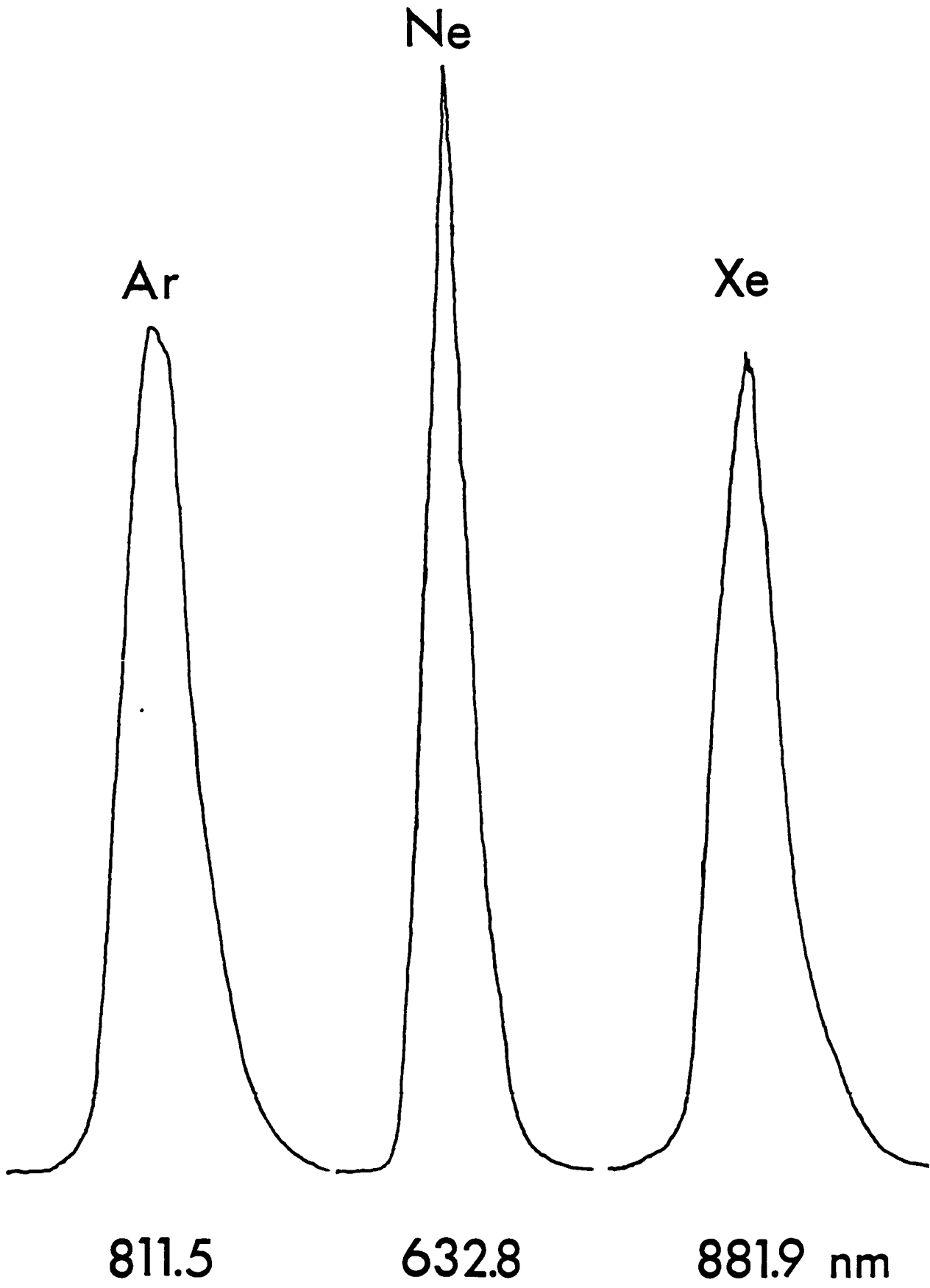
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Figure 2I.8 - Photoelectric scans of the Ar 811.5 nm line (left) and Xe 881.9 nm line (right) from the pen Ray lamps. For comparison, the Ne 632.8 laser line is shown at the center. FWHH's are 0.060 nm for the Argon line, 0.054 nm for the Xe line and a 0.028 nm for the laser line. The instrument used was a McPherson 2018 1m Monochromator (linear dispersion, 0.8 nm/mm; resolution, better than 0.015 nm). Slits were 5μ. In order to fully illuminate the grating, light from the laser was scattered by a piece of soft white paper in front of the beam.

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to be higher than the $Ar(^3P_2)$ concentration, the same value of $\alpha(3.5)$ and therefore the same absorption law (Equation 2I.6) should be valid for the purpose of obtaining relative concentrations. This approximation is particularly justified for $A_\alpha < 0.25$, see above.

g. Measurement of Absolute Metastable Concentrations. Combining equations 2I.4 and 2I.5 the concentration can be expressed as[25]

$$N = 7.15 \times 10^{13} \left(\frac{T}{M}\right)^{1/2} \frac{k_o \ell}{f\lambda} \qquad (2I.7)$$

where T is the absolute temperature of the absorbing species, M the atomic weight in amu, $k_o \ell$ the optical depth for the transition with oscillator strength f. The wavelength λ is expressed in $\mathring{A}$ units. Since $k_o \ell$ values can be obtained from the measured absorption A_α (using Figure 2I.8), only accurate oscillator strengths are needed. For argon, f-values have been measured[2] for all the transitions of interest and appear to be reliable. Measurement of the $Xe(^3P_2)$ concentration however, is more difficult because the transition probabilities in the $Xe(6s,^3P_2)$ - Xe(6p) manifold are not as reliable. Disagreement between the scarce experimental data and calculated values amount in some cases to 100% (see Section 1.e). For example, Allen, et. al.,[26] measured the lifetimes of the $Xe(6p[1/2]_o)$ and $Xe(6p(3/2]_2)$ as (40 $\pm$ 12) and (33 $\pm$ 20)ns respectively, whereas Verolainen and Osherovich[27] obtained (30 $\pm$ 4) and (48 $\pm$ 6)ns for the same states (for Xe states above the Xe(6p) manifold, particularly Xe(6p'), Xe(7p), Xe(7d) and Xe(8p), results from these two sources differ by $\sim$30%). The calculated transition probabilities of Chen and Garstand[28] give reasonable agreement for the $6p[3/2]_2$ state, 29ns; however, considering only the $6p[1/2]_0$ - $6s[3/2]_1$ ($^3P_1$) transition, the calculated transition probabilities give 28ns for the lifetime of $6p[1/2]_o$, it is expected that the other allowed transition (to the $Xe(6s'[3/2]_1)$ state) has a negligible contribution ($\tau_i = \sum_j (A_{ij})^{-1}$). Using the same calculated A_{ij}'s, the f-values

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Table 2I.2 Metastable Atom Concentrations from Atomic Absorption

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Table 2I.2 Metastable Atom Concentrations from Atomic Absorption

M	Transition	λ(nm)	f	A_α	$k_o\ell$[a]	[M*][b]; 10^{10} atoms/cm^3
Ar	$^3P_2$ - 4p[5/2]$_3$	811.5	0.380[c]	0.193	0.965	6.1
	$^3P_2$ - 4p[5/2]$_2$	801.5	0.105[c]	0.036	0.140	3.3
	$^3P_2$ - 4p[3/2]$_2$	763.5	0.223[c]	0.100	0.420	4.8
Xe	$^3P_2$ - 6p[5/2]$_2$	881.9	0.4897[d]	0.118	0.510	1.3
	$^3P_2$ - 6p[3/2]$_2$	823.2	0.234[d]	0.052	0.203	1.1
Ar	$^3P_2$ - 4p[5/2]$_3$	811.5	0.380[c]	0.187	0.922	5.9
	$^3P_2$ - 4p[5/2]$_2$	801.5	0.105[c]	0.049	0.190	4.4
	$^3P_2$ - 4p[3/2]$_2$	763.5	0.223[c]	0.105	0.460	5.3
Xe	$^3P_2$ - 6p[5/2]$_2$	881.9	0.4897[d]	0.120	0.528	1.3
	$^3P_2$ - 6p[3/2]$_2$	823.2	0.234[d]	0.062	0.248	1.4

a. $k_o\ell$ was obtained from Figure 2I.8.

b. $[M^*] = 7.15 \times 10^{13}(T/M)^{1/2}\ (k_o\ell/f\lambda)$, λ in Å.

c. Oscillator strengths were taken from Ref. 2.

d. Calculated from transition probabilities of reference 28 using the relationship $f_{ik} = (A_{ki})\ 1.499 \times 10^{-16}\ \lambda^2\ (g_k/g_i)$, λ in Å.

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for the 823.2 and 881.9 nm Xe lines are 0.23 and 0.49, in good agreement with the experimental results of Malakov,[29] 0.29 and 0.5, but the calculated transition probabilities are in the average (for 14 transitions terminating in the Xe(6s) and Xe(6s') states) a factor of five larger than the experimental values of Miller, et. al.[30]

Using the calculated values of Chen and Garstang[28] to obtain oscillator strengths for Xe, the concentrations listed in Table 2I.2 were measured in the emission studies apparatus (see Chapter 2, Part II). These Xe concentrations must be a factor of ∿2 too low according to the evidence to be presented in Section 4.b. One possible reason for the low estimate of the [Xe($^3P_2$)] from the absorption measurements is that the transition probabilities used are incorrect. If this is the case, both apparently are in error by the same amount. It is also possible that the large number of Xe isotopes and the hyperfine structure of the emission and absorption lines complicate the interpretation of the absorption measurements.[31]

h. Optimal Xenon flow. The minimum Xe flow necessary to completely eliminate argon metastable atoms from the flow before they reached the mixing zone was determined by using the following technique. With an Ar lamp as a source, Xe was added to the flow before the discharge until no Ar($^3P_2$) could be detected by absorption of the 811.5 nm line at the first observation point. Since Ar($^3P_2$) constitutes ∿87% of the total Ar metastable concentration[2] and Ar($^3P_0$) reacts with Xe faster[2] than Ar($^3P_2$), complete removal of Ar($^3P_2$) metastables ensures complete removal of Ar($^3P_0$) as well. At this point, the Ar lamp was replaced by a Xe lamp and the Xe flow increased until no further increase in the absorption of the 881.9 nm Xe line could be detected. Xe flows of the order of 160 μMol min^{-1} were sufficient to completely remove Ar metastables. Increasing the Xe flow

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beyond ∿200 μMol min^{-1} actually decreased the Xe metastable concentration.

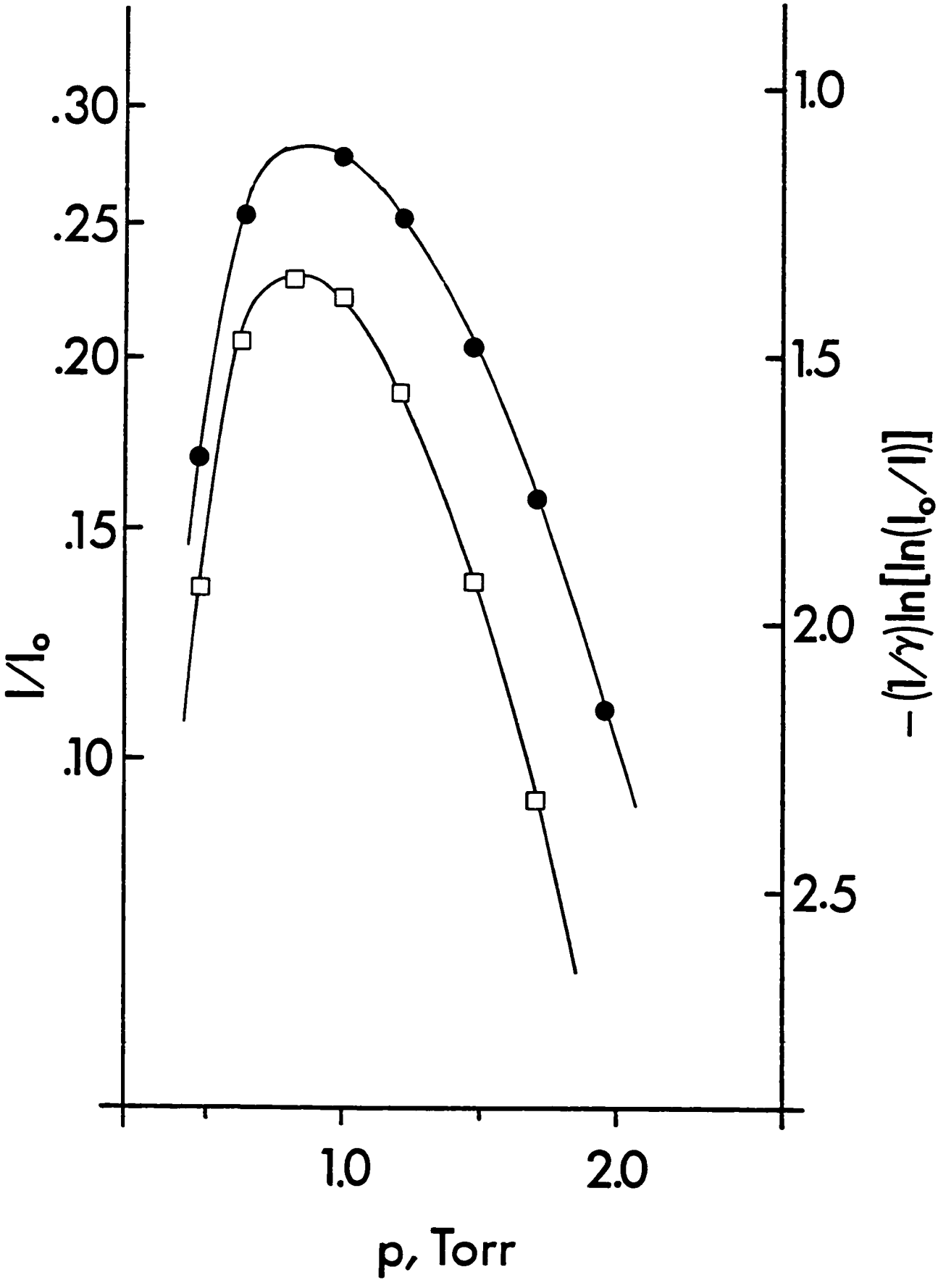
Several attempts were made to detect Xe($^3P_0$) metastables. However under no circumstances could absorption of the Xe 764.2, 820.6, 796.7, 682.7, 682.7, 582.4 or 539.3 nm lines be observed. Xe($^3P_0$) if present, must constitute less than 0.5% of the total Xe metastable concentration.[6] Similarly, Xe($^3P_1$) atoms were undetectable under the same experimental conditions.

Once the optimal Xe flow was found, it was necessary to determine the best compromise between buffer gas flows and bulk velocities. By throttling the total flow, the decay of Xe($^3P_2$) as a function of reactor pressure could be studied; results are shown in Fig. 2I.9 for two different Xe flows. 120 μMol min^{-1} and 180 μMol min^{-1}. In both cases, maximum Xe($^3P_2$) concentration is achieved at ∿7.6 x 10^3 cm sec^{-1} for the bulk flow velocity.

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Figure 2I.9 - Absorption of the Xe 881.9 nm line (for three traversals of the light across the flow tube) as a function of Ar pressure in the reactor for two Xe flows: ● ∿180 μMoles min^{-1}, □ ∿120 μMoles min^{-1}. Increasing the Xe flow beyond 180-200 μMoles min^{-1} does not increase absorption. These data were taken at the first observation point used to measure the decay of $Xe(^3P_2)$, 25 cm downstream from the discharge (13.4 cm downstream from the reagent inlet).

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PART II EMISSION STUDIES

Apparatus & Method

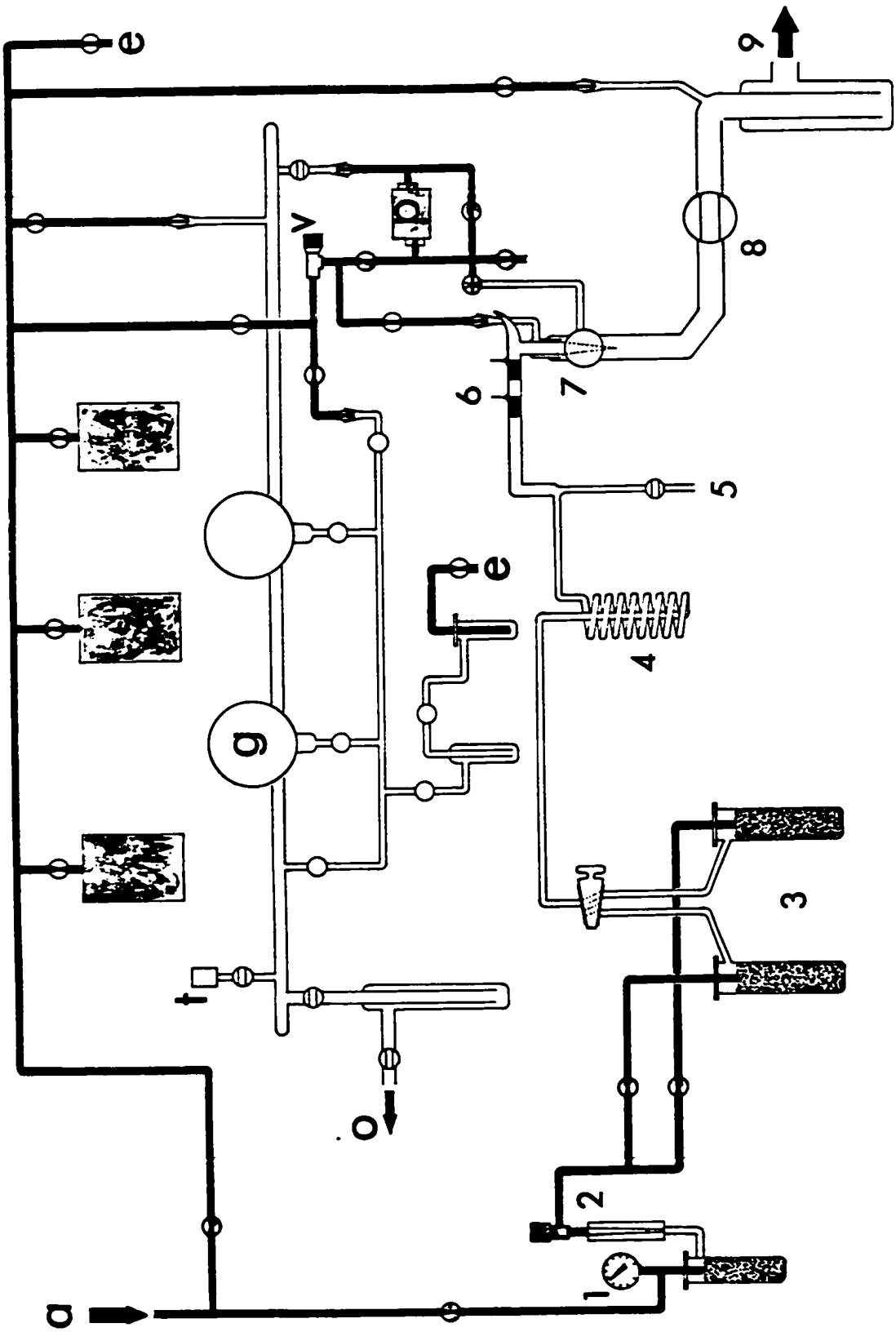
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a. Description of the system. The U.V. and visible emission arising from the quenching of the metastable states of the rare gases was studied in a flow system especially designed[1] to handle corrosive chemicals and fluorinated reagents that are very difficult to handle in convertional glass systems. A diagram of the apparatus is shown in Figure 2II.1. The apparatus consisted of three sections, a grease-free glass section, a stainless steel section and the flowing afterglow reactor (in practice, any one of these sections could be effectively isolated from the others), and each will be described separately. The grease-free glass section was used to handle and store reagents that did not attack glass but could not be quantitatively handled in flurine passivated stainless containers, i.e., HBr, Br_2, etc. Two 5 liter storage bulbs were available each independently connected to the high vacuum line and to separate loading ports. Pressure could be measured with a spiral gauge or with the pressure transducer. High vacuum Teflon valves were used throughout. All other gases were handled and stored in the ss section which had five storage vessels (it was constructed with 1/4" ss tubing for the connections and the containers were made out of ∿ 10" x 4" O.D. ss pipe with silver-soldered ss end plates). The stainless section contained the pressure transducer (Celesco Industries PD7, range ± 10 PSI) which served the dual purpose of measuring reagent flows and monitoring pressure of the reaction zone or gas handling section. All metal sections were passivated by filling with fluorine at 500 Torr for ∿60 hours. As an additional measure, the storage vessels were heated to 150°C for 12 hours before the fluorine was pumped out. This procedure was repeated with a 10% mixture of fluorine in argon after every 6 to 10 runs. The third section consisted of argon tank, a molecular sieve trap and a positive/negative pressure gauge relative to atmospheric pressure (at the high pressure side of the main flow meter), a flow meter, a needle valve,

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Figure 2II.1 - Schematic diagram of the apparatus used for the emission studies. Heavy lines indicate metal tubing. (1) Pressure gauge and molecular sieve trap. (2) Argon flow meter and flow metering valve (3) low pressure molecular sieve traps (4) glass coil (5) Xe inlet (6) Hollow cathode discharge (7) reaction zone (8) flow throttling valve (large stop-cock) (9) to mechanical pumps. (a) Argon inlet (e) loading ports (g) glass storage vessels (o) to oil diffusion pump (p) Pressure transducer (S) Stainless steel storage vessels (t) thermocouple pressure indicator (v) fine metering valve.

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two molecular sieve traps in a parallel arrangement, hollow cathode discharge, reagent inlet, and observation window. The reactor was separated from the 1000 liter min^{-1} pump by a large stopcock (25 mm bore) which was used as a throttling valve for the high pressure experiments. A detailed picture of the reactor is shown in Figure 2II.2. Xenon metastable atoms were produced (see Section 2I.a) by adding ∿0.2 mMol min^{-1} of Xe before the discharge maintained between two rolled tantalum foil electrodes located inside a 12 mm O.D. glass tubing with the mixing zone 5.5 cm downstream from the end of the cathode. The reagent was introduced coaxially[2,3] through a 18 mm O.D. tube into the 25 mm O.D. reactor. The reaction flame was observed through 2.5 cm in diameter spectrosil windows with center located ∿2 cm below the mixing point and fixed in place by epoxy resin. It was found that these dimensions provided very stable signals for reactor pressures from 0.5 to ∿40 torr. In order to better control the discharge characteristics at the higher pressures three 10-turn variable resistance pots (3kΩ, 15kΩ and 1MΩ) were added in series with the discharge.

All the glass-metal connections consisted of an inner ss 14/35 standard taper joint machined in the Department's shop and a matching glass outer Standard Taper; tensions were relieved and vibrations dampened by liberally using 1/4" flex tube in all the lines leading to glass sections.

b. Calibration of apparatus. Pressures in the emission studies apparatus were measured with the pressure transducer, except for a few instances in which a spiral gauge was used. Both, the spiral gauge and PT were calibrated against oil and mercury manometers. Figure 2II.3 shows the PT as installed in the system. The reference side of the PT was kept at 10^{-5} Torr using a combination of cold trap, oil diffusion and mechanical pumps. The calibration procedure has been fully explained in section 2I.b. Data gathered during

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Figure 2II.2 - Photograph of the reactor. The concentric inlets are clearly visible. Notice also that the standard taper used for the glass-metal interface at the reagent inlet. The reactor is shown in actual size.

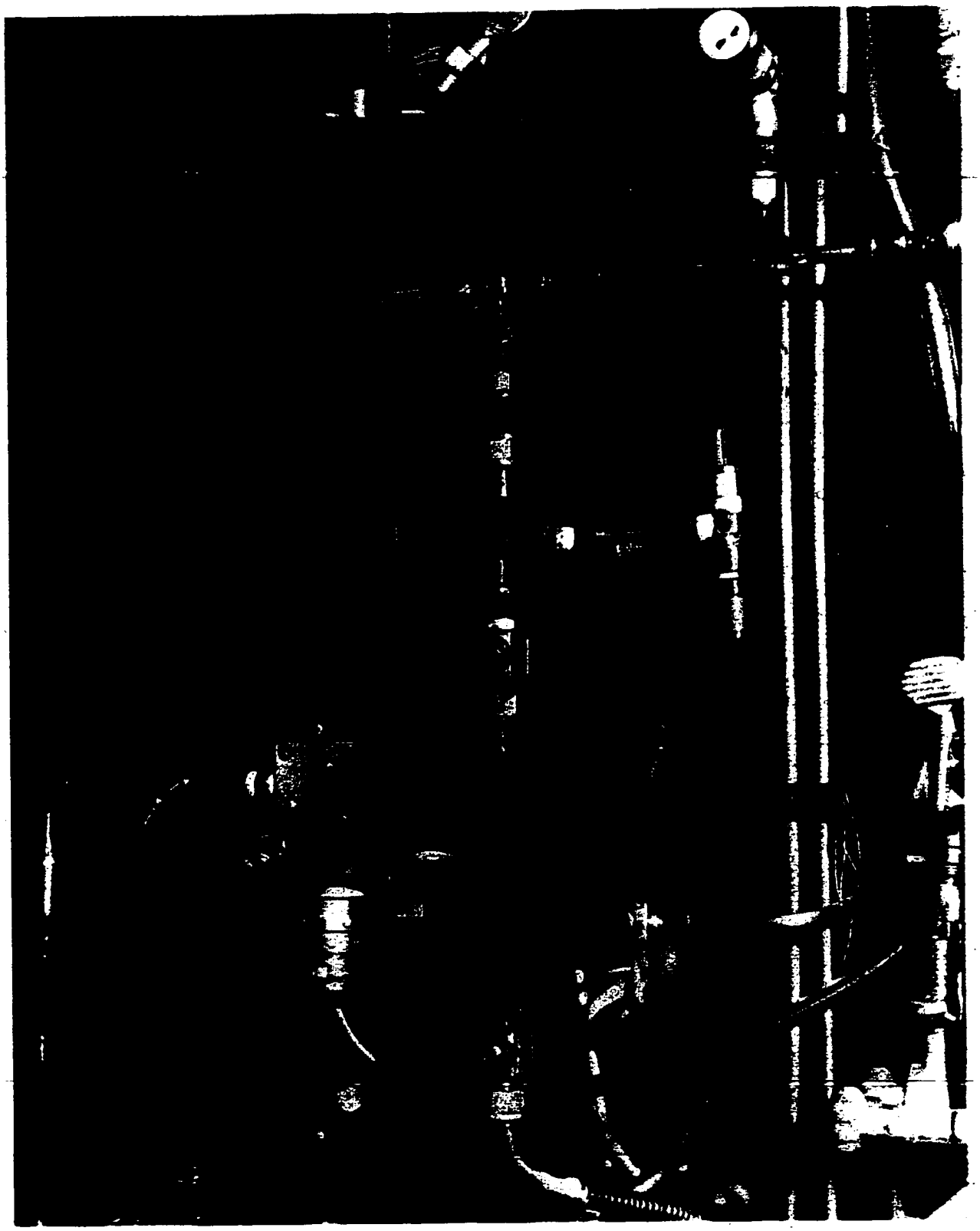
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Figure 2II.3 - Photograph of the pressure transducer as installed in the emission studies apparatus. The standard volume consisted only of the connecting ss tubing. Notice the "home-made" connecting T's.

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a calibration session are listed in Tables 2II.1 and 2II.2, for the oil and mercury manometers respectively. A set of pressure vs. indicator reading plots are shown in Figure 2II.4, while Figure 2II.5 shows a plot of the reciprocal of the pressure for full scale deflection vs. span setting. Since the transducer covers a range of rather high pressures, a small standard volume was used in order to be able to measure flows of the order of 1.0 μMole min^{-1} (see Equation 2I.1), and therefore, only the internal volume of the connecting lines was used. An average of 42 determinations with a gas burette gave V = (15.48 $\pm$ 0.03) ml. With this arrangement and by appropriately setting the span, flows in the range 5 mMoles min^{-1} to 0.5 μMoles min^{-1} could be measured. One drawback of using this transducer was that reagent gases had to be stored at relatively high pressures to avoid curvature in the pressure vs. time output of the recorder when the pressure drop across the needle valve approached zero. The great ease with which gases could be handled with the help of the transducer, however, amply made up for this inconvenience.

In the latter part of this investigation it was necessary to measure flows with back pressures as low as 5 torr. For those experiments, the diaphragm of the transducer ($\pm$ 10 PSI) was replaced by a more sensitive one, $\pm$ 0.1 PSI, and flows down to 0.05 μMoles min^{-1} were reliably measured.

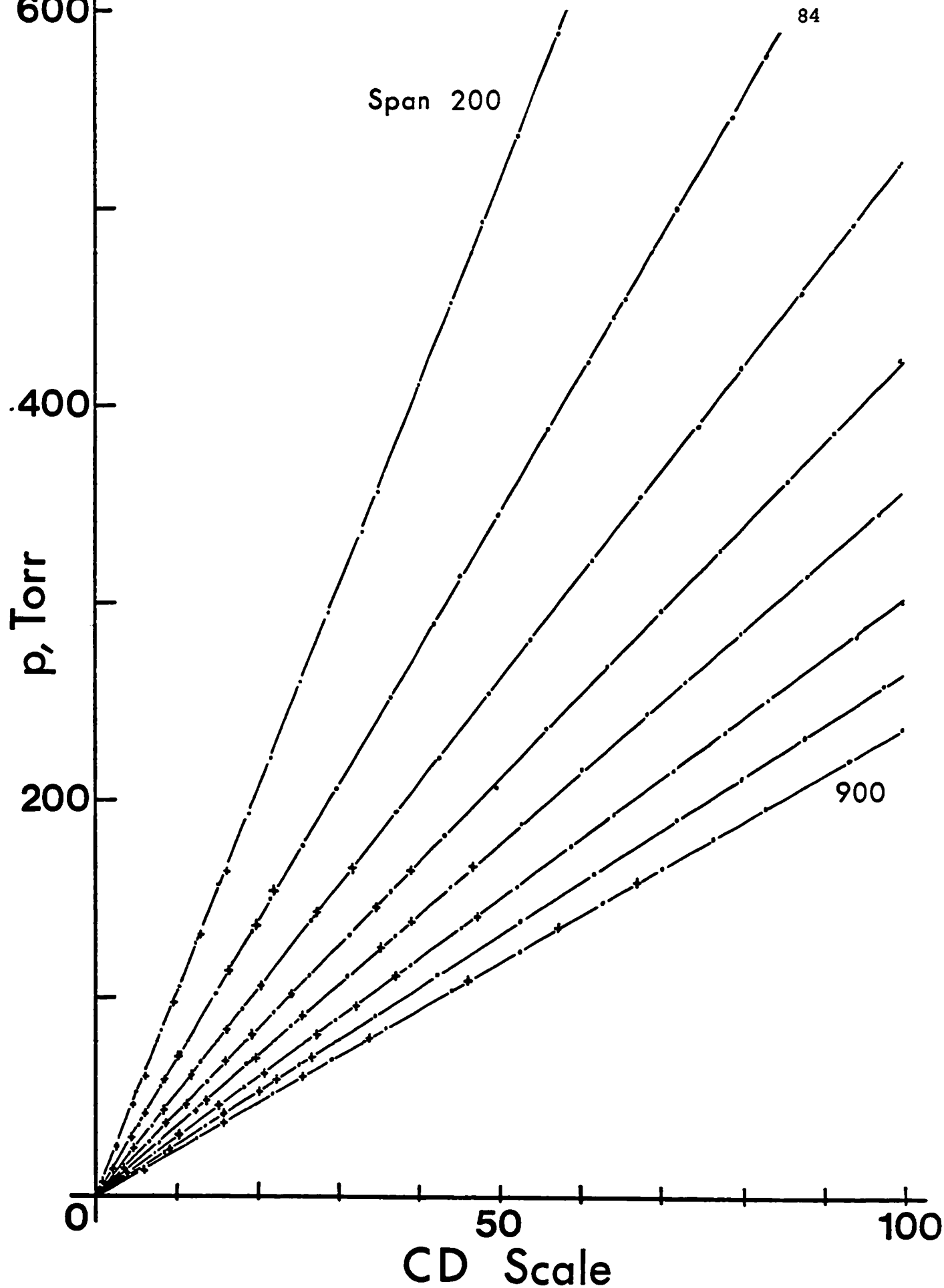
The main argon flow was measured with a Gilmont No. 12 commercial flow meter. A calibration curve for argon at 298°K with a back pressure of 1.16 atm (5 inches of mercury above atmosphere) calculated using the data supplied by the manufacturer for the linear range is shown in Figure 2II.6. The positive pressure of 5 inches of mercury was maintained to eliminate the possibility of leaks in the high pressure side of the argon line.

Under normal experimental conditions, the Ar flow was 43.7 mmoles min^{-1} and therefore the linear flow velocity,

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Figure 2II.4 - Pressure vs. Scale reading plots for the calibration of the pressure transducer. (+) represent data obtained with the oil manometer as reference, while (·) refer to to the mercury manometer.

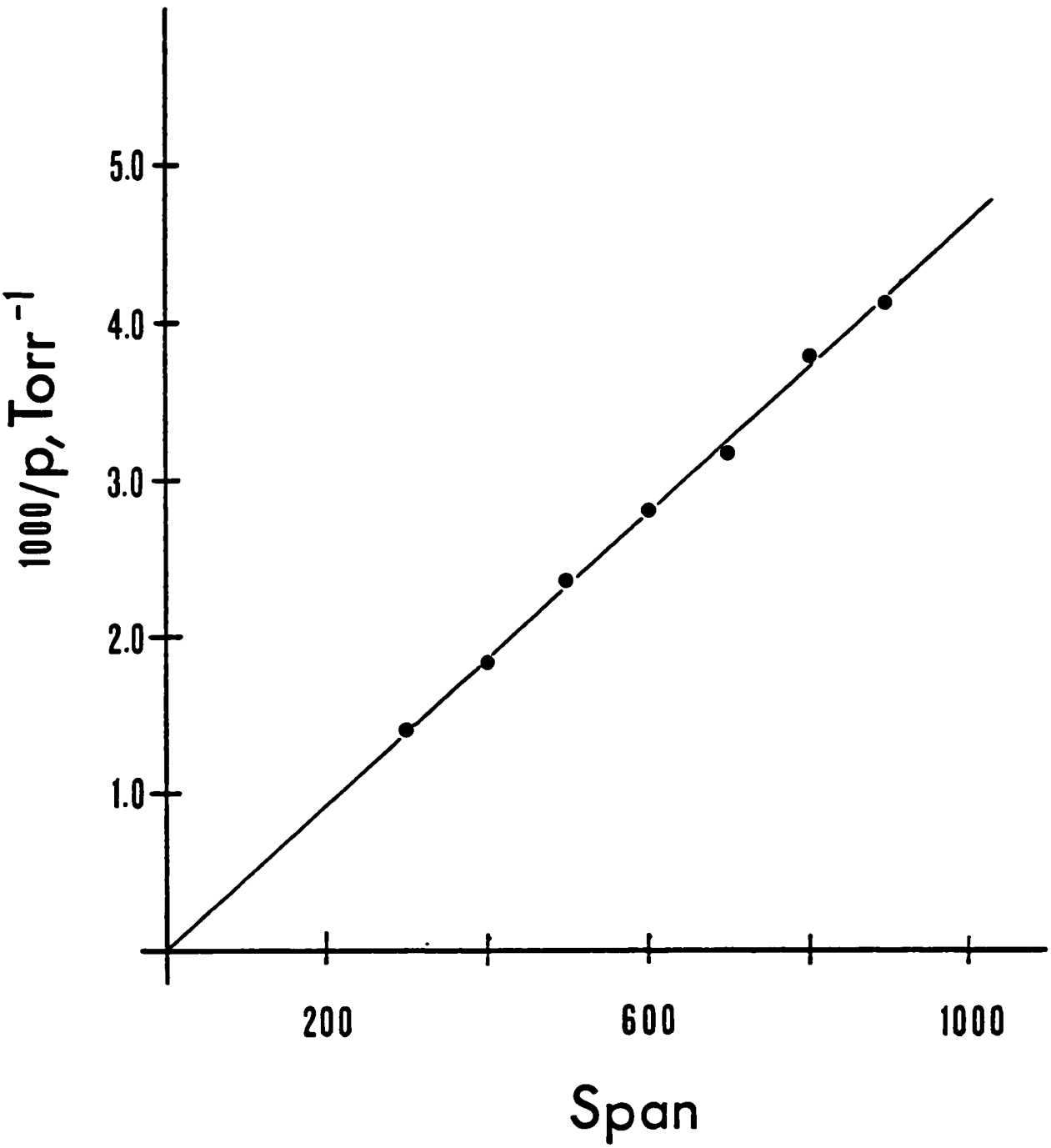
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Figure 2II.5 - Plot of the inverse of pressure for full scale deflection vs. span. Linearity is very good, and a least squares fit of the data gives $(1000/P)_{FS} = (4.635 \pm 0.004) \times 10^{-3} S$, where S is the span value.

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Table 2II.1 - Data for the calibration of the low pressure range of the transducer. The pressures were measured with an oil manometer.

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Span	Pressure (Torr)	P.T.	Span	Pressure (Torr)	P.T.
200	162.7	15.6	300	152.8	21.9
	132.0	12.9		136.3	19.7
	98.5	9.6		113.0	16.6
	61.6	6.5		71.5	13.4
	46.1	4.4		59.7	8.7
	25.0	2.0		41.9	6.1
	7.4	0.7		29.7	4.3
				13.5	1.8
				4.2	.5
400	165.4	31.2	500	163.9	38.1
	143.5	27.2		149.3	35.4
	106.8	20.6		102.5	24.5
	84.0	16.2		81.8	19.6
	60.7	11.9		69.2	16.7
	43.5	8.4		46.8	11.2
	24.4	4.1		36.4	8.8
	4.1	0.8		14.8	3.4
600	166.9	46.3	700	140.9	46.5
	138.1	39.0		110.8	37.0
	125.0	35.6		96.5	32.4
	90.3	26.1		82.0	27.7
	69.9	20.2		62.8	21.2
	48.3	14.1		45.9	15.6
	12.4	3.6		30.5	10.4
				10.1	3.5
800	70.0	26.5	900	159.9	66.6
	58.6	22.2		135.5	57.2
	53.1	20.2		108.4	46.1
	41.6	15.9		80.4	34.3
	24.2	9.5		60.3	25.7
	15.8	6.3		37.2	15.8
	2.1	1.2		13.8	5.5

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Table 2II.2 - Data for the calibration of the high pressure range of the transducer. Pressures were measured with a mercury manometer.

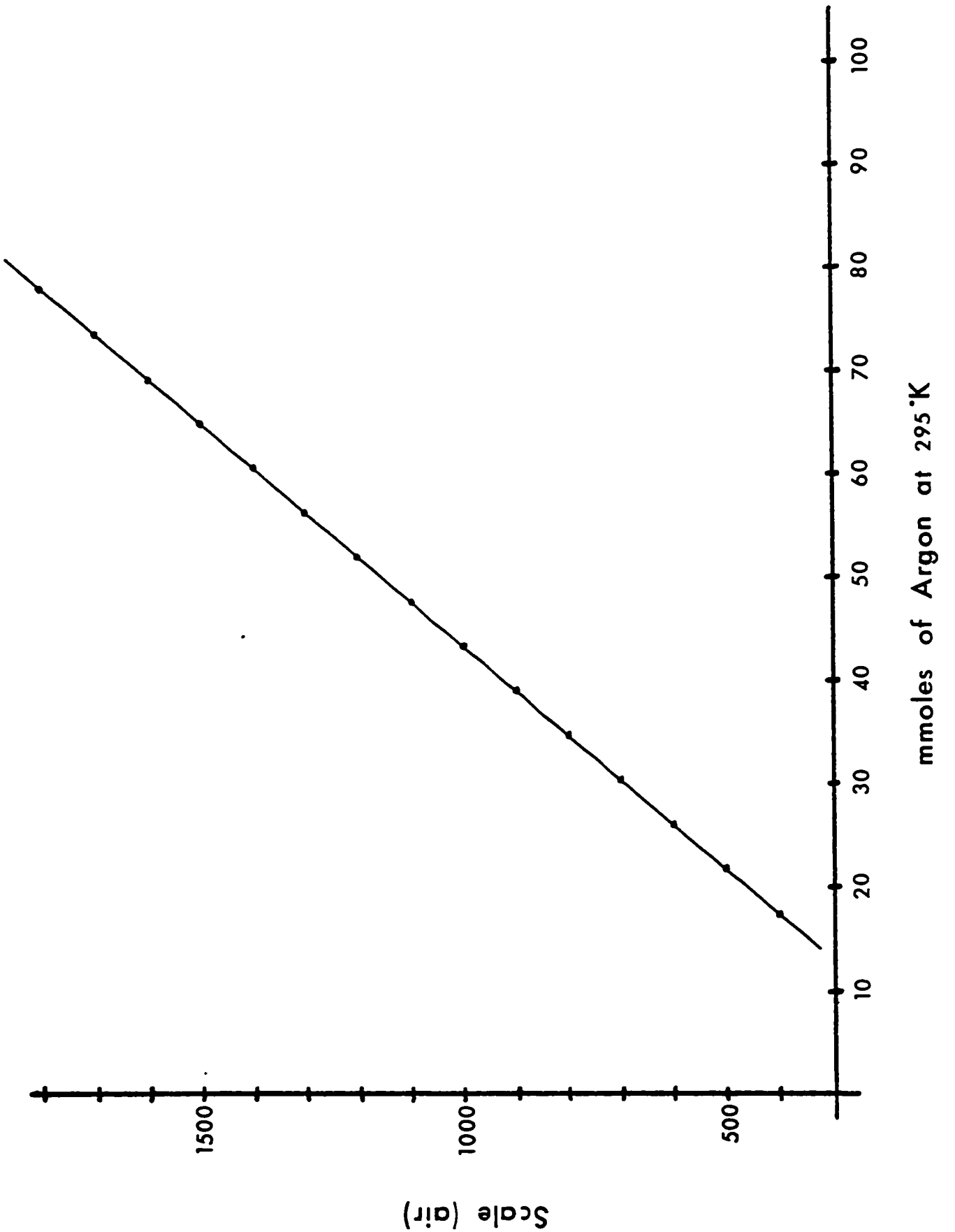
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Span	Pressure (Torr)	P.T.	Span	Pressure (Torr)	P.T.	Span	Pressure (Torr)	P.T.	Span	Pressure (Torr)	P.T.
900	237.0	99.8	800	22.6	8.7	700	300.8	99.9	600	27.2	7.7
	220.1	92.5		35.7	13.6		283.0	94.2		42.6	12.2
	196.8	82.9		66.8	25.4		250.7	83.5		65.7	18.6
	180.1	75.7		91.0	34.6		235.0	78.4		85.8	23.7
	165.5	69.6		112.4	42.7		215.2	7 .7		111.4	31.0
	149.8	62.6		138.1	52.3		193.1	64.4		131.7	36.8
	124.0	52.2		163.1	61.7		176.6	58.9		156.5	43.5
	104.2	43.6		190.0	71.8		156.5	52.1		188.4	52.7
	85.0	35.3		211.4	79.7		135.7	45.5		215.8	60.5
	68.8	28.9		232.8	88.0		116.7	38.4		244.6	68.3
	51.0	21.4		258.5	97.5		89.2	29.9		260.1	73.4
	29.8	12.4					55.4	18.5		285.0	79.6
							22.6	7.5		314.7	88.7
										346.0	96.7
500	423.4	99.4	400	35.2	6.6	300	577.0	83.0	200	52.5	5.0
	387.1	91.6		67.8	12.7		546.8	78.6		83.0	8.0
	362.0	81.1		94.5	18.2		498.3	71.4		105.2	10.1
	327.1	77.6		139.3	26.8		452.1	67.5		127.4	12.2
	296.3	70.0		194.1	37.2		445.0	63.8		156.9	15.2
	269.0	63.8		222.3	42.5		422.7	60.4		192.5	19.0
	236.2	56.1		254.6	48.7		387.1	55.5		222.7	21.3
	207.2	49.5		282.0	53.6		344.8	49.4		260.1	25.1
	181.9	43.5		322.8	62.0		313.6	45.0		295.0	28.5
	154.0	36.7		353.1	66.9		290.5	41.7		335.6	32.9
	132.2	31.4		389.6	74.4		251.5	36.2		356.8	34.5
	111.0	26.5		418.5	79.6		206.0	29.8		410.3	40.0
	92.7	22.0		457.0	87.2		177.3	26.1		450.8	43.6
	57.6	13.6		490.6	93.0		150.2	22.0		492.7	47.5
	27.2	6.5		522.2	99.5		133.1	19.2		537.2	52.2
							102.8	15.0		588.6	57.6
							72.4	10.5			
							55.0	7.5			
							35.2	5.0			

Figure 2II.6 - Calibration curve for Gilmont #12 flow meter. The fluid is argon at 295°K with a back pressure of 1.157 atm. The calibration curve only includes the linear range.

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$$v = \frac{dx}{dt} = \frac{1}{A}\frac{dV}{dt} = \frac{1}{\pi a^2}\frac{RT}{p(60)}\frac{dn}{dt} \qquad (2II.6)$$

where V = volume

A = area

a = radius of the tube,

was 25.5 m sec^{-1} at 300°K and 1.7 Torr pressure. The factor of 1/60 in equation 2II.6 converts min^{-1} to sec^{-1}. The internal radius of the tube carrying the metastable atoms (1.0 cm) was used in this calculation, because the diffusion to the total radius of the reactor does not occur until several cm (∿10 cm) downstream from the observation window. The total argon concentration at the reaction zone was 5.5×10^{16} molec cc^{-1}; whereas, the metastable concentration varied from $\sim 2 \times 10^{10}$ to 6×10^{10} metastables cc^{-1}. Typical quencher flows were ∿2 μmoles min^{-1} which corresponds to a concentration of 2.5×10^{12} molecules cc^{-1}.

c. Gas handling and gas purity. The reagent gases used in these experiments were in general the same ones used in the total rate constant determination experiments and were handled and purified as described in Section 2I.d. In addition, other gases for which the total rate constant was not measured, i.e., XeF_2, ICl, BrF_5, etc., were also studied. With exception of XeF_2 which was prepared essentially pure (> 99.7%, see section 4.b), these reagents, due to their reactivity and the fact that in many cases exist only in equilibrium with the elemental components, had purities ranging from ∿80% to >95% in favorable cases. Since extensive purification procedures were not possible in the ss line, the operational procedure was to load the reagent gas into a reservoir and let it sit for several hours after which it was pumped out and the final load stored in the container. This serves the purpose of seasoning the surface for each reagent. A rough idea of the amount of impurities and their identity could be normally derived from analysis of the

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emission spectrum after excitation by Ar* or Xe*. Typically, the reagent mole fraction was between 0.25 and 0.45 with argon diluent to a total pressure of ∿350 torr.

High purity argon (>99.995%) was used in this work and it was further purified by passing through the low pressure molecular sieve trap at 77°K. Even with the precautions used here, with only argon flowing and the discharge on, strong OH(A → X) and somewhat weaker N_2(C → B) emissions could be detected. The H_2O problem was later greatly reduced by the introduction of a glass coil, kept at liquid nitrogen temperature, in the argon line right before the discharge. Apparently, for the Ar flows used, the molecular sieve at the center of the traps never became cold thereby decreasing the trapping efficiency. The emission from the impurities was not a problem when Xe was added to the discharge, however, because Xe($^3P_2$) does not have enough energy to excite these emissions. As in the case of the total rate constant measurements, trapping of the reagents before they reached the pumps was not attempted. With some reagents, OF_2 in particular, small "explosions" were noted in the pumps, when evacuating a 20% mixture of OF_2 in Ar from the reservoir. With regard to these highly reactive and toxic chemicals, the need for extreme precautions when handling them can not be overstated.

d. Detection System. The detection system consisted of a computer interfaced 0.75 m Jarrell-Ash (Czerny-Turner) combination spectrometer and spectrograph equipped with a cooled (0°C) E.M.I. 9558QB photomultiplier tube wired for photon counting. The effective aperture of the instrument was f/6.5 and the 1180 grooves/mm grating blazed at 300.0 nm provided 1.1 nm mm^{-1} dispersion at the exit slit. The entrance slit was positioned against the spectrosil window of the observation zone and parallel to the

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reaction flame. The signal from the photomultiplier tube was processed by an 1100 series Solid State Radiations, Inc. (SSR) photon counter consisting of a Model 1120 Amplifier-Discriminator and a Model 1105 Data Converter Console, from which the data was available as either, digital for further computer processing or analog for display on a strip chart. After an initial cooldown period of at least 24 hours, the phototube dark count was 100-150 counts sec^{-1}. Signals were typically between 10K and 3M counts sec^{-1}, although normally reagents were diluted enough to keep signals below 300K to 1M counts sec^{-1}, to avoid problems that might result from the possible non-linearity of PM tube response to these strong signals.

Emission spectra were also photographed on 4" x 5" plates Polaroid Positive/Negative film type 55 using the Jarrel-Ash. Higher resolution spectra (see Section 5.e) were obtained at Dalhousie University (Nova Scotia, Canada) with 1m 1704 x Spex Spectrometer and Kodak Spectroscopic Analysis Film. Two gratings were available with this instrument: 1200 line/mm ruled grating blazed at 1.0μ (plate factor ∿0.17 nm/mm in the 4th order at 300 nm) and a 316 lines/mm Echelle grating blazed at 62° (plate factor ∿0.07 nm/mm in the 21st order at 300 nm). In all cases, the output of a Fe hollow cathode lamp was used as the wavelength calibration standard.

e. Calibration of Spectral Response. The spectral response of the detection train was calibrated using a standard Deuterium Lamp (Optronic Labs) from 190 to 350 nm and with a standard Quartz-Iodine lamp from 300 to 850 nm. Slight differences from a previous calibration[4] that used the Branching Ratio method[5-7] were found in the 190 to 300 nm region. The resulting calibration curve, shown in Figure 2II.7, was fitted piece-wise to five (3 and 4 degree) polynomials using a non linear least squares computer program. The polynomials and their range of validity are listed in Table 2II.3.

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Figure 2II.7 - Reciprocal of correction factor vs. wavelength for the 0.75 m Jarrell-Ash Spectrometer and E.M.I. 9558QB photomultiplier tube. A standard Deuterium Lamp was used from 190 to 350 nm and a Quartz-Iodine lamp from 300 to 850 nm. The curve through the experimental points is the computer fitted polynominal. Beyond 340 nm, the correction factors are the same as those of Reference 4.

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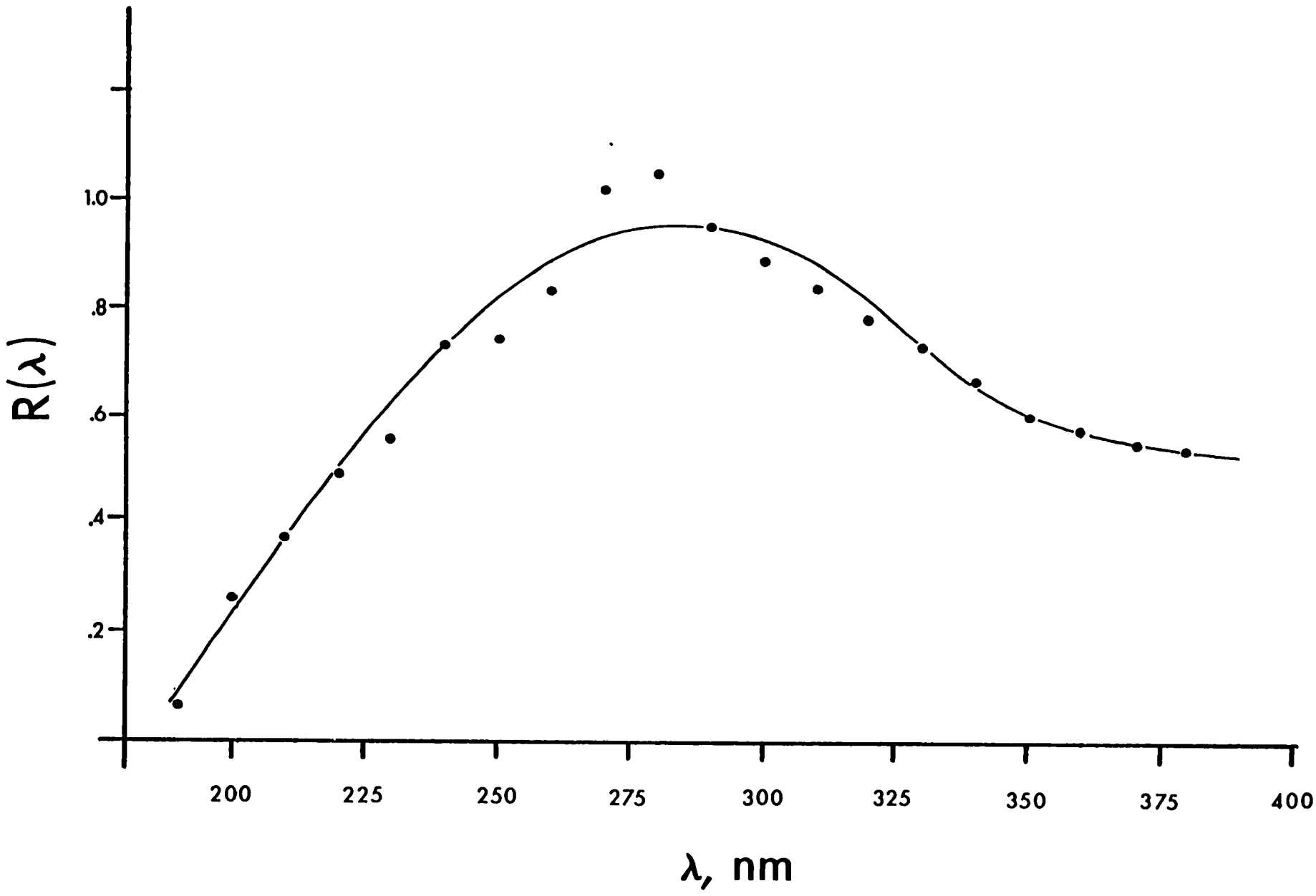


Table 2II.3 - Least squares-fitted polynomials to the reciprocal of correction factor vs. wavelength curve. The polynomial is $R(\lambda) = B_0 + B_1\lambda + B_2\lambda^2 + B_3\lambda^3$.

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Table 2II.3

Range, nm	B_o	B_1	B_2	B_3
190–340	-0.36±0.31 (1)	0.15±0.20 (-1)	0.58±0.20 (-4)	-0.20±0.10 (-6)
340–600	0.28±0.61	0.70±0.40 (-2)	-0.25±0.09 (-4)	0.21±0.06 (-7)
600–640	-0.32±0.07 (3)	0.16±0.03 (1)	-0.27±0.06 (-2)	0.15±0.03 (-5)
640–740*	0.41±0.16 (3)	-0.23±0.09 (1)	0.49±0.20 (-2)	-0.47±0.19 (-5)
740–850	0.63±0.03 (0)	-0.14±0.01 (-2)	0.81±0.04 (-6)	

* This range was best fitted by a 4th order polynomial. The coefficient $B_4 = (0.16 \pm 0.06) \times 10^{-8}$. The numbers in parenthesis correspond to the power of ten which multiplies the coefficients.

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The derived calibration curve is, in reality, the average of several calibration experiments that differed from one another in the distance between the lamp and the entrance slit of the spectrometer. For distances between 20 and 60 cm, the resultant variation in the calibration curve was less than 5% indicating that the non-complete illumination of the grating did not have a very important effect. However, considering that under actual experimental conditions the illumination of the grating is essentially completely incoherent, an error of $\pm$ 10% is associated with the response calibration curve from 200 to 800 nm.

f. Data acquisition. The Jarrell-Ash monochromator and the photon counter were interfaced to a PDP/8e computer that controlled the monochromator scan and data acquisition. Both, the interface (designed and built by Dr. M. van Swaay, Kansas State University) and the computer software for data acquisition have been described in Ref. 4 and only the highlights will be mentioned here. After calling the program JADRT1.SV the user inputs to the computer general parameters for the experiments, i.e., starting wavelength for scan, scan rate, extent of scan, number of data point to be taken per unit wavelength etc., after which the computer sends this information to the interface where the requested starting λ is compared to the monochromator's counter reading. Depending on the result of this comparison, the interface will drive the monochromator to the requested wavelength or indicate to the computer that it is ready to acquire data. With the slits shutter closed, the dark count is then taken and this number stored in the computer to later be substracted from all the other data counts. At this point the computer informs the user that it is ready to proceed with the scanning of the spectral region desired. If the go-ahead is given, the computer signals the interface to begin the scan at the same time that enables the photon counter. The photomultiplier tube signal processed by the SSR

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is then taken by the interface from the digital output of the 1105 Data Converter console and the pulses are continuously counted by the internal counter of the interface that simultaneously stores this information in an output buffer from where the computer retrieves the signal at a rate controlled by the computer's internal clock. Every time the computer retrieves signal from the interface, the counter is reset. These individual information pulses are stored in the computer memory as double precision (16-bit words) binary numbers. Up to 3712 such points can be stored in the computer's memory. If the memory is filled, the computer will stop the scan, write all of the information on magnetic tapes and then resume the scanning and data acquisition. When the interface signals to the computer that the end of the requested scan has been reached, the computer halts the scanning, writes the information on tape, closes the file and informs the user that the experiment has been completed.

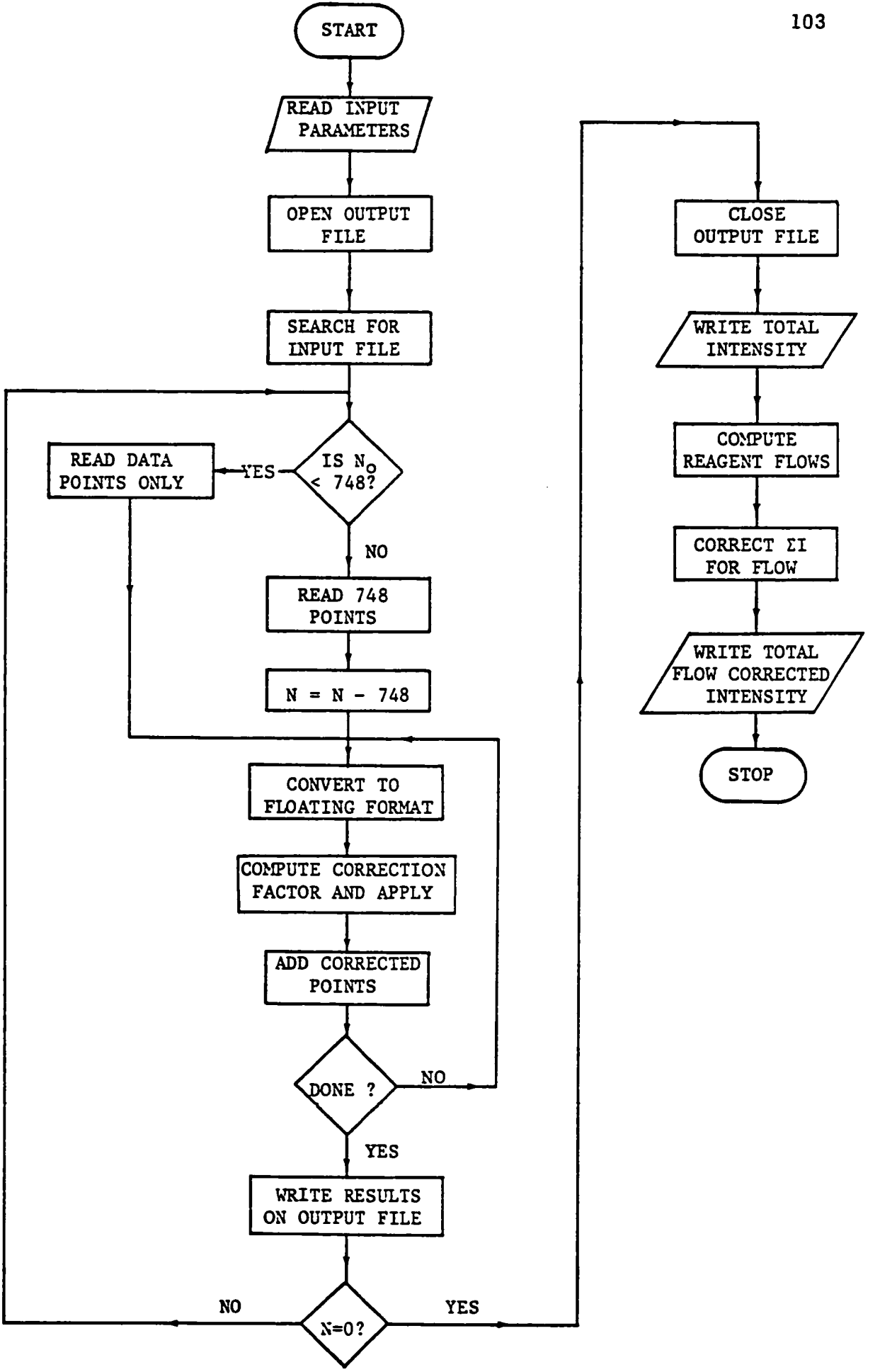
In order to obtain the total integrated intensities, a new program was written. A flow chart for INTEG3.SV is shown in Figure 2II.8. This program uses the data stored in magnetic tapes by JADRT1. The double precision binary numbers are first converted to FORTRAN II compatible floating point format and then individually corrected for the spectral response of the detection system using the polynominals listed in Table 2II.3. The total intensity is obtained by simple addition of all the data points in the file. Finally, the reagent flow is calculated and the integrated intensity corrected for reagent flows.

Usually, data points were taken at 0.1 nm intervals for the low pressure spectra and at 0.05 nm or less for the high pressure spectra. When scanning over atomic lines normally 100 or more points per nm were taken. It was found that the integrated areas for atomic lines varied by as much as 15%

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Figure 2II.8 - Flow chart for INTEG3.SV. The program is used to correct the data for the spectral response of the detection system and obtain the total integrated intensity.

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in back-to-back scans. This problem was traced back to wear in the monochromator screw drive mechanism. For longer scans however, this problem was completely absent.

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CHAPTER 3 TOTAL QUENCHING RATE CONSTANTS

"... whoever does not know the first
principles of things is even less
able to understand the end."

Vannuccio Biringuccio
"De la Pirotechnia" 1540

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a. Master Equation for Xe* Decay. In absence of diffusion, the decrease in the metastable concentration along the flow reactor, -d[Xe*]/dt, due to interactions with the buffer gas and the added quencher reagent is given by

$$\frac{-d[Xe^*]}{dt} = -v\,\frac{d[Xe^*]}{dz} = k_2[Xe^*][Ar] + k_3[Xe^*][Ar]^2 + k_Q[Xe^*][Q] \qquad (3.1)$$

where z is the distance down the flow tube, v the flow velocity, k_2 and k_3 the two and three body deexcitation rate coefficients, and k_Q is the rate constant for quenching of Xe* by molecule Q present at a concentration [Q].

Diffusion will add an extra loss term of the form[1]

$$D_{Xe^*}\left(\frac{1}{r}\right)\frac{\partial}{\partial r}\left(r\,\frac{\partial[Xe^*]}{\partial r}\right) + \frac{\partial^2[Xe^*]}{\partial z^2} \qquad (3.2)$$

where D_{Xe^*} is the diffusion coefficient of Xe* in Ar and r is the (radial) distance from the center of the tube. If the azimuthal variations ($\partial^2[Xe^*]/\partial z^2$) are small, and if only diffusion is considered, the metastable concentration decays according to[2]

$$[Xe^*] \propto \sum_{i=1} J_o(\lambda_i\, r/a)\, \exp[-\lambda_i^2\, D_{Xe^*}\, z/a^2 v] \qquad (3.3)$$

where $J_o(\lambda_i\, r/a)$ is the ith mode of the zero order Bessel function and λ_i is the ith root of J_o. The values of λ_i increase very rapidly with i, and therefore, an initial distribution of modes will rapidly relax to the fundamental mode[2] (i=1). Assuming that diffusion is controlled by the fundamental mode, and that all metastables are destroyed on collision with the walls,[3] Equation 3.1 can be integrated in the plug flow approximation[4] to give

$$\ln([Xe^*]/[Xe^*]_o) = -\{(D_o/\Lambda^2[Ar]) + k_2[Ar] + k_3[Ar]^2 + k_Q[Q]\}\,(z/v) \qquad (3.4)$$

Here the o subscript in D_o is employed to emphasize that the fundamental mode is dominant in the diffusion, in which case the characteristic diffusion

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length Λ is given by[5]

$$\Lambda^{-2} = (\pi/d)^2 + (2.405/a)^2 \qquad (3.5)$$

where d is the length of the reactor and a its radius. Finally, substitution of Equations 2I.6 and 2I.4 in Equation 3.4 yields the final equation that relates the absorption of Xe* to the different destruction processes:

$$(1/\gamma)\ln[\ln(I_o/I)] = C - \{D_o/\Lambda^2 p) + k_2 p + k_3 p^2 + k_Q[Q]\} \quad (z/v) \qquad (3.6)$$

In Equation 3.6, [Ar] has been replaced by the total reactor pressure, p, and C is a term that collects several constant terms.

At constant pressure, k_Q can be measured in (at least) two ways. Under pseudo first order conditions ([Q] >> [Xe*]), the metastable absorption is followed as a function of distance, and the resulting linear plot has a slope equal to $k_Q[Q]/v$; this process is repeated for a number of different quencher concentrations, and finally the slopes of these plots, multiplied by the bulk flow velocity, are plotted vs. [Q]. The resulting straight line with slope equal to k_Q and an intercept that depends on the Xe* decay in argon buffer gas (see Section 3.d). Alternatively, metastable absorption is measured as a function of quencher concentration at a fixed point z. The slope of this plot gives the decay constant $\Gamma = k_Q z/v$. In this case however, it is difficult to accurately determine the distance z because, since the mixing of metastables and quencher is not instantaneous, the quencher inlet can not be taken as zero five. In order to circumvent this problem, decay constants are measured at several distances down the flow tube and then plotted vs distance. The resulting straight line intercepts the z axis at a point z' which is taken as the virtual point of mixing[4] or virtual origin. The

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distances of the observation points are then referred to this virtual origin[6] to extract k_Q. Since Equations 3.4 and 3.6 are derived for plug flow, the rate constants derived by either method must be corrected for flow characteristics. The correction factor is 1.15, as discussed in Section 2I.e.

b. Example of k_Q Determination. The first method to obtain k_Q as described in the previous Section, is applied here to obtain the total rate constant for quenching of $Xe(^3P_2)$ by C_2H_6. The first three columns in Table 3.1 list actual experimental data for this reaction (for one ethane concentration). With the detection system positioned at the distance listed in the first column, the intensity of the metastable diagnostic line (881.9 nm for $Xe(^3P_2)$) is monitored with the discharge off (no metastables present) to obtain I_o and with the discharge on (metastables present) to obtain I. The distance is converted to time by dividing by bulk flow velocity,

$$v = \frac{dx}{dt} = \frac{RTf_B}{p\pi a^2} \tag{3.7}$$

This procedure is repeated for several different quencher flows and the resulting decay curves are shown in Figure 3.1. The quencher concentrations are calculated from Equation 3.8.

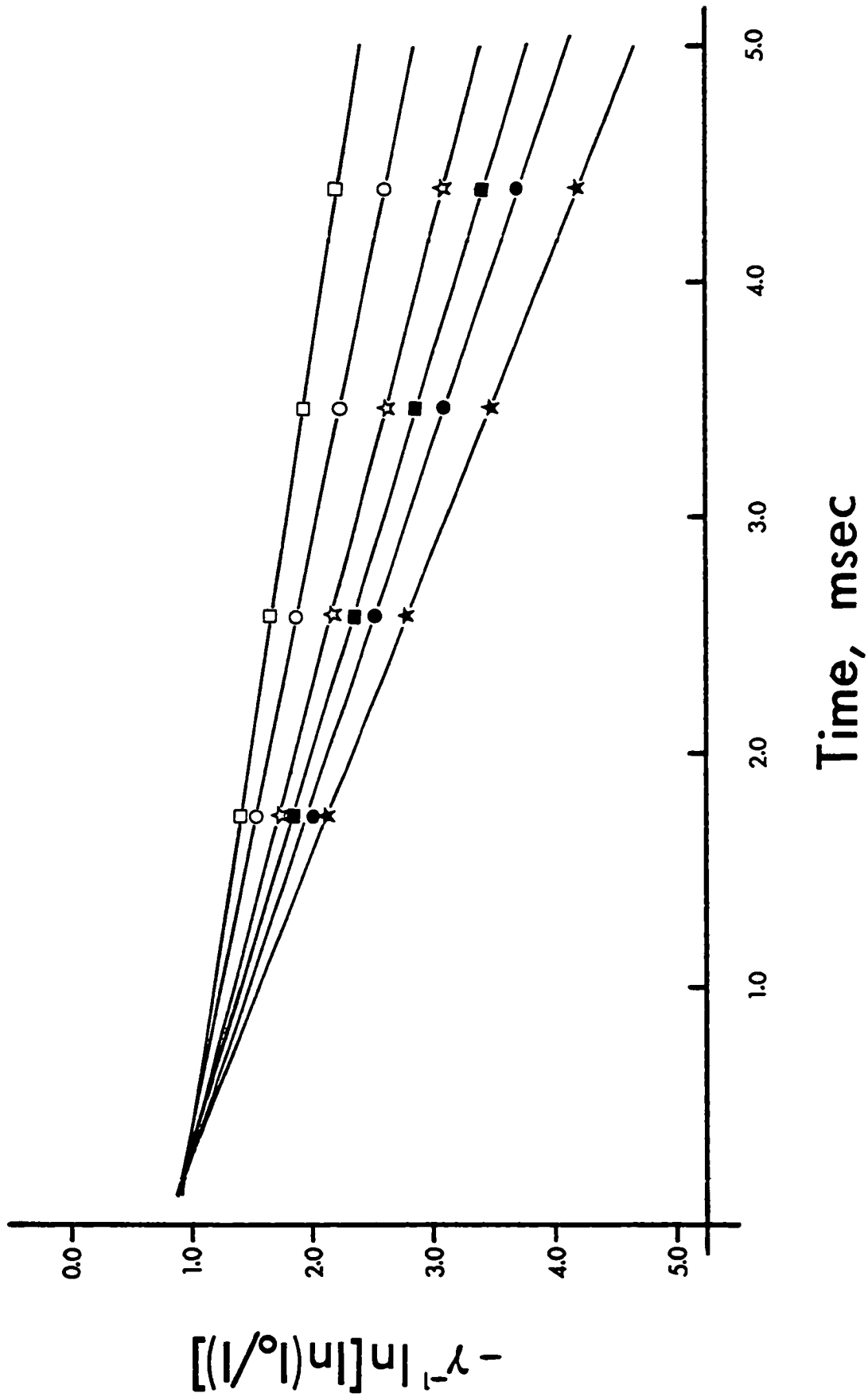
$$[Q] = [Ar]\ (f_Q/f_B) = (p/RT)(f_Q/f_B) \tag{3.8}$$

Finally, the slopes of the decay curves of Figure 3.1 (the pseudo-first order decay constants, K) are plotted vs [Q] as shown in Figure 3.2, and the slope of the resulting line is k_Q. Table 3.2 lists the pseudo-first order rate constants and quencher concentrations used in Figure 3.2 as well as the least squares fit parameters for this set of values. The flow corrected rate constant is 1.15 k_Q, see above.

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Figure 3.1 - Variation of [Xe($^3P_2$)] along the reactor axis for Xe* + C_2H_6, at four observation points downstream of the reagent inlet: 13.4, 20.0, 26.8 and 34.0 cm. The slopes of the lines are the decay constants, K. The flow speed was 77.2 m/sec and the pressure 0.81 Torr. The C_2H_6 concentrations were (from top to bottom): 1.22, 2.66, 6.02, 7.26 and 9.43 in units of 10" molecules ml^{-1}.

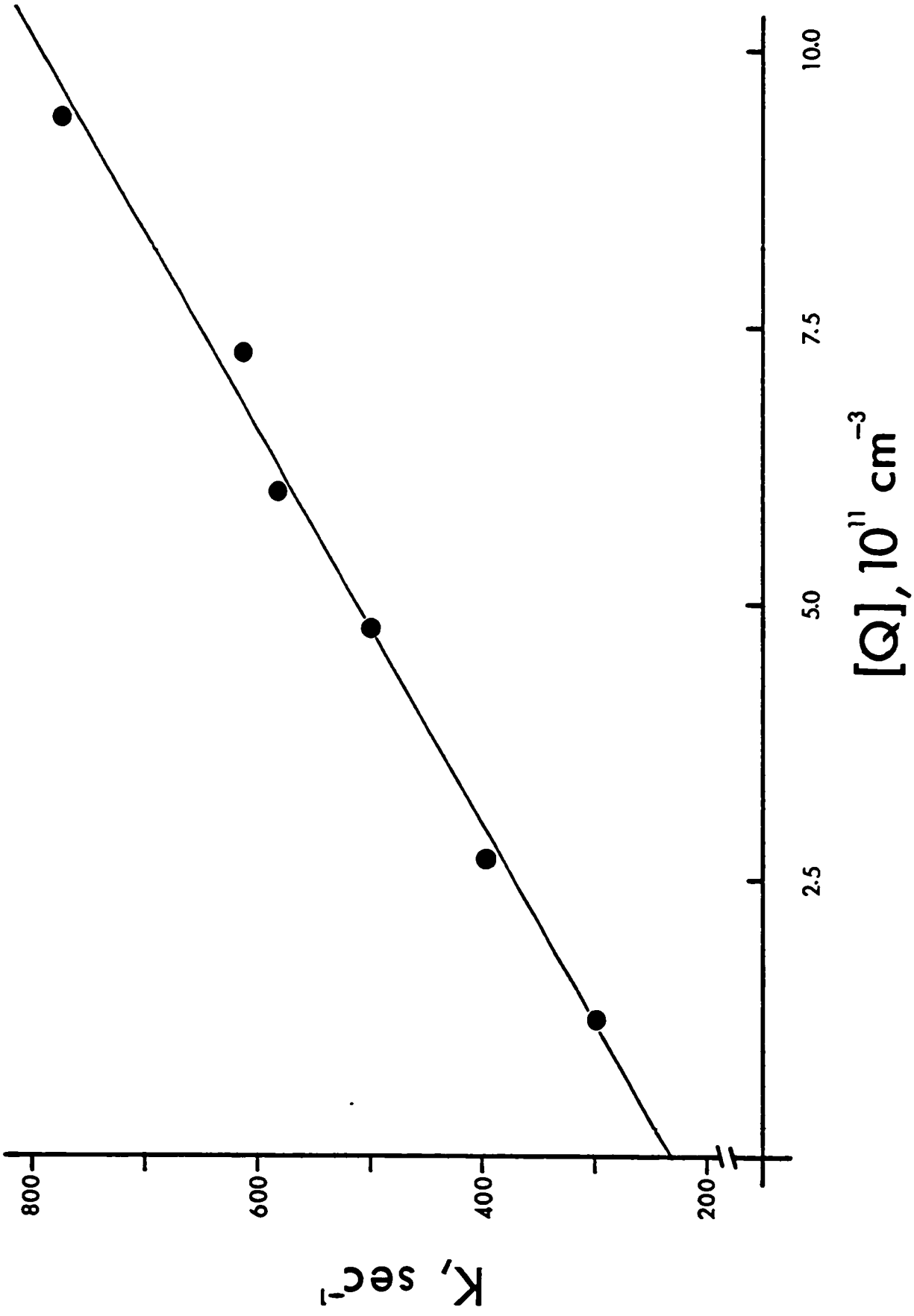
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Figure 3.2 - Decay constants vs. $[C_2H_6]$. The least squares fit parameters are listed in Table 3.2.

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Table 3.1 - Experimental data for $Xe(^3P_2) + C_2H_6$,

Table 3.2 - Decay constants for different C_2H_6 concentrations, the uncertainty listed correspond to one standard deviation.

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Table 3.1

z, cm	I_o, mV	I, mV	t, msec	$\frac{1}{\gamma}\ln[\ln(I_o/I)]$
13.40	14.78	11.38	1.736	-1.409
13.40	14.79	11.38	1.736	-1.406
13.40	14.82	11.41	1.736	-1.408
13.40	14.86	11.47	1.736	-1.419
20.00	13.73	11.20	2.590	-1.671
20.00	13.72	11.17	2.590	-1.661
20.00	13.75	11.18	2.590	-1.654
20.00	13.75	11.18	2.580	-1.954
26.80	12.85	10.98	3.471	-1.942
26.80	12.89	11.01	3.471	-1.940
26.80	12.92	11.03	3.471	-1.936
26.80	12.94	11.03	3.471	-1.926
34.00	13.55	12.00	4.404	-2.213
34.00	13.58	12.01	4.404	-2.202
34.00	13.59	12.02	4.404	-2.202
34.00	13.60	12.04	4.404	-2.210

Least squares fit parameters: $(1/\gamma)\ln[\ln(I_o/I)]$ vs. time

Slope = $-(300.0 \pm 1.9)$ sec^{-1}

Intercept = $-(0.888 \pm 0.006)$

Table 3.2

K, sec^{-1}	[Q] x 10^{-11} molec/cc
300.0 ± 2.0	1.223
399.2 ± 2.1	2.657
500.5 ± 1.4	4.757
583.4 ± 3.5	6.016
610.0 ± 4.1	7.257
773.2 ± 2.8	9.428

$k_Q = (55.19 \pm 2.82) \times 10^{-11}$ cm^3 $molec^{-1}$ sec^{-1}

Intercept = (239.4 ± 16.6) sec^{-1}

Flow corrected $k_Q = (63.5 \pm 3.2) \times 10^{-11}$ cm^3 $molec^{-1}$ sec^{-1}

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c. Results and error analysis. Thermal (∿ 300°K) rate constants for quenching of $Xe(^3P_2)$ at ∿0.85 Torr by a variety of reagents are listed in Table 3.3. Also listed are the average thermal quenching cross sections, $\sigma_Q = k_Q/\langle v \rangle$ with $\langle v \rangle = (8kT/\pi\mu)^{1/2}$, the mean Boltzman speed. The listed values were obtained by computer analysis of the data; a summary of the pseudo-first order rate constants and quencher concentrations for all k_Q's is presented in Appendix B. The rate constants as presented in Table 3.3 are, in general, the result of one determination. Tests for reproducibility were performed for several reagents and the agreement was better than 90%. Three measurements with the same Cl_2/Ar mixture gave 70.4, 64.8 and 66.3 x 10^{-11} cm^3 molec sec^{-1} for the quenching rate constant; whereas, in two determinations with different Br_2/Ar mixtures done several months apart, gave 60 and 63 x 10^{-11} cm^3 molec sec^{-1} for k_{Br_2}. No literature values for the rate constants are available for comparison.

In order to measure the total rate constant for quenching of $Xe(^3P_2)$ by Hg, a mercury boiler was added to the system before the reagent inlet to the reactor. The reservoir containing triple-distilled mercury was heated to 350°C with a heating mantle, and a flow of argon carried the mercury vapor. The Ar/Hg mixture was passed through a condenser at room temperature before entering the reaction zone. To a good approximation, the Hg concentration can be calculated from the carrier gas flow, the pressure and the temperature at the condenser, assuming that the Ar/Hg mixture reach thermal equilibrium at the condenser. No appreciable quenching was observed even for the highest Ar/Hg flows for which traces of N_2 in the Ar became the dominant quencher. The value listed in Table 3.3 was estimated from the experimental conditions in the fast flow reactor, and later checked in the emission system using a comparison method (see Section 4.d).

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Table 3.3 Experimental results for the total Xe($^3P_2$) quenching rate constants are cross sections. Numbers in parenthesis correspond to one standard deviation in the value of the rate constants.

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Table 3.3 Quenching rate constants for $Xe(^3P_2)$

Quenching Molecule	k_Q [a]	σ_Q [b]
H_2	1.6 (0.1)	0.9
N_2	1.9 (0.2)	3.7
CO	3.6 (0.3)	7.0
CO_2	45 (3)	103
O_2	22 (1)	44
NO	30 (2)	59
N_2O	55 (4)	100
HBr	61 (4)	173
HCl	56 (4)	119
Cl_2	71 (3)	193
SF_6	28 (2)	94
CH_4	33 (3)	49
C_2H_6	64 (3)	125
C_2H_4	58 (3)	110
C_2H_2	70 (3)	130
C_3H_8	54 (2)	123
C_4H_{10}	68 (1)	170
C_5H_{12}	72 (4)	192
CH_3F	44 (2)	91
CH_2F_2	41 (2)	99
CHF_3	0.2 (0.01)	0.6
CF_4	<0.03	<0.1
Hg	<0.001	<0.004
F_2	75 (4)	161
Br_2	60 (2)	202

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Table 3 continued.

Quenching Molecule	k_Q [a]	σ_Q [b]
ICl	50 (5)	171
OF_2	57 (3)	139
CF_3OF	47 (3)	143
SeF_6	65 (3)	246
TeF_6	63 (4)	230
NF_3	9.0 (0.8)	23
NOF	39 (8)	93
NOCl	51 (1.5)	135
CF_3I	52 (7)	184
PBr_3	19[c] (1.5)	72[c]
$SOCl_2$	71 (8)	224

a) k_Q units are 10^{-11} cm^3 $molec^{-1}$ sec^{-1}. Numbers in parenthesis correspond to one standard deviation.

b) $\sigma_Q = k_Q \langle v \rangle$, $\langle v \rangle = (8kT/\pi\mu)^{1/2}$ is the mean Boltzman speed. The units are $(\mathring{A})^2$.

c) These values are probably too low. See Chapter 6.

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In all cases, the computer fitted "decay curves" (Fig. 3.1) had standard deviations of less than 5%, typically $\sim \pm$ 2%, whereas for the K vs [Q] plots (Fig. 3.2) the standard deviations were $\sim \pm$ 10% for extreme cases. The relative uncertainty of the measured k_Q's, therefore, is estimated as $\pm$ 10% except for the very reactive quenchers for which, due mainly to problems in the purification of the samples, relative uncertainties of 20 to 25% are possible. Absolute uncertainties are much more difficult to estimate due to the large number of parameters involved. From Table 3.1, it can be seen that the scatter of the $(1/\gamma)\ln[\ln(I_o/I)]$ values is in general less than 1% and can be ignored. The reaction time on the other hand, has uncertainty arising from conversion of the distance measurements to times. In addition to the flow characteristics, there is some uncertainty associated with the experimental measurements. The contribution to error in the determination of the linear flow velocity from the latter can be estimated from

$$\frac{\Delta v}{v} = \frac{\Delta f_B}{f_B} + \frac{\Delta T}{T} + \frac{\Delta p}{p} \qquad (3.9)$$

This expression is obtained by applying logarithms to both sides of Equation 3.7 and by approximating differentials by increments. The three variables on the right hand side of Equation 319 are not really independent because variations in the buffer gas flow will produce variations in the reactor pressure and similarly for changes in the temperature. However, for the present purposes they are treated as independent variables and the largest experimentally observed change (during a "typical" run) are used. Buffer gas flows could be measured to $\pm$ 1% (see Section 2I.c.i); whereas, the temperature changed by $\sim$ 3°K during the 4 hours necessary to collect data for one run. The pressure was 0.85 $\pm$ 0.02 Torr. Therefore, $(\Delta v/v) = 0.01 + 0.01 + 0.02 = 0.04$, or a 4% uncertainty. Since distances

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could be measured to within 2 mm, the total uncertainty in the reaction time is of the order of 5%. From Equation 3.8, the fractional uncertainty in the quencher gas concentration is

$$\frac{\Delta[Q]}{[Q]} = \frac{\Delta p}{p} + \frac{\Delta T}{T} + \frac{\Delta f_Q}{f_Q} + \frac{\Delta f_B}{f_B} \qquad (3.10)$$

Under favorable conditions, i.e., for the reagent gases that could be handled and purified easily, quencher/argon mixtures were prepared with less than 2@ uncertainty in the mole fraction. For the more "untractable" ones, the mole fraction was as much as 10 to 15% uncertain. In addition quencher flow measurements were reproducible to within $\pm$ 2% or better, depending on flow range, back pressure etc. Using the same fractional uncertainties as before for T, p and f_B, Equation 3.10 gives $0.08 \leq (\Delta Q/Q) \leq 0.25$. Finally, errors due to flow characteristics (see Section 2I.e) including pressure drop etc., were estimated as $\sim \pm$ 25%. The maximum absolute uncertainties for the rate constants measured in the present study are therefore estimated to be between 40% and 60%; the spread is mainly governed by sample preparation.

d. Decay of Xe* in Ar. Preliminary studies were also conducted to determine the two and three body rate coefficients of $Xe(^3P_2)$ in pure argon, as well as the diffusion coefficient, D_o^{Xe*}. In the absence of added quencher gas, equation 3.6 reduces to

$$(1/\gamma)\ln[\ln(I_o/I)] = C-[(D_o/\Lambda^2 p) + k_2 p + k_3 p^2]t \qquad (3.11)$$

$$\frac{d}{dt}\left(\frac{1}{\gamma}\right)\ln\left[\ln\left(\frac{I_o}{I}\right)\right] = -K = -\frac{D_o}{\Lambda^2 p} - k_2 p - k_3 p^2 \qquad (3.12)$$

Therefore, the pseudo-first order rate constants, K, measured as a function of argon pressure can be fitted to the right side of equation 3.12 to obtain the coefficients. Alternatively the following procedure can be employed to

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evaluate D_o, k_2 and k_3. Three sets of experimental data are listed in Table 3.4. Multiplying both sides of Equation 3.12 by p yields

$$pK = \frac{D_o}{\Lambda^2} + k_2 p^2 + k_3 p^3 \tag{3.13}$$

and the plot of pK vs p^2 gives a curve with a p=0 intercept of D_o/Λ^2 determined this way, equation 3.13 can be rearranged to

$$\frac{pK - (D_o/\Lambda^2)}{p^2} = k_2 + k_3 p. \tag{3.14}$$

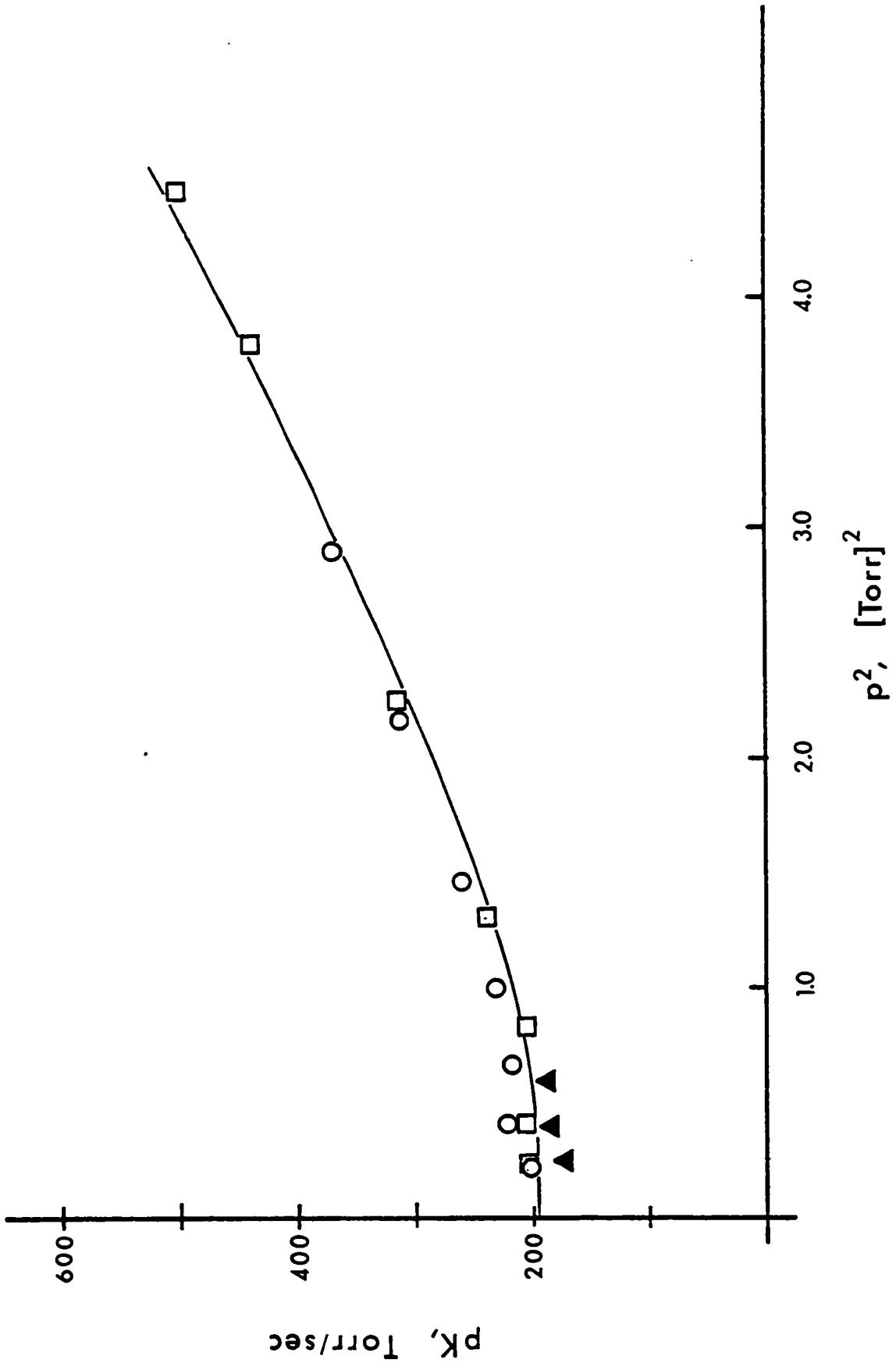
A plot of the left hand side of equation 3.14 vs. p should be straight line slope equal to k_3 and intercept equal to k_2. This plot is shown in Figure 3.4. The data shows considerable scatter but follow the general form of the rate law.

A non-linear computer fit[7] of the data in Table 3.4 to equation 3.12, shown in Figure 3.5 is the preferred way to assign the values of the coefficients. These are listed in Table 3.5; the correction factor of 1.15 due to flow characteristics has been included. Preliminary results of new experiments at lower Xe flows[16] (concentrations) gave a diffusion coefficient that agrees very well with the value reported here. However, the two and three body rates obtained in this reinvestigation are much smaller than the values listed in Table 3.5. Apparently, the pressures reached in the present study were not high enough to sample the two- and three-body regimes; this can be seen in Figure 3.5 where the preliminary results of Kolts[16] are also plotted. Agreement between the two sets of data is good up to ∿2 torr, but Kolts' data does not show a minimum even at ∿4 torr. The points above 2 torr were obtained by throttling the flow, whereas, the flow was throttled above ∿1.2 torr in the new work. For Kolts' data, the points near the onset

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Figure 3.3 - Plot of pK vs. p^2 for Xe($^3P_2$) in Argon buffer gas. The intercept gives D_o/Λ^2. The different symbols represent the three different sets of experimental points.

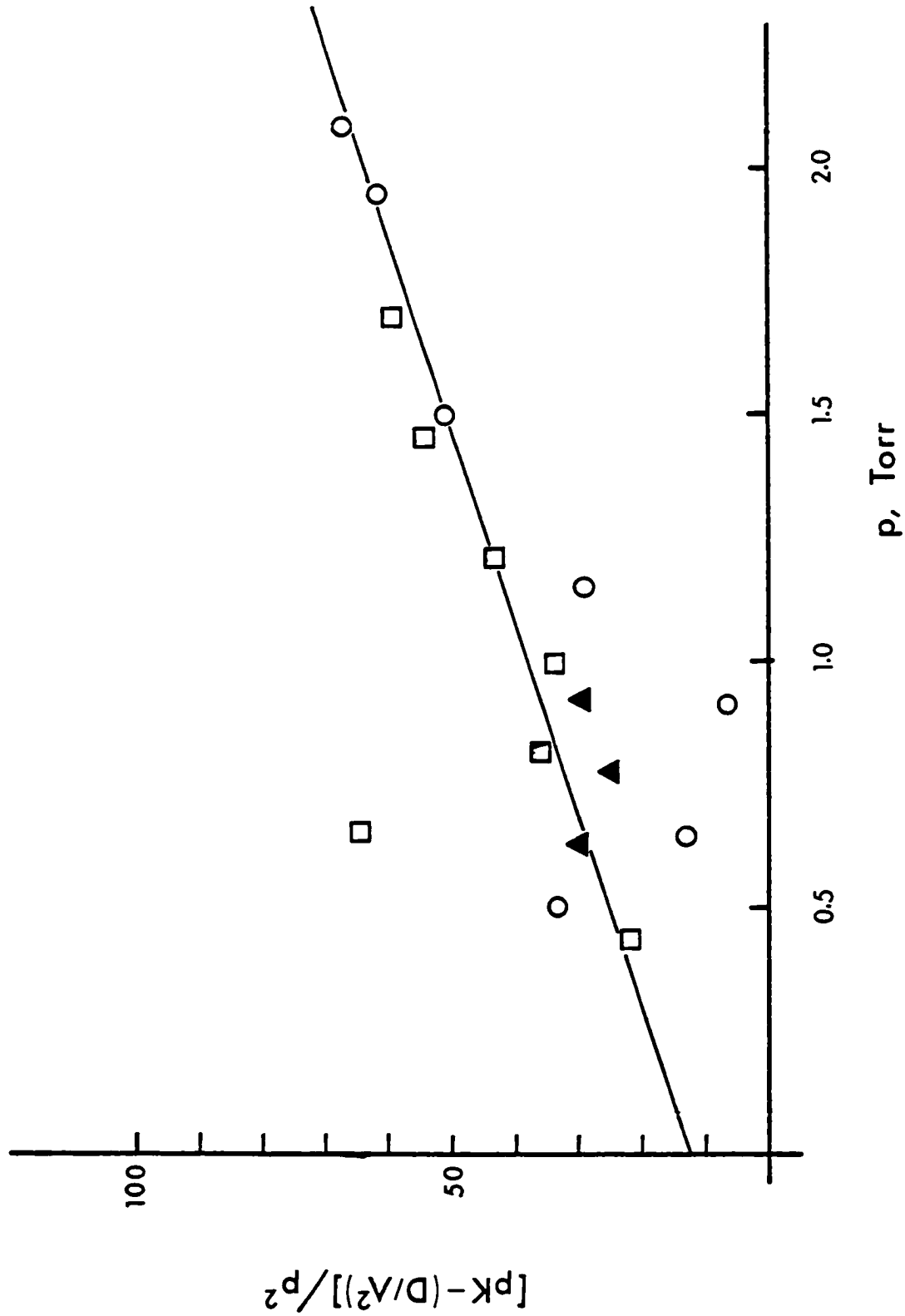
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Figure 3.4 - Plot of $\frac{pK - (D_o/\Lambda^2)}{p^2}$ vs. p, for the same data of Figure 3.3.

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Table 3.4 - Decay constants for $Xe(^3P_2)$ as a function of total reactor pressure.

Table 3.5 - "Best values" for D_o, k_2 and k_3 obtained from a computer fit of the data in Table 3.4 to Equation 3.12.

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Table 3.4

Pressure,* Torr	K, sec^{-1}	Pressure, Torr	K, sec^{-1}	Pressure, Torr	K, sec^{-1}
0.47	425	0.5	407	0.5	340
0.64	347	0.64	320	0.63	285
0.81	270	0.91	225	0.78	235
1.00	229	1.15	208	0.93	207
1.21	213	1.50	210		
1.47	213	1.95	225		
1.70	216	2.10	237		
		2.44	260		

* This is the argon buffer pressure. The $[Xe(^3P_2)]$ is $\sim 2 \times 10^{10}$ cc^{-1}.

Table 3.5

D_o	$= (2.7 \pm 0.1) \times 10^{18}$ cm^{-1} sec^{-1}
k_2	$= (3.6 \pm 7.0) \times 10^{-16}$ cm^3 $molec^{-1}$ sec^{-1}
k_3	$= (2.2 \pm 0.9) \times 10^{-32}$ cm^6 $molec^{-2}$ sec^{-1}

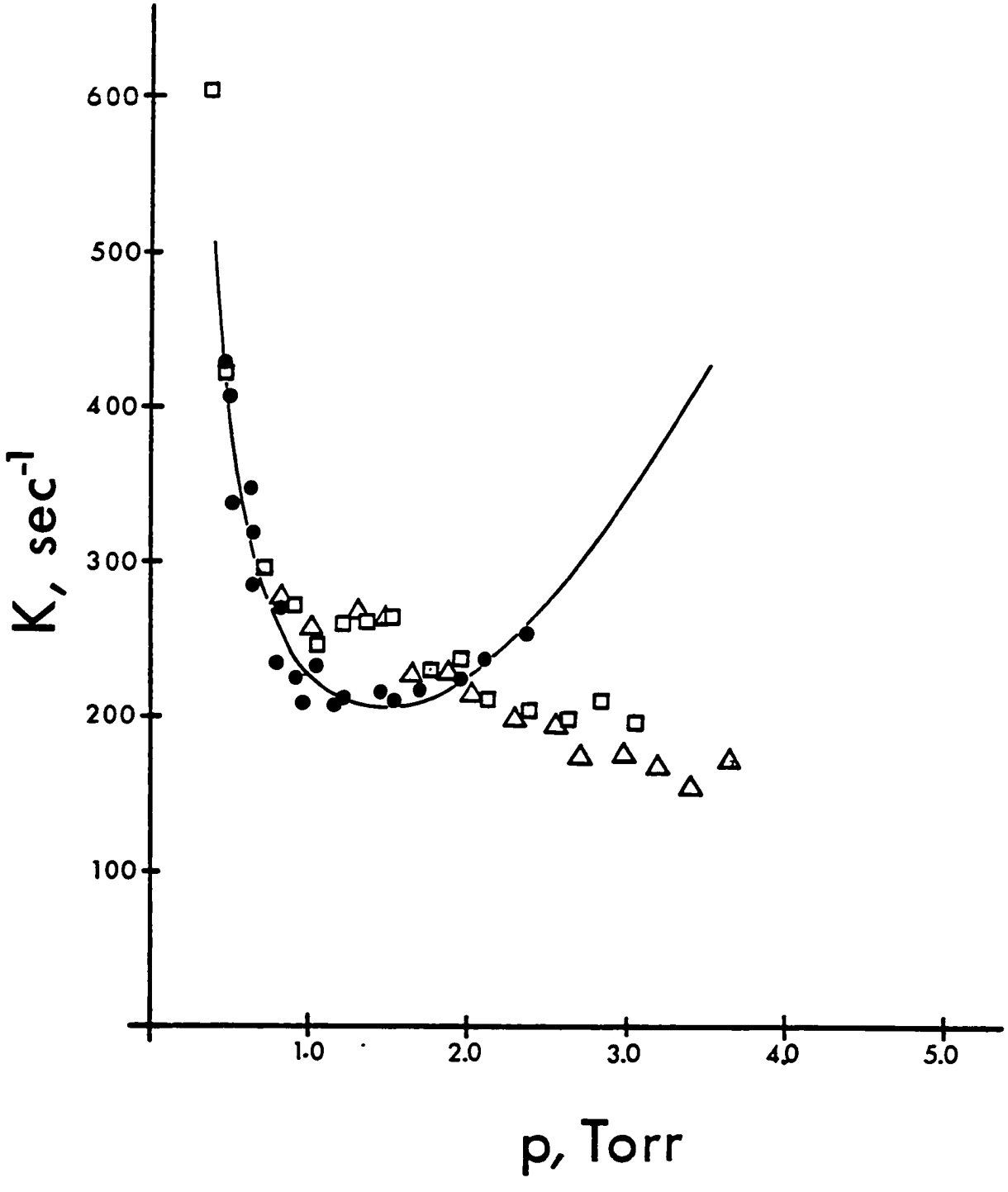
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of throttling are higher than expected (for the two runs). The reason for this is not fully understood, although it may be related to change in the flow characteristics. It is possible that the three points above 2 torr in the present study are similarly affected by the throttling, and since the two- and three-body collision rate coefficients are essentially determined by these points, the different results from the two studies can be rationalized. Replacement of the present flow meter by a larger one will permit a study of the Xe* decay at higher pressures without the need for throttling. The decay of $Xe(^3P_2)$ in argon has not been reported in the literature. However there are several studies of the decay of $Xe(^3P_2)$ in pure xenon[8-11] as well as individual determinations of the three body rate coefficient in xenon[12,13] (see section 1.d for a review of Xe* decay processes in Xe, Kr and Ar). The diffusion coefficient measured in the present study is a factor of five larger than Wieme's,[9] as expected since argon has a smaller collision diameter. The two body rate coefficient is an order of magnitude smaller than the reported values for $Xe(^3P_2)$ + Xe[8-11]. This is what one would expect since two-body quenching of the metastable states of the rare gases is thought to involve collision induced emission.[14] Similar trends were observed in $Ar(^3P_2)$ decay studies in Ne.[15] The three-body rate coefficient for $Xe(^3P_2)$ + 2Ar is $\sim$ half the literature values for $Xe(^3P_2)$ + 2Xe. The low two and three body coefficients are consistent with the presumed lack of bound molecular states correlating with Ar + $Xe(^3P_2)$[17c].

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Figure 3.5 - Decay constant for $Xe(^3P_2)$ as a function of argon pressure. The dark symbols were obtained in the present study. Open symbol are recent data obtained by J. Kolts.[16] The solid curve is the computer fitted decay equation to the dark experimental points.

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c. R. Atzmon, O. Cheshnovsky, B. Raz and J. Jortner, Chem. Phys. Lett., 29, 310 (1974).

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CHAPTER 4 PRODUCT CHANNELS AND SPECIFIC RATE CONSTANTS

"... more is owing to what we call chance, that is, philosophically speaking, to the observation of events arising from unknown causes, than to any proper design, or pre-conceived theory in this business."

Joseph Priestley
"Experiments and Observations on Different Kinds of Air" Vol. II, 1775.

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a. Product channels. Collisional quenching of the metastable states of the rare gases M*, by a molecule AB, gives different products depending on the relative importance of various possible exit channels, i.e., Penning ionization, associative ionization, ion-pair formation, excitation transfer, dissociation, dissociative excitation or reactive quenching:

$$M^* + BC \xrightarrow{k_{PI}} M + BC^+ + e \quad (4.1)$$

$$\xrightarrow{k_{AI}} (MBC)^+ + e \quad (4.2)$$

$$\xrightarrow{k_{IP}} M + B^+ + C^- \quad (4.3)$$

$$\xrightarrow{k_{ET}} M + BC^* \quad (4.4)$$

$$\xrightarrow{k_{D}} M + B + C \quad (4.5)$$

$$\xrightarrow{k_{DE}} M + B + C^* \text{ (or } M + B^* + C) \quad (4.6)$$

$$\xrightarrow{k_{RQ}} MB^* + C \quad (4.7)$$

For $Xe(^3P_2)$, Penning and associative ionization (process 4.1 and 4.2) are energetically forbidden for the quenchers used in the present study. The formation of MB* by reactive quenching (process 4.7) is probably the most significant finding of this investigation: it is the first example of similar (classical) chemical behavior between the metastable states of the heavier rare gases and the alkali metal atoms. Normally, production of MB* is followed by radiative transition to the repulsive (or weakly bound) ground state,

$$MB^* \xrightarrow{\tau^{-1}} M + B + h\nu \quad (4.8)$$

and this property has been exploited to produce new powerful and efficient lasers in the u.v. In some cases, predissociation to M + B* can be expected.

The "Branching Fraction" for formation of a given product is defined as

$$BR_i \equiv \frac{k_i}{\sum_j k_j} = \frac{\sigma_i}{\sum_j \sigma_j} \quad (4.9)$$

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where k_i or σ_i is the specific rate constant or cross section for the process leading to the desired product, and $\sum_j k_j$ and $\sum_j \sigma_j$ are the total quenching rate constant, k_Q, and cross section, σ_Q (see Chapter 3). Attempts were made to identify the products of excitation transfer (process 4.4) for all quenchers, but branching ratios for this process were not determined. The branching ratios for reactive quenching (process 4.7) however, were of particular interest in this work and the specific rate constants k_{RQ} were determined using the comparison technique[1] described below.

b. The comparison method. Recently Piper, et. al.[2] assigned absolute $Ar(^3P_{0,2})$ excitation rate constants for excitation of Kr to several states in the Kr(5p) manifold and proposed the use of Kr as an internal standard to obtain other excitation cross sections by simply comparing the relative emission intensities of Kr* and the other species. This method was successfully used in the vacuum ultraviolet by Gundel, et. al.,[1a] in their analysis of the products resulting from the interaction of $Ar(^3P_{0,2})$ with a series of chlorinated reagents. In the present work, the method was extended to cover excitation by $Xe(^3P_2)$ (and also $Kr(^3P_2)$).

The product channels for $Ar(^3P_2)$ excitation of Kr are[3] 89.5% formation of $Kr(5p[3/2]_2)$ and 10.5% $Kr(5p[3/2]_3)$, with a total rate constant[3] of 6.2×10^{-12} cm^3 $molec^{-1}$ sec^{-1}; the channels for $Ar(^3P_0)$ excitation have not been fully identified;[2] although, it is expected that some $Kr(5p[3/2]_{2,1})$ is formed. The contribution of $Ar(^3P_0)$ to $Kr(5p[3/2]_{2,1}$ production, however, can be neglected (see below) because $Ar(^3P_0)$ constitutes only ~10% of the total Ar metastable concentration, and the total rate constant for $Ar(^3P_0)$ quenching by Kr is only 2.3×10^{-12} cm^3 $molec^{-1}$ sec^{-1}, see Ref. 3.

The intensity of the fluorescence for the $Kr(5p[3/2]_2 \rightarrow 5s[3/2]_2)$ transition at 760.2 nm can be written as

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$$I(Kr^*)_{760} = (0.76)(0.895)\ k_2^{Kr}[Kr][Ar(^3P_2)] \qquad (4.10)$$

where the first factor (0.76) is the measured[2] radiative branching ratio for this transition and the second is the fraction of the total excitation transfer ending in the $5p[3/2]_2$ level. If the contribution from $Ar(^3P_0)$ excitation is taken into account, Equation 4.10 should be written

$$I(Kr^*)_{760} = (0.76)(0.895)k_2^{Kr}[Kr][Ar(^3P_2)] + R_o k_o^{Kr}[Kr][Ar(^3P_0)] \qquad (4.11)$$

where R_o is the (undetermined) product of collisional and radiative branching ratios.[1a] Similarly, for any other reagent BC, excited by $Ar(^3P_{0,2})$ to states that can radiate (or to states that predissociate producing fragments that can radiate) and in the absence of quenching or radiationless transitions, the fluorence intensity is

$$I(BC^*) = k_2^{BC^*}[BC][Ar(^3P_2)] + k_o^{BC^*}[BC][Ar(^3P_0)] \qquad (4.12)$$

If a mixture of Kr and BC is reacted with $Ar(^3P_{0,2})$, the ratio of the BC* fluorescence to the Kr* 760.2 nm fluorescence is

$$\frac{I(BC^*)}{I(Kr^*)_{760}} = \frac{k_2^{BC^*}[BC]\ \{1 + (k_o^{BC^*}/k_2^{BC^*})([Ar(^3P_0)]/[Ar(^3P_2)]\}}{k_2^{Kr}[Kr]\ \ \{(0.76)(0.895) + R_o(k_o^{Kr})([Ar(^3P_0)]/[Ar(^3P_2)])\}} \qquad (4.13)$$

The second term in the denominator, as discussed above, is less than (0.3)(1/3)(1/10)[1a,2] and can be safely dropped. Depending on the experimental conditions and on the nature of BC, the second term in the numerator of Equation 4.13 may or may not be dropped. Since in general, for efficient quenchers $k_2^{BC^*} \gtrsim k_o^{BC^*}$ as a rule,[3] this second term in the numerator might contribute 10% of the total BC* fluorescence intensity (up to 20% in some cases). Equation 4.13 can now be written as

$$\frac{I(BC^*)}{I(Kr^*)_{760}} = \frac{\beta k_2^{BC^*}[BC]}{(0.68)\ k_2^{Kr}[Kr]} \qquad (4.14)$$

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and finally, the <u>total</u> excitation rate constant becomes

$$k^{BC*} = \beta k_2^{BC*} = \frac{I(BC^*)[Kr]}{I(Kr^*)_{760}[BC]} \quad (0.68)(6.2 \times 10^{-12}) \qquad (4.15)$$

with $1.0 < \beta < 1.2$. The units are cm^3 $molec^{-1}$ sec^{-1}.

In order to extend this method to $Xe(^3P_2)$ excitation, the absolute rate constant for $Ar(^3P_{0,2})$ excitation of the Cl_2^* 256 nm continuum $(E^3\Pi_{0+g} \rightarrow B^3\Pi_{0+u})$[4] was first measured using the krypton standard and this continuum was then used as a secondary standard against which all other emissions excited by $Xe(^3P_2)$ were compared. This Cl_2^* continuum was selected for the following reasons: (1) Most of the $Xe(^3P_2)$ excited emissions studied here are extended continua.[1b,5] It was therefore desirable to compare the emission intensities against a continuum rather than an atomic line. (2) The Cl_2^* continuum occurs in the same spectral region as most emissions excited by Xe*. (3) Cl_2 can be handled and purified relatively easily

Experimental values for the Kr(760.2 nm) line and Cl_2^*(256 nm) continuum integrated intensities are listed in Table 4.1, for three different Kr/Cl_2 concentration ratios: 1%, 15% and 50%. We assumed that $Ar(^3P_0)$ also excites the Cl_2^* 256 nm emission with a rate constant similar to $Ar(^3P_2)$ excitation. For these experiments the concentration of $Ar(^3P_0)$ was ∿15% of the total and therefore $\beta \sim 1.15$. The average of 35 determinations gives $k_{Ar}^{Cl_2^*} = (2.0 \pm 0.2) \times 10^{-11}$ cm^3 $molec^{-1}$ sec^{-1}. The main source of scatter were the variable areas from scans over the atomic line.

Comparison of the Ar excited Cl_2^* continuum with another emission excited by $Xe(^3P_2)$ is straightforward . For example, the ratio of the fluorescence intensities now takes the form

$$\frac{I(Cl_2^*)}{I(XeCl^*)} = \frac{k_{Ar}^{Cl_2}\,[Cl_2][Ar(^3P_0)]}{k_{Xe}^{XeCl^*}[Cl_2][Xe(^3P_2)]} \qquad (4.16)$$

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Table 4.1 Experimental results for the comparison of the Kr* 760.2 nm line and the Cl_2^* 256.0 nm continuum excited by Ar*($^3P_{2,0}$).

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Table 4.1

Emission	Q flow μmol/min	(ΣI)/Q flow, 10^{12} counts/mol	$\Sigma I_{Cl_2^*}/\Sigma I_{Kr^*}$	$k_{Ar}^{Cl_2^*}$
Kr*	0.365	0.244	5.4	2.2[a]
Cl_2^*	2.995	1.322		
Kr*	0.365	0.277	4.8	2.0
Kr*	0.318	0.261	5.5	2.3
Cl_2^*	2.610	1.447		
Kr*	0.194	0.274	6.2	2.4
Cl_2^*	1.589	1.702		
Kr*	2.185	0.654	4.3	1.8
Cl_2^*	2.971	2.745		
Kr*	2.185	0.818	3.4	1.5
Cl_2^*	2.971	2.802		
Kr*	1.796	0.595	4.1	1.7
Cl_2^*	2.441	2.418		
Kr*	2.854	1.214	4.6	2.0
Kr*	2.654	1.191	4.7	2.0
Cl_2^*	2.654	5.544		
Kr*	1.796	0.750	3.4	1.5
Cl_2^*	2.441	2.557		
Kr*	2.103	0.724	4.2	1.8
Cl_2^*	2.103	3.072		
Kr*	1.632	0.843	3.8	1.6
Cl_2^*	1.632	3.171		
Kr*	1.824	0.720	4.1	1.7
Cl_2^*	1.824	2.985		
Kr*	2.470	1.077	5.0	2.0
Kr*	2.470	0.982	5.4	2.2
Cl_2^*	2.470	5.343		

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Table 4.1 continued

Emission	Q flow μmol/min	$(\Sigma I)/Q$ flow, 10^{12} counts/mol	$\Sigma I_{Cl_2^*}/\Sigma I_{Kr^*}$	$k_{Ar}^{Cl_2^*}$
Kr*	1.764	0.236	4.1	1.7
Cl_2^*	1.764	0.965		
Kr*	1.764	0.230	4.2	1.8
Kr*	0.360	0.251	3.9	1.6
Cl_2^*	2.040	0.972		
Kr*	0.360	0.268	3.7	1.5
Kr*	0.976	0.066	3.7	1.5
Cl_2^*	1.354	0.235		
Kr*	0.976	0.056	4.2	1.8
Kr*	1.874	0.036	5.6	2.3
Cl_2^*	1.874	0.201		
Kr*	1.874	0.037	5.4	2.2
Kr*	1.476	0.038	4.6	2.0
Cl_2^*	1.476	0.174		
Kr*	1.476	0.036	4.8	2.0
Kr*	1.332	0.036	4.5	1.9
Cl_2^*	2.050	0.161		
Kr*	2.037	0.044	4.9	2.1
Cl_2^*	1.805	0.214		

a. The units for the entries in this column are 10^{-11} cm^3 $molec^{-1}$ sec^{-1}.

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for the comparison of the Cl_2^* 256 continuum and the XeCl* emission (see Figure 4.1), but the absolute metastable atom concentrations are needed before $k_{Xe}^{XeCl^*}$ can be extracted. When the $[Xe(^3P_2)]/[Ar(^3P_2)]$ experimental ratio derived from the values of Table 2I.2 ($\sim$ 0.25) was used in Equation 4.16, a total $Xe(^3P_2)$ excitation rate constant for Cl_2 $\sim$2 times larger than the measured total quenching rate constant (see Section 3.d) was obtained. This is untenable and required further effort to determine the ratio.

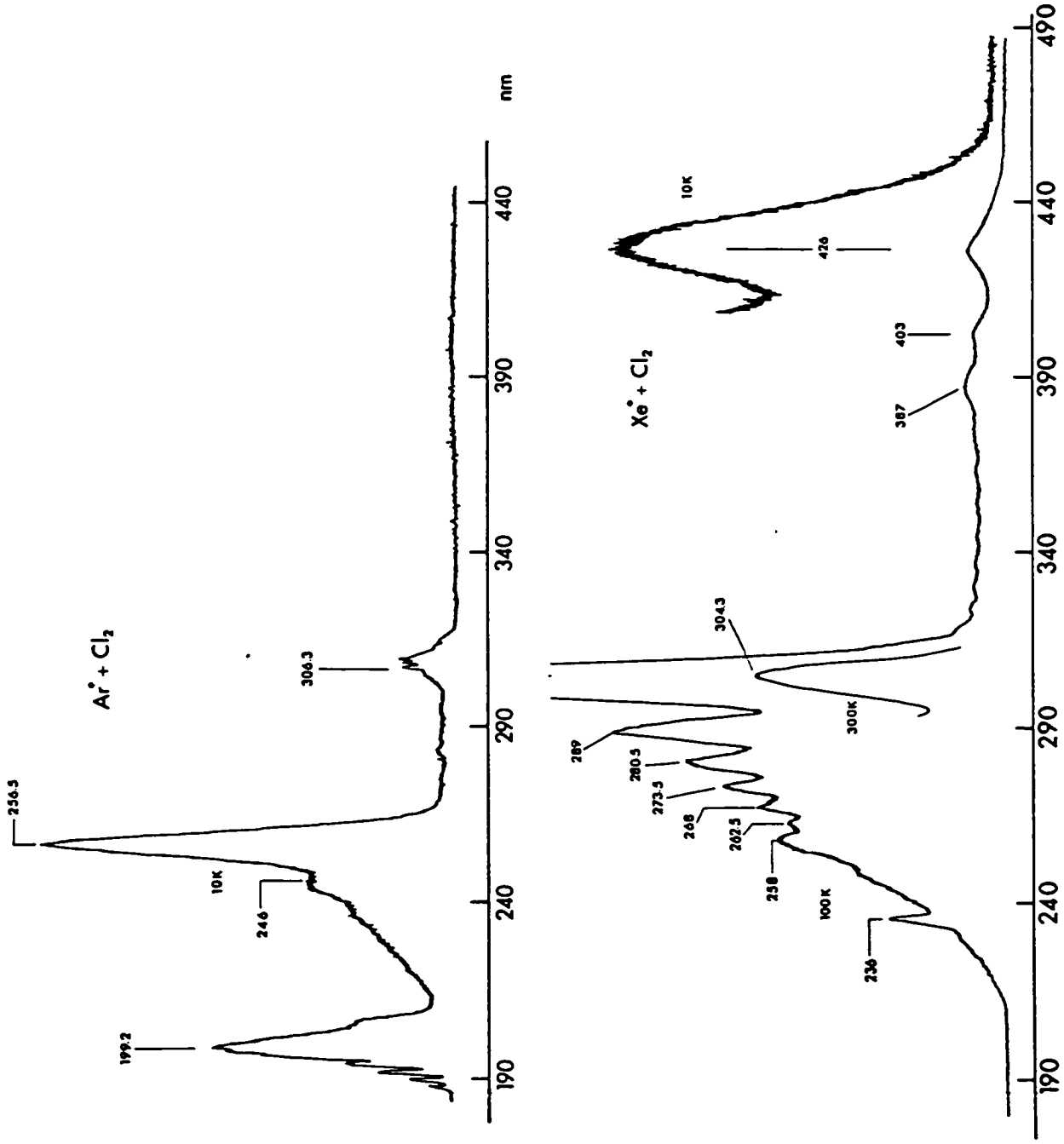
Similar absorption experiments with $Kr(^3P_2)$ were done to check the method, and gave metastable $Kr(^3P_2)$ concentrations that were fully consistent with the expected results based upon the excitation transfer from $Ar(^3P_2)$ to Kr(5p) and the radiative branching of the Kr*(5p) states[2] to the Kr(5s, $^3P_2$) and Kr(5s, $^3P_1$) states. The calculated transition probabilities of Murphy[6] were used to evaluate Kr(I) f-values, and the experimental results are shown in Table 4.2. When the rate constants for KrCl* formation were determined from Equation 4.15, the values agreed with the measured k_Q.

These results were an indication that the oscillator strengths used to calculate $[Xe(^3P_2)]$ were probably in error. In order to estimate the metastable Xe concentration, the following experiments were done under constant pressure, temperature, chlorine/krypton mixture and main argon flows. The $[Ar(^3P_2)]$ was first measured and then the Cl_2/Kr mixture was allowed into the reaction zone. $I(Cl_2^*)$ and $I(Kr^*)_{760}$ were recorded. Krypton was then introduced before the discharge until no $Ar(^3P_2)$ was present and I(KrCl*) was also recorded. After diverting the Cl_2/Kr flow from the reaction zone, $[Kr(^3P_2)]$ was measured, and finally the Kr flow was replaced by Xe and $[Xe(^3P_2)]$ and I(XeCl*) were determined. The excitation rate constants for Cl_2^* 256 nm continuum and KrCl* emission as well as

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Figure 4.1 - Comparison of the Ar excited Cl_2^* 256 nm continuum and the XeCl* emission for the same chlorine flow. The ratio of the integrated emission intensities is proportional to the ratio of the formation rate constants. Note the change in photon counter sensitivity. Slits, 120μ; reactor pressure, 1.6 Torr; chlorine concentration, ~ 4.7×10^{12} molecules cm^{-3}.

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Table 4.2 - Krypton and Argon metastable concentrations from Atomic Absorption.

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Table 4.2

M	transition	λ(nm)	f	A	$k_o\ell$[a]	10^{10} [M*][b] atoms/cm^3
Ar	$^3P_2$-4p[5/2]$_3$	811.5	0.380	0.168[c]	0.802	5.1
	-4p[3/2]$_2$	763.5	0.223	0.085[c]	0.354	4.1
Kr	$^3P_2$-5p[3/2]$_2$	760.2	0.2575	0.063[d]	0.250	1.7
	5p[5/2]$_2$	810.4	0.1026	0.028[d]	0.106	1.7
	5p[5/2]$_3$	811.3	0.4839	0.126[d]	0.550	1.9
Ar	$^3P_2$-4p[5/2]$_3$	811.5	0.380	0.205[c]	1.05	6.7
	4p[5/2]$_2$	801.5	0.105	0.061[c]	0.241	5.6
	4p[3/2]$_2$	763.5	0.223	0.122[c]	0.530	6.1
Kr	$^3P_2$-5p[3/2]$_2$	760.2	0.2575	0.069[d]	0.275	1.9
	5p[5/2]$_2$	810.4	0.1026	0.030[d]	0.111	1.8
	5p[5/2]$_3$	811.3	0.4839	0.141[d]	0.638	2.2

a. $k_o\ell$ was obtained from Figure 2I.7

b. $[M^*] = 7.15 \times 10^{13} (T/M)^{1/2} (k_o\ell/f\lambda)$, λ in Å

c. Ref. 3.

d. Calculated from transition probabilities of Ref. 6, using the relationship $f_{ik} = 1.499 \times 10^{-16}\lambda^2(g_k/g_i)A_{ki}$, λ in Å.

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the [Ar($^3P_2$)] and [Kr($^3P_2$)] agreed with previously found values. The ratio I(XeCl*)/I(KrCl*) was 2.4. However, since the total quenching rate constants for Kr($^3P_2$) and Xe($^3P_2$) by Cl_2 are very similar[1b,3] and since the only observed products from Kr($^3P_2$) + Cl_2 are KrCl* and Cl_2* 256 nm (trace), and XeCl* for the interaction of Xe($^3P_2$) with Cl_2,[1b] the value obtained for the I(XeCl*)/I(KrCl*) ratio can only be explained if [Xe($^3P_2$)] ∿ 2[Kr($^3P_2$)]. Such a large Xe metastable concentration on the other hand, is not expected from excitation transfer from Ar($^3P_2$) to Xe followed by radiative cascade to the Xe*(6s) states.[6] Direct production of Xe($^3P_2$) in the discharge was therefore postulated. This hypothesis was confirmed by measuring the I(XeCl*)/I(KrCl*) ratio when Xe and Kr were added before and after the discharge. The ratio changed from 2.45 for addition before the discharge to 1.15 for addition after the discharge (the latter corresponds to Xe($^3P_2$) and Kr($^3P_2$) formation by excitation transfer from Ar($^3P_2$).

The results of the Xe($^3P_2$) absorption measurements suggest that the f-values of Chen and Garstand[8] used to calculate the Xe($^3P_2$) concentrations of Table 2I.2 are a factor of ∿2 too large as pointed out in Section 2I.g. Detailed consideration of the hyperfine structure in Xe might conceivably alter the interpretations of the absorption measurements, although the large number of Xe isotopes render this effect a very difficult one to evaluate.

For the present work, the [Xe($^3P_2$)]/[Ar($^3P_2$)] ratio was taken as ∿0.55 and the [Kr($^3P_2$)]/[Ar($^3P_2$)] ratio was set to the measured value of 0.36. The branching ratio for reactive quenching (or excimer formation) of Xe($^3P_2$) by Cl_2:

$$Xe(^3P_2) + Cl_2 \xrightarrow{k_{XeCl}} XeCl^* + Cl \qquad (4.17)$$

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was assigned the value of 1.0 because: (i) comparison experiments with $Kr(^3P_2)$ for which the absolute metastable concentration could be measured reliably, showed that the branching ratio for excimer formation, $k_{KrCl}/k_{Cl_2}^{Kr}$, was 0.90.[1b] (ii) the ratio of the total rate constants for quenching of Kr* and Xe* by Cl_2 is 1.0, whereas, the ratio of the excimer formation rate constants, k_{XeCl}/k_{KrCl}, is > 1.0. (iii) No other emission was observed[23] for the reaction of Xe* with Cl_2.

Under conditions such that Equation 4.12 is valid, i.e., no electronic quenching and no non-radiative transitions, the rate constant for reactive quenching of $Xe(^3P_2)$ by Q can be obtained from comparing unknown emissions to the XeCl* emission.

$$k_{RQ} = k_{XeCl} \frac{I_{XeX}^{Q}/[Q]}{I_{XeCl}^{Cl_2}/[Cl_2]} \qquad (4.18)$$

where I_{XeX}^{Q} = fluorescence intensity of excimer product

$I_{XeCl}^{Cl_2}$ = fluorescence intensity of XeCl from Xe* + Cl_2

$k_{XeCl} = 72 \times 10^{-11}$ cm^3 molec^{-1} sec^{-1}.

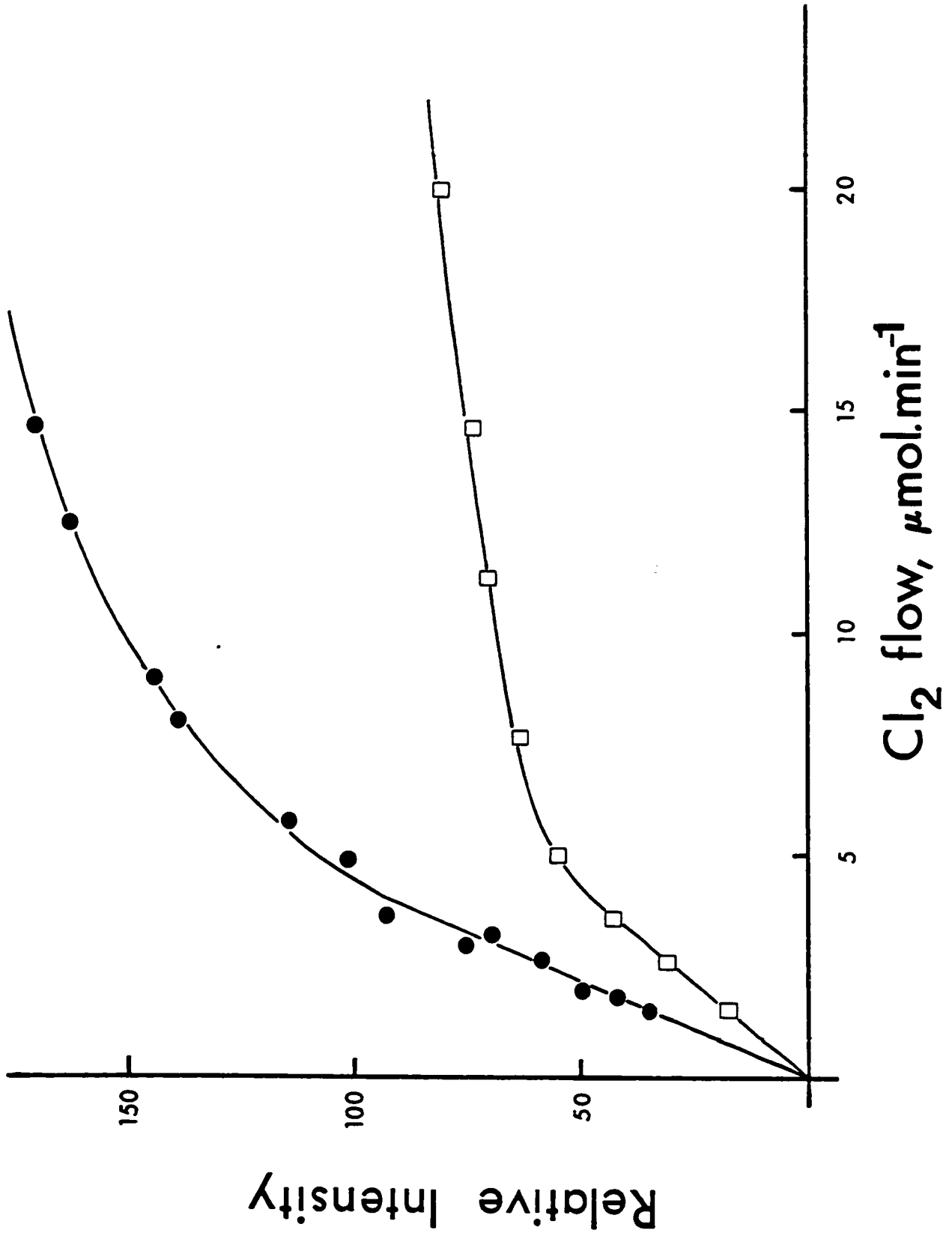
Ideally the absolute emission intensities should be employed, this is not necessary in actual practice providing the same fraction (geometric) of the total intensity (corrected for the wavelength dependence of the detector efficiency) be measured for each reaction.

Equation 4.18 is valid only under strictly first order conditions, and consequently, it was necessary to determine the range of quencher flows for which the intensity of the metastable excited emission is linear with quencher flow. Experiments were done to monitor the XeCl* and KrCl* emission as a function of Cl_2 flow. Results are shown in Figure 4.2. In both cases it was found that for Cl_2 flows larger than $\sim 5 \times 10^{-6}$ moles min^{-1}, the

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Figure 4.2 - Excimer emission intensity as a function of added Cl_2 flow: ● XeCl*, monochromator slits = 150μ; ▢ -KrCl*, monochromator slits = 120μ. For the relative rate constant measurements, reagent flows never exceeded 3 μMoles min^{-1} which corresponds to 6.3×10^{12} molec. cc^{-3}.

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intensity was not linear due to the depletion of the metastable concentration. A quencher flow of 4×10^{-6} moles min^{-1} was set as the upper limit for emission intensity comparisons; but, quencher flows of the order of 2×10^{-6} moles min^{-1} were typically employed, corresponding to a quencher concentration of 2.5×10^{12} molecules cm^{-3}. Figure 4.3 shows the flames produced by the reaction of $Xe(^3P_2)$ with XeF_2, N_2 and Cl_2 respectively.

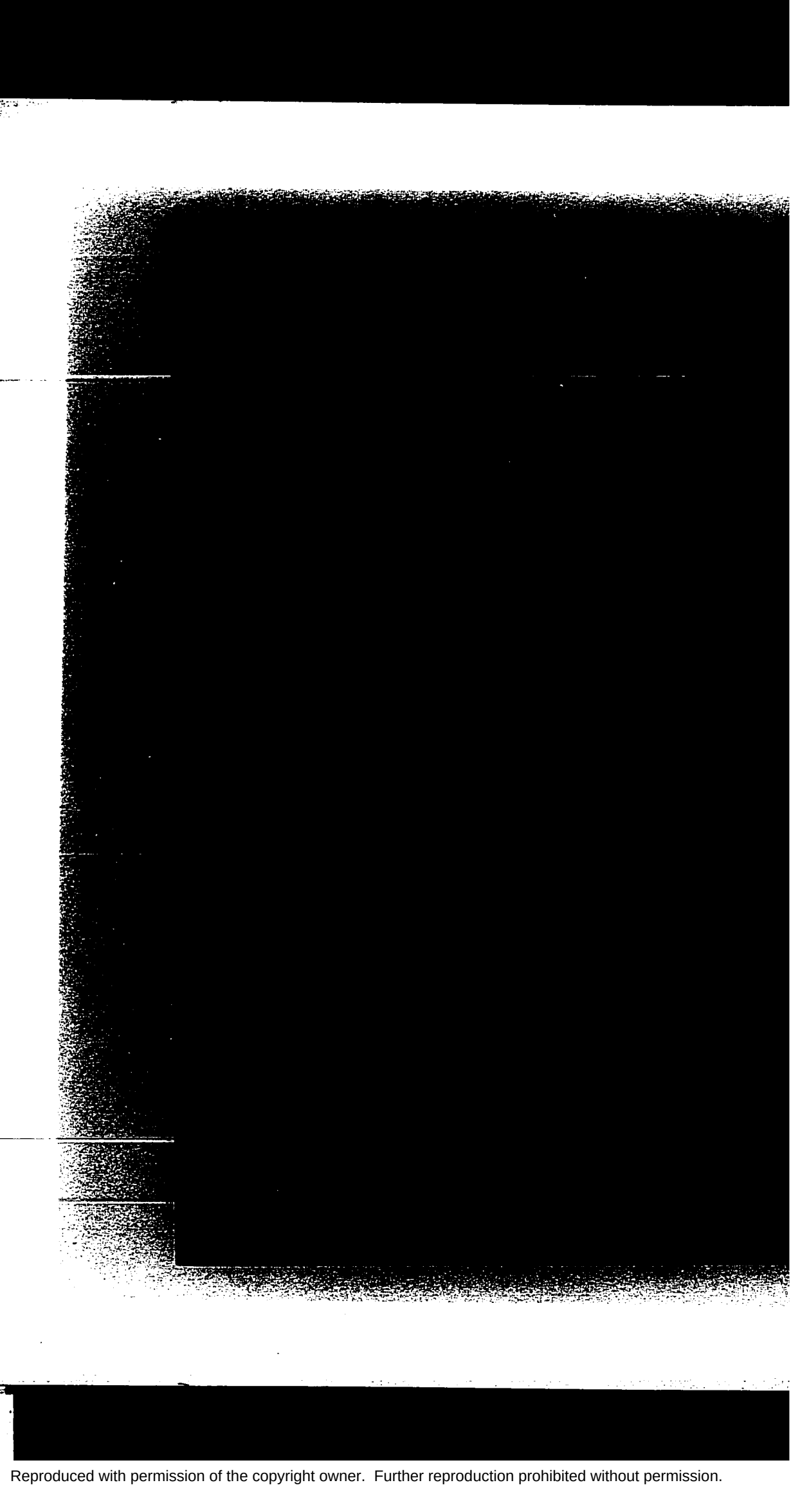
c. Halogenated Donors. The reactive quenching of the rare gas metastable states by halogens and halogen-containing molecules, has found immediate application in the production of new lasers. An efficient commercial laser however, requires a donor that can be handled safely (i.e., preferably non-toxic and non-corrosive) and leads to excimer formation with high probability. The results to be presented in this Section, specific rate constants and branching fraction for excimer formation for a large number of halogenated molecules constitute a solid data base for the selection of good donors for laser applications.

Emission spectra from the reactive quenching of $Xe(^3P_2)$ by some halogenated donors RX (X = F, Cl, Br, I) are shown in Figures 4.4 and 4.5. A detailed discussion of the characteristics of these spectra as well as of the states involved is presented in Chapter 5. For the present purposed four excimer states can be identified: the ground state, $X^2\Sigma^+$ that correlates with $Xe(^1S) + X(^2P)$ and three excited states, $B^2\Sigma^+$ and $C^2\Pi_{3/2}$ correlating with $Xe^+(^2P_{3/2}) + X^-(^1S)$, and $D^2\Pi_{1/2}$ that correlates with $Xe^+(^2P_{1/2}) + X^-(^1S)$. The measured branching ratios for reactive quenching include contributions for formation of all, B, C and D excimer states, although in general the observed emission from the D state is much weaker ($\sim$1%) than the B $\rightarrow$ X emission.

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Figure 4.3 - Photographs of the reaction flames. Clockwise, from upper left, $Xe(^3P_2)$ + XeF_2, N_2, Cl_2 front view and Cl_2 side view. Unfortunately, the film used was significantly more sensitive to the red than to the blue, and the Xe* + Cl_2 flame can not be fully appreciated.

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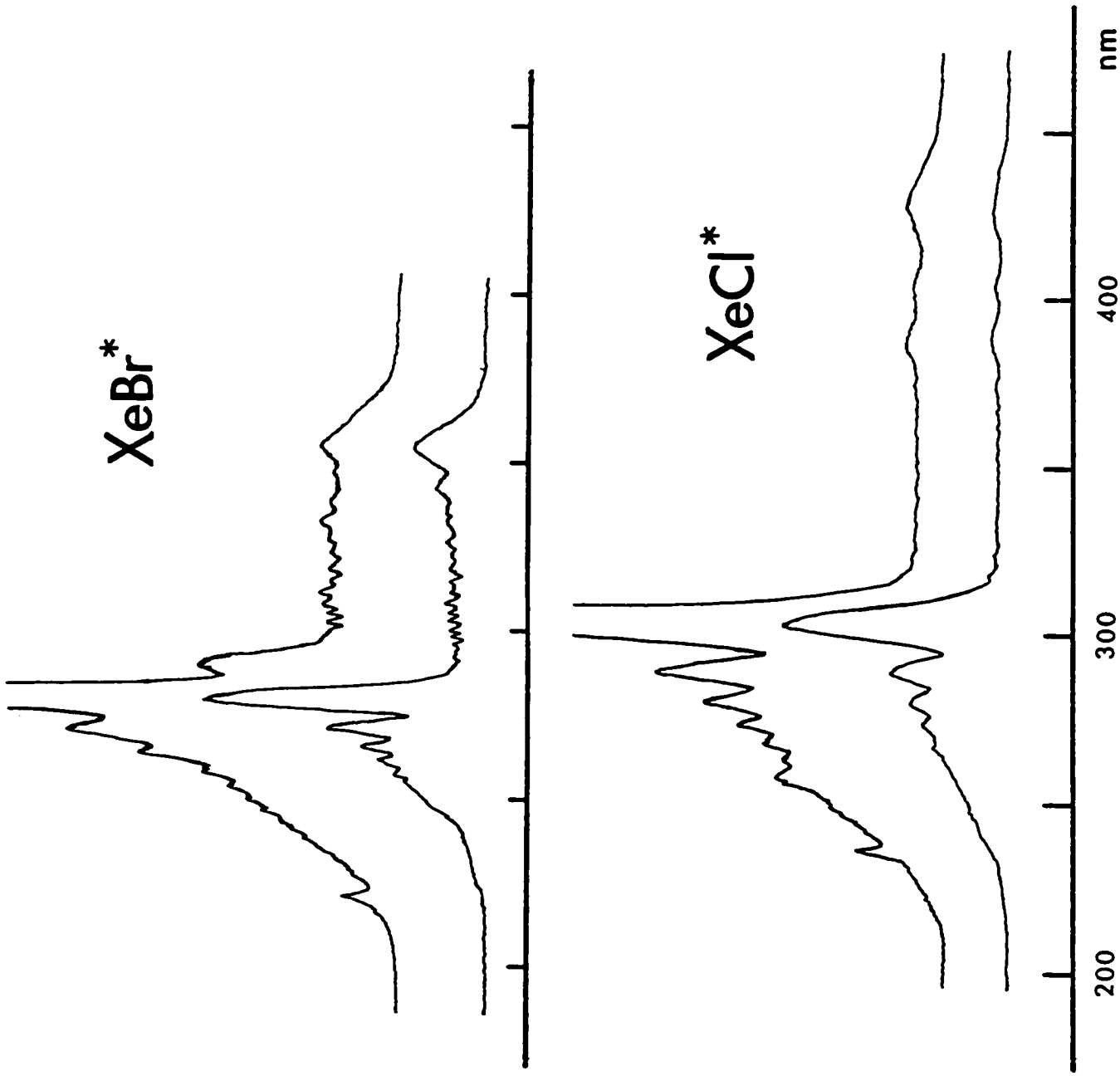
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Figure 4.4 - a. XeBr* emission from the reaction of $Xe(^3P_2)$ with Br_2 (upper) and PBr_3 (lower).

b. XeCl* emission from the reaction of $Xe(^3P_2)$ with Cl_2 (upper) and $SOCl_2$ (lower).

In all cases, slits were 120μ and the scan speed 12.5 nm/min. Reactor pressure was 1.0 Torr.

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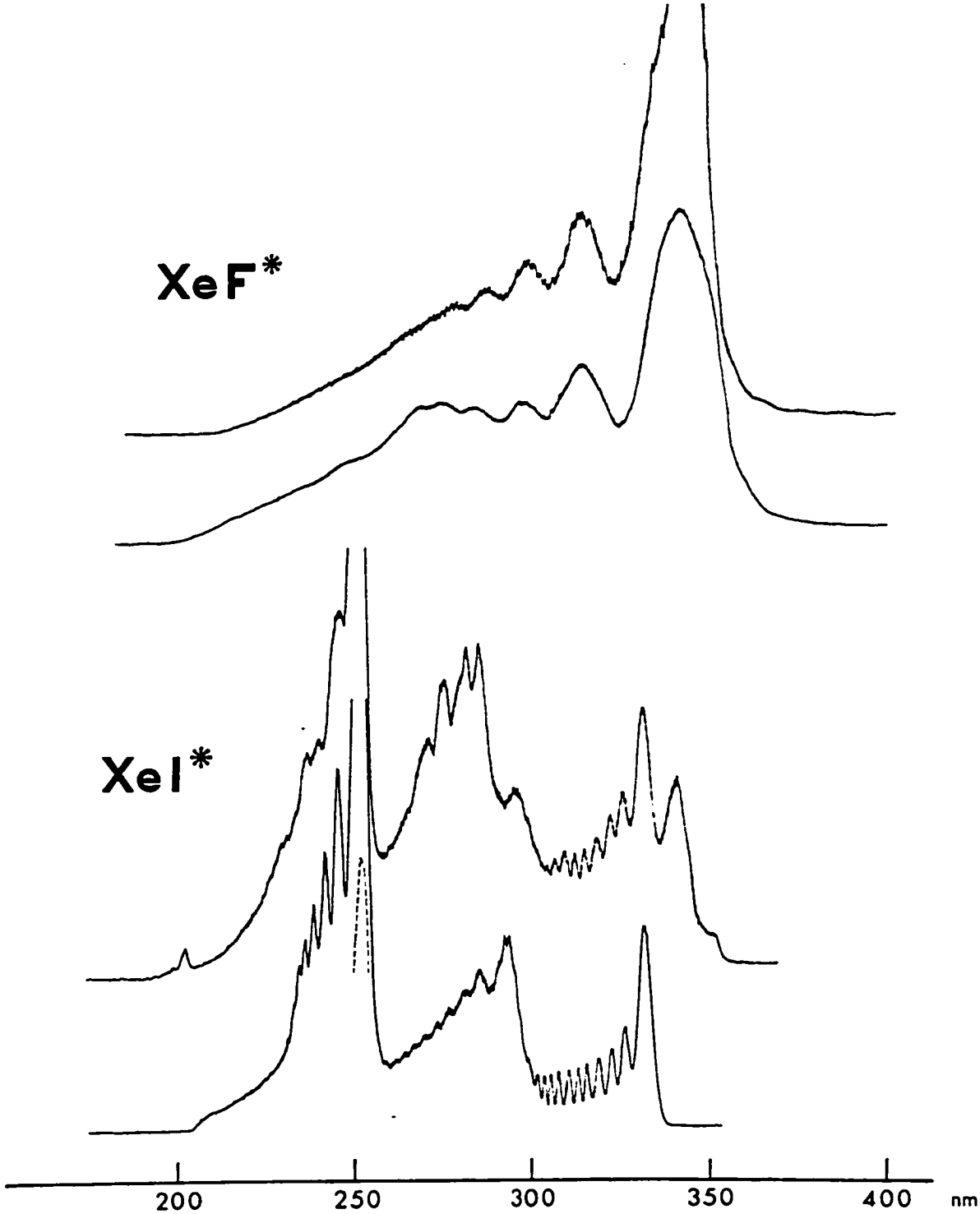
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Figure 4.5 - a. XeF* emission from the reaction of $Xe(^3P_2)$ with CF_3OF (upper) and F_2 (lower).

b. XeI* emission from the reaction of $Xe(^3P_2)$ with I_2 (upper) and CF_3I (lower).

In all cases, slits were 120μ and the scan speed 12.5 nm/min. Reactor pressure was 1.0 Torr.

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The results from the specific rate constant determinations are listed in Table 4.3 as absolute excimer formation rate constants (k_{XeCl} = 7.2 x 10^{-10} cm^3 $molec^{-1}$ sec^{-1}), and Table 4.4 as branching fractions for excimer formation. Also listed in Table 4.3, are the dissociation energies for the halogen-donor bond and finally, λ_{min}^{obs} and λ_{min}^{cal} the observed and calculated (λ_{min}^{cal} = $[E(Xe(^3P_2)) - D_o^\circ(R\text{-}X)]^{-1}$) short wavelength limit for the onset of the XeX emission. There is generally good agreement between the observed and calculated λ_{min}, indicating that the exoergicity of the reactions appears as vibrational excitation of the excimer. The dynamic and mechanistic implications of this and other observations are discussed in Chapter 6.

In all cases, the listed k_{RQ} values are the average of at least two measurements, with a Xe* + Cl_2 comparison run immediately before and/or after each Xe* + RX experiment. When more than one excimer emission was observed for a given donor (as in the case of the interhalogens), the individual contributions were obtained by manual deconvolution of the spectra. This procedure was also employed to determine the relative intensities of the XeF(D-X) and XeF(B-S) emissions from the reaction of $Xe(^3P_2)$ with XeF_2.

Agreement between the intensities for back-to-back scans of the same emission was better than 95% (for the same gas mixture), whereas the relative rate constants were reproducible to within 10%. This error was largely due to the instability of the discharge with the concomitant variation in the Xe metastable concentration over the ~4 hour period required to obtain the scans for Cl_2 and RX. Reagent purity was the main source error in the specific rate constant measurements, and particularly for the more corrosive halogen donors, might have contributed up to 20% to the uncertainty. In particular, analysis of the short wavelength onset for emission in the reactive quenching of $Xe(^3P_2)$ by the

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Table 4.3 Specific rate constant for excimer formation.

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Table 4.3 Rate constants for excimer formation

Group	Reagent	Excimer[a]	k_{RQ} $10^{-11}cm^3sec^{-1}$	D° kcal/mole	$\lambda_{min}^{obs} - \lambda_{min}^{calc}$, nm[b]	Class
VIII	XeF_2	XeF(total)	75.5	61[c]	217-219	B
		XeF(B)	66			B
		XeF(D)	9.5			C
VII	F_2	XeF	73	37.8[d]	<190-186	A
	Cl_2	XeCl	72	58	210-214	A
	ClF	total	62	61.4	220-219	B
		XeF	14			C
		XeCl	48			C
	ClF_3	total	23	41.7[e]	207-190	B
		XeF	14			C
		XeCl	9			C
	BrF_5	XeBr	17	44.7[e]	193-194	B
	IF_5	XeI	4	64.2[e]	205-224	B
	ICl	total	8[f]	50	212-202	C
	Br_2	XeBr	63	46.1	195-196	A
	I_2	XeI	Large[f]	36.1	<190-184	
	IBr	total	Large[f]	42	192-191	
VI	OF_2	XeF	50	40[g]	202-188	A
	CF_3OF	XeF	19	44[h]	205-194	A
	SF_6		0	>78		
	SOF_2	XeF	~0.5	~86[i]	308-271	C
	SO_2F_2		0	~84[i]		
	S_2Cl_2	XeCl	8.5	44.2[j]	210-194	B
	$SOCl_2$	XeCl	14	~55[i]	207-209	B
	SF_4		0	~83[k]		
	SeF_6		0			
	TeF_6		0			
V	NF_3	XeF	10	57.1[l]	245-212	A
	N_2F_4	XeF	20	71[l]	230-237	A
	ONF	total[m]	29	55.4[l]	220-209	B
		XeF	27			B

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Table 4.3 continued.

Group	Reagent	Excimer[a]	k_{RQ} $10^{-11}cm^3sec^{-1}$	D° kcal/mole	$\lambda_{min}^{obs} - \lambda_{min}^{calc}$, nm[b]	Class
V	ONCl	total[m]	34	38[n]	210-186	B
		XeCl	33			B
	PF_3		0	119[o]		
	PCl_3	XeCl	5	76[p]	250-247	B
	PBr_3	XeBr	~10[q]	62[p]	218-220	C
	N_2O	XeO	~5 x 10^{-4}	40	204-188	CC
IV	CF_4		0	125[n]		
	CF_3Cl	XeCl	<0.5	56.8[s]	230-212	C
	CF_2Cl_2	XeCl	~0.5		217-	C
	CHF_2Cl	XeCl	<0.3		230-	C
	$CHFCl_2$	XeCl	<0.1			C
	$CFCl_3$	XeCl	5	73[t]	240-241	B
	CCl_4	XeCl	8	70[n]	225-235	B
	CF_3Br	XeBr	6	70.6[u]	215-236	B
	CF_3I	XeI	14	53.5[v]	206-207	A
	$COCl_2$	XeCl	12	65.6[s]	225-227	A
	SiF_4		0	110[n]		
III	BF_3		0	134[n]		
	AlF_3		0	130[n]		
	BCl_3		0	128[n]		
I	HF		0	135.8[n]		
	HCl	XeCl	<4	103[n]	220-322	CC
	HBr	XeBr	<6	87.5[n]	205-274	C
	HI	XeI	~4	71.3[n]	225-237	C

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a. When more than one emitter is formed, the total rate constant is given first followed by the individual contributions obtained by a manual deconvolution procedure.

b. $\lambda_{min}^{calc} = (67068\ cm^{-1} - D°(R-X)cm^{-1})^{-1}$; The energy of $Xe(^3P_2) = 67068\ cm^{-1}$.

c. Calculated from $\Delta H_f°(XeF_2) = 64.6$ kcal/mole and D(Xe-F) = 3.5 kcal/mole, see Ref. 10.

d. Bond dissociation energies for halogenated donors, unless indicated otherwise, were taken from Ref. 11.

e. Average bond energy.

f. Flows could not be measured accurately.

g. Average of the values reported in Ref. 12.

h. Ref. 13

i. Average bond energy calculated from heats of formation.

j. Ref. 14.

k. Ref. 15.

l. Ref. 16.

m. In addition to excimer emission, NO(γ) (for NOF) and NO(γ and β) (for NOCl) bands are also observed.

n. Ref. 17.

o. Average bond energy taken from Ref. 18.

p. Average bond energy taken from Ref. 19.

q. Estimated value from earlier data (J. E. Velazco, 1974).

r. This is the estimated absolute value of k_{RQ}.

s. Calculated from heats of formation; data taken from NSRDS-NBS 37 (1970).

t. Ref. 20.

u. Ref. 19 and 21.

v. Ref. 22.

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Table 4.4 Experimental Branching Fractions for XeX* formation.

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Table 4.4 Branching fractions for excimer formation[a]

Reagent	Branching Fraction
F_2	1.0
Cl_2	1.0
ICl	0.16[b]
Br_2	1.0
OF_2	0.88
CF_3OF	0.40
$SOCl_2$	0.20
NF_3	1.1
N_2F_4	0.63[c]
ONF	0.69
ONCl	0.65
PBr_3	0.53[d]
N_2O	$\sim 10^{-5}$ [e]
CF_3I	0.27
HBr	< 0.10[e]
HCl	< 0.05

a. Unless indicated otherwise, uncertainties are $\pm$0.20.

b. This value appears to be low.

c. The total quenching rate constant was taken as the average of the rate constants for quenching $Ar(^3P_2)$ and $Kr(^3P_2)$.

d. Error limits: +0.10, -0.30. See text.

e. See text for uncertainties in k_{RQ}.

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hydrogen halides (Group I), indicates the presence of an unidentified emission at wavelengths shorter than the calculated limit set by the H-X bond strength. Since excimer emission is very weak in these reactions, the short wavelength emission could be due to traces of Cl_2 present in the reactor (and/or reagent lines), or to emission from HX* produced by excitation transfer.[1a] The listed rate constants for reactive quenching are therefore upper limits. In the case of the chlorofluorocarbons (Group IV), the specific rate constants are given as upper limits because in order to obtain reasonable signal levels, the quencher flows employed were close to or at the limit of the I vs. [Q] linear range; however, there was no evidence for impurity contributions. The largest uncertainty is associated with k_{RQ} for N_2O. The XeO signal was very weak and background contributions became significant. The value reported in Table 4.3 is probably good only to an order of magnitude.

The results in Table 4.3 are labeled as Class A, B or C depending on whether the uncertainty in the relative rate constants is $\leq$ 10%, $\leq$ 20% or $\geq$ 20%. It should be emphasized however, that these are <u>estimated</u> uncertainties based on the manufacturer's stated gas purity and on the reproducibility of the rate constants when using different reagent/argon mixtures. In addition, the specific rate constants, k_{RQ}, and the branching ratios have larger errors due to the contributions of the uncertainties in k_{XeCl} ($\sim$5%, relative) and k_Q(5 to 10% typically, relative).

Emissions other than XeX* were not observed for these donors, except for NOF, NOCl and PCl_3. In the case of NOF and NOCl, emission from NO(A $\rightarrow$ X) - the NO(γ) bands - was also observed and constituted approximately 4% of the total emission. The reaction of $Xe(^3P_2)$ with PCl_3 also excited an unassigned banded emission in the 300 to 400 nm region.

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d. Other Results. The product channel distribution for the quenching of $Xe(^3P_2)$ by Hg, N_2, CO, NO and O_2 were also studied, but assignment of specific rate constants was attempted only for Hg. The results of these studies are presented and discussed in Chapter 6 in terms of the mechanisms for formation.

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(23) Actually very weak Cl_2^* emission was observed at 256 nm, see Figure 4.1. However, this contribution (<1%) can safely be neglected.

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CHAPTER 5 RARE GAS-HALOGEN EXCIMERS

"Systems in physical science, ..., [are] approximate methods which put us on the path to the solution of the problem; these are the hypotheses which, successively modified, corrected and changed in proportion as they are found false, should lead us infallibly one day, by a process of exclusion, to the knowledge of the true laws of nature."

Antoine Laurent Lavoisier
"Memoires" 1777

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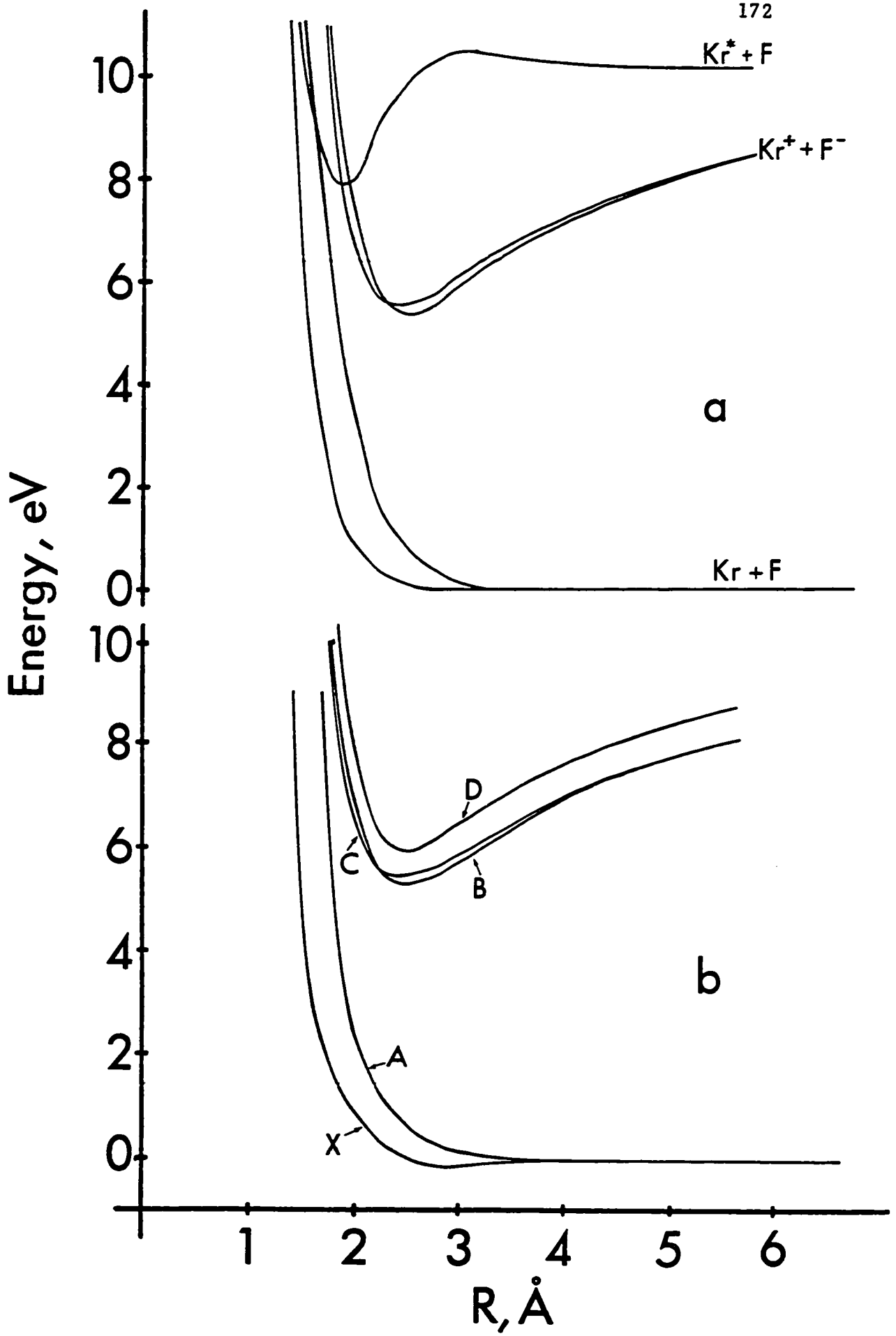
a. Electronic states for rare gas-halogen molecules. When the rare gas-halogen excimers were first reported,[1,2] it was proposed that the observed emission was associated with a transition originating from an upper, Coulombic state, and terminating on a repulsive or flat ground state. This model based on simple considerations,[1-3] has been confirmed by direct analysis of the spectra[4] and by quantum-mechanical calculations.[5] The lower (molecular) electronic states correlate with the ground state atoms $M(^1S) + X(^2P)$, while the excited states correlate with ground state ions, $M^+(^2P) + X^-(^1S)$. In coupling case a, interaction of ^1S and ^2P atomic states gives $^2\Sigma^+$ and $^2\Pi$ molecular states, as shown in Figure 5.1a for the calculated[5] potential energy curves for the electronic states of KrF. The two valence states $1\,^2\Sigma^+$ and $1\Pi^2$ are expected to be repulsive, but since the $^2\Sigma^+$ has only three $p\sigma$ electrons, it should be less repulsive (or even slightly bound) than the $^2\Pi$ state with four $p\sigma$ electrons, and, therefore, it is designated as the gound state (X) while the $^2\Pi$ state is designated A. The upper states are strongly bound due to the Coulombic attraction.

Although coupling case a is probably correct for the valence states of the fluorides, the increase in the spin-orbit splitting for the heavier halogens will make case c coupling and designations more appropriate. In case c, the ground state is X(1/2) and the A state is split into two components, A(3/2) and A(1/2). The effect of the spin-orbit splitting of the heavier halogens on the lower states of the excimer is shown in the diagram of Figure 5.2, where the lower state of XeF(a) and XeI(b) are compared. The large spin-orbit splitting in the ground states of the rare gas ions ($\Delta(^2P_{3/2}-^2P_{1/2}) \sim 1.3$ eV for Xe), implies strong case c coupling for these states. Thus, the nearly degenerate $2\,^2\Sigma^+$ and $2\Pi^2$ in case a, are split into three curves, 3/2 and 1/2, correlating with $M^+(^2P_{3/2}) + X^-(^1S_0)$, and 1/2 that correlates with $M^+(^2P_{1/2}) + X^-(^1S_0)$. The two 1/2 states are strong

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Figure 5.1. - Calculated potential curves for KrF. The spin-orbit coupling in Kr^+ is included in (b). The results were taken from reference 5.

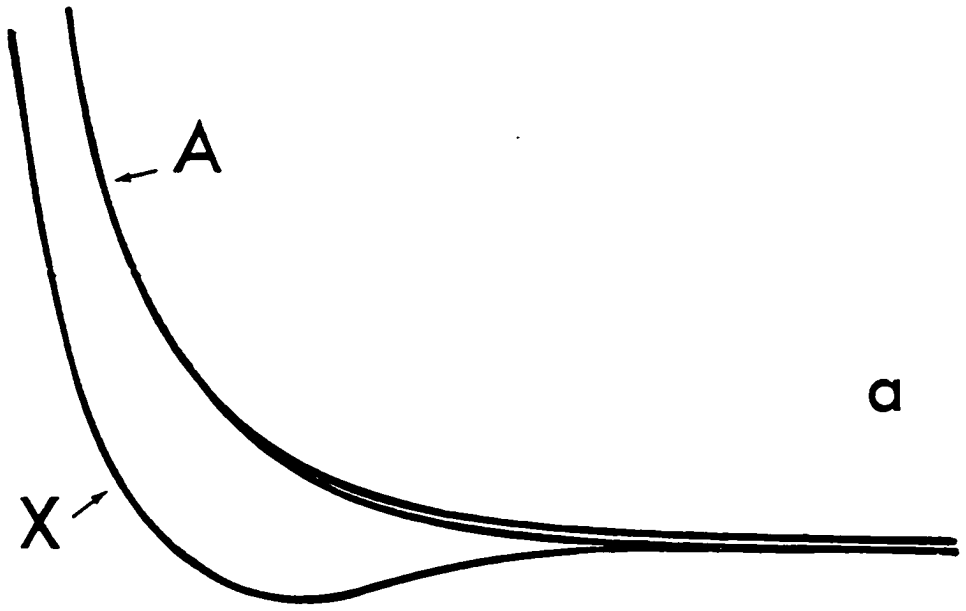
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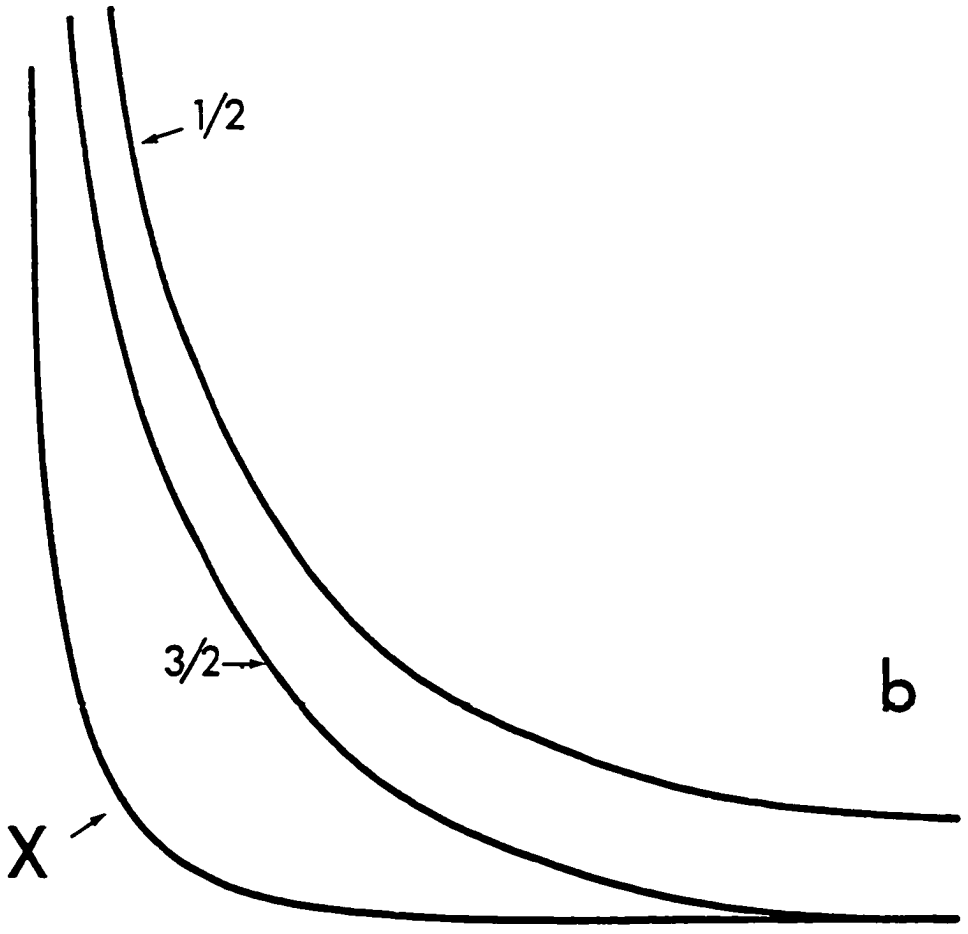


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Figure 5.2. - Potential curves correlating with the ground state atoms for XeF(a) and XeI(b) to illustrate the effect of the spin-orbit coupling in the halogen atom. The drawings are only schematic.

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mixture of the $2\,^2\Sigma^+$ and $2^2\Pi$, whereas, the 3/2 state is composed of only the $2^2\Pi$ state.[5] As a result, the 3/2 and the lower 1/2 states are nearly degenerate, see Figure 5.1b, and the relative ordering of these states is not straight forward.[4d,b] Dunning and Hay[5] use B for the 3/2 state and C for the lower of the 1/2 states. However, preliminary results of the rotational analysis[7] in XeF suggest that the 1/2 state is lower than the 3/2, and, therefore, following Tillinghuisen et al.[4d] we label the lower 1/2 state B, and the higher 3/2 and 1/2 states, C and D, respectively.

In addition to the states just described, there is the possibility of other bound states - Rydberg states - converging on MX^+ which is expected to be strongly bound by analogy to the isoelectronic interhalogen (i.e., KrF^+ is isoelectronic with BrF, $XeCl^+$ with ICl etc.); these states are likely to play an important role in laser performance, particularly for the heavier halogens for which the number of possible states becomes very large, and the laser frequency might match a transition between the C or B states and the higher Rydbergs. On the other hand, it is also possible that these states provide the means to channel products of neutral recombination to the ion-pair curves. Figure 5.1a shows the lowest of these Rydberg states for KrF; namely, the Rydberg that correlates with $Kr(^3P_2)$ + F. For iodine with Ar, Kr or Xe, low lying Rydbergs that correlate with I* may be responsible for the predissociation of MI*. In fact, with the exception of the fluorides, XeCl* and possibly KrCl*, there are halogen excited states below the metastable energy limits that provide competitive exit channels <u>via</u> predissociation of MX* (for appropriate ranges of vibrational energy), as shown by Gundel et al.[1c] for $Ar(^3P_2) + Cl_2$.

The (continuous) low pressure excimer emission spectra are similar for all the rare gas-halogen combinations (see Figures 4.4 and 4.5). They

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show a gradual increase in intensity from the onset (determined by the exoergicity of the reaction, see Section 4.c) up to a strong, broad ("main") peak; this portion of the spectrum is termed "primary emission" or "Spectrum I", and generally displays a broad oscillatory pattern. To the red of the main peak the intensity drops sharply, but emission continues until a distinct red cut-off. This emission is called "secondary emission" or "Spectrum II"; the general appearance of the secondary emission varies from flat and weak, without a sharply defined red cut-off for XeF*, to relatively strong and highly structured for XeI*. For the heavier halogens there are indications of two different maxima in the secondary emission (clearly visible for XeI*). Finally, superimposed on the short wavelength end of the primary emission, there is a third sharp feature, normally weak and not present for all donors, labeled as "Spectrum III".

The primary emission is assigned to B(1/2) → X(1/2) ($2\ ^2\Sigma_{1/2} \rightarrow X\ ^2\Sigma^+$) although contributions from C(3/2)→ X(1/2) ($2\ ^2\Pi_{3/2} \rightarrow X\ ^2\Sigma^+$) can not be ruled out, at least at low pressure. The secondary emission which is understood less well (see below), is thought to involve the C(3/2) → A(1/2,3/2) ($2\ ^2\Pi_{3/2} \rightarrow 1\ ^2\Pi_{1/2,3/2}$) transitions, with possibly some contributions from B(1/2) → A(1/2,3/2) ($2\ ^2\Sigma^+ \rightarrow 1\ ^2\Pi_{1/2,3/2}$). Finally, from the observed separations between the main peak and Spectrum III, the latter can be assigned as D(1/2) → X(1/2) ($2\ ^2\Pi_{1/2} \rightarrow 1\ ^2\Sigma^+$). Table 5.1 summarizes the observed wavelengths for these transitions and the energy of the states.

The D(1/2) → B(1/2) ($2\ ^2\Pi_{1/2} \rightarrow 2\ ^2\Sigma^+$) transitions should be fully allowed, and in principle, strong; however, attempts to observe this (bound to bound) emission were not successful,[4c] and the intensity ratio of (B-X)/(D-B) has been experimentally set as $\sim$ 1000.[4c] This ratio is one order of magnitude larger than the theoretical ratio of the transition probabilities.

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Table 5.1 Observed emissions for XeF, XeCl, XeBr and XeI, and calculated energies of the B and D states.

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Table 5.1. Observed emissions for the xenon halides[a]

Transition	XeF[b]	XeCl[c]	XeBr[c]	XeI[d]
$\lambda(B \rightarrow X)$	346.6	307.8	281.8	254.0
E_B	3.577	4.028	4.400	4.881
$\lambda(D \rightarrow X)$	262.3	235.8	221.3	203.0
E_D	4.710	5.258	5.603	6.108
$\lambda(C \rightarrow A)$[e]	-	428.0	357.0	326(293)
$\Delta E(D \rightarrow B)$	1.133	1.230	1.203	1.227

a. The wavelengths are in manometers and the energies in electron volts. No energy is given for the C state because the position of the A curve is not known.

b. The wavelengths correspond to the (0,0) transitions.

c. The wavelength of the B → X emission is for the (0,0) transition in XeCl and for the observed laser emission in XeBr, while that for the D → X corresponds to the peak in the low pressure spectra.

d. Wavelengths corresponds to the peak of low pressure emissions.

e. Listed wavelengths correspond to the redmost maxima in the low pressure spectra; for XeI the two distinct maxima in Spectrum II are listed.

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Comparison of the different rare gas-halogen excimer emissions show that the main structural differences are the secondary emission. As pointed out earlier, the intensity and presence of structure increase with increasing weight of the halogen. On the other hand, for a constant halogen, secondary emission intensity increases with decreasing mass of the rare gas. The increase in structure for the heavier halogens follows from the larger spin-orbit splitting that separates the A(1/2,3/2) curve into two components and by the shape of the difference potentials. The correlation of the changes in intensity with the nature of the upper states however, is more difficult to rationalize. Also,[1c] the intensity ratio of the primary to the secondary emission varies with donor, which implies that different donors yield different intial population ratios for the excited states. Table 5.2 summarizes the behavior of the secondary emission relative to the primary emission as a function of halogen donor for some systems. Since the calculated[5b] transition probabilities for the C → A transitions are nearly constant[5b] for the different rare gas-halogen combinations, the observed variations in the II/I ratio lead to the conclusion that the chemical reaction must be selective in the formation of the upper states. Within the framework of the reaction model for these type of systems (see Chapter 6), selectivity in the entrance channel of the collision can be determined by conservation of symmetry throughout the collision process and/or by the relative strengths of the couplings between the entrance and exit channels. These possibilities are explored in detail in Section 6.d.

The production of the D(1/2) state presents other interesting questions. In principle, formation of this state depends only on the energy available from the exoergicity of the reaction; however, it is found that Cl_2(D° = 58 kcal/mole) gives the D(1/2) state, whereas NOCl(D° = 38 kcal/mole) does not. F_2(D° = 38 kcal/mole) does not give the D(1/2) state, but NOF(D° = 55.4

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Table 5.2 Experimental II/I ratios for the rare gas-chlorides and xenon-halides.

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Table 5.2

Excimer	donor	I[a]	II[a]	I/(I + II)	II/(I + II)	II/I
ArCl	Cl_2	-	-	-	-	0.50[b]
KrCl	Cl_2	903	363	0.71	0.29	0.40
XeCl	Cl_2	529	157	0.77	0.23	0.30
XeF	F_2	295	51.5	0.85	0.15	0.17[c]
XeCl	Cl_2	284	94.6	0.75	0.25	0.33
XeBr	Br_2	564	243	0.70	0.30	0.43
XeI	CF_3I	192	144	0.57	0.43	0.75
XeF	F_2	304	48.6	0.86	0.14	0.16[c]
	OF_2	184	36.4	0.83	0.17	0.20c
	NF_3	207	50	0.81	0.19	0.24[c]
	N_2F_4	406	122	0.77	0.23	0.30[c]
XeCl	Cl_2	147	50	0.75	0.25	0.33
	CCl_4	645	249	0.72	0.28	0.39
	PCl_3	106	80	0.57	0.43	0.76
	S_2Cl_2	104	59	0.64	0.36	0.56
	$SOCl_2$	296	119	0.71	0.29	0.40
	CF_2Cl_2	100	60	0.63	0.37	0.59
	$CFCl_3$	788	284	0.73	0.27	0.36
	$COCl_2$	893	379	0.70	0.30	0.43
XeBr	Br_2	167	74	0.69	0.31	0.44
	PBr_3	675	371	0.64	0.36	0.56

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Table 5.2 continued

Excimer	donor	I[a]	II[a]	I(I + II)	II/(I + II)	II/I
XeI	CF_eI	192	144	0.57	0.43	0.75
ArCl	Cl_2	-	-	-	-	0.50[b]
	CCl_4	-	-	-	-	1.00[b]

a. The listed intensities are relative, and may vary by orders of magnitude in absolute value for different donors.

b. Reference 1c.

c. This value should be an upper limit; the secondary emission in XeF is weak and extends over ∿ 150 nm, and therefore residual noise may contribute to the integrated signal.

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kcal/mole) does. These observations suggest once more that finer details of the collision process might be responsible for the formation of the D state, see Section 6.d.

b. Discussion of bound-free transitions. Bound-free transitions have been observed previously in molecular iodine[8] and other halogens,[8c] hydrogen,[9] molecular helium,[10] and more recently in alkali-metal-noble gas molecules.[11] Mulliken[12] examined the general problem of bound-to-free transitions and derived the quantum-mechanical expressions that describe the transition from a high v' level of a bound state to a continuum. This analysis was later applied by Tellinghuisen[13] to explain the "McLennan" bands of I_2 and, in a more qualitative fashion, by Golde[14] to explain the rare gas-halogen excimer emissions. Mulliken's[12] derivation is based on the fact that, according to the Frank principle, the probability of an electronic transition is maximized when the position and momenta (or kinetic energy) of the nuclei are conserved. Commonly, this is taken to mean that the probability of a transition reaches a maximum at the classical turning points of the oscillation, since at those points, the kinetic energy is zero ("the oscillator spends more time at the turning points"). However, for states with high vibrational energies, conservation of momentum indicates that transitions can occur with relatively large probability, at R values (R is the internuclear distance) where the kinetic energy is maximum. The main aspects of the model are described below.

The probability per unit time for a transition (in emission) is given by

$$(dp/dt) = C\,\nu^3[\int\psi'(q,R)\chi'(R)\mu_{op}\psi''(q,R)\chi''(R)dpdR]^2 \qquad (5.1)$$

where ψ and χ are the electronic and nuclear wave functions, μ_{op} is the

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dipole moment operator and q and R refer to electronic coordinates and internuclear separation, respectively. Making use of the Born-Oppenheimer approximation, integration over the electronic coordinates gives

$$(dp/dt) = CQ_{el}\ \nu^3\ [\int \chi'(R)\chi''(R)dR]^2; \tag{5.2}$$

the integral is simply the Frank-Condon integral. Consider the situation of Figure 5.3a, which represents the transition from a high v' level of a harmonic oscillator potential ($U(x) = (kx^2/2)$) down to a flat lower state.

The position of the oscillator is given by $(R-R_e) = x = A \sin(2\pi ft)$, where A is the amplitude of the oscillation and $f = (1/2\pi)\ (k/\mu)^{1/2}$ the frequency of the oscillator of reduced mass μ. The kinetic energy K' of the vibration is

$$K' = \frac{k}{2}\ (A^2-x^2) \tag{5.3}$$

Since we deal with only one v' level, $\chi'(R)$ in Equation 5.2 is fixed; expanding this wave function in terms of the lower state wave functions

$$\chi'(R) = \sum_i c_i \chi_i'' \tag{5.4}$$

$$\sum_i c_i^2 = 1 \tag{5.5}$$

and summing over all possible χ'' or classically, integration over all accessible E" reduces Equation 5.2 to

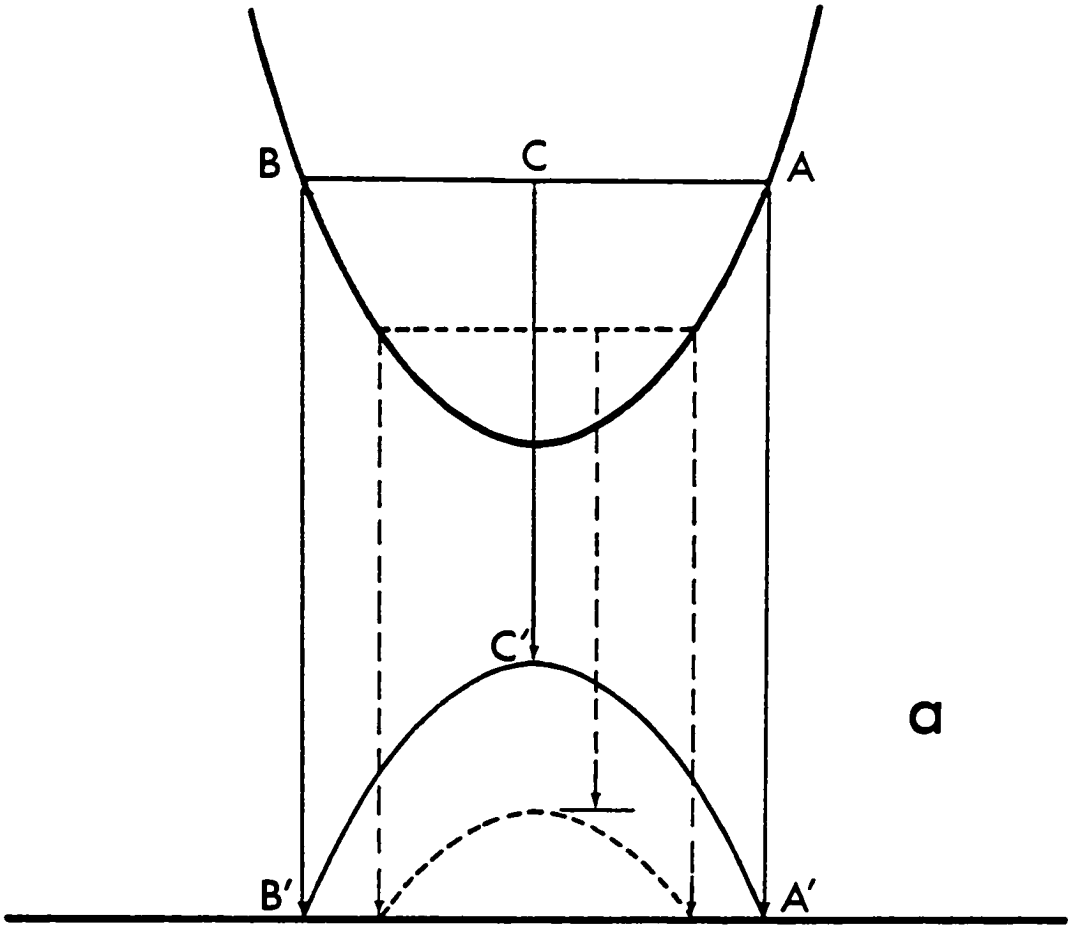
$$(dP/dt) = CQ_{el}^2\ \langle\nu^3\rangle \tag{5.6}$$

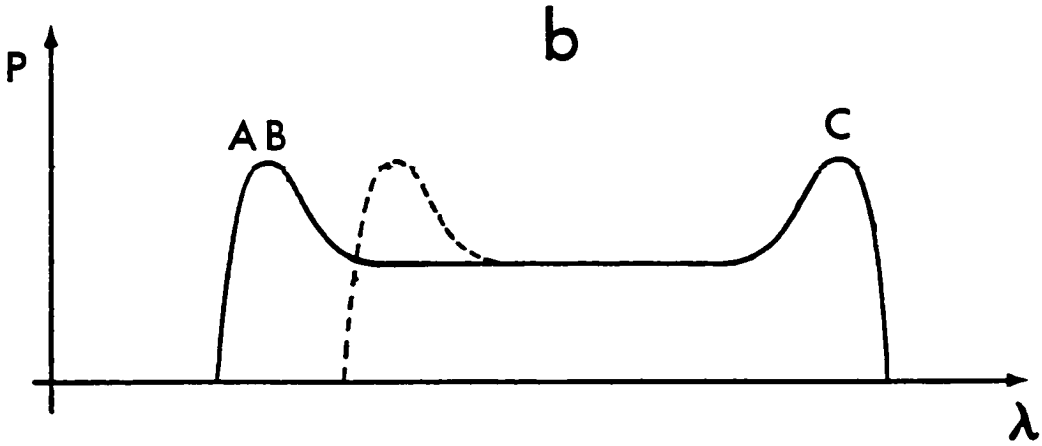
where dP/dt corresponds now to the total probability per unit time for an electronic transition from the upper state v'. The requirement that the kinetic energy be conserved in the electronic transition, implies that K" is also given in Equation 5.4, and therefore it follows that a E" = K" versus x curve will be a mirror image of the upper potential U'(x), as shown in Figure 5.3a. The observed frequencies in the emission spectrum

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Figure 5.3 (a) Schematic diagram for transitions from high v' levels of a harmonic oscillator to a flat lower potential. The effective lower potential is due to conservation of momentum. (b) Expected transition probability for the case shown in (a).

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vary between $\nu_{max} = E'_A h^{-1}$ and $\nu_{min} = (E'-E'')_C h^{-1}$, where the subscripts A and C indicate the position of the oscillator when the transition occurs (see Figure 5.3a). Transitions from points equidistant from x = 0 will be at the same frequency, and in contrast to ν_{max}, ν_{min} is independent of v' (and E') and is equal to $E_o h^{-1}$, where E_o is the energy at the bottom of the upper curve.

The (total) probability of transition as a function of ν is

$$\frac{1}{h}\frac{dP}{d\nu} = \frac{1}{h}\frac{dP}{dt}\frac{dt}{dx}\frac{dx}{d\nu} = \frac{C\,Q_{el}^2 \langle \nu^3 \rangle}{[(2\pi f\cos(2\pi ft)][kA\sin(2\pi ft)]} \tag{5.7}$$

or, as a function of wavelength,

$$\frac{1}{h}\frac{dP}{d\lambda} = \frac{CcQ_{el}^2\,\lambda^{-5}}{[2\pi f\,A\cos(2\pi ft)][k\,A\sin(2\pi ft)]} \tag{5.8}$$

According to this function, the semi-classical spectrum for a transition between states of energies E' and E'' would appear as in Figure 5.3b, with maxima at E'/h and E_o/h, but with substantial spectral intensity in the middle range of classical motion.[12] Transitions from upper states of different E' are also indicated in Figure 5.3b (dashed curve).

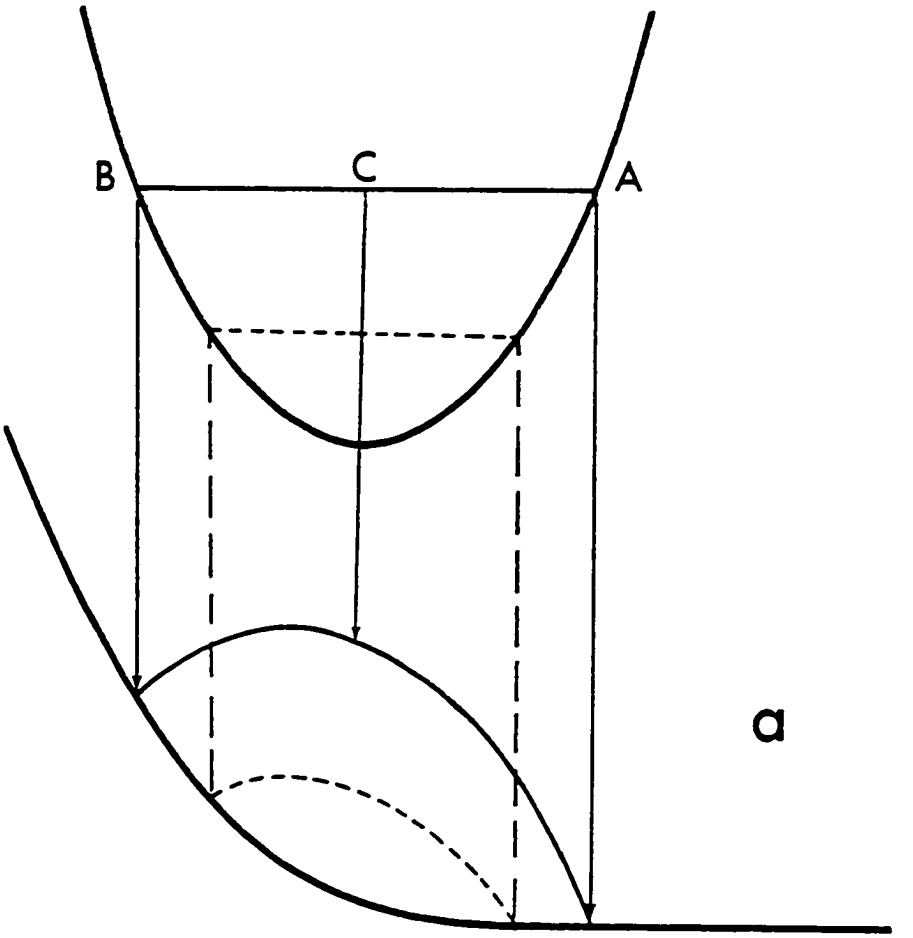
In the more general case in which U''(x) ≠ 0, see Figure 5.4a, the same method predicts a spectrum with intensity maxima at three different frequencies, corresponding to transitions from A, B or C; this is indicated in Figure 5.4b. The relative position of the peaks corresponding to B and C - assuming only one upper state, thus fixing A - will depend on the position and slope of the lower potential. Thus, Figure 5.3a can be considered as an extreme case where the repulsive wall of the lower potential sets in at $R'' \ll R_e'$ in which case peaks A and B are coincident. If the lower potential is now moved so that $R'' \sim R_e'$, peak B moves towards peak C and eventually, when U'(B)-U''(B) = U'(C)-U''(C), peaks B and C

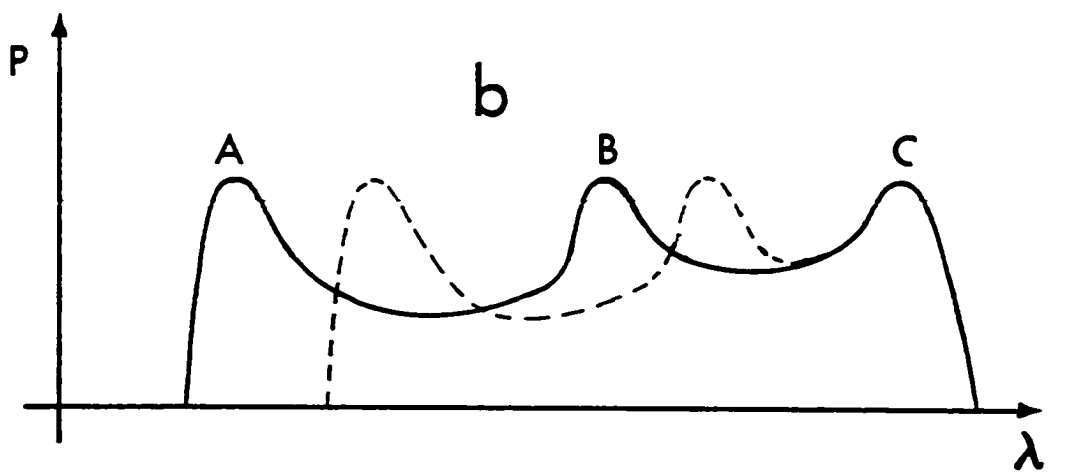
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Figure 5.4 (a) Schematic diagram for the transitions from high v' levels of a harmonic oscillator to an exponential (repulsive) lower state.

(b) Expected transition probability. Note that transitions from A and B no longer occur at the same energy (compare to Figure 5.3(b)).

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will coincide. Further shifts in R" (or increases in the slope) will move peak B past peak C (see Figure 5.5).

The semiclassical picture gives a smooth continuum of frequencies between ν_{min} and ν_{max}. However, rigorous application of Equation 5.3 should give an overall intensity pattern like the ones in Figures 5.3b and 5.4b, but modulated by the specific behavior of $[\int \chi'\chi'' \, dx]^2$. This effect was analyzed by Mulliken in terms of the equality of the deBroglie wavelengths implied by the conservation of momentum:

$$\lambda'' = \lambda' = h/|\mu\dot{x}| = h/(2\mu K)^{1/2} \tag{5.9}$$

Mulliken's analysis however, does not take into account the variation of λ'(or λ'') with x, or the effect of a finite lower potential. Golde[14] also uses a flat lower potential, and he concludes that interference will result between two transitions equidistant from x, and also, due to the relative phase differences of the waves in the upper and lower states; in the first case broad oscillations result, whereas, the phase difference give fast oscillations.

Recently, K. Tamagake[15] of this laboratory has completed a more realistic analysis of the behavior of the Franck-Condon integral of bound-free transitions using a momentum representation and a variety of forms for the lower repulsive potential. According to his results, the origin of the oscillatory structure is twofold: (1) the oscillatory nature of the upper state wave function (in the p-representation) provides broad oscillations in the observed spectrum. (2) The phase properties of the lower state wave function are responsible for the faster oscillations observed. The net effect of this second source of oscillatory structure is extremely dependent on the position of the wall relative to R_e', and to the slope of the lower potential. Results of numerical calculations using a Morse upper potential and a repulsive exponential for the

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Figure 5.5 Effect of the position of the repulsive wall of the lower potential on the expected emission spectrum. The solid line calculated spectrum under each pair of potential curves is the result of quantum mechanical calculation, while the dashed curve is the classical spectrum. Note the relative positions of the A, B and C peaks. The author thanks Dr. K. Tamagake for making these diagrams available.

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lower state are shown in Figure 5.5, to illustrate the change in the observed spectrum due to shifts in the lower potential relative to the upper curve. The "classical" spectra are shown as dotted lines.

For a fixed lower potential, the positions of the maxima in the broad oscillations initially change rapidly with increasing v', but for higher v' levels the relative shift of the maxima becomes slower. Superposition of all these contributions has the net effect of producing a structureless continua with a build up of intensity around "peak C", where all of the transitions have their maximum intensity. For high enough v' ($v' \gtrsim 50$), the maxima in the oscillations become coincident, giving the observed broad oscillations of the primary emission of the rare gas-halogen excimer spectra.

As a result of Dr. Tamagake's calculations, it became apparent that the assumption of constant transition moment vs Re was a poor one, and in order to simulate the observed low pressure excimer emissions, a transition moment was required that rapidly falls in value at the turning points to eliminate the "A" type band. This coordinate dependence of the transition moment with internuclear distance agrees with the calculated transition moment of Dunning and Hay.[5] Best agreement between the simulated and experimental low pressure "primary" spectra of KrF from Kr + F_2 was obtained for an "inverted ramp" vibrational distribution, peaking close to the highest v' levels allowed by the thermochemistry of the reaction and steadily decreasing with decreasing v'. It was not possible to define a unique distribution, because the similarity of the higher v' wave function allows considerable leeway in the assignment of the peak population; however, the shape of the distribution and $\langle f_v \rangle$ can be defined by simulation of the spectra.

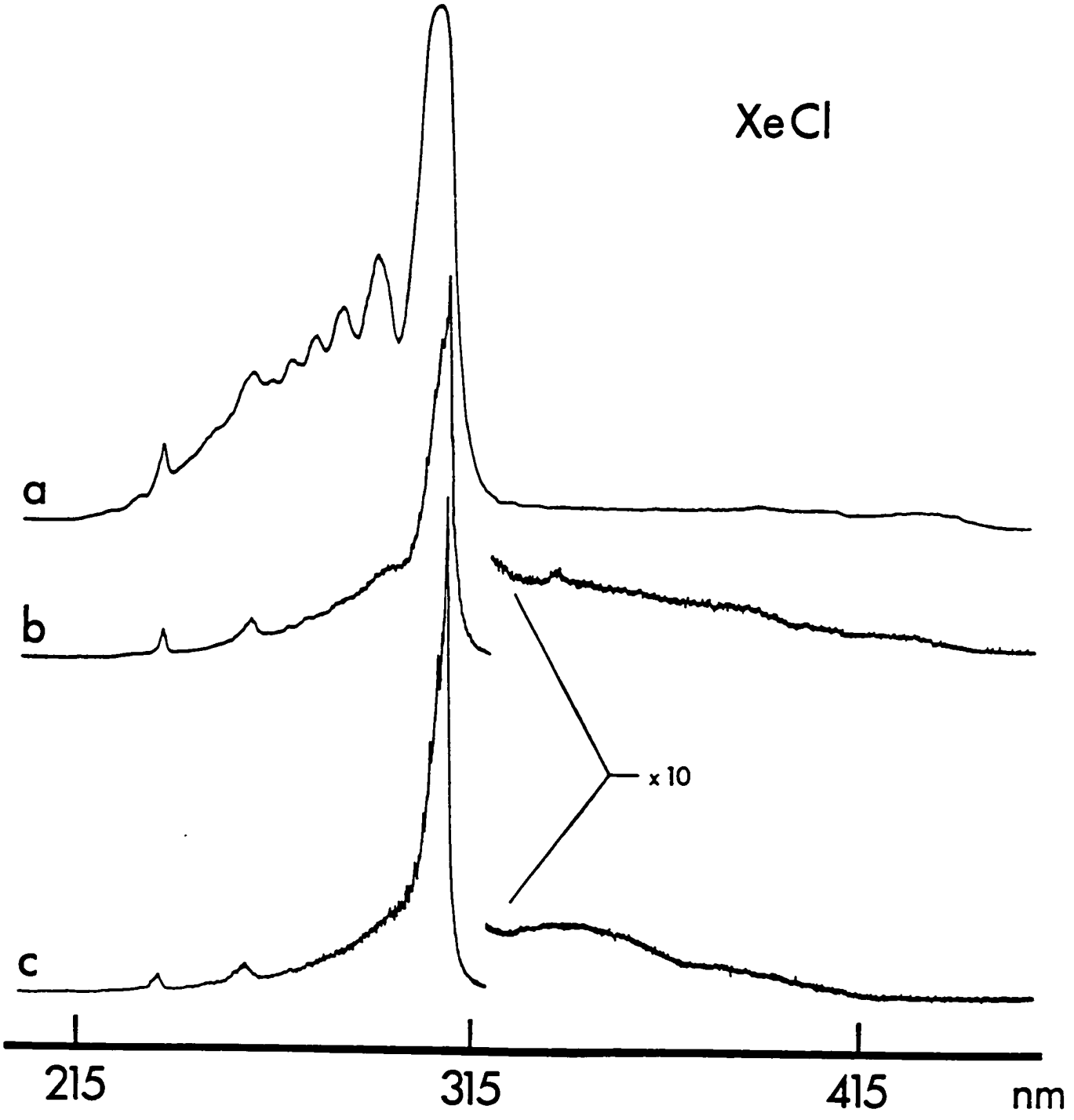
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c. Pressure dependence of the bound-free emission. The pressure dependence of the excimer emission can provide valuable clues regarding the nature of the states involved in the transitions, the lifetimes of the upper state, and the type of interaction between the excited molecule and the buffer gas (i.e., relaxation or quenching). Figure 5.6 shows the XeCl* emission spectrum for various buffer gas (argon) pressures from Xe + Cl_2. At 1.7 Torr, Figure 5.6a, the onset of the emission is located at 210 nm and the red cut off at 458 nm; the overall spectrum exhibits all of the characteristics of the bound-free transitions. A relatively modest increase in pressure (7 Torr) however, produces dramatic changes; the onset moves to 225 nm (Figure 5.4b), the broad oscillations in the primary emission disappear almost completely, the main peak increases in intensity and in sharpness, and discrete structure becomes evident. The red cut-off on the contrary, is shifted towards the blue, to ∿450 nm, and the intensity of the secondary emission is decreased. The D(1/2) → X(1/2) emission peak is still present, and its decrease in intensity could be attributed in part, to the decrease of the background B(1/2) → X(1/2) emission. Increasing the pressure further, to ∿26 Torr (Figure 5.4c), does not produce a significant change in the onset, but the broad oscillatory structure in the primary emission is completely washed out, and faster oscillations appear near the base of the main peak, where discrete vibrational structure is clearly visible. The D(1/2) → X(1/2) emission slightly decreases in intensity, while the red cut-off is shifted to ∿420 nm and the intensity of the secondary emission decreases and concentrates near the main peak. The variation of the II/I intensity ratio with pressure is illustrated by the data in Table 5.3, for XeCl*, KrCl* and XeI*.

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Figure 5.6 Emission spectra of XeCl* at various pressures. (a) 1.7 Torr, (b) 6 Torr, (c) 26 Torr. For all cases the slits were 55μ and scan speed was 12.5 nm min^{-1}. At the highest pressure discrete bands from individual v' levels can be identified at λ = 307.7; 305.7; 304.0; 302.3; 300.7; 298.9, which correspond to the 0-0, 1-0, 2-0, 3-0, 4-0 and 5-0 transitions[4a]. The XeCl(D → X) emission is identified. The unlabeled band at 256 nm is Cl_2* emission.

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These observations are consistent with the assignment of three different transitions; the shift of the onset of the B(1/2) → X(1/2) emission towards longer wavelengths indicates that the X(1/2) state is not repulsive in the Franck-Condon region, and further, the presence of vibrational structure implies a potential well that supports several v" levels. On the other hand the shift of the secondary emission towards shorter wavelengths is consistent with the transitions to a lower state that is more repulsive than the upper Coulombic State. The decrease in intensity of the secondary emission relative to the primary emission could be explained as a combination of vibrational and electronic relaxation[14,1b] of the C(3/2) state that shifts the population to the B(1/2) state. This postulate, however, brings up questions that cannot be easily answered regarding the relative probabilities of the B-A and C-A transitions.[16]

The pressure dependence of the XeBr* and XeI* primary emission and the XeBr* secondary emission follow the same general trend; the secondary emission in XeI* however, is peculiar in that, while the peak closest to the main peak (II') decreases in intensity with increasing pressure, the peak on the red end (II") shows the opposite behavior, see Table 5.3. Furthermore, the decrease of the II'/I ratio and the constancy (within experimental error) of the II"/I ratio, suggest that emissions I and II' originate on the same upper state.

The observation of extensive vibrational relaxation in the B-X emission between 2 and 7 torr suggests that the lifetime of the B state in the higher vibrational levels is 50 ns, since the time between gas kinetic collisions is ∿50 ns at 5 torr. The lifetime of the D state might be shorter than this estimate as suggested by (i) the fact that the D-X

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Table 5.3 Experimental II/I ratios for XeCl, XeI and KrCl as a function of pressure.

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Table 5.3

Excimer	donor	P(Torr)	I	II	I/(I + II)	II/(I + II)	II/I	II'	II"	II'/II"	II'/I	II"/I
XeCl	Cl_2	1.7	187	56	0.77	0.23	0.30					
		7.6	925	259	0.78	0.22	0.28					
		12.5	782	203	0.79	0.21	0.26					
		16.2	800	176	0.82	0.18	0.22					
		22	632	120	0.84	0.16	0.19					
		26	550	82.5	0.87	0.13	0.15					
KrCl	Cl_2	1.7	903	363	0.71	0.29	0.40					
		7.4	781	231	0.77	0.23	0.30					
		15	478	135	0.78	0.22	0.28					
		28	450	120	0.79	0.21	0.26					
XeI	CF_3I	1.7	192	144	0.57	0.43	0.75	84	60	0.72	0.44	0.31
		4.0	207	131	0.61	0.39	0.65	68	63	0.93	0.34	0.31
		8.5	346	213	0.62	0.38	0.62	97	116	1.20	0.28	0.34
		12.4	297	182	0.62	0.38	0.61	78	104	1.32	0.26	0.35

emission does not show the sharpening of the B-X, and (ii) the near constancy of the D-X onset above ∿7 Torr in spite of the weakening of primary emission to the red of the D-X peak. The estimated lifetime of the B-X transition is a factor of ∿3 larger than the value obtained for XeF* by Searles and Eden[17] for the v' = 0-3 levels and an order of magnitude larger than the calculated value of Dunning and Hay[5a] for KrF* (v' = 0).

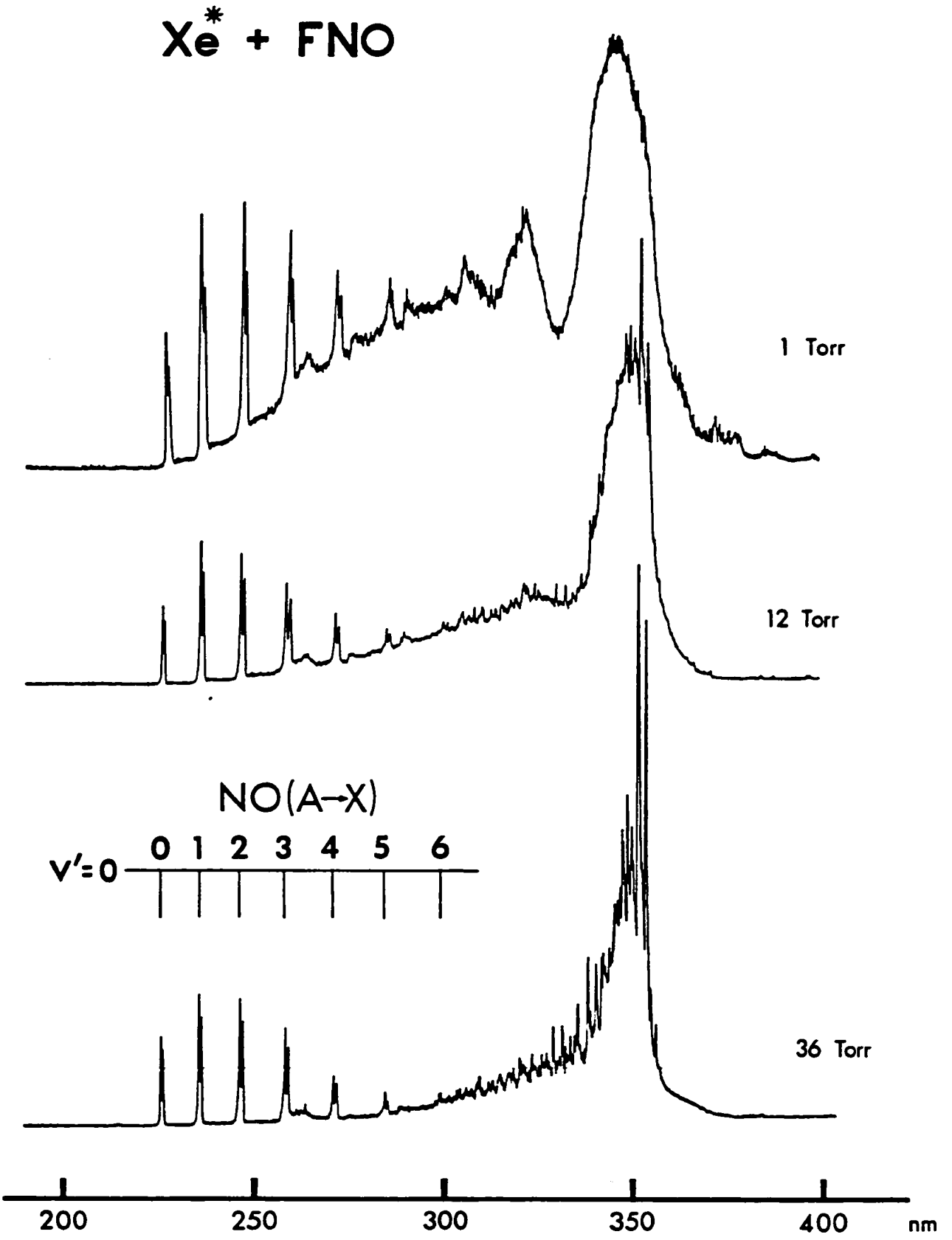
In order to establish if Ar electronically quenches the rare gas-halide excimers, the pressure dependence of the XeF* emission produced by the reaction of $Xe(^3P_2)$ with NOF was studied. This reaction, in addition to the excimer emission, also excited the NO(γ) bands. Since argon does not quench the NO(A) state, the NO(A-X) emission was used as an internal standard to which the excimer emission could be compared. The results shown in Figure 5.7, indicate that vibrational relaxation occurred without any significant electronic quenching for pressures up to 40 Torr (Although the XeF continuum decreases in intensity, the bound-bound emission gains intensity, and the ratio of the total integrated intensities for NO(A-X) and XeF(B-X) reamined unchanged).

d. <u>High resolution spectra of the bound-bound XeF* and XeCl* emission</u>. High resolution (∿0.1 nm mm^{-1}) spectra were obtained for the XeF*(B → X) and XeF*(D → X) transitions excited by the reaction of $Xe(^3P_2)$ with XeF_2. Figure 5.8 shows the vibrational assignments to the B-X spectrum,[18] and Table 5.4, contains the Deslandres scheme derived from the measurements. Similarly, the results for the XeF*(D-X) transition are presented in Figure 5.9 and Table 5.5. The listed results and assignments for the (D-X) transition are tentative, and several features remain unassigned. Final results will be published elsewhere.[19]

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Figure 5.7 Emission spectra from Xe($^3P_2$) + NOF at various pressures. Notice the extensive relaxation of the main band over the pressure range. The emission from XeF(D) excimer state at 264 nm is still observable at 36 Torr.

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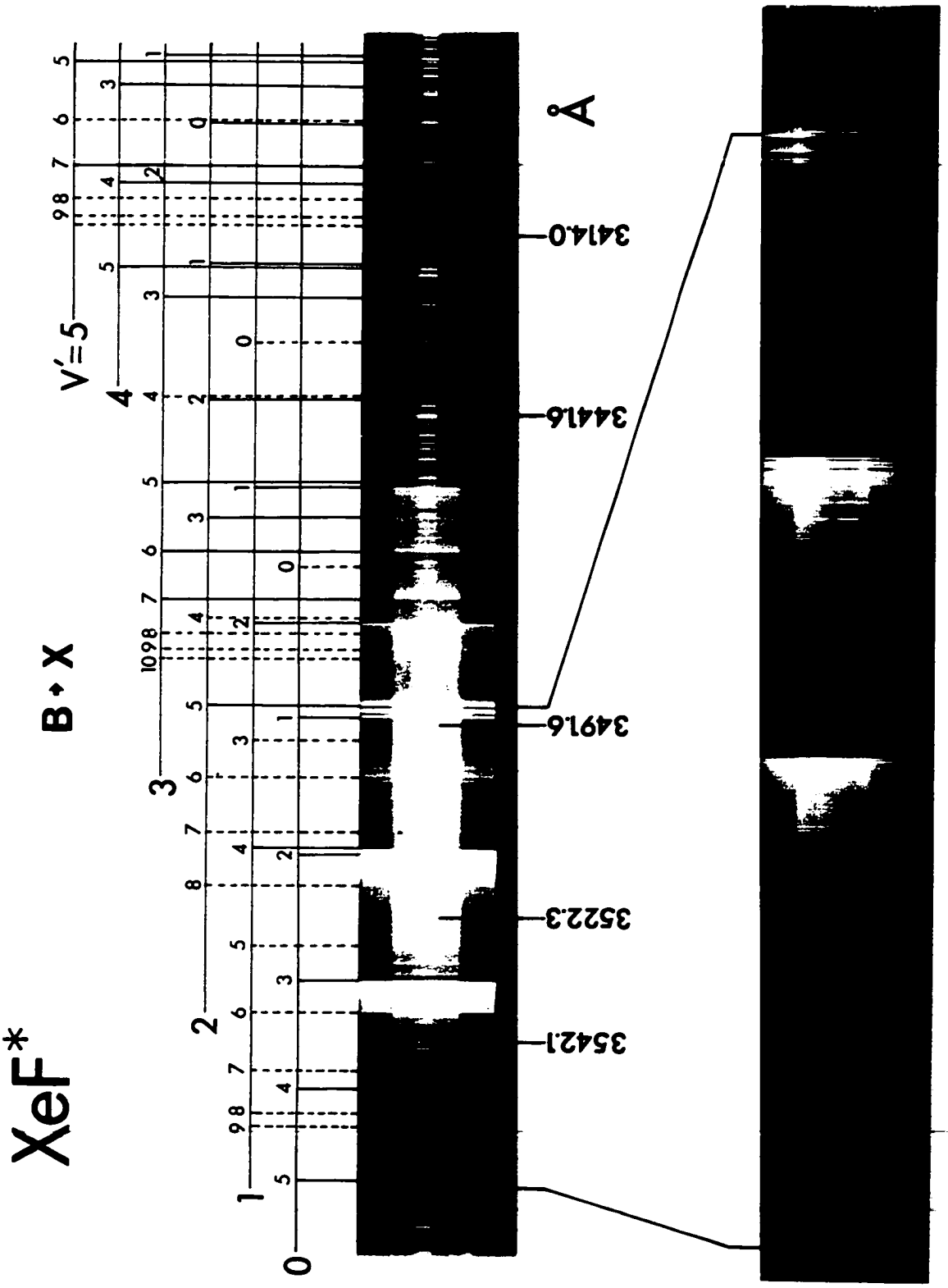
The XeCl* spectra (Figure 5.10a and 5.10b) were obtained using the $Xe(^3P_2)$ + Cl_2 reaction in a flow reactor at high pressures (~40 Torr); the results are listed in Table 5.6.

Finally, Figure 5.11 shows a detail of the complex rotational structure of the XeF*(B → X) (0,2) and (1,4) bands. The lower portion of the figure corresponds to a 10X enlargement of the original negative and the broadening of the rotational lines due to the many isotopes in Xe can be clearly seen. Spectra obtained with isotopically pure xenon will be needed before the rotational structure of this molecule can be completely analyzed.

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Figure 5.8 - Observed XeF(B → X) emission spectra. Top, 3rd order (1200 lines/mm grating); slits, 15μ; exposure time, 1 hour 40 min. Bottom, 18th order (316 lines/mm echelle grating); slits, 15μ; exposure time, 1 hour 15 min. In both cases, the emission source was the fluorescence from the reaction $Xe(^3P_2) + XeF_2$. The film used was Kodak Spectroscopic Analysis.

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Table 5.4 Deslandres Table for the observed XeF(B → X) emission.

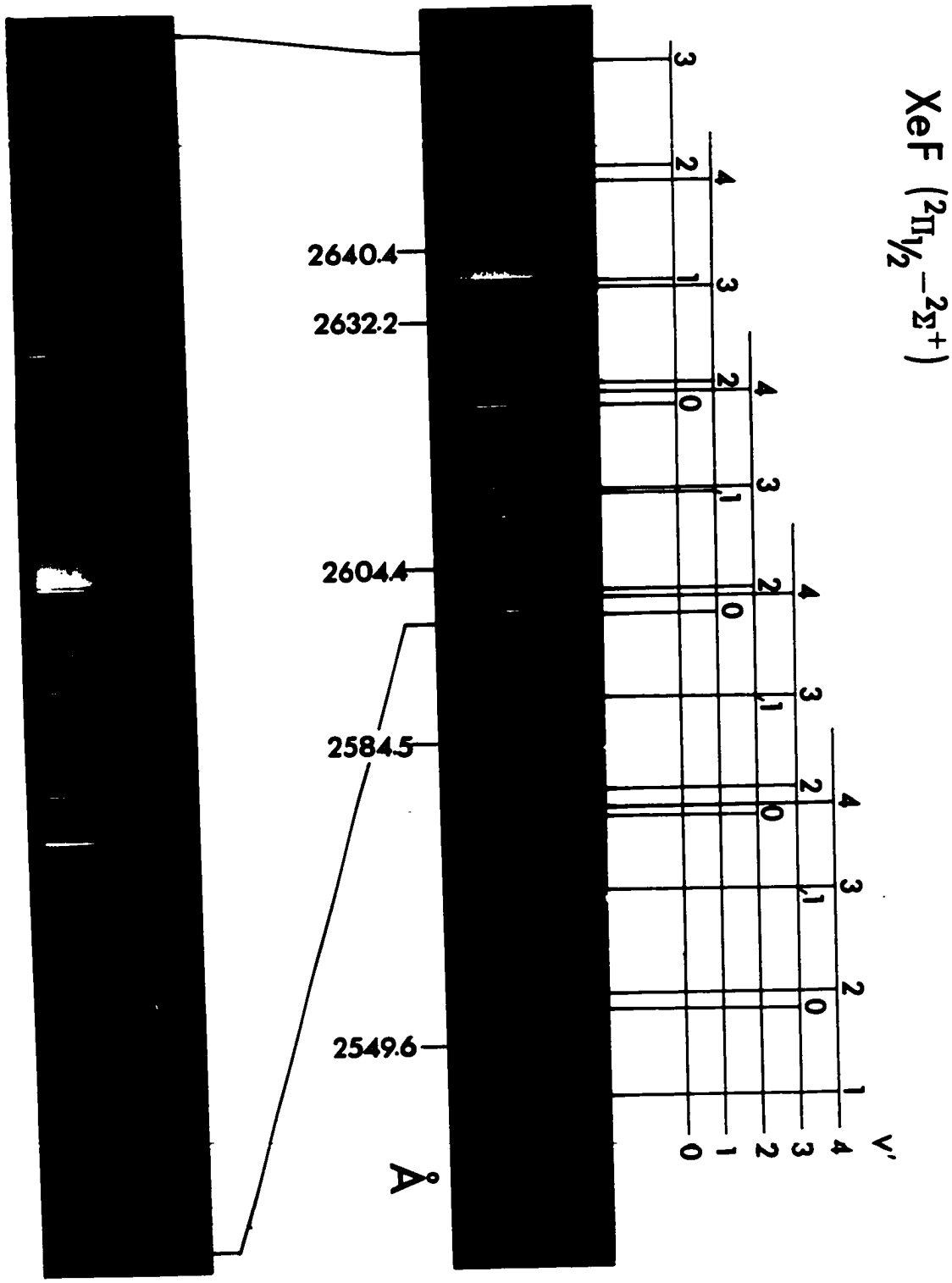
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v'/v"	0		1		2		3		4		5		6		7
0			28652 304	184	28468 305	161	28307 306	137	28170 205	116	28054	93	27961 310	64	27897 308
1	29161* 303	205	28956 305	183	28773 304	160	28613 302	138	28475				28271 302	66	28205
2	29464 300	203	29261 300	184	29077 298	162	28915* 300				28663 299	90	28573 300		
3	29764 297	203	29561 295	186	29375	160	29215 297				28962 296	89	28873	65	28808
4	30061 293	205	29856 295				29512 295				29258 294				
5	30354 292	203	30151 292				29807*	139	29668* 291	116	29552	89	29463 291	64	29399
6	30646 288	203	30443 289	185	30258 290				29959				29754		
7	30934 283	202	30732	184	30548						30130				
8	31217												30326		
9															
10	31781 275				31391								30888		
11	32056														
12															
13															

Figure 5.9 Observed XeF(D → X) emission spectra. Top, 4th order (1200 lines/mm grating); slits, 25μ; exposure time, 2 hours. Bottom, 21st order (316 lines/mm echelle grating); slits, 15μ; exposure time, 1 hour 30 minutes. In both cases, the emission source was the fluorescence from the reaction $Ar(^3P_2) + XeF_2$. The spectra were recorded on Kodak Spectroscopic Analysis Film with a 1m 1704 x Spexs Spectrograph. The wavelength calibration lines correspond to a Fe hollow cathode lamp.

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Table 5.5 Deslandres Table for the observed XeF(D → X) emission.

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Table 5.5

v'/v"	0		1		2		3		4		5		6
0	38118 347.7	201.8	37916.3 345	181.1	37735.2 345.7	165.6	37569.6 346.7	141.1	37428 347.6				
1	38465.7 342.5	204.4	38261.3 343.2	180.4	38080.9 342.5	164.6	37916.3 342	140.7	37775.6 342.5	113.1	37662.2 342.1	89.9	37573.6 340.6
2	38808.2 338.8	203.7	38604.5 333.2	181.6	38422.9 338.4	164.7	38258.2	140.2	38118.1 338.4	113.5	38004.6 339.3	91.4	37913.2
3	39146	208.3	28937.7	181.4	38756.3				38343.9	89.1	38254.8		
4													

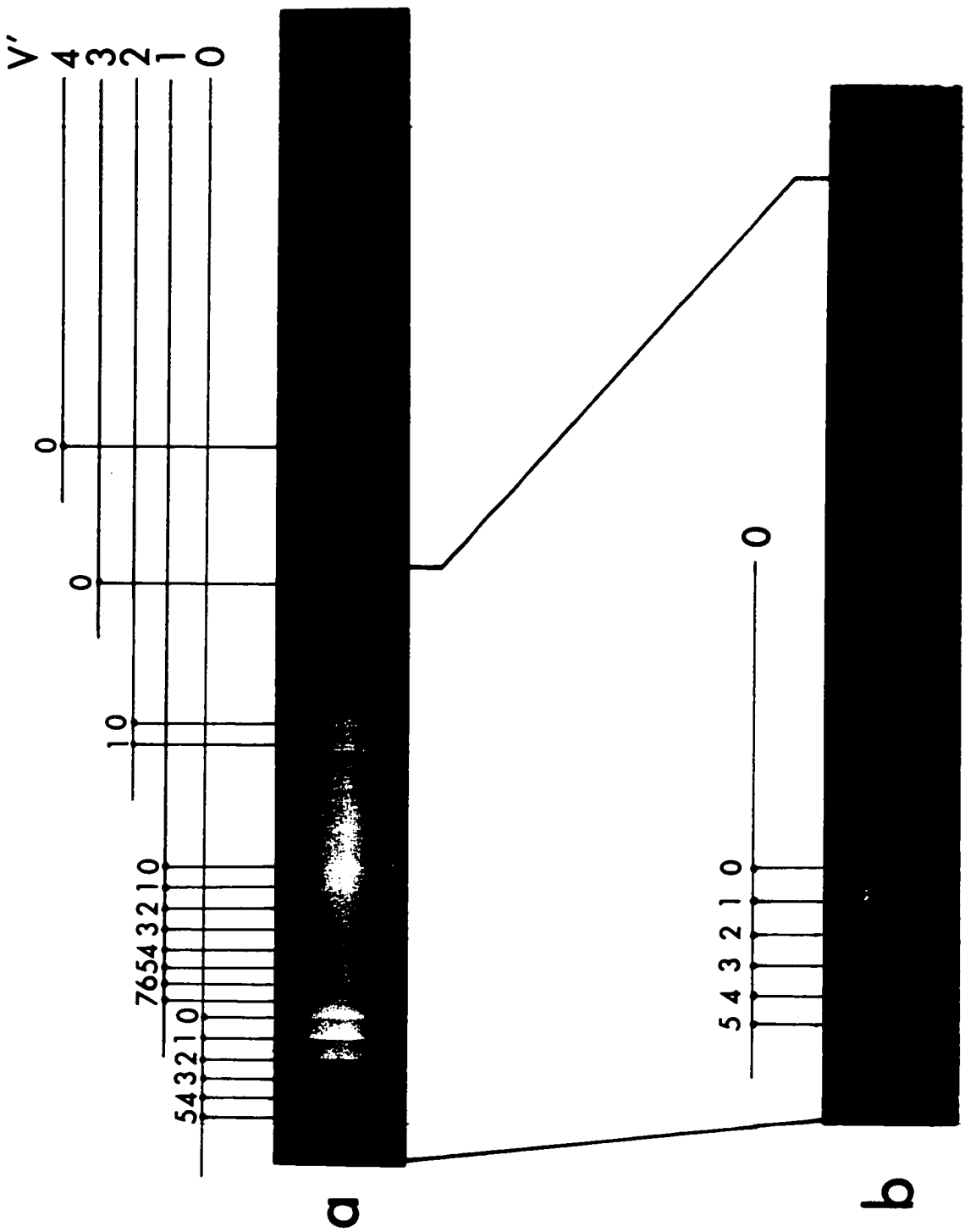
Figure 5.10 Observed XeCl(B → X) emission spectra.

(a) 3rd order (1200 lines/nm grating); slits 30μ; exposure time, 2 hours 30 minutes.

(b) 17th order (316 lines/mm echelle grating); slits 20μ; exposure time, 2 hours.

In both cases the emission source way the fluorescence from the reaction of $Xe(^3P_2) + Cl_2$. The spectra were recorded on Kodak Spectroscopic Analysis Film with a 1 m 1704 x Spexs spectrograph. The wavelength calibration lines correspond to a Fe hollow cathode lamp. The assignments are those of Tellinghuisen, et. al.[4a]

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Table 5.6 Deslandres Table for the observed XeCl(B → X) emission.

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Table 5.6[a]

v"/v'	0		1		2		3		4
0	32490.6 (26.9)	(194.3)	32684.9 (27.1)	(192.4)	32877.3 (26.1)	(192.6)	33045.3[b] 33069.9	(191.1)	33261.3
1	32463.7 (26.6)	(194.1)	32657.8	(193.4)	32851.2				
2	32437.1 (26.0)								
3	32411.1 (24.9)								
4	32386.2 (23.6)								
5	32362.6	(194.3)	32556.9 (23.8)						
6			32533.1 (20.3)						
7			32512.8						

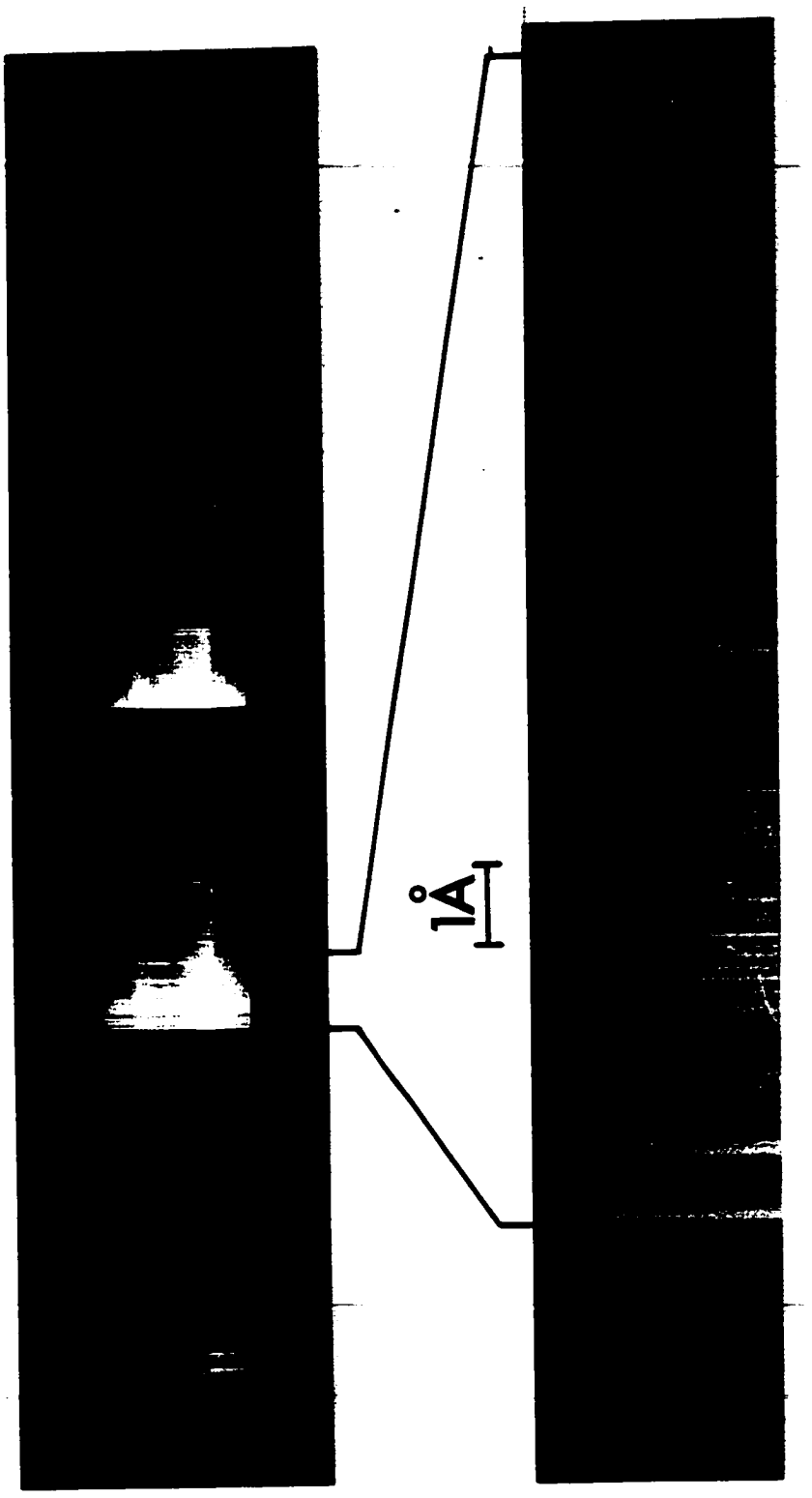
a. Vibrational assignments are from Ref. 4a.

b. This is a ^{37}ClXe band.

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Figure 5.11 - Rotational structure in the (0,2) and (1,4) bands of the XeF(B → X) emission. The original negative shown in the lower portion of Figure 5.8 was enlarged by 10X. Note the broadening of the rotational lines due to the large number of isotopes in natural xenon.

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CHAPTER 6 DISCUSSION OF QUENCHING MECHANISMS

"But I should perchance forgive the Hypothesis I have been all this while examining, if, though it reaches but a very little part of the world, it did at least give us a satisfactory account of those things to which 'tis said to reach."

Robert Boyle
"The Sceptical Chymist",
London 1661

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a. Quenching Mechanisms. Electronic quenching "reactions" and, in general, electronic-to-electronic (E → E) energy transfer processes have been under study since the early part of this century.[1] Although the volume of bulk data gathered up to the present is considerable, the mechanism governing these processes is poorly understood, partly because of the difficulties in the theoretical treatment of excited state potentials and also because some reactions proceed by more than one mechanism. The very detailed experimental data needed relating individual reactant and product states have only recently become available albeit for only a few systems. Broadly speaking, the several postulated mechanisms can be divided into three general classes, namely, Orbiting, "Golden Rule" and Curve Crossing. The most important features of each one of these classes of mechanisms are presented below, while Section 6.b discusses these mechanisms in the light of the experimental results presented in Chapters 3, 4 and 5.

Mechanisms in the first class, assume that a fraction, ω, of those incoming trajectories which surmount the centrifugal barrier lead to quenching collisions. In the simple case of two particles interacting through a C_6/r^6 attractive potential, the effective potential will be given by Equation 6.1,

$$V_{eff} = -\frac{C_6}{r^6} + \frac{Eb^2}{r^2} \qquad (6.1)$$

where the second term on the right hand side is the centrifugal repulsion; the position of barrier (at r_o) as well as the critical impact parameter b_o at collision energy E, can be determined from the (simultaneous) relations $(V_{eff})_{r_o} = E$ and $(dV_{eff}/dr)_{r_o} = 0$. The cross section is then given by

$$\sigma_Q = \omega\pi \; b_o^2 = (3/2)\omega\pi(2C_6/E)^{1/3} \qquad (6.2)$$

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where b_o is the critical impact parameter and the factor (3/2) in the rightmost expression relates b_o^2 to the square of the distance of closest approach, r_o^2. For a generalized (attractive) long-range potential C_n/r^n, this model predicts[2] that $\sigma_Q \propto (C_n/E)^{2/n}$, and the quenching cross section will depend on the type of interaction between the particles; however, dispersion interactions (r^{-6}) are commonly considered to be the most important.† This model has been applied with limited success to describe the Penning ionization by helium metastables[3b] and the quenching of Na*($3p\ ^2P$).

The mechanisms falling under the "Golden Rule" class require that reactants and products be coupled, normally <u>via</u> a continuum of product states, and are more dependent upon electronic (rather than nuclear) motions during the transition from reactant to products. The long range dipole-dipole mechanism[4-8] is representative of this class, while Penning ionization[9] - though based on the same formalism - describes the ionization process with the electronic transition occurring at short range.

The basic assumption in the dipole-dipole mechanism[4-8,10] is that the rate of transition between the initial state Ψ_i (M* + Q) and the final state Ψ_f (M + $Q^\dagger$), M is an atom and Q a molecule, is given by Fermi's Golden Rule, where $\rho(E)$ is the density of available product states and

$$k(t) = (2\pi/h)\rho(E)\left|(\Psi_i\ V\ \Psi_f)\right|^2. \qquad (6.3)$$

---

†For example, for quenching molecules with permanent dipoles, the dominant interaction is dipole-quadrupole which varies r^{-4}. This interaction however, will average to zero over one rotational period of the quencher molecule, except for symmetric top molecules with excited rotations about the axis of symmetry.[3a] However, see the next section for a fuller discussion of the intermolecular interactions.

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The interaction potential V is normally given by the electric dipole terms in the multipole expansion.[10] Using the Born-Oppenheimer approximation for Q, the matrix elements in Equation 6.3 can be evaluated to give

$$k(t) = (2\pi/h)\rho(E)q_{ij}(M_M^2 M_Q^2)/r^6 \qquad (6.4)$$

where q_{ij} are the Franck-Condon factors for the $Q \rightarrow Q^\dagger$ transitions and M_M^2 and M_Q^2 are the electronic transition moments for $M^* \rightarrow M$ and $Q \rightarrow Q^\dagger$, respectively. The transition probability is given by time integration of Equation 6.4, and the quenching cross section by the integral of the transition probability over impact parameter. For a Boltzmann distribution of molecular velocities the cross section is explicitly given by Equation 6.5.

$$\sigma(v) = \sigma_o(v) + [\pi^{7/2}/6h(2kT)^{1/2}]\rho q(M_M^2 M_Q^2/r_c^3)\mu^{1/2} \qquad (6.5)$$

In this expression, $\sigma_o(v)$ is the contribution of small impact parameters to the total cross section, k is the Boltzmann constant, μ the reduced mass, ρq replaces the more explicit $\rho(E)q_{ij}$ form, and r_c is some critical impact parameter which defines the range of validity of the integration of the transition probability over impact parameters.[10]

Thayer and Yardley[7] parametized Equation 6.5 to the form

$$\sigma_Q \propto \mu^{1/2} M_Q^2/r_c^3 \qquad (6.6)$$

while Steinfeld gives

$$\sigma_Q \propto \mu^{1/2} I\,\alpha/r_c^3 \qquad (6.7)$$

In both cases, it is assumed that $\rho(E)q_{ij}$ does not vary considerably from molecule to molecule and that the contributions from small impact parameters can be neglected. Furthermore, in Equation 6.7 the transition moment M_Q^2 has been replaced by the product of the ionization potential I and polarizability α of the quencher molecule.

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In order to accommodate the contributions from small impact parameters explicitly in the final expression, Gedenken et. al.,[8] and others,[4,5] assume that the probability of a transition is small - less than some small number γ - so the integration of the transition probability over impact parameter can be carried out for all values of b. The resulting expression for σ_Q is given by Equation 6.8.

$$\sigma_Q = (2/3)(\pi^2/\gamma^{3/5})[\pi/2h)\rho q(M_M^2\ M_Q^2/v)]^{2/5} \qquad (6.8)$$

This last expression can be written parametrically as

$$\sigma_Q \propto (\mu^{1/2}\ M_Q^2)^{2/5} \quad \text{or} \quad \sigma_Q \propto (\mu^{1/2}\ \sigma_a)^{2/5} \qquad (6.9)$$

indicating that a plot of the logarithm of σ_Q vs. the logarithm of $\mu^{1/2}\ \sigma_a$, where σ_a is the quencher photoabsorption cross section at the wavelength corresponding to the transition $M^* \rightarrow M$, should have a slope of 2/5.

The validity of the parametrized versions of the dipole-dipole theory, particularly Equations 6.6 and 6.7, have recently been questioned by Beyer and Lineberger,[11] who found that these correlations hold for systems for which the dipole-dipole mechanism should be inapplicable. Nevertheless, this formulation has been applied with varying degrees of success to quenching[6] and excitation transfer processes[12] and even to quenching of metastable states of the rare gases.[10] In principle, since M_M^2 vanishes for the metastable atoms, this mechanism should not apply to the quenching of these states; however, on the basis of limited data, the correlation of σ_Q with the photoabsorption cross section for the quencher molecule[10] (Equation 6.9) suggests that some form of collisional mixing of resonance and metastable states[10] might occur (also, see next section).

In the curve crossing mechanism, quenching proceeds by direct interaction of the potential surface for the initial state (M* + Q) and

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one or more potential surfaces leading to the final state(s) $(M + Q^+)$. When the entrance and exit channels are coupled directly (see figure 6.1a), and the collision energy is sufficient to reach the crossing point, the cross section is given by[13a]

$$\sigma_Q = 2\pi r_x^2 P_x \exp(\Delta E/kT) \qquad (6.10)$$

provided the probability of crossing from entrance to exit channels, P_x, is small; r_x is the crossing point and ΔE is the energy defect between the entrance and exit channels. The Boltzmann averaged transition probability is commonly derived from the Landau-Zener formula,[13b] for the probability of curve crossing with velocity v.

$$\exp[-(2\pi/h)H_{12}^2/v|(dH_{11}/dr)-(dH_{22}/dr)|_x] \qquad (6.11)$$

H_{12} is the interaction matrix element between the intersecting potentials H_{11} and H_{22}, and is often an adjustable parameter due to the lack of accurate excited state potentials. Quenching cross sections, according to this model, are not expected to be much larger than $\sim 10\mathring{A}$, as can be seen by assuming $r_x = 5\mathring{A}$, $\Delta E = -kT$ which gives $\sigma_Q = 60P_x(\mathring{A})^2$ with P_x not much larger than 0.1. Larger cross sections could be obtained if the entrance and exit channels are nearly resonant or if the entrance channel crosses a large number of exit channels. More sophisticated treatments[14] of the same problem allow for the vibrational levels in both the reactant and product surfaces, and also use more refined expression for the coupling or inter-potential interaction (see Section 6.d).

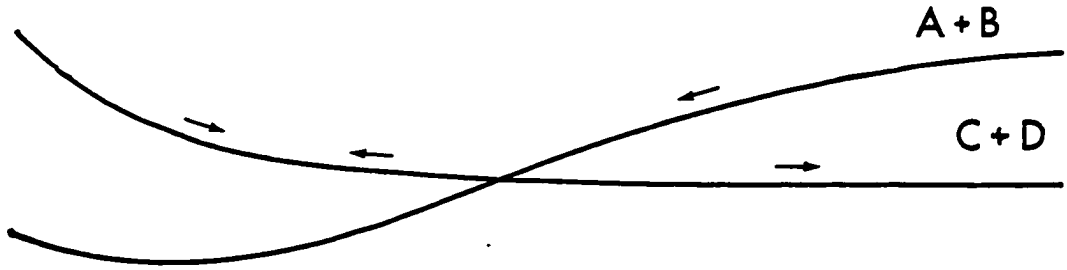
Figure 6.1b depicts the case where the input and exit channels do not interact directly, but rather, are coupled by an intermediate potential surface, which in this case is an ion-pair curve. If the entrance channel is flat to distances close to the crossing point, the value of r_x can be

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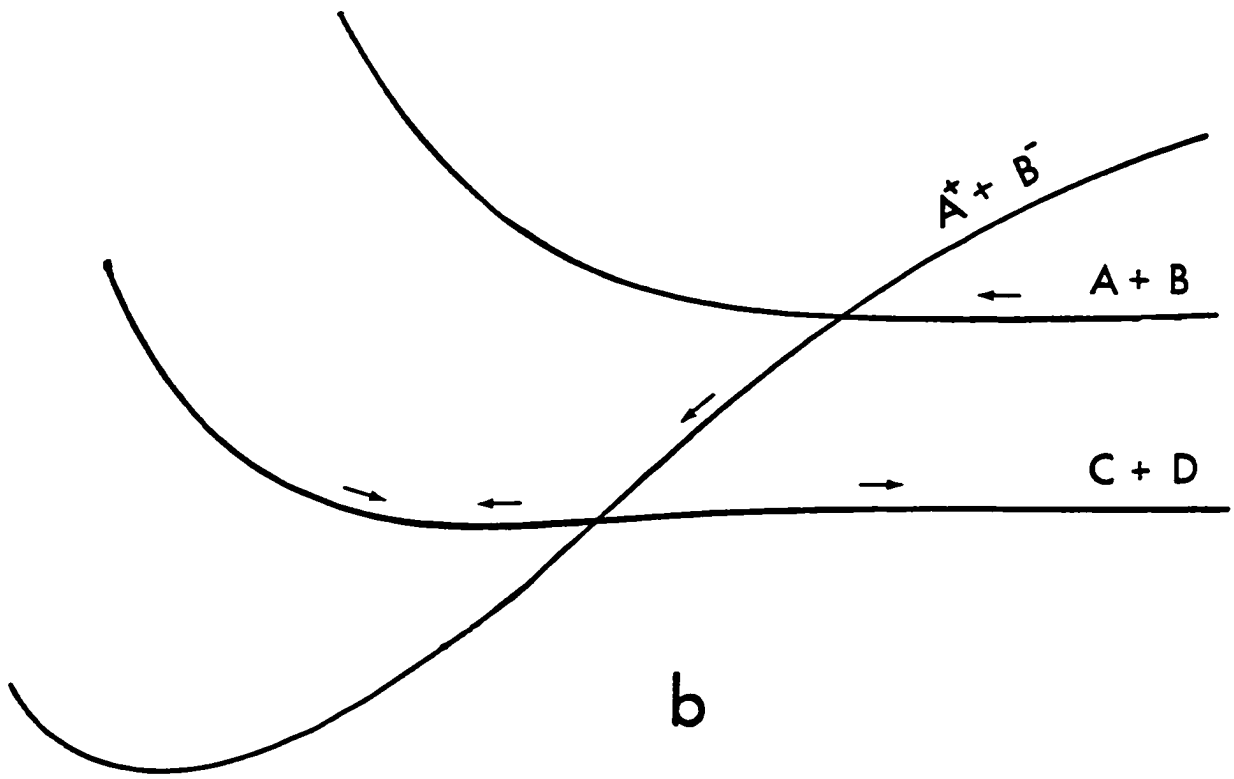
Figure 6.1 - a. Schematic diagram of the simple curve crossing between the entrance (A+B) and exit (C+D) channels.

b. Curve crossing with ionic intermediate. If the second crossing does not occur, the reaction produces A^+B^-.

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a



b

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determined from the condition $(H_{11} = H_{22})_{r_x}$, or $(e^2/r_x) = IP(M^*) - EA(Q)$, which gives

$$r_x = e^2/[IP(M^*) - EA(Q)] \qquad (6.12a)$$

$$r_x(\mathring{A}) = 14.41/(IP-EA) \qquad (6.12b)$$

In the last expression, the ionization potential and electron affinity are expressed in eV. Since at the crossing point an electron transfer occurs, this mechanism can only be of importance for systems in which one of the collision partners has a low ioninzation potential and the other, a large (positive) electron affinity. Once the system crosses over to the ionic curve, it is then coupled to a variety of exit channels, as the ion-pair curve crosses them. Of course, the distribution of products will be determined by the relative strength of the couplings between the ionic curve and the various product channels. In the special case when the second crossing does not occur, either because the interactions are very weak or because no other exit channels exist, the collision retains the ionic character and leads to reactive quenching,[15] in analogy to the reactions of the alkali-metal atoms with halogen and halogen-containing molecules.[16] Several specific features of reactive curve crossing are discussed in more detail in sections 6.c and 6.d.

b. Mechanism(s) for quenching of Metastable States.

The cross sections measured in the present work are "thermal averages", i.e., $\sigma_Q = k_Q/\langle v \rangle$, where $\langle v \rangle$ is the mean Boltzman speed, while the expressions derived with the orbiting and curve-crossing models are cross sections for a specific velocity; the final form of the dipole-dipole derivation gives velocity averaged cross sections. Although the distinction is of fundamental importance, it does not have any practical effect on the discussion that follows.

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The correlation of the cross sections for quenching of $Ar(^3P_2)$ with the polarizability of the quencher molecule has been noted previously,[17] The orbiting model was tested by Piper et. al.[10] for the "efficient" quenchers, $\sigma_Q > 30$ $\mathring{A}^2$, using the attractive part of the Van der Waals dispersion forces ($-C_6/r^6$) as the interaction potential; the plot of ln σ_Q vs. ln C_6 had unity slope[10] instead of the 1/3 slope predicted by the Orbiting Model (see Section 6.a). A similar plot is shown in Figure 6.2 for a larger number of quenchers. Although this correlation plot dismisses the simple orbiting model with the C_6/r^6 potential (with unit probability of quenching for trajectories crossing the orbiting barrier) as a possible mechanism for the quenching of the metastable states, it reinforces the suggestion that the attractive part of the interaction potential plays an important role in determining the total quenching cross section. For the less efficient quenchers, $H_2(D_2)$, N_2, CO and CF_4, the cross section appears to be independent from the polarizability.[10,17] Although this model gave a satisfactory description of the energy dependence of the quenching cross sections for $Na^*(3^2p)$,[3a] a plot of the cross sections for quenching vs the C_6 coefficient (not shown) fails to give the expected relationship; this can be seen easily, since low polarizability quenchers such as N_2, $H_2(D_2)$ and CO have large cross sections,[18] whereas, the saturated hydrocarbons, with larger polarizabilities, have very small cross sections.[18]

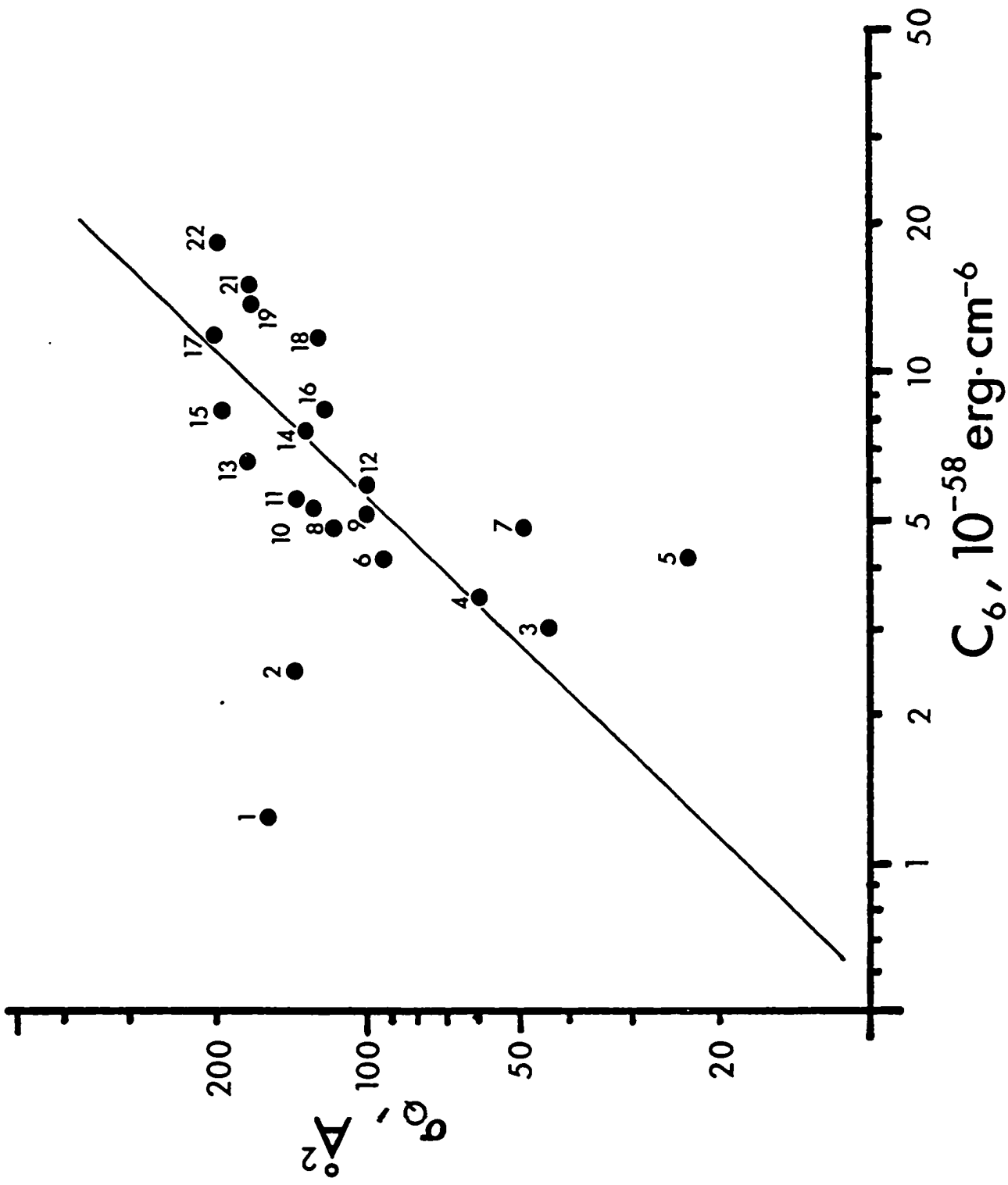
Table 6.1 summarizes the available photoabsorption cross sections at 149 nm and 107 nm for molecules of known cross sections towards the quenching of $Xe(^3P_2)$ or $Ar(^3P_2)$. A plot of σ_Q for $Xe(^3P_2)$ vs. the photoabsorption cross section at 149 nm is shown in Figure 6.3. If the experimental values for O_2 and NO are ignored, the data correlate best with a line of slope ~3/20 (=0.15) and not with a slope of 2/5 (=0.4), indi-

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Figure 6.2 - Correlation diagram between the cross sections for quenching of $Xe(^3P_2)$ and the C_6 coefficient of the Van der Waals interaction. The C_6 coefficients were calculated using the Slater-Kirkwood approximation. The solid reference line has unit slope.

1, F_2; 2, OF_2; 3, O_2; 4, NO; 5, NF_3; 6, NOF; 7, CH_4; 8, HCl; 9, CO_2; 10, C_2H_2; 11, CF_3OF; 12, N_2O; 13, HBr; 14, NOCl; 15, Cl_2; 16, C_2H_6; 17, Br_2; 18, C_3H_8; 19, ICl; 21, C_4H_{10}; 22, C_5H_{12}.

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Table 6.1 Summary of photoabsorption cross sections at 107 nm and 149 nm.

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Table 6.1

Molecule	σ_a(Mb)[a,b] 107 nm	σ_a(Mb)[a,b] 149 nm	σ_Q(Å^2)[c] Ar($^3P_2$)	σ_Q(Å^2)[c] Xe($^3P_2$)
O_2	3.27[d]	16.36[d]	35	44
CO_2	20.45[d]	0.61[d]	97	103
N_2O	36.81[d]	2.54[d]	81	100
NO	8.18[d]	4.09[d]	36	59
Br_2	95.0 [e]		65	
HCl	5.0 [f]	3.0 [f]	61	119
H_2O	4.0 [g]		61[h]	
H_2S	40.0 [g]	2.0 [g]	140[h]	
NF_3	>8.0 [i]	3.0 [i]	28	23
CH_4	28.63[d]	0.007[d]	45(81)[h]	49
C_2H_6	41.43[i]	1.0 [i]	109	125
C_3H_8	54.5 [j]	4.0 [j]	134	123
C_4H_{10}	81.0 [j]	12.0 [j]	149	170
C_5H_{12}	113.74[l]	25.0 [l]	200[k]	192
C_2H_2	27.13[l]		89	
C_6H_6	63.7 [l]		160[h]	
CH_3F	19.21[m]	0.4 [m]	120[h]	
CH_2F_2	8.16[m]	0.2 [m]	[illegible]	91
CHF_3	6.6 [m]	0.0 [m]	64	99
CF_4	~0.4 [n]	0.0 [n]	8	0.6
CH_3OH	30.78[l]		100	<0.1
CH_3Br	61.48[l]		180[h]	
CH_3CHO	35.83[l]		113[k]	
H_2	~0.01[d]	~0.01[d]	3.6	0.9
N_2	~0.01[d]	0.01[d]	5.8	3.7
CO	~0.01[d]	0.01[d]	2.3	7.0

a. The absorption cross section are for T = 298°K and 1 atm.

b. 1 Mb = 10^{-18} cm^2

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Table 6.1 ref. cont.

c. Unless noted otherwise, the quenching cross sections were measured at Kansas State University.

d. From Reference 19.

e. J. H. Carver and J. H. Gardner, J. Quant. Spectrosc. Radiative Transfer, 12, 207 (1972).

f. J. A. Myer and J. A. R. Samson, J. Chem. Phys., 52, 266 (1970).

g. K. Watanabe and A. S. Jursa, J. Chem. Phys., 41, 1650 (1964).

h. M. Bourene and J. LeCalve', J. Chem. Phys., 58, 1452 (1973).

i. S. R. LaPaglia and A. B. F. Duncan, J. Chem. Phys., 34, 1003 (1961). These authors measured σ_a down to 125 nm and observed significant increase in σ_a below this wavelength, which they attributed to Rydberg transitions. The listed absorption cross section at 107 nm was extrapolated from Figure 1 of this reference and should be considered a lower limit.

j. H. Okabe and D. A. Becker, J. Chem. Phys., 39, 2549 (1963).

k. H. J. DeJong, Chem. Phys. Lett., 25, 129 (1974).

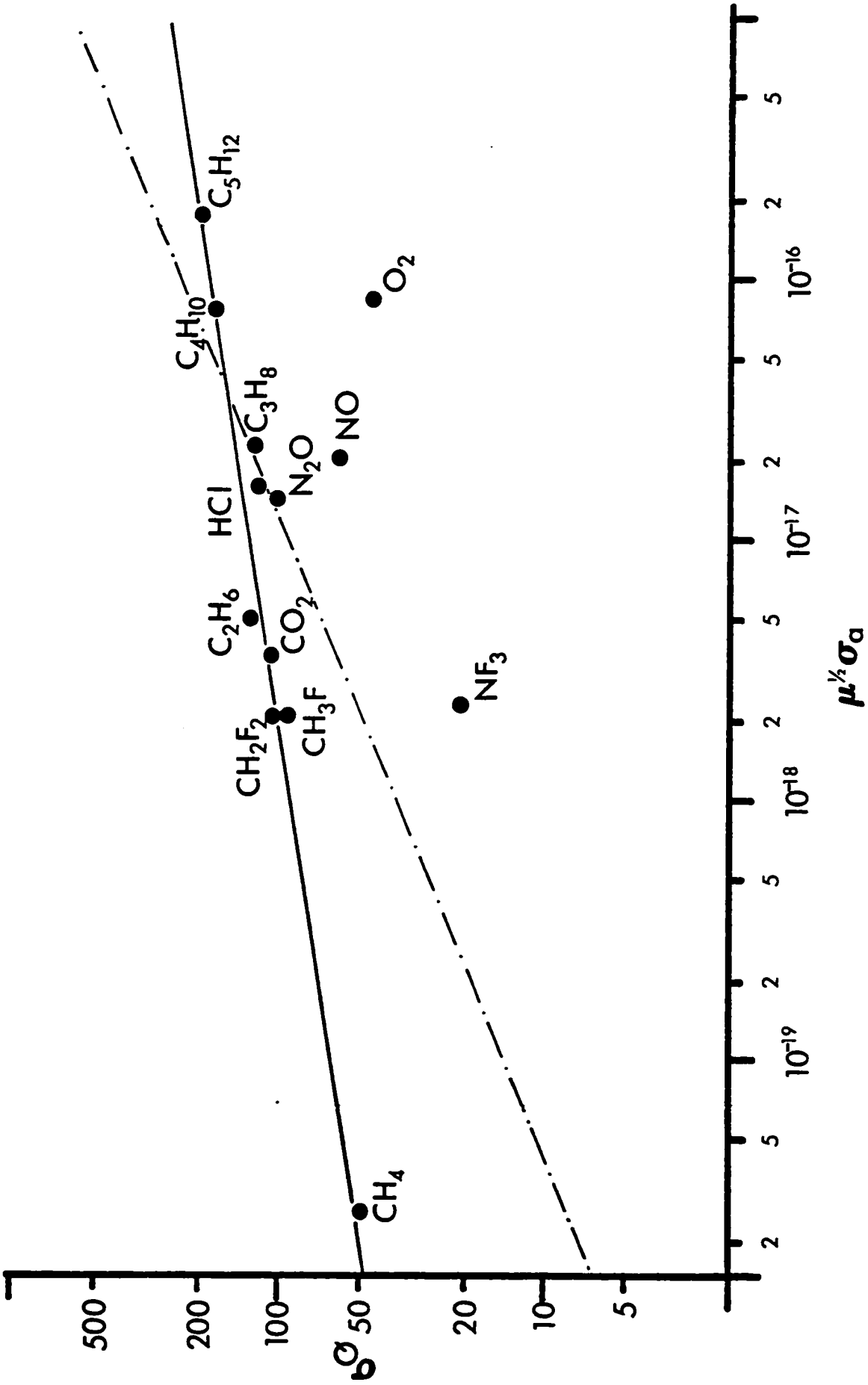
l. J. C. Person and P. P. Nicole, ANL Radiation Physics Division, Annual Report, 7/69-6/70, ANL-7760, Part I and references therein.

m. L. Edwards and J. W. Raymonda, J. Amer. Chem. Soc., 91, 5937 (1969). The values reported by these authors were scaled up so their measurement for CH_4 would agree with the photoabsorption cross section of CH_4 at 107 nm as measured by Watanabe, et. al. (Ref. 20).

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Figure 6.3 - Plot of the Xe($^3P_2$) quenching cross sections vs. the photoabsorption cross section of the quencher molecule at 149 nm. The broken line has a slope of 2/5 as predicted by the dipole-dipole moment. The solid line has a slope of 0.15.

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cating clearly that the dipole-dipole mechanism is not operative. This test of the dipole-dipole mechanism is more stringent than the previous one[10] made for the argon metastable quenching using the quencher optical absorption coefficients at 107 nm (shown in Figure 6.4) because the variations in σ_a are much more pronounced in the Xe* energy range than in the Ar* range. This can be seen more clearly from the data in Table 6.2, which list the calculated and experimental ratio $\sigma_Q^{Xe}/\sigma_Q^{Ar}$. From Equation 6.8, this ratio is given by

$$(\sigma_Q^{Xe}/\sigma_Q^{Ar}) = [(\mu_{Xe}/\mu_{Ar})^{1/2} (M_{Xe}^2/M_{Ar}^2) (M_{Q,Xe}^2/M_{Q,Ar}^2)]^{2/5} \quad (6.13)$$

assuming that the change in $\rho(E)q_{if}$ is not appreciable. The ratio of the transition moments for the atomic excited states was calculated using the radiative lifetimes of the respective resonance states.[19] The mass factor varies slightly, from 1.0 to 1.11 for quencher masses from 2 to 80, while the second factor is a constant and equal to 2.17; since most quenchers absorb more strongly at the argon resonance line than at the xenon resonance line, the last factor is less than 1.0 except for oxygen. Therefore the variations in this factor amounts to only a factor of 2 for an order of magnitude change in either of the absorption coefficients (provided the ratio remains less than unity), and the test in this region is not very sensitive to the relatively small uncertainty in σ_a obtained from different sources. In the case of oxygen, absorption is stronger at the xenon transition - 400 cm^{-1} vs 80 cm^{-1} than at the argon resonance transition[20] - and the calculated σ_{Xe}/σ_{Ar} ratio is a factor of 2 larger than the experimental value. On the other hand, the absorption coefficient of CH_4 is only[20] 0.18 cm^{-1} at 149 nm as compared to $\sim$700 cm^{-1} at 107 nm,[20] giving a theoretical ratio twenty times smaller than the experimental value.

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Figure 6.4 - Plot of the $Ar(^3P_2)$ quenching cross sections vs. the photoabsorption cross section of the quencher molecule at 107 nm. The solid line has a slope of 2/5. For comparison, the broken line has a slope of 0.15 (see Figure 6.3).

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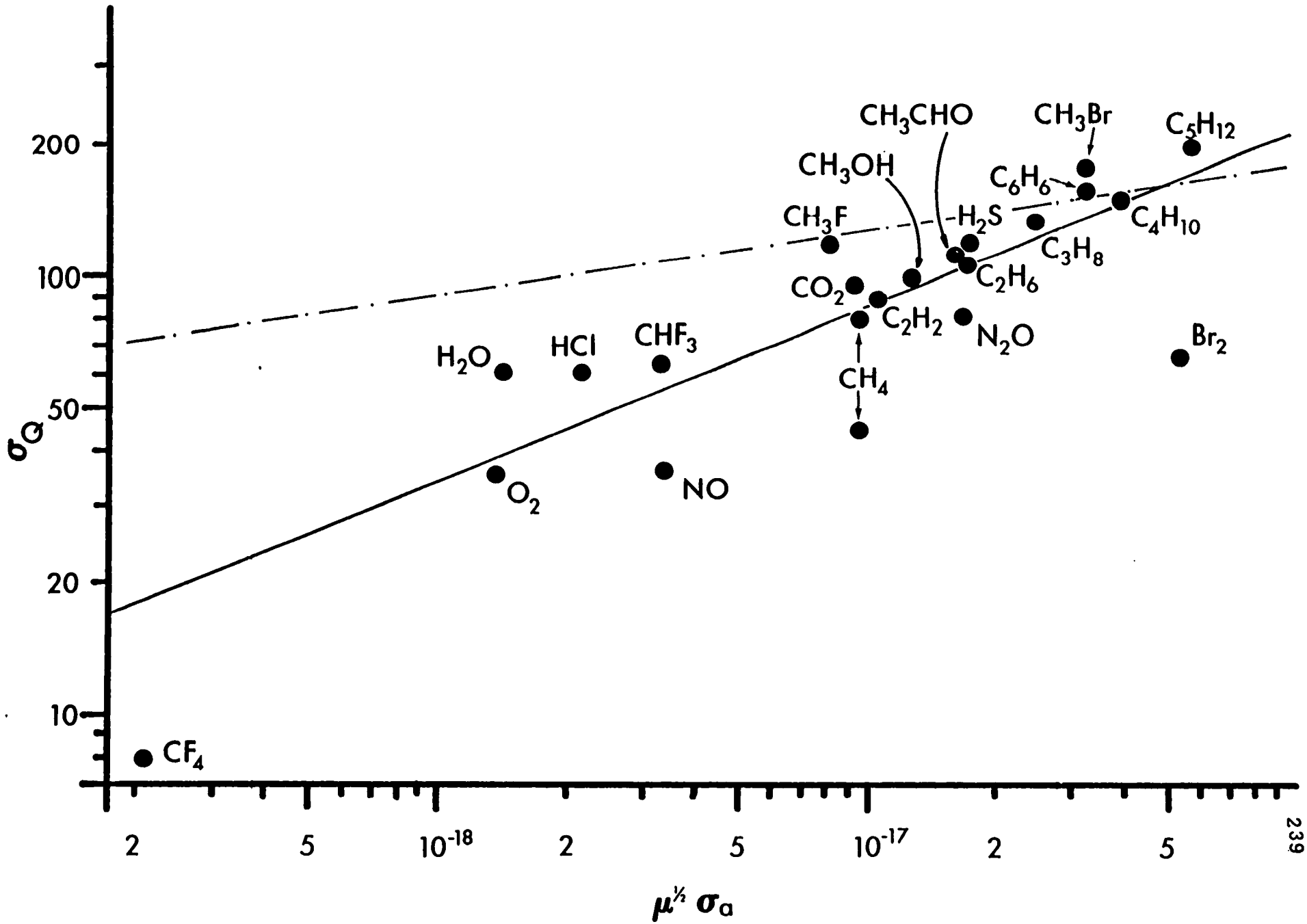


Table 6.2 Comparison of experimental and calculated (using Equation 6.13) values of $\sigma_Q^{Xe}/\sigma_Q^{Ar}$.

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Table 6.2

Quencher	Calc.[a]	Exp.
O_2	2.80	1.26
CO_2	0.34	1.06
N_2O	0.51	1.23
NO	1.11	1.64
CH_4	0.05	1.09(0.6)
CH_3F	0.31	0.76
C_2H_6	0.33	1.15
C_3H_8	0.52	0.92
C_4H_{10}	0.71	1.14
C_5H_{12}	0.84	0.96
HCl	1.20	1.95
NF_3	1.03[b]	0.82
H_2	1.40	0.25
N_2	1.46	0.67
CO	1.46	3.0
CF_4	0	0.0012
CHF_3	0	0.009

a. The ratio $\sigma_Q^{Xe}/\sigma_Q^{Ar}$ was calculated using Equation 6.13.

b. See footnote i of Table 6.1.

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The weak quenchers, H_2, N_2 and CO have vanishingly small (but comparable) absorption coefficients in the regions of interest.[20] Assuming the last fraction in Equation 6.13 to be nearly unity, the cross sections for quenching $Xe(^3P_2)$ should be ~1.5 times larger than for $Ar(^3P_2)$, when in fact they are smaller for H_2 and N_2. The absorption coefficient of CO at the frequency of the $Xe(^3P_2 \rightarrow {}^1S_0)$ transition is probably larger than at the $Ar(^3P_2 \rightarrow {}^1S_0)$ frequency because of the proximity of the strong $CO(X'\Sigma;\ v''=0) \rightarrow CO(A^1\Pi,\ v'=1)$ absorption at 149.6 nm.

The agreement between the more quantitative predictions of the dipole-dipole mechanism and the experimental observations is, in general, poor, although the magnitudes of the cross sections for quenching of the metastable states of the rare gases do increases with increasing magnitude of the quencher absorption coefficients. The correlation of σ_Q with $(\mu^{1/2}\ \sigma_a)^{2/5}$ for $Ar(^3P_2)$ is probably fortuitous.

The quenching of the rare gas metastable states by halogenated donors has been shown to proceed by the curve-crossing mechanism via an ionic-covalent interaction (see Section 6.d for a fuller discussion of this mechanism) and from the discussion above, neither the orbiting model nor the dipole-dipole scheme in their present forms should be expected to account for the specific behavior of the quenching cross sections. Both mechanisms however, correctly predict the general trends, i.e., the dependence of σ_Q on the interaction potential (orbiting) and small quenching cross sections for reagents with small (or zero) absorption coefficients at the energy of the metastable state (dipole-dipole). In both mechanisms, the long range interaction between the excited atom and the quencher molecule is assumed to be governed by the Van der Waals attraction (London's "dispersion forces") with an interparticle coordinate dependence of r^{-6}. This is probably not a good approximation.

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In his original Quantum Mechanical treatment of the interatomic (or molecular) interactions, London[21a] used perturbation theory ("uncharge resonanz") to obtain the average or effective perturbation energy. The first step is to determine the classical interaction: for two atoms at a distance r from each other, the instantaneous (classical) potential energy can be expressed as a power series of (1/r). The leading term in this series in $(1/r^3)$ for the interaction of neutral particles; for ions, the series starts with (1/r), while for ion-molecule, the leading term is $(1/r^2)$. Thus, for neutral molecules the main interactions are dipole-dipole $(1/r^3)$, dipole-quadrupole $(1/r^4)$ and quadrupole-quadrupole $(1/r^5)$:

$$V = \frac{A_1}{r^3} + \frac{A_2}{r^4} + \frac{A_3}{r^5} \tag{6.14}$$

where, for instance[21b]

$$A_1 = e^2 \sum_{ij} (2z_i^a a_j^b - x_i^q x_j^b - y_i^a y_i^b). \tag{6.15}$$

In Equation 6.15, the superscripts a, b are used to label the nuclei, and the z, x, y refer to the individual electronic coordinates centered at a particular nucleus. The coefficients A_2 and A_3 have cubic and quartic dependence on the electronic coordinates, respectively.[21] Since the actual behavior of atoms or molecules depends on the average of the instaneous potential over the electronic motions, the next step in London's derivations was to find the quantum mechanical mean of V by perturbation calculations.

If the unperturbed states of the atoms (or molecules) are non-degenerate, the perturbation energy is given by perturbation theory as[21]

$$\Delta E = \Delta E_1 + \Delta E_2 + ... = V_{kk} + \sum_{i \neq k} \frac{|V_{ki}|^2}{E_k - E_i} + ... \tag{6.16}$$

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In the case of a degenerate unperturbed state, ΔE is no longer given by Equation (6.16) but instead, by the n roots of the determinant

$$|V_{ij} - \Delta E\ \delta_{ij}| = o \tag{6.17}$$

in which the V_{ij} are the matrix elements $\langle \Psi_i | V | \Psi_j \rangle$ that couple the n components (Ψ_i) of Ψ. If both atoms are in S states, the first order perturbation, $\Delta E_1 (= V_{kk})$ vanishes[21] and the interaction energy is given by

$$\Delta E_2 = \sum_{i \neq k} \frac{|V_{ik}|^2}{E_k - E_i} = \frac{1}{r^6} \sum_i \frac{|C_{ik}|^2}{E_k - E_i} \tag{6.18}$$

where E_k is the sum of the ground state energies of the two atoms and E_i the sum of the excited states energies. Since $E_k < E_i$, ΔE_2 is always negative resulting in an attractive force between the two atoms. This term is commonly written as (C_6/r^6), explicit forms for C_6 having been derived by London[21a,23] and Slater and Kirkwood[24] among others. In the interaction of saturated molecules with one another, the induction effect, and, if the molecules are polar, the orientation effect are added to the dispersion effect, both causing additional Van der Waals attraction. The origin of these forces is the same,[21b] and the distinction is mostly historical. The dispersion effect is obtained if, in the second order perturbation formula (Equation 6.18), the summation is extended only over matrix elements corresponding to large (electronic and vibrational) energy differences; the orientation or alignment effect is obtained by summing only over small (rotational) energy differences. If the sum of one molecule is over large differences and for the other molecule over small differences, the induction effect is obtained.[21,23b] These effects are normally additive.

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For atoms in normal states other than ^1S states, Knipp[25] has shown that a first order perturbation results, on account of the quadrupole-quadrupole interaction which varies as $1/r^5$. For interatomic distances of twice the atomic radii, this interaction gives binding energies that amount to a sizeable fraction of an electron volt,[25] which indicates the possibility of molecule formation by "L-valences."[21]

If the atoms are alike but in different electronic states, the degeneracy at large intermolecular separation corresponding to the exchange of excitation energy leads to a first order perturbation which varies as $1/r^3$, that is, there is a Van der Waals attraction at much greater separations than for atoms in their ground state. This interaction is called "symmetric" resonance force. Consider the case of two atoms, one in its normal state (with L=0) and the other in an excited state (L = 1).[21b] There will be be resonance degeneracy between the functions

$$\Psi_1 = \psi_o(a)\psi_1(b) \text{ and } \Psi_2 = \psi_1(a)\psi_o(b) \tag{6.19}$$

retaining for simplicity only the dipole-dipole part of the interaction potential, it is apparent that the matrix elements between Ψ_1 and Ψ_2 are

$$\begin{aligned} V_{11} &= V_{22} = 0 \\ V_{21} &= V_{12} = (e^2/r^3)(2Z^2-X^2-Y^2) \end{aligned} \tag{6.20}$$

$V_{21} = V_{12}$ can be written in terms of the oscillators strengths as [21b]

$$V_{12} = V_{21} = \frac{3}{2r^3} \frac{(he)^2}{m(E_1-E_o)} \frac{1}{(2\pi)^2} (2f^z - f^x - f^y) \tag{6.21}$$

The secular equation (Equation 6.17) is for this case

$$\begin{vmatrix} -\Delta E & V_{12} \\ V_{12} & -\Delta E \end{vmatrix} = 0 \tag{6.22}$$

with solutions $\Delta E = \pm V_{12}$.

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This analysis can be extended to more complicated cases; the main features of resonance forces are[21b]: (i) they are proportional to $(1/r^3)$, (ii) their strength is proportional to the transition probabilities connecting the states, (iii) numberically, they are ∿100 times as strong as the dispersion forces near the distance of closet approach, (iv) the mean over all m values vanishes. In addition to resonance forces, like-atoms in different electronic states also experience dispersion forces $(1/r^6)$ but with different and interesting properties; first, these forces can be either attractive or repulsive, and secondly, they are anisotropic. The anisotropic nature of the dispersion forces for complex molecules has been discussed by Boer and Heller.[26] Finally, quadrupole-quadrupole interactions of the type described by Knipp[25] also contribute, although to be sure, to a much lesser extent.

The discussion so far has been limited to collisions between like atoms, and in the case of collisions between excited and ground state atoms, to symmetric resonances. It remains to consider the general features of energy transfer collisions in which there is a close approach to energy resonance, though the collision is between two different kinds of atoms ("unsymmetrical or accidental resonance"). The problem is of considerably more complexity, and it is treated by time-dependent perturbation theory using the asymptotic forms of the wave functions.[27,28] If the transitions are associated either with dipole or quadrupole moments, the behavior is essentially the same whether the resonance is symmetrical or merely accidental.[27] The reason being, that, since the interaction is of long range, the cross section is independent of the behavior at close range, and depends only on the asymptotic form of the interaction potential, the energy discrepancy and the relative velocity of the collision.[27] The absolute value of the cross sections in accidental resonance however, is expected to be smaller than for symmetric resonance.[28]

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Massey[27] has used semi-classical scattering theory to estimate the order of magnitude of the cross sections for the case when the transitions in the collision of like partners are of the dipole-dipole type and for quadrupole-quadrupole. In the first case, the expected cross sections are $\sim 10^{-14}$ cm^2 (100 $\mathring{A}^2$) while the quadrupole-quadrupole cross sections are expected to be $\sim 10^{-15}$ cm^2 (10 $\mathring{A}^2$).

In the derivation of the dispersion forces, it is implicitly assumed that the first excited state is far above the state under consideration so no interaction between these levels takes place. When this condition is not met, the r^{-6} law does not hold, and strong attraction may result, as in the case of BeH or MgF.[29] Obviously, this breakdown of the dispersion law should be quite possible for the interaction of excited atomic states with any collision partner.

From the foregoing analysis, the following conclusions emerge.

(a) The use of only C_6/r^6 as the attractive interaction potential in collisions involving electronically excited states is probably an oversimplification.

(b) A more realistic interaction potential should depend on the optical properties of the quencher. In addition to the dependence on polarizability, transition probabilities to specific states should be involved.

(c) Dipole-dipole, dipole-quadrupole and quadrupole-quadrupole transition will lead to cross sections with a magnitude spread of a factor of ~ 10.

The correlation presented in Figures 6.3, 6.4 and Table 6.2 show that the dipole-dipole mechanism can not explain the experimental observations; modifications in the interaction potential as discussed above, will not change the predictions of this model, simply because the model starts

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with the instantaneous (classical) potential which is later incorporated in the quantum-mechanical formulation. The orbiting model, on the other hand, depends on the total long range attractive quantum-mechanical average potentials and hence, modifications of these potentials will likely change some of the features of the model. Thus, the observed correlation σ_Q with the photoabsorption cross section follows naturally, since the realistic potential depends explicitly on the oscillator strength of the transitions involved. Also, the similarity in the magnitudes of the quenching cross sections for the resonance and metastable states of Ar observed previously[10] can be understood in terms of the magnitudes of the accidental resonance forces. Unfortunately, at present, there are no explicit forms for a "realistic" potential with all the properties required by the preceeding analysis. It is interesting nevertheless, to see the effect of a more attractive potential on the position of the centrifugal barrier, the magnitude of the critical impact parameter and the crossing radius in the case of curve crossing mechanisms (see Section 6.d.ii).

In the simple case of a $V^- = -C_6/r^6$ attractive potential, the location of the centrifugal barrier is given by[2]

$$r_o = (2C_6/E)^{1/6} \tag{6.23}$$

If an attractive potential $-C_3/r^3$ is included in V^-, the conditions $V^-(r_0) = E$ and $(dV^-/dr)_{r_o} = 0$ give

$$2E\, r^6 - C_3 r^3 - 4C_6 = 0; \tag{6.24}$$

a simple change of variables $r^3 = x$ transforms Equation 6.24 into the standard quadratic form with solution

$$r_o = \{(C_3/4E) + [(C_3^2/16E^2) + (2C_6/E)]^{1/2}\}^{1/3} \tag{6.25}$$

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The values for r_o obtained from Equation 6.25 will be greater than those obtained from Equation 6.23 if $C_3 > 0$. In a similar fashion, it can be shown that if the additional attractive force is of the form $-C_4/r^4$, r_o will be given by

$$r_o = \{[(C_6/E) + [(C_6^2/E^2) - (C_4^3/27E^3)]^{1/2}]^{1/3} + [(C_6/E) - [C_6^2/E^2) - (C_4^3/27E^3)]^{1/2}]^{1/3}\}^{1/2} \qquad (6.26)$$

and in general, r_o given by Equation 6.26 will be larger than the value given by Equation 6.23, provided C_4 is finite. These results can be generalized to any attractive force, but closed-form expressions can not be derived easily and, in fact, in the majority of the cases it is impossible to obtain a closed-form. Numerical solutions of the relevant equations are then necessary to obtain r_o.

The crossing radius r_c, is obtained from the condition $H_{11} = H_{22}$, which in the simple case of $H_{11} = \Delta - e^2/r$; $H_{22} = C_6/r^6 + Eb^2/r^2$, gives

$$\Delta r^6 - e^2 r^5 - Eb^2 r^4 + C_4 r^2 + C_6 = 0, \qquad (6.28)$$

and since the coefficients of the common terms in Equations 6.27 and 6.28 are the same, it follows that r must be smaller in Equation 6.28, so the leading term is smaller. Therefore, a more attractive covalent potential will shift the crossing radius inwards.

The reaction cross section in the "orbiting controlled" curve crossing mechanism is simply given by $\sigma_Q = \pi\, b_o^2$, where b_o is the critical impact parameter. Consequently, although b_o and r_o are closely related, it is informative to derive an explicit expression for b_o^2 corresponding to r_o of Equation 6.25. For that case, the critical impact parameter is given by

$$b_o^2 = (3C_6/Er_o^4) + (3C_3/2E\, r_o) \qquad (6.29)$$

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If Equations 6.25 for r_o is replaced in Equation 6.29,

$$b_o^2 = \frac{6C_6 + 3C_3\{(C_3/4E) + [(C_3^2/16E^2) + (2C_6/E)]^{1/2}\}}{2E\{(C_3/4E) + [(C_3^2/16E^2) + (2C_6/E)]^{1/2}\}^{4/3}} \qquad (6.30)$$

This equation can be parametrized to give a log σ_Q vs. log C_6 plot with unit slope. A somewhat similar expression is obtained for b_o^2 when the additional attractive force is $-C_4/r^4$ and r_o is given by Equation 6.26. Assuming that the $-C_3/r^3$ potential has the same magnitude as the $-C_6/r^6$ potential at r = 6 Å, the C_3 coefficient is approximately equal to 0.005 C_6 and Equation 6.30 gives a critical impact parameter ∿6% larger than the one expected in the absence of this additional force, which translates into a cross section ∿12% larger. For C_3 = 0.1 C_6, the magnitude of the r^{-3} potential is 20 times larger than the r^{-6} potential at r = 6 Å, but b_o increases by ∿80%.

The addition of a more attractive interaction potential therefore, not only predicts (correctly) larger cross sections than the simple C_6/r^6 interaction, but can also explain the strong correlation between σ_Q and C_6. The problem remains however, of deciding the exact form of the potential. Since the metastable states can be considered as the first terms in the Rydberg series for the rare gases, one possibility is to include the ion-molecule term, $-C_4/r^4$; this probably is a good approximation when the outer electron is in a p-Rydberg and the molecule approaches the core perpendicular to the p-orbital, but for the rare gas, the outer electron is in a s-orbital and the approximation should be a poor one, except for the highest s-Rydberg. The other possibility, is to evaluate the potential terms arising from the near-resonant intereaction between excited atoms and molecules,[20-21] as discussed above.

The different power dependence of σ_Q for Ar($^3P_2$) and Xe($^3P_2$) quenching on the photoabsorption cross sections of the quencher molecule does

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not necessarily follow from this qualitative treatment of the interaction forces. Explicit forms of the interaction potentials are necessary before these details can be tested. It is also possible that the reaction proceeds by more than one mechanism, and that in going from $Ar(^3P_2)$ to $Xe(^3P_2)$ the predominance of one mechanism over the other becomes apparent, because of the larger interactions expected in the slower collisions of $Xe(^3P_2)$ as compared to $Ar(^3P_2)$.

C. Non-halogenated donors.

(i) Analysis of exit channels. The products of the quenching of $Xe(^3P_2)$ by a variety of molecules were identified by the emission excited by the interaction between metastable and quencher. For the majority of the molecules studied - halogen donors - only excimer formation (via reactive quenching) was observed in emission. For the non-halogenated quenchers Hg, N_2, CO, NO and O_2, the emission indicated that excitation transfer was the dominant process, except for O_2, which gave no emission in the 200 to 800 nm range (see below). The intensity of these emissions was significantly weaker than for the halogenated donors, although for N_2, CO and NO the emission extended into the vacuum u.v. and could not be detected with the instrument used. The quenching rate constants for $Xe(^3P_2)$ are small for Hg, N_2 and CO and since the intensity of the observed emission was weak, the branching ratio for excitation transfer probably is nearly unity. On the other hand, NO formation probably has a branching fraction significantly less than unity with dissociation being the competing channel, unless the bulk of the emission occurs in the vacuum u.v. The main exit channel for O_2 is probably excitation transfer followed by dissociation since no emission is observed.

The mechanisms for excitation transfer to non-halogenated donors is discussed below, while the reactive quenching by halogen-containing

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molecules is treated in Section 6.d.

(ii) $Xe(^3P_2) + Hg$. The reaction of $Xe(^3P_2)$ with Hg is in principle a very promising system for laser applications. As shown in Figure 6.5, $Xe(^3P_2)$ atoms have enough energy to excite the 7s levels of mercury, which in turn, are coupled to the lowest excited multiplet of Hg by strong transition probabilities, providing the three level system necessary for laser action. On the other hand, $(HgXe)^+$ is a known bound system; therefore, bound Rydberg states converging on this ion are expected to correlate with excited states of the atoms, raising the possibility of the existence of bound-free transitions similar to the ones observed in the rare gas-halogen excimers. Although HgXe* could not be formed in a binary Xe* + Hg collision, the bound potentials may affect the exit channel distribution of the inelastic collision.

The total quenching rate constant was too small to be measured in the fast flow reactor (see Section 3.c); therefore, the rate constant was estimated from a comparison of Hg* emission intensities excited by $Xe(^3P_2)$ and the Hg 253.7 nm emission excited by the reaction of Hg + $N_2(A^3\Sigma_u^+)$. The rate constant for formation of the $Hg(^3P_1)$ state by the reaction with $N_2(A)$ has been measured,[31] and for the present analysis the value $k_{254}^{N_2} = 7.6 \times 10^{-11}$ cm^3 $molec^{-1}$ sec^{-1} was adopted.[31a] Since the most important mode of decay for $Hg(^3P_1)$ is radiation to the ground state, the intensity of the 253.7 nm line excited by reaction with $N_2(A)$ can be written as:

$$I_{254} = k_{254}^{N_2}[N_2(A)][Hg] \qquad (6.31)$$

whereas, emission from the 7s states - that can only be excited by $Xe(^3P_2)$, since $N_2(A)$ does not have enough energy - will be given by

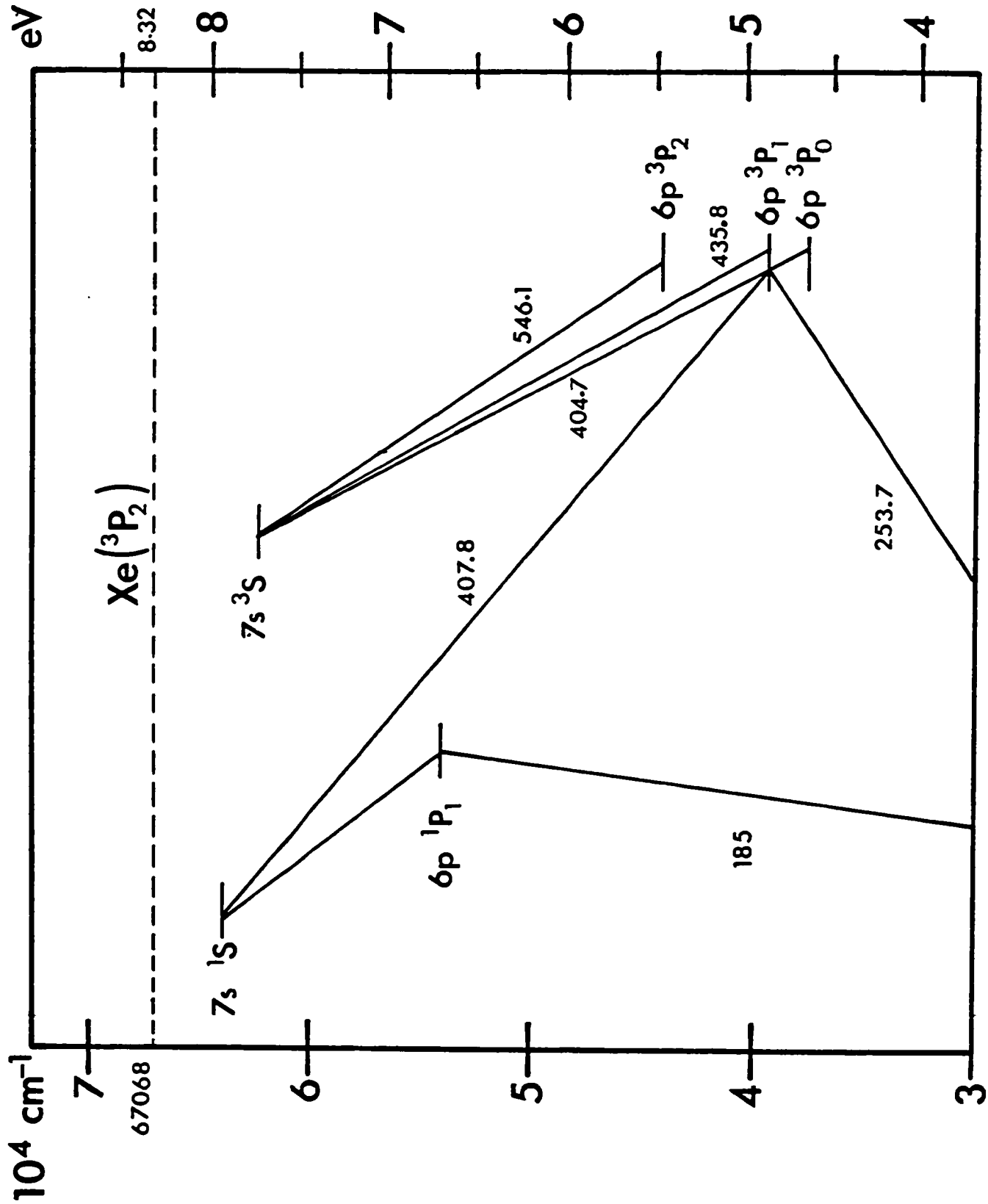
$$I_{407} = (0.31)\ k_s^{Xe*}[Xe*][Hg] \qquad (6.32)$$

$$I_{436} = (0.43)\ k_t^{Xe*}[Xe*][Hg] \qquad (6.33)$$

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Figure 6.5 - Mercury energy levels below 8.3 eV. The ground state is not shown.

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where k_s^{Xe*} and k_t^{Xe*} are the rate constants for formation of the 7s singlet (^1S) and triplet (^3S) states respectively. The numerical factors in front of Equations 6.32 and 6.33 are the calculated radiative branching ratios for the transition being monitored (407.8 nm for 7d ^1S and 435.8 nm for 7s ^3S, see Figure 6.5). Transition probabilities of Anderson, Lee and Lin[32] were used to calculate these factors. Summing Equations 6.32 and 6.33 Equation 6.34 is obtained,

$$(0.31)^{-1} I_{407} + (0.43)^{-1} I_{436} = k_T^{Xe*}[Xe*][Hg] \qquad (6.34)$$

which relates the total rate constant for formation of Hg(7s) to the observed emission. Finally, dividing Equation 6.34 by Equation 6.31 and rearranging gives

$$k_T^{Xe*} = k_{254}^{N_2} \frac{(0.31)^{-1} I_{407} + (0.43)^{-1} I_{436}}{I_{254}} \frac{[N_2(A)]}{[Xe*]} \frac{(0.49)^{-1}}{(0.75)^{-1}} \qquad (6.35$$

where the last fraction is the ratio of detector sensitivity factors at 420 nm and 254 nm. The intensity of the 407.8 and 435.8 nm line was first recorded and then N_2 was introduced in the reactor through a separate inlet to produce $N_2(A)$ by the reactions

$$Xe(^3P_2) + N_2(X) \rightarrow Xe(^1S) + N_2(B) \qquad (6.36)$$

$$N_2(B) \rightarrow N_2(A) + h\nu \qquad (6.37)$$

The Hg emission was observed as far downstream from the mixing zone as possible ($\sim$10 cm) and assuming complete conversion of $Xe(^3P_2)$ to $N_2(A)$ values between 1.4×10^{-13} and 1.2×10^{-14} cm^3 $molec^{-1}$ sec^{-1} were obtained for k_T^{Xe*} in a series of experiments. These values should be considered as upper limits because: (1) the comparison were made between weak (407.8 nm gave $\sim$100 counts sec^{-1}, while the 435.8 nm line was essentially undetectable) and very strong signals (253.7 nm $\sim 10^6$ counts sec^{-1}), and P.M. response

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can not be expected to be perfectly linear over this range; (2) the assumption of equal concentrations for $Xe(^3P_2)$ and $N_2(A)$ is based on ideal and instantaneous mixing of the reagents, and in the premise that no other destruction mechanisms deplete the $N_2(A)$ (or $N_2(B)$) population.

The exit channels for the reaction, V(Xe, Hg[7s, ^{1}S]) and V(Xe, Hg) [7s, ^{3}S]) are 0.39 eV and 0.58 below the V(Xe[6 $^3P_2$], Hg) entrance channel. To a first approximation, the entrance channel can be replaced by Cs + Hg, while the exit channel probably resembles Fr + Xe. From molecular beam scattering data,[33] it is known that the alkali-mercury curves are considerably more attractive than alkali-rare gas, and therefore a crossing of the entrance and exit channels will occur generating a mechanism for the population of the product states. The situation is similar to $Ar(^3P_2)$ + Kr, studied by Piper, et. al..[34] In the Landau-Zener theory,[13b] the probability of crossing from one potential surface to another depends only on the interaction matrix element V_{if} and on the differences of the potential gradients at the crossing point; in the case of Ar* + Kr, the main exit channels are 0.002 eV and 0.025 eV below the entrance channel; therefore, the slopes of the potentials at the crossing points are not expected to differ greatly. In the Xe* + Hg case however, see Figure 6.6a the asymptotic energy defect of the product states dictate a much greater difference in the slopes at the crossing point and thus, assuming comparable interaction matrix elements, the probability for crossing in the Xe* + Hg system is much smaller than in the Ar* + Kr case, as reflected by the smaller rate constant.

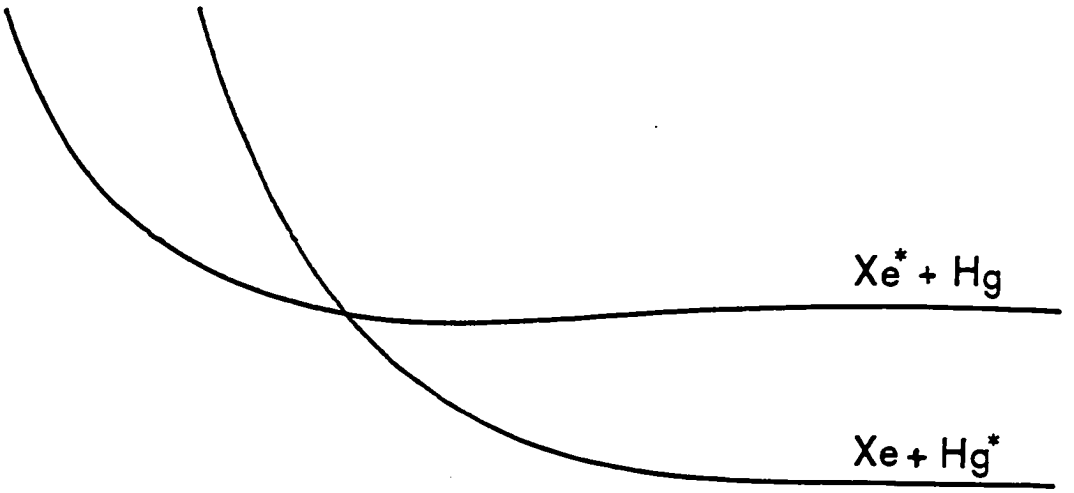
iii) $\underline{Xe(^3P_2) + N_2}$. Xenon metastable excitation of N_2 produces the N_2 ($B^3\Pi_g \rightarrow A^3\Sigma_u^+$) emission (First positive, 1+) as shown in Figure 6.7. Emission is observed from $N_2(B^3\Pi_g, v' \leq 5)$, although production of $N_2(B, v' = 5)$ is endoergic by 475 cm^{-1}; it had been previously suggested[35]

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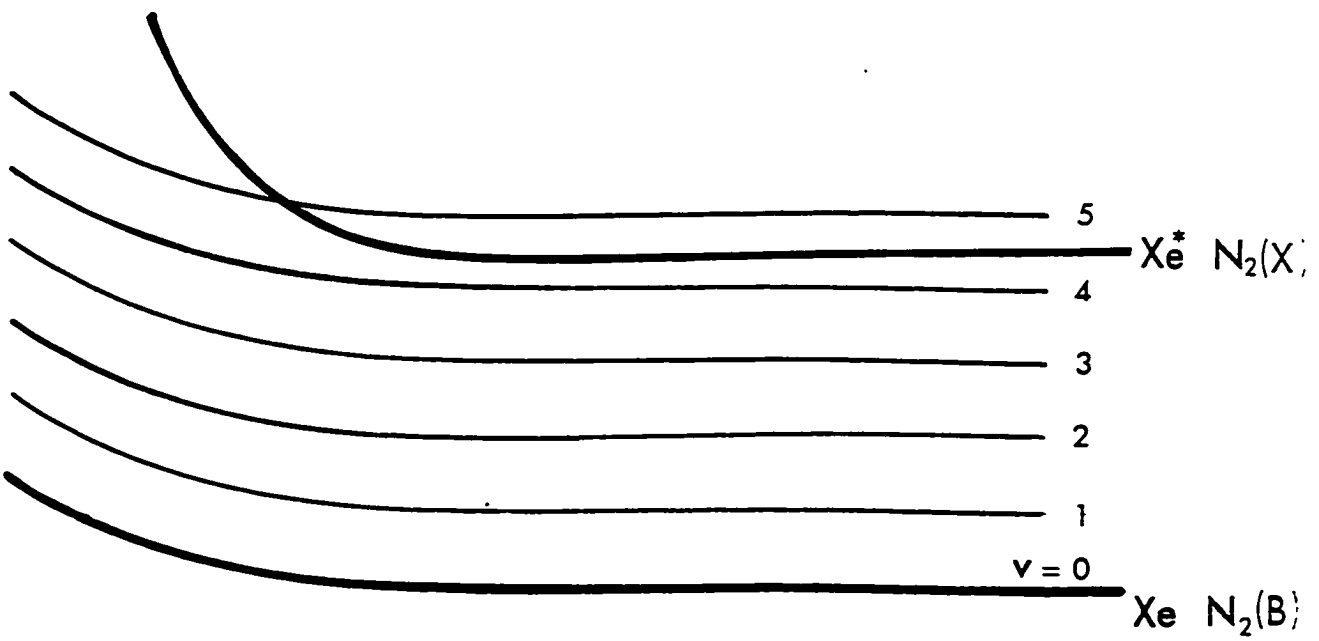
Figure 6.6 - a. Schematic diagram for the curve crossing between $Xe(^3P_2)$ + $Hg(^1S_0)$ and $Xe(^1S_0)$ + Hg[7s, ^3S]. The asymptotic separation between the two curves is ~0.4 eV.

b. Schematic diagram for the curve crossing between $Xe(^3P_2)$ + $N_2(X, v=0)$ and $Xe(^1S)$ + $N_2(B, v \leq 5)$.

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a

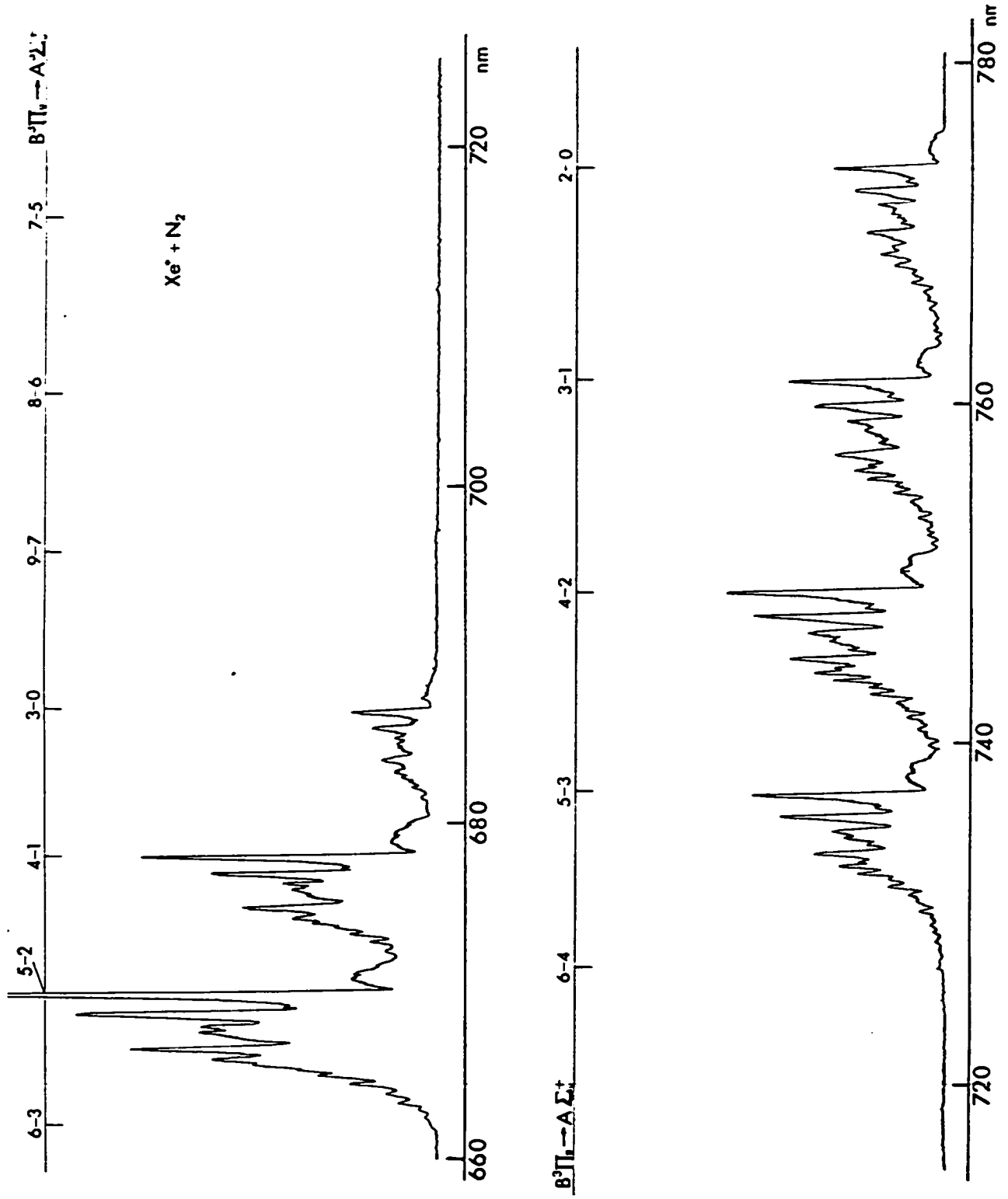


b

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Figure 6.7 - $N_2(B^3\Pi_g \rightarrow A^3\Sigma_u^+)$ excited by the reaction $Xe(^3P_2) + N_2(X)$. No other emissions were observed. The spectrum was recorded using a 0.75m Jarrell-Ash spectrometer and a single photon counting detection train. Monochromator slits, 80μ; scan speed, 12.5 nm/min.

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that $Xe(^3P_1)$ might be responsible for formation of this state, but the work of Nguyen et. al.[36] conclusively demonstrates that only $Xe(^3P_2)$ is involved in the production of $N_2(B, v'=5)$, which implies that the kinetic energy of the collision partners is required to make up for the energy defect.[37] This endoergic aspect correlates well with the slow nature of the reaction.

These results can be qualitatively understood in terms of the curve crossing scheme depicted in Figure 6.6b where the $Xe(^3P_2) + N_2(X^1\Sigma_g^+)$ is shown as slightly more repulsive than the $Xe(^1S) + N_2(B^3\Pi_g, v'=5)$. Collisions with low relative kinetic energy will move on the $Xe^* + N_2$ surface but will be reflected by the repulsive wall. Higher energy collisions however, will be able to climb the repulsive wall and eventually cross over to the $Xe + N_2$ $(B^3\Pi_g, v'=5)$ surface. The ionic curve $Xe^+ + N_2^-$ is not considered in this analysis because the $EA(N_2)$ ~ 0 and the crossing is expected to occur at very close range (~ 3.0 Å), which corresponds to the strongly repulsive part of the $Xe^* + N_2$ potential. Attempts were made to analyze the system in a more quantitative fashion, but since the problem involves features in the potential in the ~ 0.06 eV range, very accurate potential curves are needed which, unfortunately, are not available.

Recently Lee and Martin[38] studied the velocity dependence of the $Ar(^3P_{0,2}) + N_2(X)$ reaction and concluded that the reaction proceeds via a bound $Ar\text{-}N_2$ complex[38] with an activation energy of 0.008 eV. The reaction of $Xe(^3P_1)$ with the isoelectronic molecule CO is claimed[39] to proceed <u>via</u> a complex formation (see below). The possibility therefore exists that the reaction of $Xe(^3P_2) + N_2$ involves complex formation, and the presence of an activation energy would explain why the emission is above the metastable energy. This investigation however, does not provide any

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additional evidence to support one mechanism over the other.

(iv) $Xe(^3P_2)$ + CO. The reaction of $Xe(^3P_2)$ with CO excites emission in the "triplet" ($d^3\Delta \rightarrow a^3\Pi$), Asundi ($a'\ ^3\Sigma^+ \rightarrow a^3\Pi$), Herman ($e^3\Sigma \rightarrow a^3\Pi$) and fourth positive ($A^1\Pi \rightarrow X^1\Sigma_g^+$) systems as shown in Figure 6.8. In all cases, the highest v' level observed exceeds the energy of $Xe(^3P_2$; 67068 cm^{-1}); the following v' levels are the highest observed in the emission:

$A\ ^1\Pi$, v'=2 at 67678 cm^{-1}

$a'\ ^3\Sigma_u^+$, v'=11 at 67531 cm^{-1}

$d\ ^3\Delta$, v'=7 at 68259 cm^{-1}

$e\ ^3\Sigma^-$, v'=4 at 67970 cm^{-1}

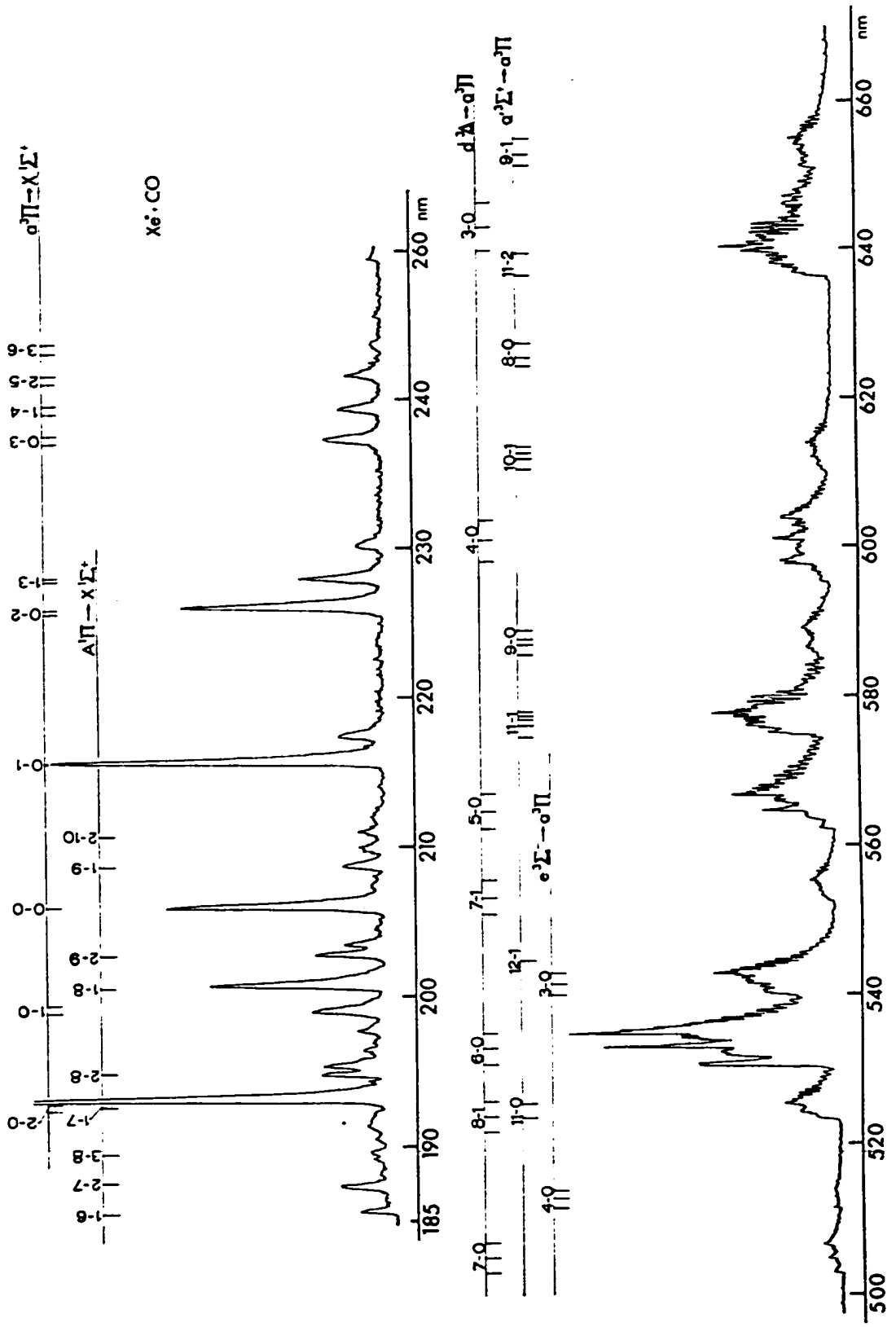
Emission intensities appear to peak for $A^1\Pi$, v'=2, $a'^3\Sigma_u^+$, v'=10; $d^3\Delta$, v'=6; and $e\ ^3\Sigma^-$, v'=3. These same emissions, with maximum intensity at the highest v' levels listed, have been observed in the $Xe(^3P_1)$ sensitized photolysis of CO at 147 nm,[40] while the unsensitized photolysis at the same wavelength[40] produced mostly $CO(d^3\Delta$, v'=7). Except for $CO(d^3\Delta$, v'=7), the other observed states could be produced by direct excitation transfer from $Xe(^3P_2)$ if the kinetic energy of the collision is also considered; the presence of $CO(d\ ^3\Delta$, v'=7) and $CO(e\ ^3\Sigma^-$, v'=4) in the emission can not be easily explained in terms of kinetic energy contributions to the excitation, since they are ∿1200 and 900 cm^{-1} endoergic, respectively. Emission from these two levels is weak, and the easiest alternate explanation would be excitation transfer from $Xe(^3P_1)$ giving $CO(d\ ^3\Delta$, v'=7) with subsequent fast cross-relaxation[41] to the $CO(e\ ^3\Sigma^-$, v'=4).

It was not possible to detect $Xe(^3P_1)$ by Atomic Absorption Spectroscopy. Considering that the limit of detection in the system is ∿1% absorption,

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Figure 6.8 - CO emission excited by the reaction $Xe(^3P_2)$ + CO(X). The spectrum was recorded using a 0.75m Jarrell-Ash spectrometer and single photon counting. Monochromator slits 100μ; scan speed, 12.5 nm/min.

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the maximum [Xe($^3P_1$)] possible was calculated assuming a 0.8% absorption of the Xe 838.0 nm line (f ~0.4). This corresponds to [Xe($^3P_1$)] $\leq 8 \times 10^8$ cm^{-3}, or [Xe($^3P_2$)]/[Xe($^3P_1$)] ≥ 50. The experimental ratio of (d $^3\Delta$, v'=7)/(d $^3\Delta$, v'=6) is roughly consistent with the atomic ratio but comparison of other bands with v levels above the Xe metastable energy (i.e. (d-a')(7,1) and (5,0)) gives a much smaller ratio, suggesting that Xe($^3P_2$) must be involved in the formation of the higher levels.

Analysis of the CO system is significantly more complicated than the N_2 case because of the large number of perturbations between the $A^1\Pi$ and the triplet states which facilitate rapid relaxation. Slanger and Black[40] have shown that collisional transfer between states is possible at 1 torr of rare gas.

The present study of the Xe($^3P_2$) quenching by CO, due to its limited scope, does not offer further insight as whether the proposed complex formation mechanism[8,10] for the quenching of Xe($^3P_1$) is also operative in the xenon metastable quenching. Further studies of this system should provide a better understanding of Xe metastable quenching vis a vis resonance state quenching, highly documented by the work of Slanger and Black.[40,41] Finally, although the majority of the states observed are triplets, no definitive statement can be made as to the validity of the Wigner Spin rule because (1) most of the emission originating from the $A^1\Pi$ state is in the vacuum u.v., and therefore was not observed, and (2) formation of the tirplet states may originate from cross-relaxation of the $A^1\Pi$ state.

(v) Xe($^3P_2$) + NO. The NO emission excited by xenon metastable occurs between 190 and 400 nm. Relative to cases with high photon yields the emission is rather weak and dissociation probably is the main channel.

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Most of the NO* Rydberg states are predissociated. No evidence was found for excited states of O or N. Transitions in the β,γ,δ and ε systems ($B^2\Pi$, $A^2\Sigma^+$, $C^2\Pi$, and $D\ ^2\Sigma^+ \rightarrow X^2\Pi$) respectively were identified. With the possible exception of $D^2\Sigma^+$, these states are not expected to be directly populated if the entrance channel is not attractive, since they are more than 2 eV below the $Xe(^3P_2)$ energy ($D^2\Sigma^+$ is ∿1.75 eV below). The $S^2\Sigma^+$ state is nearly resonant with the metastable energy and it may be directly populated by Xe*. Transitions involving this state have been observed only in absorption but emission to the $D^2\Sigma^+$ and/or $C^2\Pi$ should occur in the 600 to 800 nm; whereas, transitions to the $A^2\Sigma^+$ are expected above 400 nm. Under certain experimental conditions (not fully characterized), emission is observed in these regions, but unfortunately the reproducibility was poor. Since there was no apparent correlation between the intensity of these emissions and the intensity of the β-bands or impurity level (see below), no assignment was attempted. A somewhat similar situation arises with the γ-bands. The relative intensity of this system was found to vary from run to run, but it was soon recognized that the variation could be attributed to N_2 impurity in the gases and, hence, to the reaction sequence

$$Xe(^3P_2) + N_2(X) \rightarrow Xe(^1S) + N_2(A) \qquad (6.38)$$

$$N_2(A) + NO(X) \rightarrow N_2(X) + NO(A);^{34} \qquad (6.39)$$

Very weak emissions in the 700 to 750 nm were tentatively assigned to intercombination Rydberg bands $o'\ ^2\Pi \rightarrow C^2\Pi$ and $o'^2\Sigma \rightarrow D^2\Sigma^+$, but they could also fit Ogawa's b and b' systems. The β-bands are the most intense in the observed emission; however, most of the δ and ε transitions are in the vacuum u.v., and therefore these were not monitored. It is possible that, similar to the CO case, the upper states in NO are also heavily perturbed particularly by the $B'\ ^2\Delta_i$ and $B\ ^2\Pi_r$, since the repulsive

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part of these potentials cross the C, D, E and possibly the S states. Further work is necessary to elucidate this point.

(vi) $Xe(^3P_2) + O_2$. No emission was observed from the reaction $Xe(^3P_2) + O_2$ in the 195 nm to 800 nm range. The $Xe(^3P_2)$ energy is approximately 1.2 eV above the dissociation limit of the $O_2(B^2\Sigma_u^-)$ state, and several repulsive states with the same asymptotic limit as the $B\ ^3\Sigma_u^-$ state ($O^3P + O^1D$) are expected in this region. This suggested the possibility of excitation transfer followed by dissociation. However, no emission from the $O(^1D \rightarrow {}^3P)$ transition at 632 nm was observed.

(vii) $Xe(^3P_2)$ + Hydrocarbons. No emission was observed from the reaction of $Xe(^3P_2)$ + hydrocarbons or $Xe(^3P_2)$ + fluorocarbons. The cross sections for quenching of $Xe(^3P_2)$ by the hydrocarbons and fluorocarbons correlates quite well with their absorption cross sections at 149.1 nm (see Section 6.b) and therefore comparison of product states from photolysis and resonance sensitization would be quite worthwhile. Such a comparison would aid in the assignment of quenching mechanism. In the case of CH_4, CH_3F and CH_2F_2 the excitation is to the first (3s) Rydberg state.[42] Edwards and Raymonda[43] assign this transitions as localized bonding-antibonding transition at the C-H bonds leading to dissociation. An extensive literature exists which attempts to assign the primary photochemical processes in hydrocarbons. Work also has been done with the same objective for determining the primary processes from sensitization reactions. Since the product states from metastable quenching are not yet known, no attempt will be made to summarize these fields.

d. Halogenated donors.

i) Analysis of product channels. The general interaction mechanism in the

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reaction of $Xe(^3P_2)$ with halogenated donors appears to be a charge transfer mechanism[16] in which the entrance channel V(M*, RX) is crossed at relatively long distance by an ion-pair potential $V(M^+, RX^-)$. Efficient transfer to $V(M^+, RX^-)$ is followed by competition between the formation of MX* + R (Reactive Quenching) and M + RX* (Excitation Transfer). Reactive quenching is the preferred channel, provided the formation of MX* is energetically allowed. Excitation transfer is observed in a few cases as either molecular fluorescence, e.g., Cl_2*, or as dissociative excitation of RX with production of R*, as in the case of $Xe(^3P_2)$ + NOF, NOCl, which gives NO* in addition to XeX*.[44] A weak, unidentified molecular emission was observed superimposed on XeBr* from the reaction of $Xe(^3P_2)$ with PBr_3. Since no other competing emissions were observed, dissociation of RX* must occur without excitation of R in contrast to other systems, such as the reaction of rare gas metastables (including Xe*) with RCN[45] where extensive dissociative excitation results.[45] The reaction of $Xe(^3P_2)$ with larger quencher molecules may also involve long lived complexes as suggested by the reactions of Xe* + SF_6, BrF_5 etc. (see Section 6.d.iv).

The dynamics of these reactions are discussed in more detail below, with the quenchers subdivided into two groups: diatomic halogens, (homonuclear and heteronuclear) and larger (polyatomic) halogen donors.

ii) $Xe(^3P_2)$ + X_2, XY. For reactions known to occur by the long range ionic-covalent, curve-crossing mechanism, a large fraction of the available energy is released as internal energy of the halide product,[46] as in the case of the alkali atom-molecular halogen reactions. The observed emission from the reactions $Xe(^3P_2)$ + X_2, XY does extend to the thermochemical limit, suggesting that these reactions procedd via the curve-crossing mechanism. Furtheremore, the magnitude of the total reactive cross sections for the rare gas metastable states with molecular halogens

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are remarkably similar to those of the alkali metals, as shown in Table 6.3. These results strongly support the same mechanism for both types of reactions, because in the "harpoon" mechanism the only parameters that identify the electron donor are the ionization potential and, to a lesser extent, the polarizability. From the parameters listed in Table 6.4, it is apparent that within the framework of this model, K, Rb and Cs are indistinguishable from $Ar(^3P_2)$, $Kr(^3P_2)$ and $Xe(^3P_2)$ respectively, and it should be possible to explicitly apply the large number of theoretical treatments of the alkali atom-molecular halogen reaction to the rare gas metastable atom reactions.

Davidovits and coworkers[47] measured the reaction cross sections for Na, K, Rb and Cs with Cl_2, Br_2 and I_2, and found that the simple "harpoon" model of Polanyi[48] and Magee[49] (see Section 6.2), though adequate to represent the gross features of these reactions, was in conflict with certain experimental evidence, namely;[47c] (i) different electron affinities are needed for a given halogen in order to reproduce the cross sections for the various alkali atoms; (ii) the experimental cross sections are always greater than the calculated ones; (iii) for a fixed alkali atom, the relative magnitudes of the cross sections do not follow the predicted trend ($\sigma_{I_2} > \sigma_{Br_2} > \sigma_{Cl_2}$); and (iv) the spread in σ for a given alkali metal atom should increase from Na to Cs, when in fact the opposite is true.

Better agreement between theory and experiment is obtained with the model developed by Anderson and Herschbach[50] and by Grice and Herschbach,[51] which modifies the original harpoon mechanism to take into account the interaction potential between the ionic and covalent diabatic curves, as well as the angular momentum of the collision. In this approach, the

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Table 6.3 Comparison of physical properties and reaction cross sections for alkali metal atoms and rare gas metastables with halogens.

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Table 6.3

Atom	α, $\mathring{A}^3$ [a]	I.P., eV [b]	Molecule	σ, $\mathring{A}^2$ [c]
K	44.3	4.34	Cl_2	154
			Br_2	151
$Ar(^3P_2)$	47.5	4.21	Cl_2	142
			Br_2	147
Rb	48.0	4.18	Cl_2	190
			Br_2	197
$Kr(^3P_2)$	50.2	4.09	Cl_2	179
			Br_2	178
Cs	61.5	3.89	Cl_2	196
			Br_2	204
$Xe(^3P_2)$	62.6	3.82	Cl_2	193
			Br_2	202

a. Taken from R. W. Molof, H. L. Schwartz, T. M. Miller and B. Bederson, Phys. Rev. (A), 10, 1131 (1974).

b. "Handbook of Chemistry and Physics", 56th Editrion, The Chemical Rubber Co., Cleveland, Ohio, 1975.

c. The cross sections for the metastable atoms were measured in this laboratory. The ones for the alkali metal atoms were taken from Ref. 47.

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two adiabatic surfaces are constructed from the diabatic ionic and covalent curves by including a coupling potential. The adiabatic potentials V^- and V^+ are found as the eignevalues of the electronic Hamiltonian matrix.

$$H = \begin{vmatrix} H_i & H_{ic} \\ H_{ic} & H_c \end{vmatrix} \tag{6.40}$$

and are therefore given by[52,53a]

$$V^{\pm} = (1/2)\{(H_i + H_c) \pm [(H_i - H_c)^2 + 4H_{ic}^2]^{1/2}\} \tag{6.41}$$

where H_i = ionic diabatic potential

H_c = covalent diabatic potential

H_{ic} = coupling potential.

The coupling potential, H_{ic}, is obtained by computing the overlap integral of H_i and H_c near the crossing point (r_c) and is normally expressed as a function of the form $A\, e^{-\alpha r}$. In general, both A and α depend on the ionization potential of the electron donor, the electron affinity of the halogen molecule and the overlap integral, but Olson[54] has obtained simple expression for H_{ic} in terms of I.P. and E.A. only. The dependence of this interaction term on the angular orientation has been analyzed by Anderson[50b] for the alkali atom-halogen reactions, but he concluded that for reactive collisions, X_2 tracks the alkali atom in a collinear configuration, and therefore, the angular dependence can be neglected. Gislason and Sachs[55], however, introduce a preexponential factor of cos η to account for the approach configuration.

Angular momentum considerations do require the introduction of a centrifugal energy term in H_c.[50,51] If r_c is less than the orbiting radius the criteria for reaction is surmounting the centrifugal barrier, i.e.,

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once the collision surmounts the barrier, the electron transfer occurs with unit probability. The reaction, therefore, is governed by "orbiting". As in the simple harpoon model, the crossing point between the two diabatic curves is given by the solution of $H_i(r_c)=H_c(r_c)$; however, the position of centrifugal barrier is determined by the simultaneous solution of $V^-(r_o) = E$ and $(dV^-/dr)_{r_o} = 0$. The effective potential curves for this improved model are shown in Figure 6.9. Numerican solutions of the relevant equations for r_c and r_o for a wide range of realistic values for the ionization potentials and electron affinities of alkali metal atoms + halogens place the centrifugal barrier $\sim$2 Å outside the diabatic crossing radius.[51,52] The position of the barrier is much less sensitive to variations in the halogen electron affinity than is the crossing radius. Thus, for K + Br_2 the crossing point is located at $\sim$4.6 Å if the electron affinity of Br_2 is taken as 1.2 eV, and moves out to $\sim$6.4 Å for $E.A._{Br_2}$ = 2.1 eV;[51] for these two cases the orbiting barrier is located at 8.5 Å and 9.2 Å, respectively.[51] At low collision energies (E < 10 eV) the reaction is expected to follow the lower adiabatic potential V^-, because the probability of non-adiabatic transitions (in the Landau-Zener formulation) is neglible. This expectation was confirmed by the trajectory calculations of Evers and De Vries[52] for K + Br_2, which gave total reactive cross sections that were in very good agreement with the measured values over the whole energy range of 0-4 eV. This "orbiting-controlled" curve-crossing mechanism provides, in general, better agreement with experiment than the simple harpoon model;[47] but, optimum agreement is obtained only after the different parameters in the theory are adjusted to values which are reasonable but not unique or independently proven. This is not a drawback of the theory since, as shown by the work of Los and co-workers,[53] bond stretching during the collision process and transitions to excited states of X_2^- can, account

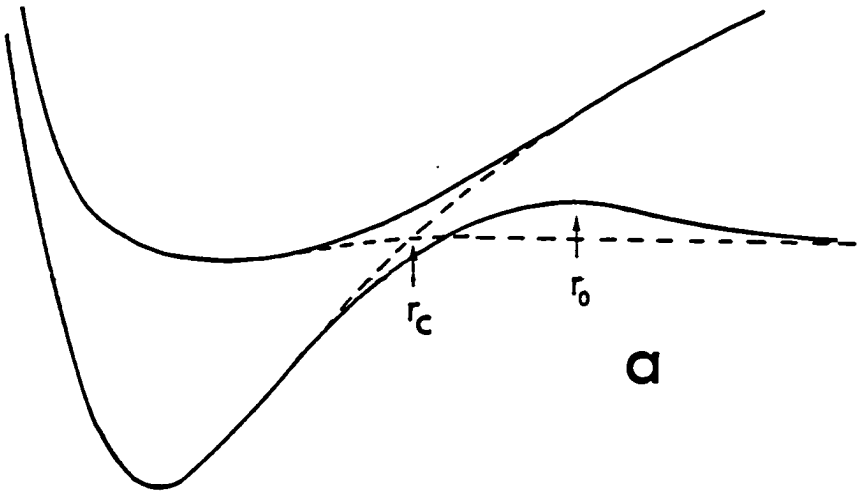
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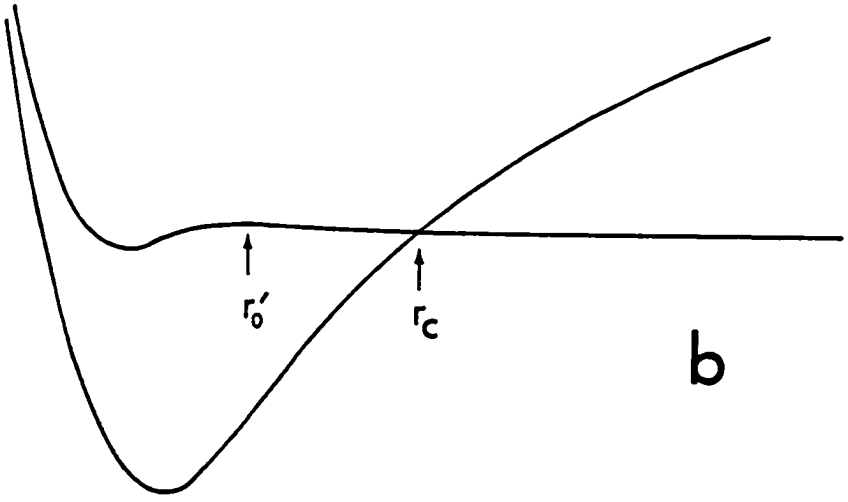
Figure 6.9 - a. Schematic diagram of the adiabatic potential curves for the "orbiting-controlled" curve crossing mechanism. The orbiting radius, r_o is normally $\sim$2 Å outside the crossing point, r_c.

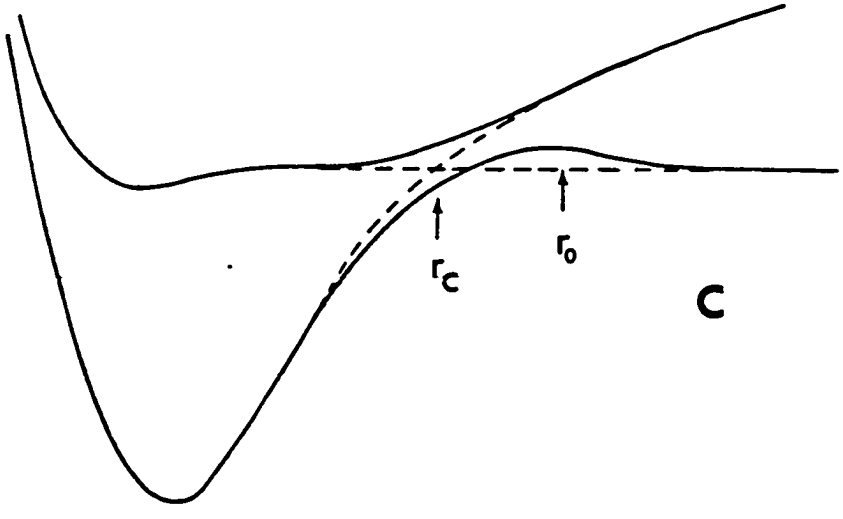
b. Simple diabatic curves for the Xe* + F_2 case with the centrifugal repulsion included in the covalent potential.

c. Approximate adiabatic curves for Xe* + F_2. The effective orbiting radius is outside the value predicted by the simple diabatic curve.

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for the electron affinities required for the best fit of the reactive cross sections. Detailed computations for the reactions of the metastable states of the rare gases using this model are not justified, since, as shown in Table 6.3, the physical parameter characterizing the metal atoms are nearly identical to those of the metastable atoms, and from the foregoing discussion, the predictions of this theory are not very sensitive to small variations in the parameters involved (which are the IP and the polarizability of the electronic donors).

The conditions for orbiting implicity define a relationship between the critical impact parameter and the interaction potential.[2] This correlation was discussed in Section 6.b, and from Figure 6.2 it is apparent that the halogenated donors, with the exception of F_2 and OF_2, follow the same correlation. Parameters for reactions involving F_2, i.e., vertical electron affinity, coupling potentials etc., are not well characterized; however, rough estimates of the crossing radius and location of the centrifugal barrier can be made assuming only a Van der Waal's interaction in the diabatic covalent curve and neglecting the coupling potentials. For the reaction $Xe(^3P_2) + F_2$, and assuming vertical electron affinities of 1.2 eV and 2.1 eV[56], crossing of the two diabatic curves occurs at 5.5 Å and 8.4 Å, respectively, while the centrifugal barrier is located at ~4.5 Å. The Slater-Kirkwood approximation was used to calculate the C_6 coefficient. Therefore, for F_2 (and probably for OF_2) the crossing region occurs outside the "conventional" orbiting radius. However, if the coupling potential is added to the diabatic potential in order to obtain the lower adiabatic potential, the effective entrance channel becomes much more attractive in nature. Therefore, the effective orbiting distance defined by this adiabatic potential will be at longer distance than the position predicted by the weak C_6 attraction in the covalent

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diabatic potential (see Figure 6.9b), and F_2 would not be expected to follow the σ_Q vs. C_6 correlation.

These results strongly suggest that the model of Anderson,[50b] Grice and Herschbach[51] is the mechanism governing the reactions of the metastable rare gas atoms with halogen molecules, provided allowance is made for the effective adiabatic potential as the entrance channel. Of course, this relatively simple model can be applied only to reactions that remain on the lower adiabatic potential V^- i.e., the relative kinetic energy of the collision partners is low enough so that the probability for non-adiabatic transitions is negligible. For higher collision energies, the probability for non-adiabatic transitions should be treated in the Landau-Zener formalism.[13]

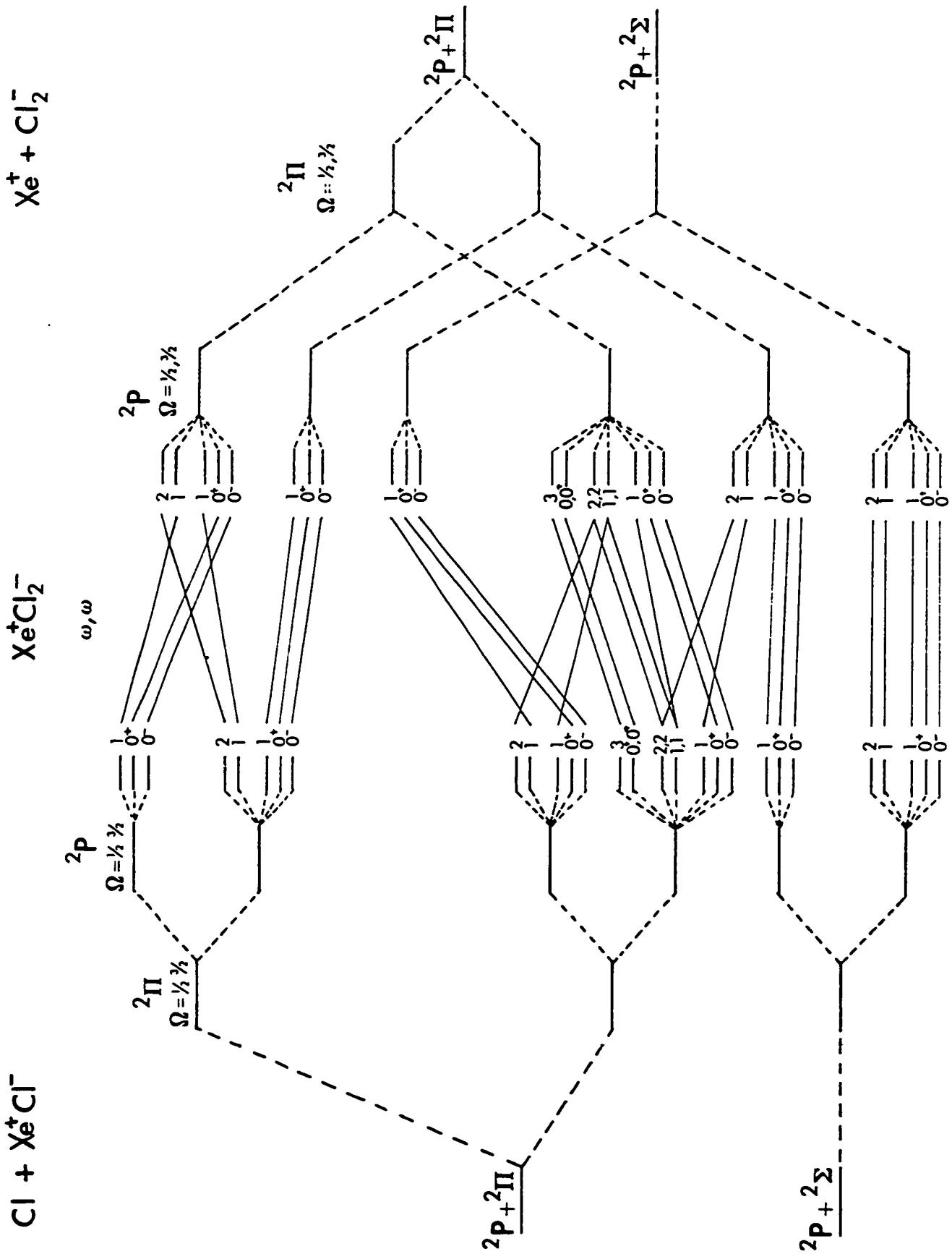
The reaction of the rare gas metastable states with the diatomic halogens is almost certainly direct with the exit channels dominated by excimer formation, as indicated by the measured branching ratios (see Chapters 4 and 5). However, as pointed out in Section 5.a, the relative populations of the different excimer electronic states varies systematically for the various metastable-halogen combinations, and this effect was atributed to selectivity in the entrance channel. This feature is discussed below in terms of state-symmetry correlation diagrams shown in Figures 6.10 and 6.11.

The correlation diagram of Figure 6.10 applies for cases in which the spin orbit coupling in the rare gas ion is much larger than in the halogen atom (e.g., XeCl). The right hand side of the diagram represents the limit $Xe^+ + Cl_2^-$, while Xe^+Cl^- + Cl is on the left hand side. The (ω,ω) coupling[29] scheme is used throughout for an assumed linear configuration of reactants and products. The experimentally observed products

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Figure 6.10 - Approximate correlation diagram for $Xe^+ + Cl_2^- \rightarrow Xe^+Cl^- + Cl$ in the (ω,ω) coupling. Only the ionic states are included.

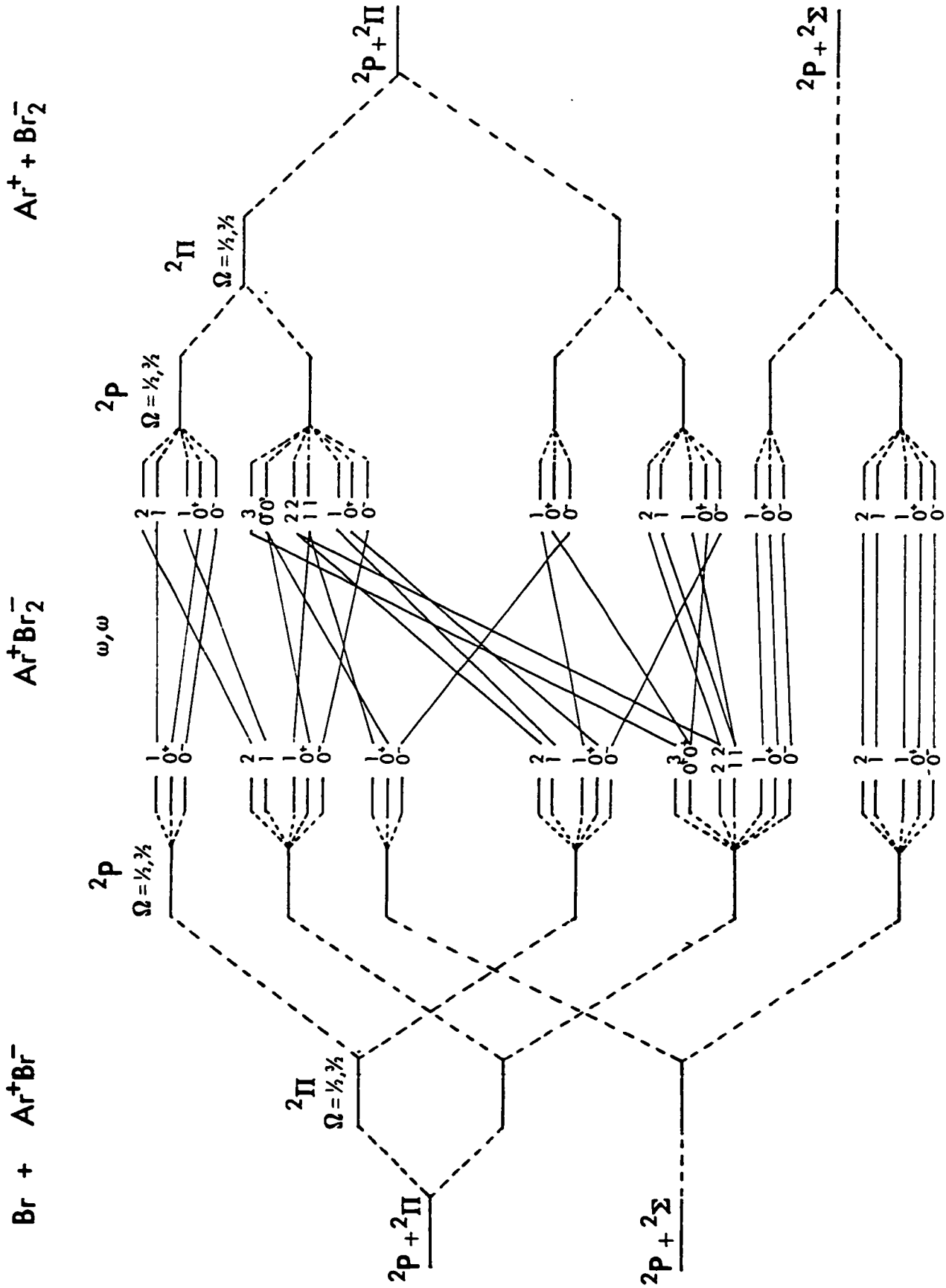
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Figure 6.11 - Approximate correlation diagram for $Ar^+ + Br_2^- \rightarrow A^+Br^- + Br$ in the (ω,ω) coupling; only the ionic states are included.

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$Cl(^2P)$ + $XeCl(^2\Sigma)$ and $Cl(^2P)$ + $XeCl(^2\Pi, \Omega = 1/2 \text{ and } 3/2)$ give rise to a total of 48 possible states for $Xe^+Cl_2^-$, but the $Xe^+(^2P)$ + $Cl_2^-(^2\Sigma, 2441)$ entrance channel produces only 12 states. Therefore, it is necessary to include the $Cl_2^-(^2\pi, 2432)$ excited state of Cl_2^- in the correlation in order to account for all the possible exit channels. Recent studies[53] of the alkali-atom-halogen molecule (homonuclear and heteronuclear) reactions confirm the role of the $X_2^-(^2\Pi)$ excited state in the collision process. The relative ordering of the 48 possible states is not straightforward. Consider the pseudo-molecules $Xe^+(^2P)$ + $Cl_2^-(^2\Sigma)$; all the possible states (3 x 2 x 1 x 2 = 12) resulting from a collinear approach arise from the (ω,ω) combinations (3/2,1/2) and (1/2,1/2). In the first case, Ω = 2,1; $1,0^+,0^-$, while the second combination gives $\Omega = 1,0^+;0^-$. These states correspond - in (Λ,S) coupling - to $^1\Pi$; $^3\Pi_{2,1,0^+,0^-}$ and $^3\Sigma_{1,0^+}$; $^1\Sigma$ respectively, but with different intra- and inter-multiplet splittings.[29] Since $Xe^+(^2P_{3/2})$ is ~1.3 eV more stable than $Xe^+(^2P_{1/2})$, the states arising from the (3/2,1/2) combination should be at lower energy than those arising from (1/2,1/2) and therefore, the lowest surface will have "Π" symmetry. The right hand side of the correlation diagram was drawn with the "π" states at lower energies than the other terms arising from a given combination of atomic and molecular states. Assuming that the electron transfer in the rare gas-halogen case produces mostly $M^+(^2P_{3/2})$, $X_2^-(^2\Sigma_{1/2})$ and $X_2^-(^2\Pi)$, the correlation diagram (Figure 6.10) predicts that the main products should be $XeCl(^2\Sigma_{1/2})$ + $Cl(^2P_{3/2})$ and $XeCl(^2\Pi_{3/2})$ + $Cl(^2P_{3/2})$. The correlation also predicts that $Cl(^2P_{1/2})$ should be formed. For the assumed entrance channels, formation of $XeCl(^2\Pi_{1/2})$ is not possible, since it requires <u>both</u>, the $Cl_2^-(^2\Pi)$ and the $Xe^+(^2P_{1/2})$ in the entrance channel. However, this state is observed for all the diatomic halogens

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(X_2 and XY) reacting with $Xe(^3P_2)$ except for F_2. This implies that (i) the reaction does not proceed on the lower adiabatic curve exclusively, i.e., that the probability of reaction after the centrifugal barrier has been surmounted is less than unity in the incoming part of the collision, since the crossing of the entrance channel $V(Xe(^3P_2), Cl_2(^1\Sigma_g^+))$ with $V(Xe^+(^2P_{1/2}), Cl_2^-(^2\Sigma_u^+))$ occurs $\sim$2.5Å deeper than the $V(Xe^+(^2P_{3/2}), Cl_2^-(^2\Sigma_u^+))$ crossing; (ii) ionization of $Xe(^3P_2)$ to the $Xe^+(^2P_{1/2})$ limit takes place readily, even though it requires a change in the electronic configuration of the core; and (iii) the $F_2^-(^2\Pi)$ excited states are probably too repulsive to have a large probability of a Franck-Condon transition from the ground state of F_2; alternatively, these reactions may remain on the lower adiabatic surface preferentially. If the correlation is assumed to proceed preferentially on the π surfaces, only 4 channels lead to XeCl $^2\Pi_{3/2}$ production, while there are 12 states converging on the XeCl $^2\Sigma$ state (Note that all the states with $\Omega > 0$ are doubly degenerate).[29] However, if a correlation diagram is constructed for the case of a larger spin-orbit splitting in the halogen atoms as compared to the spin-orbit splitting in the rare gas ion (e.g., ArBr) as shown in Figure 6.11, there are 7 paths leading to $ArBr(^2\Pi_{3/2})$ as compared to 8 for the formation $ArBr(^2\Sigma)$. This means that in ArBr, the ratio of the integrated intensities for the (B-X) and (C-A) emissions, (B-X)/(C-A), should be smaller than for XeCl. These expectations can be generalized to mean that the probability of $^2\Pi_{3/2}$ excimer state formation increase with larger spin-orbit splitting in the halogens, while the opposite is true for the rare gases, in agreement with the experimental observations (see Section 5.a). Of course, the correlation diagrams of Figures 6.10 and 6.11 are highly simplified and represent extreme cases

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where separation of the various states might be possible. In addition, it is assumed that entrance channels involving the $^2P_{1/2}$ rare gas ionic state do not contribute significantly to the population of the different excimer states, except of course, to the $^2\Pi_{1/2}$. For Ar, this may not be a good assumption since the spin-orbit splitting is small and the separation may be negligible at the crossing, splitting into the two branches at closer range. In the more general cases where the spin-orbit slittings in the rare gas ion and halogen are not drastically different, $M^+X_2^-$ molecular states correlating to different halogen and excimer spin-orbit states will very likely overlap or interact strongly, since the symmetries may be the same.

The $X_2^-(^2\Pi)$ excited states merit special attention, because of the role they play in the production of the $^2\Pi$ states of the excimers. Note that the participation of the $X_2^-(^2\Pi)$ states requires either, nonadiabatic behavior or some sort of effective transfer from $X_2^-(^2\Sigma)$ back to $X_2^-(^2\Pi)$ at shorter range. From Figure 6.10 and 6.11, it is clear that production of the $^2\Pi_{1/2}$ excimer states requires the simultaneous contribution of both, the rare gas ($^2P_{1/2}$) ion and the halogen ($^2\Pi$) states. As noted above, with the exception of the reactions of F_2, the $D(^2\Pi_{1/2})$ excimer state is weakly observed in all the reactions involving $Xe(^3P_2)$ + X_2, XY. The lack of D state emission from the reaction of $Xe(^3P_2)$ with larger molecules (see next section) must be related to the symmetry of the ground state molecular negative ion, since it does not appear very likely that the probability of an electron jump at the first crossing is unity for the larger molecules, that in general, have smaller electron affinities than the diatomic halogens (see next Section). In the reaction of alkali metal atoms with heteronuclear halogens XI, the formation of the iodide product is associated with the $XI^-(^2\Pi_{1/2})$ excited state,[57] because the ground ($^2\Sigma^+$)

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and first excited state ($^2\Pi_{3/2}$) dissociate to $I(^2P_{3/2}) + X^-(^1S_0)$ in the asymptotic limit.[57] On this basis some XeI(D → X) emission would be expected from the reaction of BrI and ClI with $Xe(^3P_2)$. Since $IX(^2\Sigma$ and $^2\Pi_{3/2})^-$ states also are formed, the other excimer states also are populated. Preliminary results on these reactions confirm this expectation for BrI + $Xe(^3P_2)$ which shows both XeBr(D → X) and XeI(D → X) emissions, but the reaction $Xe(^2P_2)$ + ClI gives only XeCl(D → X) emissions in addition to XeCl(B → X) and (C → A), and XeI(B → X) and (C → A). The Xe* + XI reactions favor XeX* formation over XeI*, as predicted by simple statistical[58] or information theoretical[59] considerations; Xe* + ClF on the other hand, gives XeCl as the preferred exit channel, in conflict with the statistical prediction, which suggests some specificity relating to the correlation diagram. These reactions are being experimentally reinvestigated, and final results will be presented elsewhere.

The cross sections for the reactions of $Xe(^3P_2) + X_2$, XY can be explained in a satisfactory fashion by an "orbiting-controlled" curve-crossing mechanism. The distribution of reaction products appears to favor a collinear approach of the ion pair and a collinear separation of the products. Based upon the correlation diagrams the excited states of the halogen molecular ions are required in order to explain the observed electronic states of the rare gas halide excimer. The inclusion of the Π states of X_2^- in the reaction of the rare gas metastable atoms with halogens contrasts the alkali-atom reactions, where the role of the halogen Π-excited states is not obvious before careful analysis of the data is attempted. Finally, formation of the $XeX(D^2\Pi_{1/2})$ states requires the presence of $Xe^+(^2P_{1/2})$ and excited X_2^- states. This explains the low probability for formation of XeX(D) for most reactions.

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iii) $Xe(^3P_2) + RX$. Excimer formation from the reactive quenching of $Xe(^3P_2)$ by most polyatomic halogen donors also proceeds by a direct reaction, since the XeX* emission extends to the thermochemical limit. However, the measured branching fractions for this exit channel are less than unity. This can be understood on account of competing exit channels and reduced probability for adiabatic behavior. In addition, other mechanisms may contribute to the quenching, e.g., complex formation and/or excitation transfer. The escape from the $V(Xe^+, RX^-)$ potential to some V(Xe, RX*) exit channel can be quite rapid, as evidenced by the formation of Cl_2* in the reaction of Cl_2 with Ar, Kr, and Xe metastables. For the polyatomic quenchers, the density of excited and dissociative states below the energy of the rare gas metastable energy increases very rapidly with the size of the molecule. Some of these M + RX* channels are expected to cross the M^+RX^- curve providing additional exit paths that can be populated at the expense of the excimer formation channel. A second effect that may reduce the excimer yield, is the increase in the probability for non-adibatic behavior, due to the smaller electron affinities and to the more stringent symmetry requirements of the collision. In the first case, if the quencher molecule has a very small electron affinity, the crossing radius will be shifted to closer range where direct interaction of the entrance and V(M,RX*) exit potentials is possible, leading to excitation transfer. The reaction of XeF_2 with argon, kripton and xenon metastable atoms constitute an example of the cases in which excitation transfer competes with the direct reaction, as evidenced by recent product comparison experiments between metastable reactions and photolysis at the wavelength corresponding to the metastable energy. Of course, similar competition may result if the coupling potential is very small, due, for instance, to

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a vanishingly small angular or symmetry factor in H_{ic}. Thus, the reaction of K + CH_3I has been shown[60] to lead to reaction almost exclusively for the K + ICH_3 approach, and to elastic scattering for the K + CH_3I collision configuration.[60]

If RX^- is a stable negative ion with a sizeable $D^\circ_0(R\text{-}X')$, the additional degrees of freedom in a large polyatomic molecule could be very efficient in stabilizing the $M^+ + RX^-$ colision complex, thus providing an additional exit channels for $[M^+, RX^-]^*$. The reactions of $Xe(^3P_2)$ + IF_5, BrF_5 appear to be examples of reactions proceeding via the formation of a collision complex, since excimer formation occurred by extraction of the central atom (however, see below).

Polyatomic donors consistantly yield larger relative populations in the excimer B $^2\Pi_{3/2}$ state than do the diatomic halogens. As noted before, this effect must be related to the nature of the ground and low lying excited states of RX^-, and to the symmetry properties of the collision. Moreover, the states of RX^- must play a very important role not only in determining the relative populations of the excimer states, but, more importantly, the nature of these negative ion states probably determine whether the XeX excimer states can be formed at all. The quenching of the rare gas metastable states by SF_6 proceeds with large cross sections (> 140 $\AA^2$), but no evidence of reactive quenching was found. The ion SF_6^- is very stable and the process $SF_6^- \rightarrow SF_5 + F^-$ has never been observed.[61] Thus the primary exit channel may be $Ar^+SF_6^-$; this ion may then predissociate to other neutral products. In contrast, IF_5^- and BrF_5^- must be very unstable ions since all attempts to identify these species experimentally have failed, although XF_4^- and XF_6^- can be readily prepared.[62] IF_5^- is isoelectronic with XeF_5 (also an unstable molecule[62]) and therefore it is expected that the extra electron

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in IF_5^- will be localized on the iodine atom, leading to the XeI* (or XeBr) excimer formation. As a final example, NF_3 attaches electrons but NF_3^-, once formed, readily dissociates into $NF_2 + F^-$. Therefore, the properties of RX^- can be valuable in predicting the products (or lack thereof) of reactive quenching. Unfortunately, the properties of molecular negative ions are, in general, unknown. However for efficient XeX formation one needs EA $\geq$ 0.5 eV with the lowest MO being located in a strongly anti-bonding orbital.

Finally, the "direct" nature of the reactive quenching allows the measurement of R-X bond energies from the observed low pressure onset of the excimer emission. With the exception of NF_3 and N_2F_4 the bond strengths derived in this fashion agree very well with the accepted values (see Table 4.3). The $D°(N_2F_3\text{-}F)$ value reported by Lawless and Smith[63] (71 kcal/mole) is probably too high, since excimer emission extends to shorter wavelengths than predicted using this value. Golde[64] arrives at the same conclusion from his analysis of the Ar* + N_2F_4 reaction. The observed emission limit in the Xe* + NF_3 reaction suggests a bond energy larger than the accepted 57 kcal/mole.[63] However, it has been noted that since this reaction yields NF_2, the stabilization energy of this radical may not be released to the XeF* product.[65]

e. Summary. The present study represents an attempt to systematically study the reactions of $Xe(^3P_2)$ with a variety of quencher molecules. Total quenching rate constants were measured for nearly 40 different molecules and atoms and exit channels were identified for these and other molecules. As a result of this study, reactive quenching of the rare gas metastable atoms by halogen containing molecules was established as a dominant mechanism in this type of reaction. This process consitutes the first

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example of the chemical similarity between the rare gas metastable states and the corresponding alkali atoms. The spectroscopic properties of the newly discovered XeX excimers (exciplexes) were studied as well as the detailed dynamics of their formation. Several proposed mechanisms for the reactions of the metastable states were tested. In light of the results obtained in this work and previous research performed at Kansas State University, two dominant mechanisms were established. For quencher molecules of low electron affinity, reaction proceeds by direct interaction of the entrance and exit channels, with excitation transfer or dissociative excitation transfer as the dominant exit channels. The reaction of the metastable states with diatomic halogens can be best described by the "orbiting-controlled" curve crossing mechanism of Anderson[50b] and Grice and Herschbach.[51] The products of these are primarily the rare gas-halogen excimers, although, as the energy of the metastable increases other competing exit channels become important. For polyatomic quenchers, in addition to direct reactions, other mechanisms are operative, such as direct interaction of the entrance and exit channels and possible complex formation. The importance of the nature and characteristics of the molecular negative ions (in the ground and low lying excited states) was emphasized with regard to the role that these negative ions may play in determining the product distributions.

The "Golden Rule" mechanism previously accepted as a possible mechanism governing the metastable quenching reactions[10] was shown to be inoperative for Xe* reactions. Analysis of the model led to the conclusion that previous agreement between predictions of this model and the experimental data for $Ar(^3P_2)$ reactions[10] was probably fortuitous. Finally, evidence was presented showing that the long range interaction potential controlling the metastable-quencher collisions is not well represented by a simple Van der Waal's C_6/r^6 attraction.

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APPENDICES

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Table A.1 - A_α for several $k_o\ell$ and α values.

$k_o\ell$ =	α = 0.0	1.00	2.00	3.00	4.00	5.00	6.00
0.00	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.10	0.0952	0.0679	0.0431	0.0305	0.0234		
0.20	0.1813	0.1305	0.0831	0.0589	0.0452		
0.25	0.2212	0.1600	0.1021	0.0724	0.0555	0.0449	0.0377
0.30	0.2592	0.1883	0.1203	0.0853	0.0655		
0.40	0.3297	0.2415	0.1549	0.1100	0.0844		
0.50	0.3935	0.2907	0.1871	0.1330	0.1022	0.0827	0.0694
0.60	0.4512	0.3362	0.2171	0.1544	0.1187		
0.70		0.3781	0.2451	0.1745	0.1342		
0.75		0.3979	0.2583	0.1840	0.1415	0.1146	0.0962
0.80	0.5507	0.4170	0.2711	0.1932	0.1487		
0.90			0.2955	0.2108	0.1622		
1.00	0.6321	0.4861	0.3182	0.2272	0.1749	0.1417	0.1189
1.10			0.3395	0.2426	0.1869		
1.20	0.6988	0.5453	0.3594	0.2571	0.1981		
1.25			0.3689	0.2640	0.2034	0.1649	0.1384
1.30			0.3780	0.2707	0.2086		
1.40	0.7534	0.5962	0.3955	0.2834	0.2185		
1.50				0.2954	0.2279	0.1847	0.1551
1.60	0.7981	0.6400	0.4272	0.3067			
1.70				0.3174			
1.75				0.3225	0.2489	0.2019	0.1696
1.80	0.8347	0.6778	0.4553	0.3275			
2.00	0.8647	0.7050	0.4801	0.3459	0.2672	0.2169	0.1821
2.20	0.8892	0.7389	0.5022	0.3624			
2.25				0.3663	0.2832	0.2299	0.1931
2.40	0.9093	0.7636	0.5219	0.3773			
2.50				0.3841	0.2972	0.2414	0.2028
2.60	0.9257	0.7851	0.5395	0.3906			
2.75				0.3998	0.3097	0.2516	0.2114
2.80	0.9392	0.8039	0.5554	0.4028			
3.00	0.9502	0.8204	0.5697	0.4138	0.3207	0.2806	0.2191
3.20	0.9592	0.8349	0.5827				
3.25				0.4262	0.3305	0.2687	0.2259
3.40	0.9666	0.8476	0.5945				

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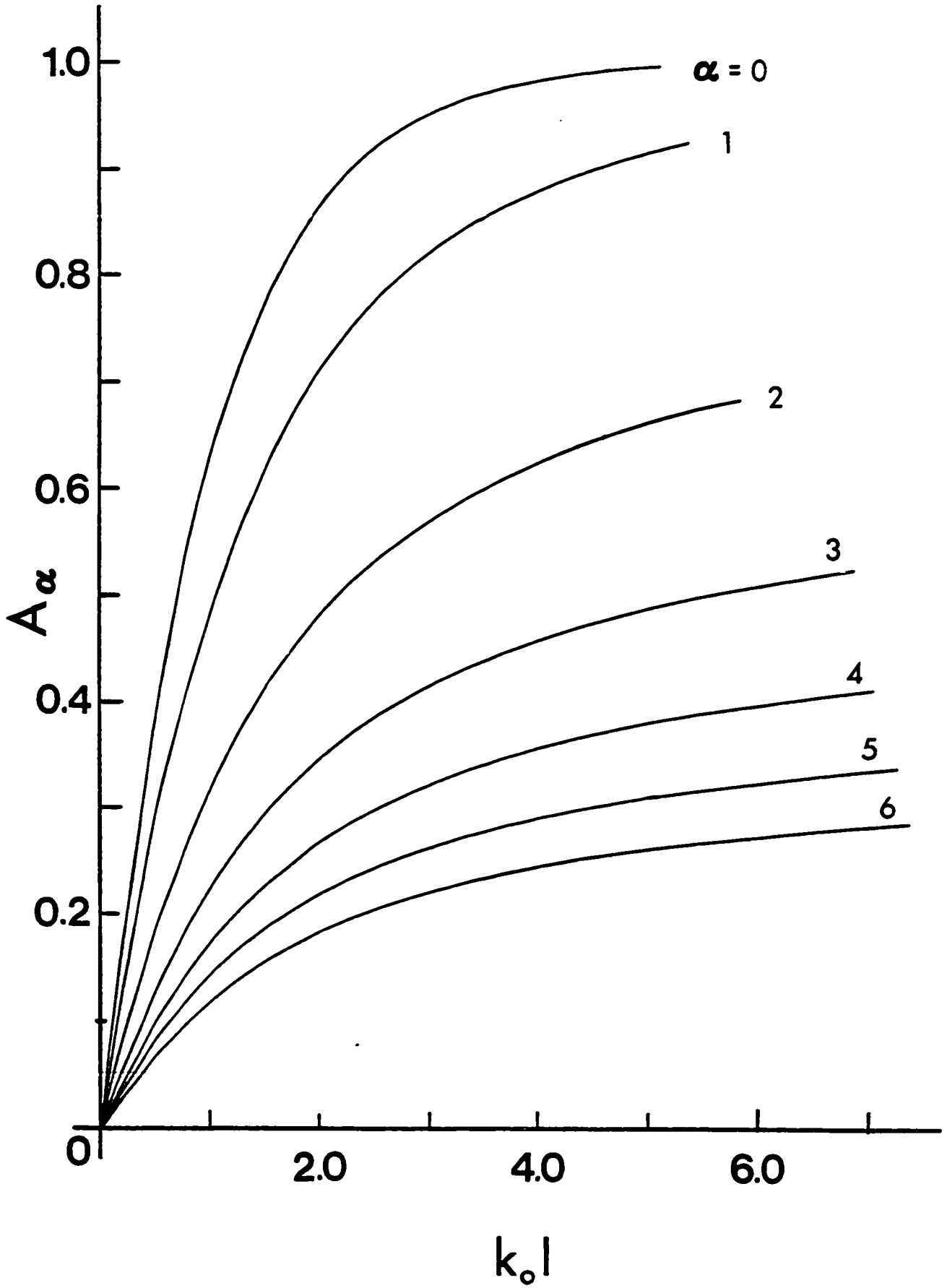
Table A.1 continued.

$k_o\ell$ =	α = 0.0	1.00	2.00	3.00	4.00	5.00	6.00
3.50				0.4373	0.3394	0.2760	0.2321
3.60	0.9727	0.8590	0.6052				
3.75				0.4473	0.3474	0.2826	0.2377
3.80	0.9777	0.8690	0.6150				
4.00	0.9817	0.8779	0.6241	0.4564	0.3547	0.2887	0.2428
4.20	0.9850	0.8859	0.6324				
4.25				0.4647	0.3614	0.2941	0.2475
4.40	0.9877	0.8930	0.6472				
4.50				0.4722	0.3675	0.2992	0.2518
4.60	0.9899	0.8994	0.6472				
4.75				0.4792	0.3731	0.3039	0.2558
4.80	0.9918	0.9052	0.638				
5.00	0.9933	0.9105	0.6600	0.4856	0.3783	0.3082	0.2595
5.25				0.4915	0.3831	0.3122	0.2629
5.50				0.4971	0.3877	0.3160	0.2661
5.75				0.5023	0.3919	0.3195	0.2691
6.00				0.5071	0.3956	0.3229	0.2719
6.25				0.5117	0.3996	0.3260	0.2746
6.50				0.5160	0.4032	0.3290	0.2771
6.75				0.5201	0.4066	0.3318	0.2796
7.00				0.5240	0.4098	0.3345	0.2819
7.25						0.3372	0.2842
7.50						0.3399	0.2865
7.75						0.3427	0.2889
8.00						0.3460	0.2917

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Figure A.1 - Calculated plots of A_α vs. $k_o\ell$ for various α values.

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Table B.1 Decay constants K(sec^{-1}), vs. quencher concentrations (10^{10} molec cc^{-3})

Xe* + N_2O		Xe* + CO		Xe* + CO_2	
K	[Q]	K	[Q]	K	[Q]
352 ± 1	28.2	325 ± 6	159.3	358 ± 1	22.1
444 ± 4	37.3	381 ± 3	296.8	467 ± 5	40.2
485 ± 9	52.6	452 ± 11	403.3	505 ± 2	61.3
575 ± 3	70.7	494 ± 7	616.9	661 ± 9	92.6
632 ± 6	88.6	568 ± 10	831.4	706 ± 4	109.8
763 ± 7	105.8	652 ± 14	1167.8		

Xe* + Cl_2		Xe* + O_2		Xe* + N_2	
K	[Q]	K	[Q]	K	[Q]
323 ± 0.6	18.3	355 ± 4	27.8	351 ± 5	245.9
365 ± 1	24.3	407 ± 13	49.3	388 ± 2	395.7
387 ± 1	31.1	453 ± 7	72.2	469 ± 5	895.6
437 ± 3	41.0	428 ± 5	64.1	492 ± 6	1158.8
505 ± 2	49.1	469 ± 9	91.5	547 ± 10	1321.2
558 ± 3	60.2	549 ± 8	123.3		
640 ± 5	66.8	581 ± 11	146.1		

Xe* + $SOCl_2$		Xe* + Cl_2		Xe* + Br_2	
K	[Q]	K	[Q]	K	[Q]
312 ± 7	6.46	312 ± 1	5.22	286 ± 4	15.0
364 ± 8	8.39	383 ± 2	14.2	330 ± 10	24.1
316 ± 18	17.4	421 ± 2	20.4	417 ± 9	36.8
453 ± 11	34.2	497 ± 5	34.9	484 ± 7	53.1
534 ± 15	48.5	578 ± 3	46.2	625 ± 8	79.9
699 ± 26	64.6				

Xe* + CF_3I		Xe* + PBr_3		Xe* + C_3H_8	
K	[Q]	K	[Q]	K	[Q]
264 ± 20	4.33	263 ± 7	0.455	308 ± 2	13.7
274 ± 8	16.5	278 ± 3	3.50	403 ± 3	31.7
375 ± 16	24.4	270 ± 3	5.35	442 ± 2	44.8
477 ± 13	38.7	289 ± 3	12.0	565 ± 4	66.6
563 ± 23	51.8	299 ± 2	17.0	621 ± 4	80.2
575 ± 3	67.8	341 ± 4	45.5		
639 ± 5	95.2				

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Table B.1 continued

Xe* + C_4H_{10}		Xe* + C_5H_{12}		Xe* + C_2H_6	
K	[Q]	K	[Q]	K	[Q]
347 ± 19	5.00	335 ± 3	15.2	300 ± 2	12.2
414 ± 2	16.2	382 ± 3	17.5	399 ± 2	26.6
473 ± 5	26.0	438 ± 5	29.6	500 ± 1	47.6
549 ± 2	38.3	525 ± 3	42.5	583 ± 3	60.2
626 ± 4	51.0	611 ± 3	57.4	610 ± 4	72.6
684 ± 29	63.1			773 ± 3	94.3

Xe* + C_2H_4		Xe* + CH_4		Xe* + SF_6	
K	[Q]	K	[Q]	K	[Q]
246 ± 3	0.000	321 ± 10	11.5	301 ± 2	21.6
369 ± 5	22.2	340 ± 2	24.7	388 ± 2	38.5
404 ± 5	27.7	367 ± 4	41.7	417 ± 1	65.3
440 ± 3	38.9	462 ± 4	63.5	511 ± 2	93.4
519 ± 5	56.3	491 ± 4	86.2	563 ± 4	124.1
611 ± 4	68.9	640 ± 8	124.5	672 ± 5	164.6

Xe* + NO		Xe* + ICl		Xe* + SeF_6	
K	[Q]	K	[Q]	K	[Q]
291 ± 3	23.4	328 ± 4	22.6	336 ± 3	8.25
335 ± 5	38.2	393 ± 4	39.5	409 ± 5	20.2
364 ± 3	53.4	426 ± 5	47.8	460 ± 4	34.5
439 ± 6	73.4	492 ± 4	55.3	550 ± 2	45.5
456 ± 4	95.1	545 ± 6	74.1	774 ± 7	85.1
544 ± 6	116.0			556 ± 8	50.4

Xe* + TeF_6		Xe* + NOCl		Xe* + NF_3	
K	[Q]	K	[Q]	K	[Q]
335 ± 15	10.2	275 ± 5	10.9	242 ± 2	19.5
400 ± 4	17.4	326 ± 3	20.2	265 ± 2	45.4
434 ± 9	25.7	362 ± 7	31.4	292 ± 2	84.2
549 ± 17	50.6	447 ± 6	50.7	328 ± 2	117.2
643 ± 28	62.4	574 ± 11	76.7	341 ± 2	156.3

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Table B.1 continued

$Xe^* + F_6$		$Xe^* + OF_2$		$Xe^* + H_2$	
K	[Q]	K	[Q]	K	[Q]
292 ± 2	9.74	336 ± 4	19.1	259 ± 20	0.00
387 ± 2	21.7	406 ± 5	31.0	294 ± 16	256.9
461 ± 3	35.0	487 ± 5	48.4	348 ± 6	601.4
619 ± 3	54.4	697 ± 5	86.0	403 ± 9	1027.5
766 ± 6	82.9	767 ± 6	108.9	458 ± 172	1541.9
				548 ± 20	2003.6

$Xe^* + C_2H_2$		$Xe^* + HBr$		$Xe^* + NOF$	
K	[Q]	K	[Q]	K	[Q]
319 ± 34	8.99	324 ± 4	22.9	380 ± 74	23.2
391 ± 8	20.4	363 ± 70	33.7	345 ± 17	54.3
458 ± 4	26.3	442 ± 3	45.3	450 ± 8	66.7
478 ± 4	36.3	524 ± 2	59.1	593 ± 21	102.0
564 ± 6	45.0	580 ± 4	76.8	687 ± 20	124.5
627 ± 8	54.2	678 ± 4	87.9		
687 ± 11	68.2				
740 ± 11	79.0				

$Xe^* + CF_3OF$	
K	[Q]
297 ± 13	4.87
318 ± 9	11.5
325 ± 37	20.7
436 ± 2	37.1
525 ± 3	55.7
636 ± 13	92.0

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VITAE

The author is the youngest of three children born to Lina Reckling and Roger Velazco. He was born in La Paz, Bolivia, on February 7, 1949, and received elementary and high school education at the Colegio La Salle of La Paz, from where he obtained the degree of Bachelor in Humanities in 1967. During the same year, he served in the Bolivian Army. In 1968, he enrolled in the Instituto de Ciencias Basicas of the Universidad Mayor de San Andres where he stayed until August, 1971, when the University was closed indefinitely by the new military government for reorganization. Forced to pursue his education elsewhere, he applied and was accepted by the Graduate School at Kansas State University to the Ph.D. program in Physical Chemistry. He started his graduate work in 1972 under the direction of Dr. D. W. Setser.

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A STUDY OF THE TOTAL AND REACTIVE ELECTRONIC QUENCHING OF THE LOWEST METASTABLE STATE OF XENON

by

JOSE ENRIQUE VELAZCO

B.S., Universidad Mayor de San Andres, 1971

AN ABSTRACT OF A DOCTOR'S DISSERTATION

submitted in partial fulfillment of the

requirements for the degree

DOCTOR OF PHILOSOPHY

Department of Chemistry

KANSAS STATE UNIVERSITY
Manhattan, Kansas

1977

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ABSTRACT

The rate constants and cross sections for electronic quenching of the lowest metastable state of xenon ($Xe(^3P_2)$) by 36 simple molecules were measured at room temperature in a fast flowing afterglow apparatus under pseudo-first order conditions. The xenon metastable atoms were produced by passing a mixture of pre-purified argon and ∿0.1% research grade xenon through a hollow cathode discharge (250V, ∿1 mA); it was established that both, electronic energy transfer from Ar* atoms and direct excitation at the discharge contributed to the production of $Xe(^3P_2)$. The decay of Xe* along the reactor was followed by Atomic Absorption Spectroscopy, and the quenching rate constants were obtained from the dependence of this decay on the quencher concentration. With the exception of H_2, N_2, CO, CF_4, CF_3H and Hg, the quenching cross sections are larger than 30×10^{-16} cm^2, and for the halogens and halogen-containing molecules the cross sections are larger than 100×10^{-16} cm^2. A linear correlation exists between the quenching cross sections and the C_6 coefficient, the coefficient of the attractive part of the Van der Walls dispersion interactions.

The distribution of product channels was established by analysis of the emissions resulting from the electronic quenching of $Xe(^3P_2)$, and specific rate constants were assigned by comparison with the absolute rate constants for excitation of Kr* emissions by energy transfer from $Ar(^3P_{0,2})$. For molecules with low (or positive) electron affinity the dominant process is excitation transfer, while for molecules with high electron affinity - halogens and halogen-containing molecules in this study - reactive quenching with the concomitant formation of a rare gas

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halogen excimer accounts for all the observed emissions, although in a few instances there is evidence for excitation transfer and/or complex formation. The branching fraction for excimer formation is unity for the halogen and interhalogen molecules, and approaches unity for some halogen-containing triatomic molecules; whereas, for larger molecules the branching fraction for excimer formation is reduced by the competition with other exit channels, particularly dissociative channels since no other emissions were observed.

Three distinct emissions were identified in the low pressure excimer spectra. "Spectrum I" ($B \rightarrow X$) involves transitions from an ionic state that correlates with the lower spin-orbit state of the rare gas ion to the ground state, whereas, "Spectrum II" ($C \rightarrow A$) terminates on other repulsive states that correlate with the ground state atoms. "Spectrum III" ($D \rightarrow X$) originates on a second ionic state that correlates with the higher spin-orbit state of Xe^+. The relative emission intensity ratio of Spectrum I to Spectrum II is donor and pressure dependent; Spectrum III is not present for all the donors used. Pressure studies on the excimers indicate that more than 40 torr of Ar are required to quench the emission and suggest a lifetime of $\sim$50 ns for the B state, with a somewhat shorter lifetime for the D state. High resolution spectra for bound-bound XeF ($B \rightarrow X$), XeF ($D \rightarrow X$) and XeCl ($B \rightarrow X$) emissions are presented.

Correlations of the cross sections with several parameters are tested and the reactive (electronic) quenching of $Xe(^3P_2)$ is discussed in terms of an electron transfer mechanism analogous to the mechanism for alkali atom-halogen reactions.

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